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AC Drives

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Revision History		
Date	Revisor	Changes
7-15-2015	M. Chapman	Added Opto-Coupler to Generic AC Drive Digital Input Picture
7-15-2015	M. Chapman	Corrected wording on Constant Torque explanation Decrease not increase
7-15-2015	M. Chapman	Added Opto Coupler to Generic Drive Digital Input section
7-15-2015	M. Chapman	Added Ramp to Hold to Stop Modes Section
7-16-2015	M. Chapman	Enhanced the Volts per Hertz and Vector Control topics
7-30-2015	M. Chapman	Rewrote the Sink Source section
8-1-2017	M. Chapman	Revised entire book for Tablets

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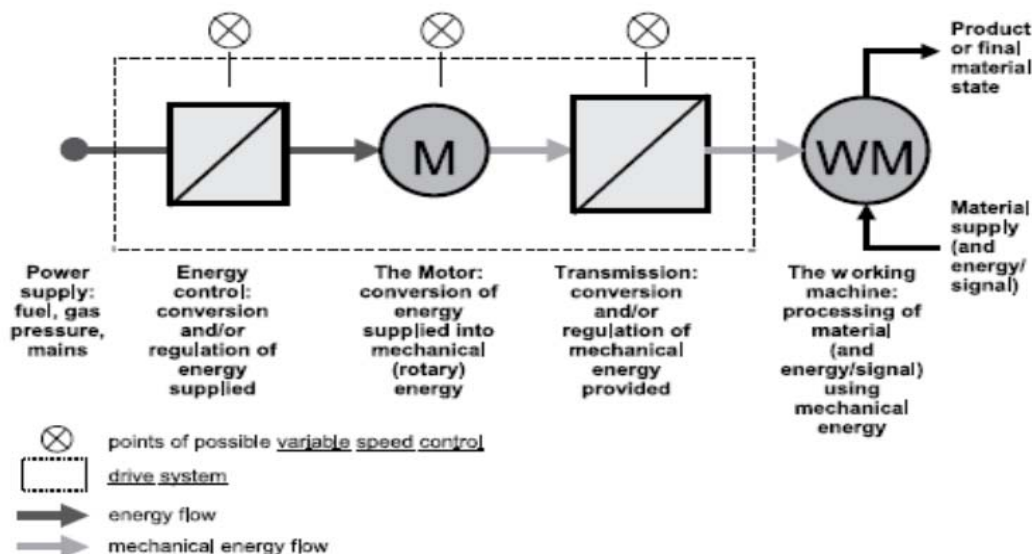
Introduction

To understand why variable speed control is necessary, we first need to understand the requirements of different processes. These processes can be divided into two main categories; material treatment and material transport, although there are many different sub-categories that come under these two basic headings. Common to both main categories, however, is the need to be able to adjust the process.

This is accomplished with Variable Frequency Drives (VFDs). For example, in HVAC applications, air flow requirements change according to the humidity and temperature in the room. These can be met by adjusting the supply and return air fans. These adjustments are carried out with VFDs. Fans are also used in power plants and the chemical industry. In both cases, the fans need to be adjusted according to the main process. In power plants, the main process changes due to varying demands for power at different times of the year, day or week. Likewise, the need for VFDs differs according to the process.

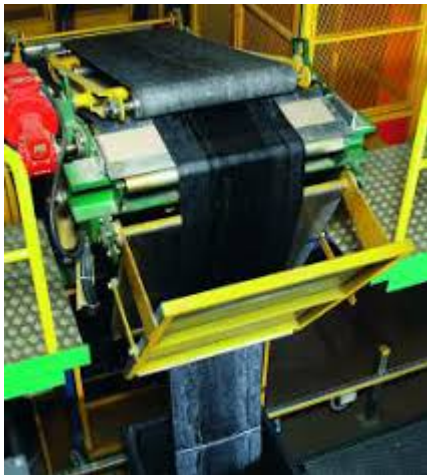
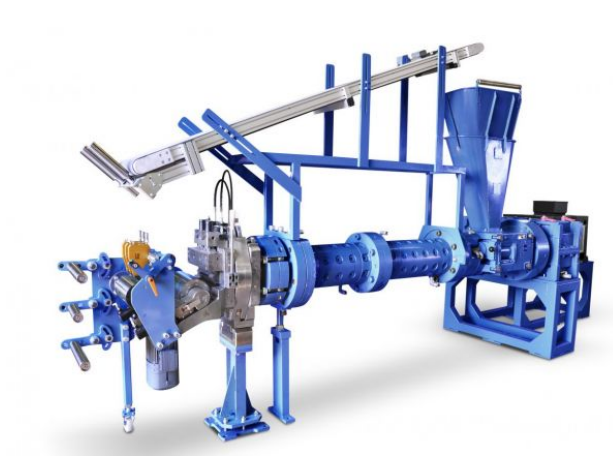
Every machine consists of four different components, shown in the diagram. These components are energy control, the motor, transmission and the working machine. Together, the first three components comprise the so called “**drive system**”. This drive system can transform a given type of energy, usually electrical, into mechanical energy, which is then used by the working machine.

Energy is supplied to the drive system from the power supply. In each of the three drive system components, variable speed control is possible. Variable speed control can be accomplished, for example, using a frequency converter as the energy control component, a Squirrel Cage motor as the motor component and a gearbox as the transmission component.



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Applications where VFDs are used

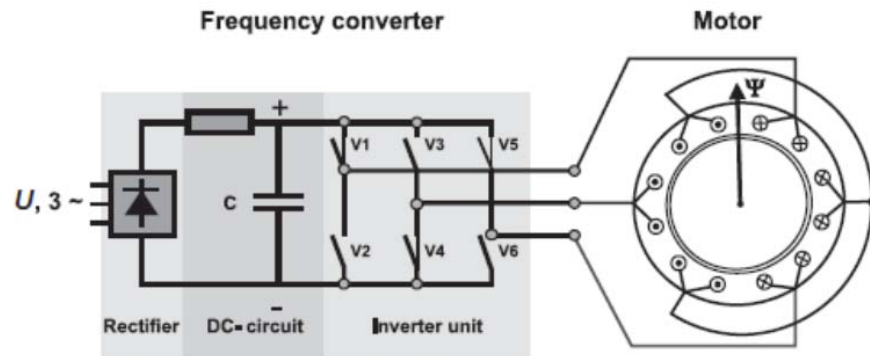


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As mentioned earlier, most machines are driven by an electric motor. Electric motors can be divided into AC and DC motors. AC motors, particularly squirrel cage motors, are the most common motors in industrial processes.



An AC motor's ability to convert electrical energy into mechanical energy is based on electromagnetic induction. The voltage in stator windings forms the current and magnetic flux. The direction of this flux can be determined using the right hand rule from the stator current. By changing the direction of the voltage in stator windings, the direction of the flux can also be changed. Changing the voltage direction in the three phase motor windings in the correct order causes the magnetic flux of the motor starts to rotate. The motor's rotor will then follow this flux with a certain slip. This is the basic principle used to control AC motors.

This control can be achieved using a frequency converter. As the name suggests, a frequency converter changes the frequency of the alternating current and voltage. A frequency converter consists of three parts. Regular 60 Hz 3-phase current is fed in to the rectifier (Converter) part, which converts it to direct current. The DC voltage is fed into the DC bus circuit, which filters the pulsating voltage.

The inverter unit then connects each motor phase either to the negative or the positive DC bus according to a certain order. To receive the flux direction shown in the diagram, switches V1, V4 and V5 should be closed. To make the flux rotate counterclockwise, switch V6 has to be closed but V5 has to be open. If switch V5 is not opened, the circuit will short circuit. The flux has turned 60° counterclockwise.

Variable Frequency Drive applications will typically have one or more of the following characteristics:

- Constant torque below base speed
- Constant horsepower above base speed
- High starting loads or overloads
- High dynamic performance
- A process (driven machine) that cannot be started or run without a variable speed control

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AC Motor Review

AC Motor Construction: Three-phase AC induction motors are commonly used in Michelin applications. This type of motor has three main parts, **rotor**, **stator**, and **enclosure**. The stator and rotor do the work, and the enclosure protects the stator and rotor.

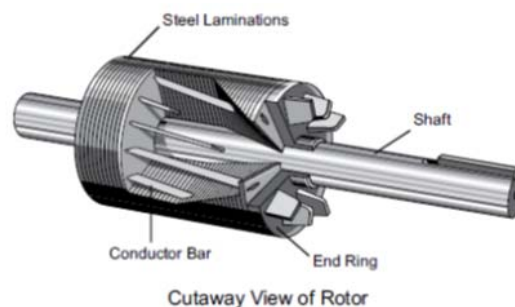
Stator Windings: Stator laminations are stacked together forming a hollow cylinder. Coils of insulated wire are inserted into slots of the stator core.



When the assembled motor is in operation, the stator windings are connected directly to the power source. Each grouping of coils, together with the steel core it surrounds, becomes an electromagnet when current is applied. **Electromagnetism** is the basic principle behind motor operation.



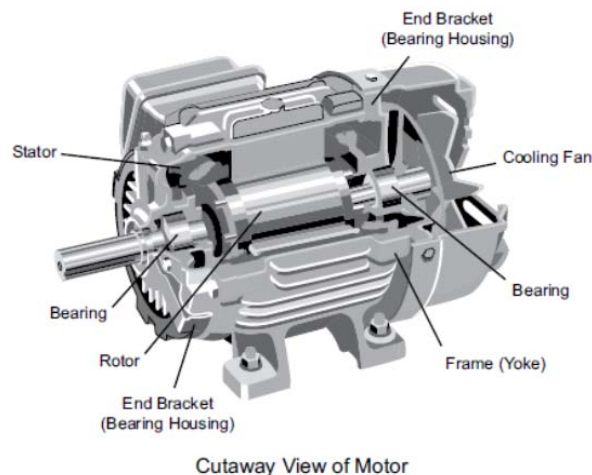
Rotor Construction: The rotor is the rotating part of the motor's electromagnetic circuit. The most common type of rotor used in a three-phase induction motor is a **squirrel cage rotor**. A squirrel cage rotor core is made by stacking thin steel laminations to form a cylinder.



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Enclosure: The enclosure consists of a **frame** and two end brackets (or bearing housings). The stator is mounted inside the frame. The rotor fits inside the stator with a slight air gap separating it from the stator. There is no direct physical connection between the rotor and the stator.

The enclosure protects the internal parts of the motor from water and other environmental elements. The degree of protection depends upon the type of enclosure. Enclosure types are discussed later in this course. **Bearings**, mounted on the shaft, support the rotor and allow it to turn. Some motors, like the one shown in the following illustration, use a **fan**, also mounted on the rotor shaft, to cool the motor when the shaft is rotating.



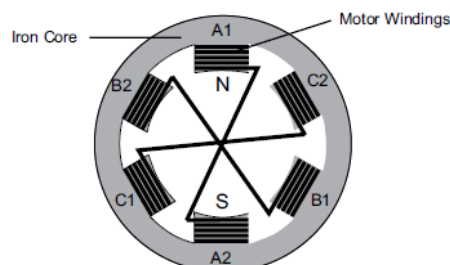
Theory of Operation

The principles of electromagnetism explain the shaft rotation of an AC motor. Recall that the stator of an AC motor is a hollow cylinder in which coils of insulated wire are inserted.

Stator Coil Arrangement

The following diagram shows the electrical configuration of stator windings. In this example, six windings are used, two for each of the three phases. The coils are wound around the soft iron core material of the stator. When current is applied, each winding becomes an electromagnet, with the two windings for each phase operating as the opposite ends of one magnet.

In other words, the coils for each phase are wound in such a way that, when current is flowing, one winding is a north pole and the other is a south pole. For example, when A1 is a north pole, A2 is a south pole and, when current reverses direction, the polarities of the windings also reverse.

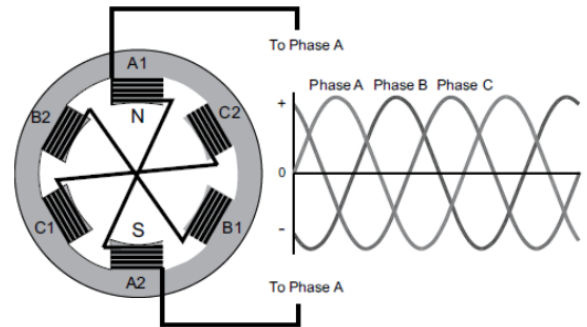


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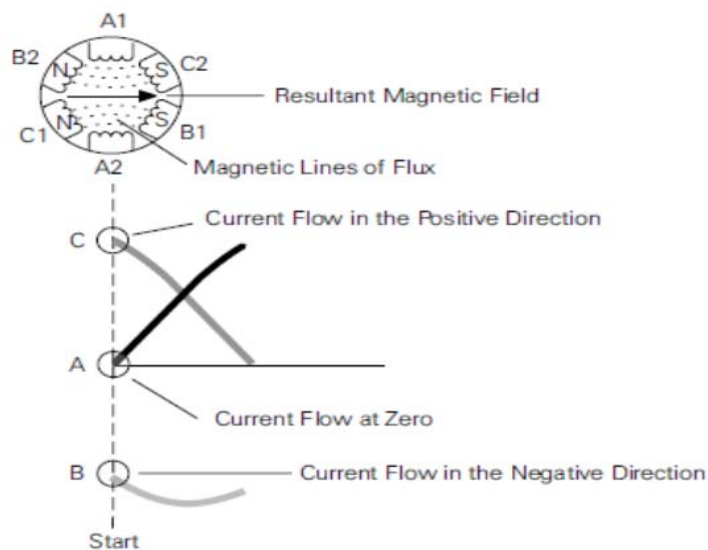
Stator Power Source

The stator is connected to a three-phase AC power source. The following illustration shows windings A1 and A2 connected to phase A of the power supply. When the connections are completed, B1 and B2 will be connected to phase B, and C1 and C2 will be connected to phase C.

As the following illustration shows, coils A1, B1, and C1 are 120° apart. Note that windings A2, B2, and C2 also are 120° apart. This corresponds to the 120° separation between each electrical phase. Because each phase winding has two poles, this is called a two-pole stator. When AC voltage is applied to the stator, the magnetic field developed in a set of phase coils depends on the direction of current flow.



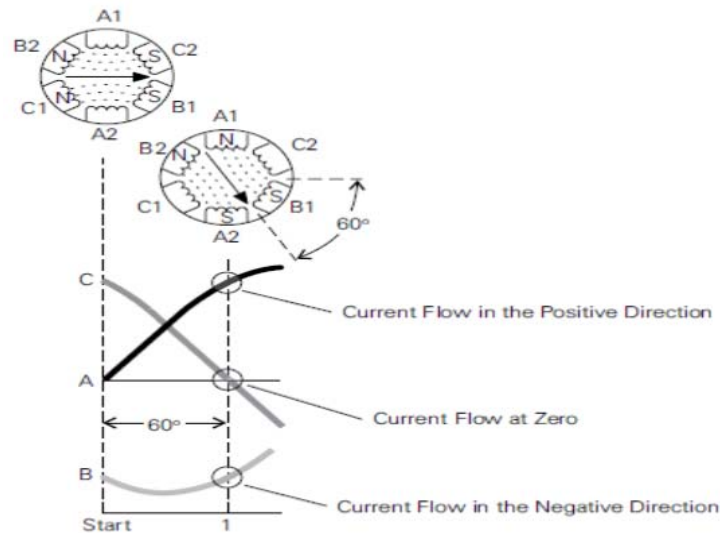
Start In the following illustration, a start time has been selected during which phase A has no current flow and its associated coils have no magnetic field. Phase B has current flow in the negative direction and phase C has current flow in the positive direction. Based on the previous chart, B1 and C2 are south poles and B2 and C1 are north poles. Magnetic lines of flux leave the B2 North Pole and enter the nearest South Pole, C2. Magnetic lines of flux also leave the C1 North Pole and enter the nearest South Pole, B1. The vector sum of the magnetic fields is indicated by the arrow.



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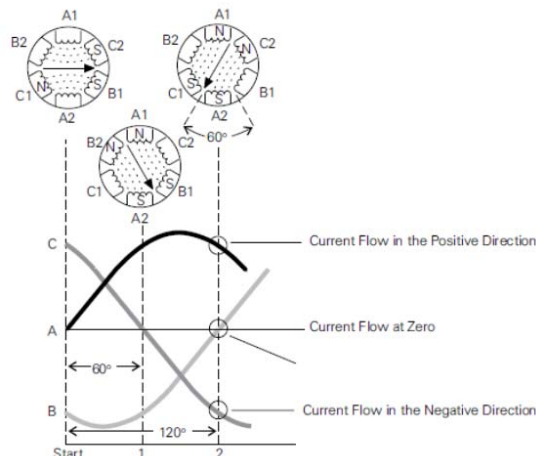
Time 1

The following chart shows the progress of the magnetic field vector as each phase has advanced 60°. Note that at time 1 phase C has no current flow and no magnetic field is developed in C1 and C2. Phase A has current flow in the positive direction and phase B has current flow in the negative direction. As the previous chart shows, windings A1 and B2 are north poles and windings A2 and B1 are south poles. The resultant magnetic field vector has rotated 60° in the clockwise direction.



Time 2

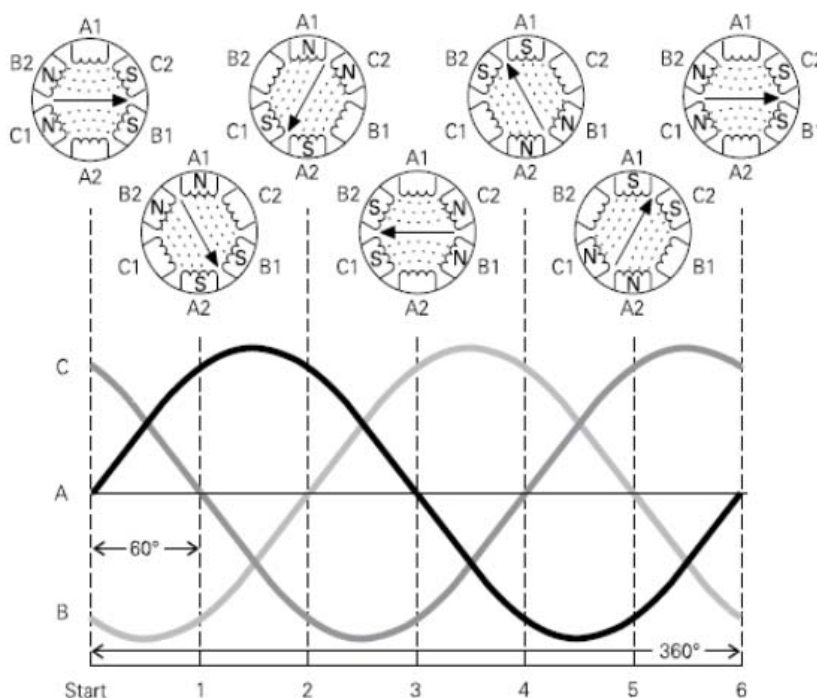
At time 2, phase B has no current flow and windings B1 and B2 have no magnetic field. Current in phase A is flowing in the positive direction, but phase C current is now flowing in the negative direction. The resultant magnetic field vector has rotated another 60°.



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360° Rotation

At the end of six such time intervals, the magnetic field will have rotated one full revolution or 360°. This process repeats 60 times a second for a 60 Hz power source.



Synchronous Speed

The speed of the rotating magnetic field is referred to as the synchronous speed (N_s) of the motor. Synchronous speed is equal to 120 times the frequency (F), divided by the number of motor poles (P). The synchronous speed for a two-pole motor operated at 60 Hz, for example, is 3600 RPM.

$$N_s = \frac{120F}{P} \Rightarrow N_s = \frac{120 \times 60}{2} \Rightarrow N_s = 3600 \text{ RPM}$$

Synchronous speed decreases as the number of poles increases. The following table shows the synchronous speed at 60 Hz for several different pole numbers.

No. of Poles	Synchronous Speed
2	3600
4	1800
6	1200
8	900
10	720

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Slip

For a three-phase AC induction motor, the rotating magnetic field must rotate faster than the rotor to induce current in the rotor. When power is first applied to the motor with the rotor stopped, this difference in speed is at its maximum and a large amount of current is induced in the rotor. After the motor has been running long enough to get up to operating speed, the difference between the synchronous speed of the rotating magnetic field and the rotor speed is much smaller. This speed difference is called slip. Slip is necessary to produce torque. Slip is also dependent on load. An increase in load causes the rotor to slow down, increasing slip. A decrease in load causes the rotor to speed up, decreasing slip. Slip is expressed as a percentage and can be calculated using the following formula.

For example, a four-pole motor operated at 60 Hz has a synchronous speed (N_s) of 1800 RPM. If its rotor speed (N_r) at full load is 1775 RPM, then its full load slip is 1.4%.

$$\begin{aligned}\% \text{ Slip} &= \frac{N_s - N_r}{N_s} \times 100 \\ \% \text{ Slip} &= \frac{1800 - 1775}{1800} \times 100 \\ \% \text{ Slip} &= 1.4\%\end{aligned}$$

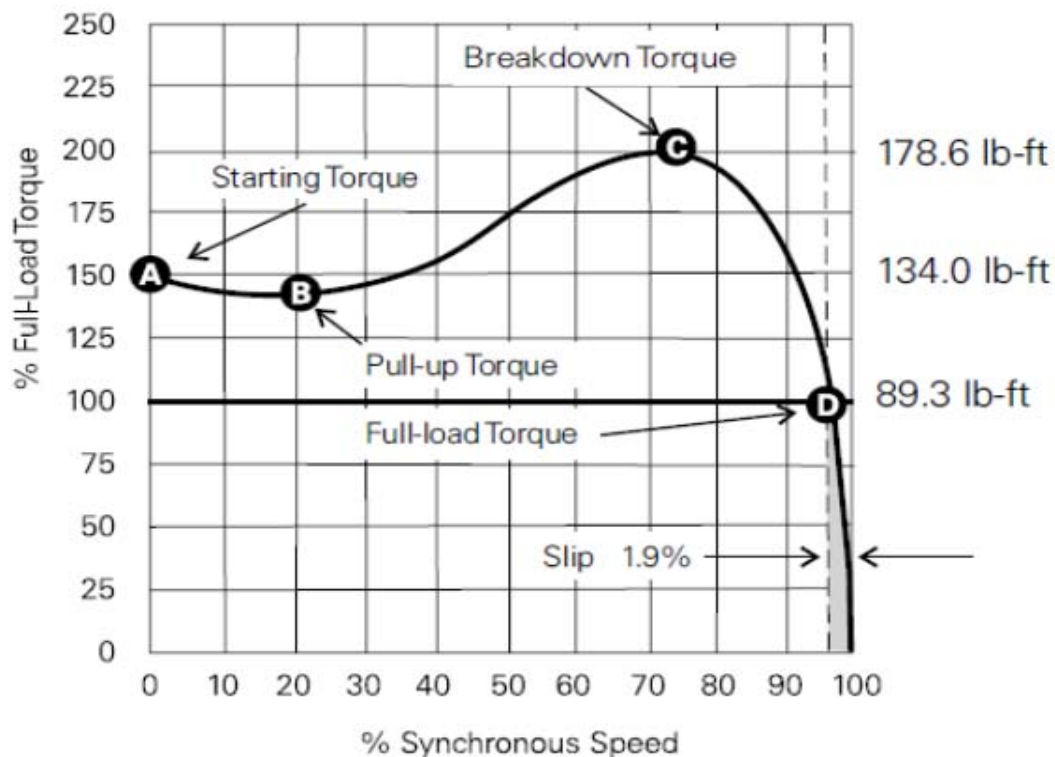
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Starting Torque: Starting torque, also referred to as locked rotor torque, is the torque that the motor develops each time it is started at rated voltage and frequency. When voltage is initially applied to the motor's stator, there is an instant before the rotor turns.

Pull-up Torque: As the motor picks up speed, torque decreases slightly until point B on the graph is reached.

Breakdown Torque: As speed continues to increase from point B to point C, torque increases up to a maximum value at approximately 200% of full-load torque.

Full-Load Torque: Torque decreases rapidly as speed increases beyond breakdown torque until it reaches full-load torque at a speed slightly less than 100% of synchronous speed. Full load torque is developed with the motor operating at rated voltage, frequency, and load. The motor in this example has a synchronous speed of 18 00 RPM and a full-load speed of 1765 RPM. Therefore, its slip is 1.9%.

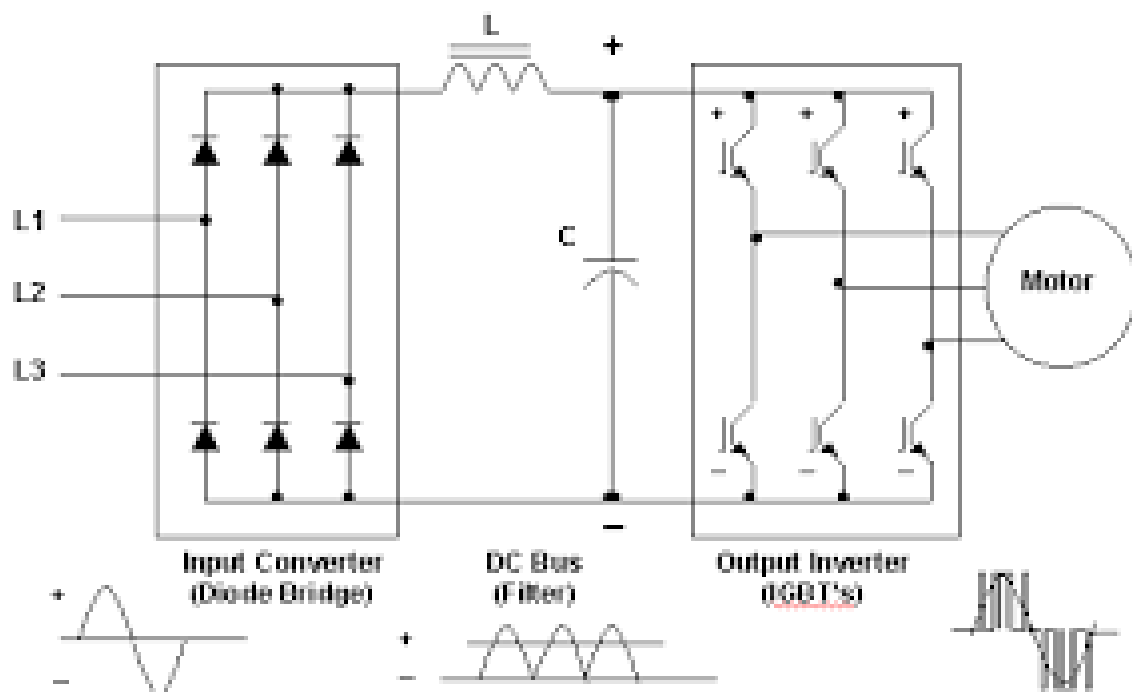


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AC Drive Controllers

“An AC Drive controller is an electronic device that can control the speed, torque, horsepower and/or direction of an AC motor.”

This control can be achieved using an AC Drive Controller (Frequency Converter). As the name suggests, a frequency converter changes the frequency of the alternating current and the 3-phase voltage to the motor. An AC Drive controller consists of three main sections and a microprocessor monitoring or controlling each part.



1. A 60 Hz 3-phase voltage is fed in to the rectifier (Converter) part, which converts it to a pulsating direct current (DC) voltage.
2. The DC voltage is fed into the DC bus circuit, which filters the pulsating voltage.
3. The inverter unit then connects each motor phase either to the negative or the positive DC bus according to a certain order. To make the flux rotate the microprocessor controls the firing of the transistors in a precise order based on the speed reference provided.

The result is a variable 3-phase voltage and a variable frequency output from the controller to the motor.

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AC Drive controllers will control the following characteristics when connected to a Squirrel Cage AC Motor:

- ***Constant Torque***
- ***Constant Horsepower***

Both will be discussed in detail later in this section.

There are two types of AC Drive controllers considered in this class:

- Variable Voltage / Variable Frequency Drives (***Volts/Hertz Drive***)
- Vector Control Drives (***Sensorless Vector Drive***)

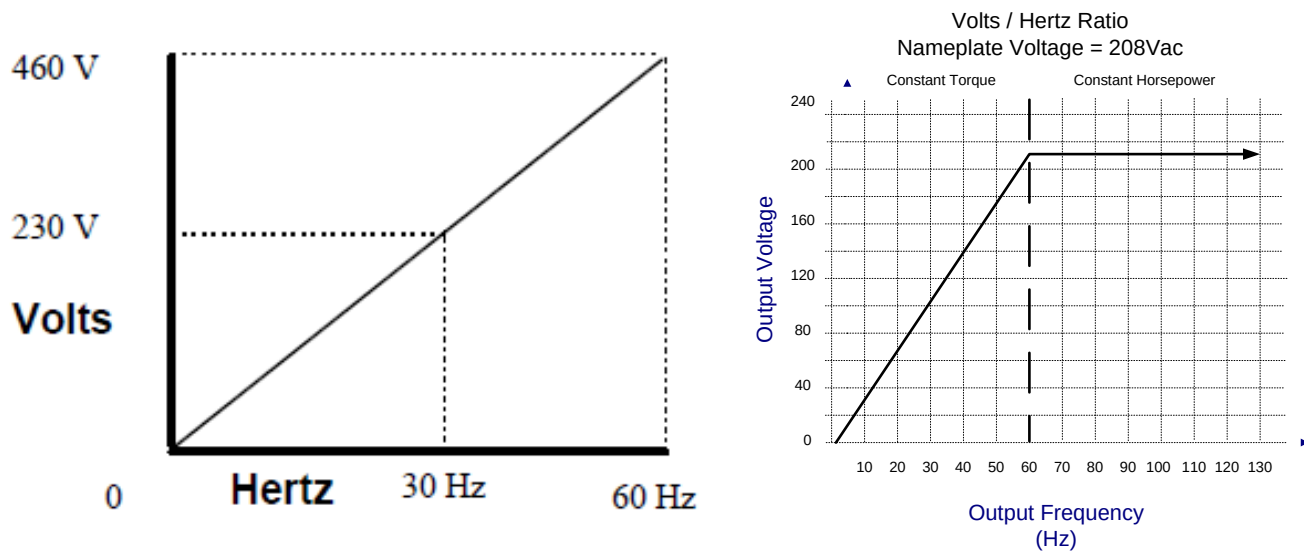
Both types perform similar functions (ie. variable voltage, variable frequency, etc...). Both drives respond to changing loads and may operate in “Open Loop” applications. “Open Loop” means that there is no feedback as to the direction or speed of the motor to the controller.

Many processes may be operated in “Open Loop”, if there are no negative effects from speed or torque variations based on the changing loads to the motor. In other words the uniformity of the product is not affected by these changes.

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Volts/Hertz Drive

When a Three Phase voltage is applied to the stator of a Squirrel Cage Induction Motor, the current generates magnetic flux and a rotating field and torque is produced. This magnetic flux must remain constant in order to produce full-load torque. This is most important when running a motor at less than full speed. Because AC drives are used to provide slower running speeds, there must be a means of maintaining a constant magnetic flux in the motor. This method of magnetic flux control is called the volts-per-hertz ratio. With this method, the frequency and voltage must increase in the same proportion to maintain good torque production at the motor.



For example, if the motor rated frequency is 60 Hz and the motor rated voltage is 460 V, then the volts per Hertz ratio (460 divided by 60) would be 7.6 V/Hz. So, at half speed on this system, the output frequency would be 30 Hertz and the voltage applied to the motor would be 230 V and the ratio would still be maintained at 7.6 V/Hz.

Although, this ratio pattern saves energy going to the motor, it is also very critical to performance. The Volts/Hertz drive maintains this ratio because as the motor speed changes, motor current can become unstable and torque can diminish. On the other hand, excessive current could damage or destroy the motor.

Whenever the drive is operating at a frequency which is less than motor nameplate frequency (typically 60 Hz), the drive is maintaining a **Constant Torque**. The Volts/Hz ratio is what allows this to happen. This ratio is trying to maintain a constant stator current, thus, maintaining a constant Torque.

Example: The drive is running a motor at 52 Hz. The output voltage should measure approximately 395 Volts ($V/Hz * \text{Output Frequency}$; $7.6 * 52 = 395$).

“The Volts/Hertz Ratio is the basis of control with the Volts/Hertz AC Drive Controller”

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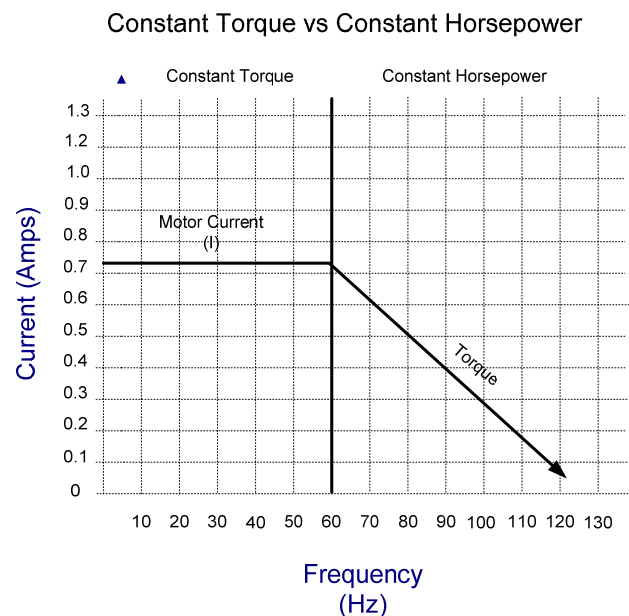
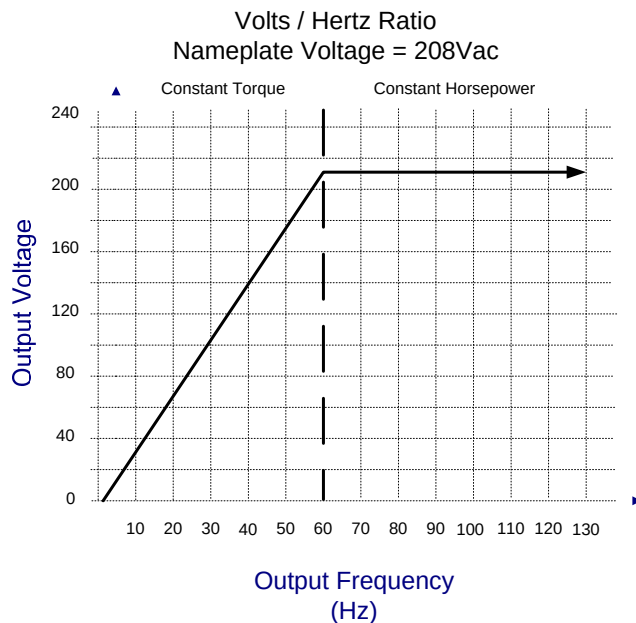
Another way to consider this is analyzing the following formulas used in AC Theory:

$$I = V / Z; \quad Z = \sqrt{r^2 + X_L^2}; \quad X_L = 2 \pi f L$$

As drive output frequency **decreases**, X_L decreases, Z decreases and I would increase **IF** Voltage stays the same. Decreasing V proportionally with the frequency causes I to remain constant. This is the essence of how the Volts/Hertz ratio operates. The voltage output to the motor is varied, along with the frequency; up to Motor Nameplate Frequency. After reaching Motor Nameplate Frequency (60 Hz), the drive will maintain a constant output voltage.

If we relate the current (I) in the motor to Torque, then we have a constant torque in theory between 0 and 60 Hz.

$$\text{Motor Horsepower} = \text{Torque} * \text{rpm}$$



On the other hand, as **frequency increases**, X_L increases ; Z increases, and I will **decrease** because V stays the same. Once again, this is the essence of how the Volts/Hertz ratio operates.

If we relate the current (I) in the motor to Torque, then we have a constant horsepower, in theory, between 60 and 120 Hz.

In summary, a standard Volts per Hertz drive outputs a Voltage and Frequency pattern designed to maintain a constant V/Hz pattern to the motor under ideal conditions. How the motor reacts to this pattern is fully dependent upon the load conditions. This drive only tells the motor what to do. If for example it provides 43Hz to the motor, and the motor spins at a speed equivalent to 40Hz, the V/Hz Drive doesn't know. You can't regulate true torque control with a V/Hz drive because the drive has no way of knowing what the motor output torque is.

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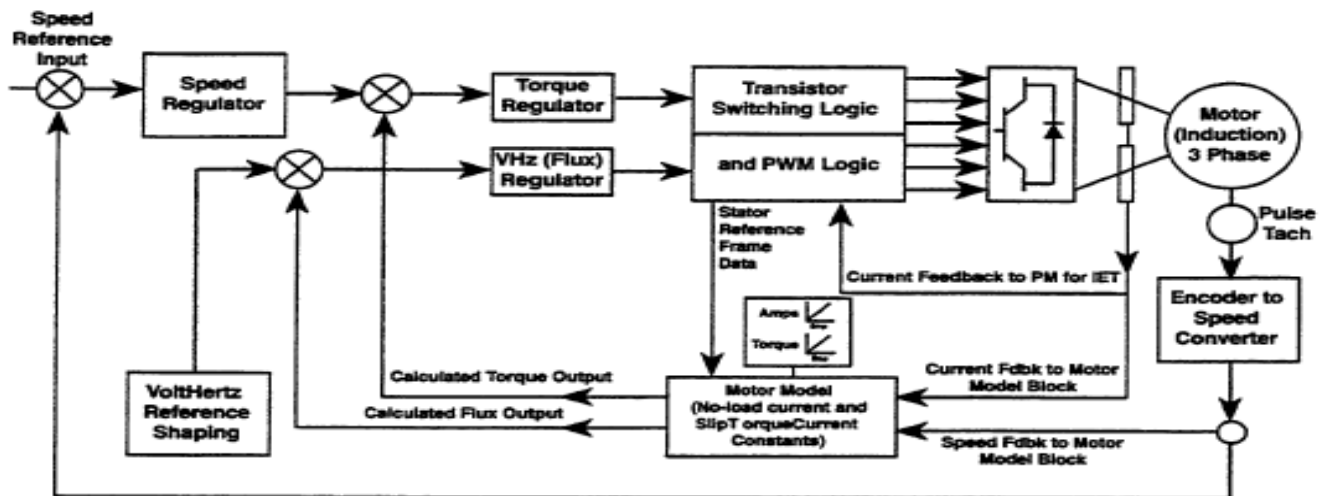
Vector Control Drives

The Vector Control Drive offers the same abilities as the Volts/Hertz Drives and:

1. The ability to control both the magnitude and direction of the magnetic flux of an AC motor through independent control loops. This allows the drive to handle abrupt changes in load, speed or direction.
2. The ability to operate in “Closed Loop” using an encoder as a feedback device providing extremely precise speed/torque regulation and positioning of the motor.
3. A high dynamic response time.
4. Smooth low speed operation, even under changing loads.
5. Linear torque control for positioning or tensioning.

What is a Vector Control Drive and how does it get this mysterious name? A Vector Control Drive (VCD) borrows its name from the fact that a vector (from math class) has both a magnitude and direction. The VCD uses a more sophisticated regulator to control both the magnitude and direction of the magnetic flux in an AC motor through independent control loops.

The diagram shown below illustrates the regulator section of a Vector Control Drive.



Two separate control loops have been added making it more complex. The control loops allow independent control of the speed and strength of the motor’s magnetic field. They may also allow the regulator to measure the actual speed of the motor (Encoder Feedback) and to calculate the amount of torque being produced.

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The “**Motor Model Block**” Secret:

The “Motor Model Block” shown below is the secret of the Vector Control Drive. It uses a microprocessor to compute torque calculations very quickly.

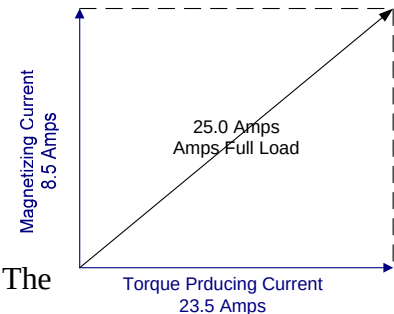
Slip, Current and Torque are related			
Load (%)	Slip (rpm)	Current (Amps)	Torque (Ft-Lbs)
0	0	8.5	0
25	11	10.3	15
50	21	14.1	30
75	33	19.1	45
100	47	25	60
Slip = Stator Speed – Rotor Speed			

The Vector Control Drive makes use of a very important principle of AC Induction Motor: Torque, Current and Slip are related. Looking at the table above, it compares the slip, current and torque versus load for an AC motor. It is easy to see that an AC motor will slow slightly and draw more current as the **load increases**.

Torque and Slip are directly related. An increase in torque requires a proportional increase in slip. However, this change is NOT linear. Fortunately, the change in current varies in a predictable way.

The total current drawn by a motor has two components. The first is no-load or magnetizing current. This portion is simply reactive current that does not represent mechanical power. The second is torque producing current that is electrically in-phase with the applied voltage to the motor. This portion delivers power to the motor and varies directly with the output torque.

The figure at right helps explain this concept. It shows a “current diagram”. A magnetizing (flux) current of 8.5 amps is represented by the Y-axis and a torque-producing current of 23.5 Amps is represented by the X-axis. The actual full load current is the vectored sum of the Magnetizing current and the Torque-producing current or 25 Amps.



The Vector drive calculates Torque producing current by knowing the actual amps and the magnetizing current

Magnetizing Current = No Load Current

The VCD can measure the magnetizing current and the total current. The regulator then must solve the following equation to determine the Torque-Producing current which is the last part of the “Motor Model Block” puzzle.

$$I_{\text{TORQUE}} = \sqrt{I_{\text{MEASURED}}^2 - I_{\text{MAGNETIZING}}^2}$$

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Vector Control Drive Summary

Open Loop is actually a misnomer because it is actually a closed loop system, but the feedback loop comes from within the Vector Control Drive itself instead of an external encoder. For this reason, they are also referred to as "Sensorless Vector" drives. The microprocessor creates a mathematical "model" of the motor operating parameters and keeps it in memory. As the motor operates, the microprocessor monitors the output current, compares it to the model and determines what the different current effects mean in terms of the motor performance. Then the microprocessor executes the necessary error corrections just as the closed Loop Vector Drive does. The only drawback is that as the motor gets slower, the ability of the microprocessor to detect the subtle changes in magnetics becomes more difficult. At zero speed it is generally accepted that an Open loop Vector Drive is not reliable enough to use on cranes and hoists. For most other applications though it is just fine.

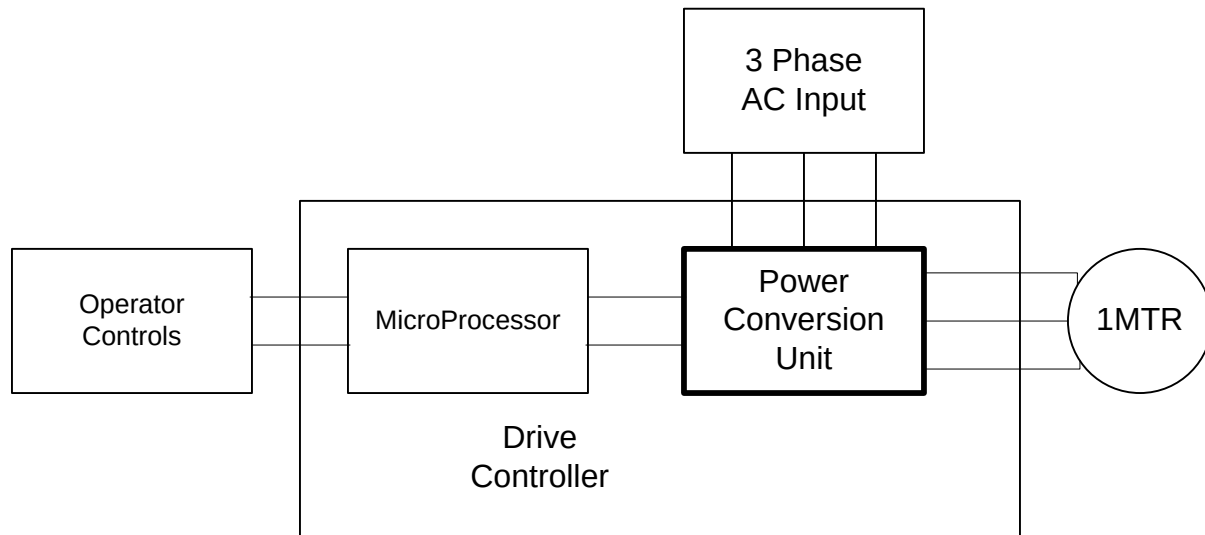
A true Closed Loop Vector Drive uses a shaft encoder on the motor to give positive shaft position indication back to the microprocessor. The microprocessor can then alter the output voltage and frequency on the fly to make up for any error. For torque control, the feedback allows the microprocessor to adjust the output pattern so that a constant level of torque can be maintained regardless of speed, i.e. a winder application where diameters are constantly changing. If the shaft moves one way or the other too much or too fast or slow, the torque requirement is wrong and the error is corrected. A true closed Loop Vector Drive can also make an AC motor develop continuous full torque at zero speed, something that previously only DC drives were capable of. That makes them suitable for crane and hoist applications where the motor must produce full torque before the brake is released or else the load begins dropping and it can't be stopped. Closed Loop is also so close to being a servo drive that some people use them as such. The shaft encoder can be used to provide precise travel feedback by counting pulses.

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Simplified AC Drive System

An AC adjustable frequency drive system typically consists of three basic blocks:

- **Operator controls**
- **AC Drive Controller**
- **AC motor**



AC Adjustable Frequency Drive System

The “**operator controls**” allow the operator to start, stop, and change direction and speed of the controller by simply turning potentiometers or other operator devices. These controls may be an integral part of the controller or may be remotely mounted or determined by PLC I/O or through network communications.

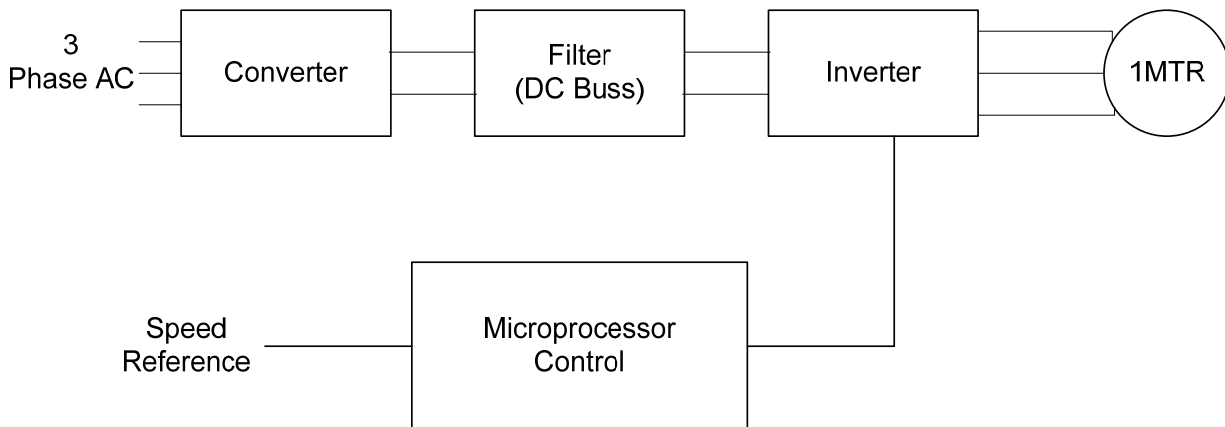
The **Drive Controller** converts a fixed AC input voltage to an AC waveform with adjustable frequency and voltage for its output. It consists of a Microprocessor and a Power Conversion Unit as seen in the figure above.

The Microprocessor oversees the operation of the drive and provides valuable system diagnostic information and communication capabilities.

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AC Drive Block Diagram

The figure below shows a block diagram of the **Power Conversion Unit** with the Microprocessor. These blocks are present in every **Manufacturer's** controller (Volts/Hertz and Vector Control Drives).



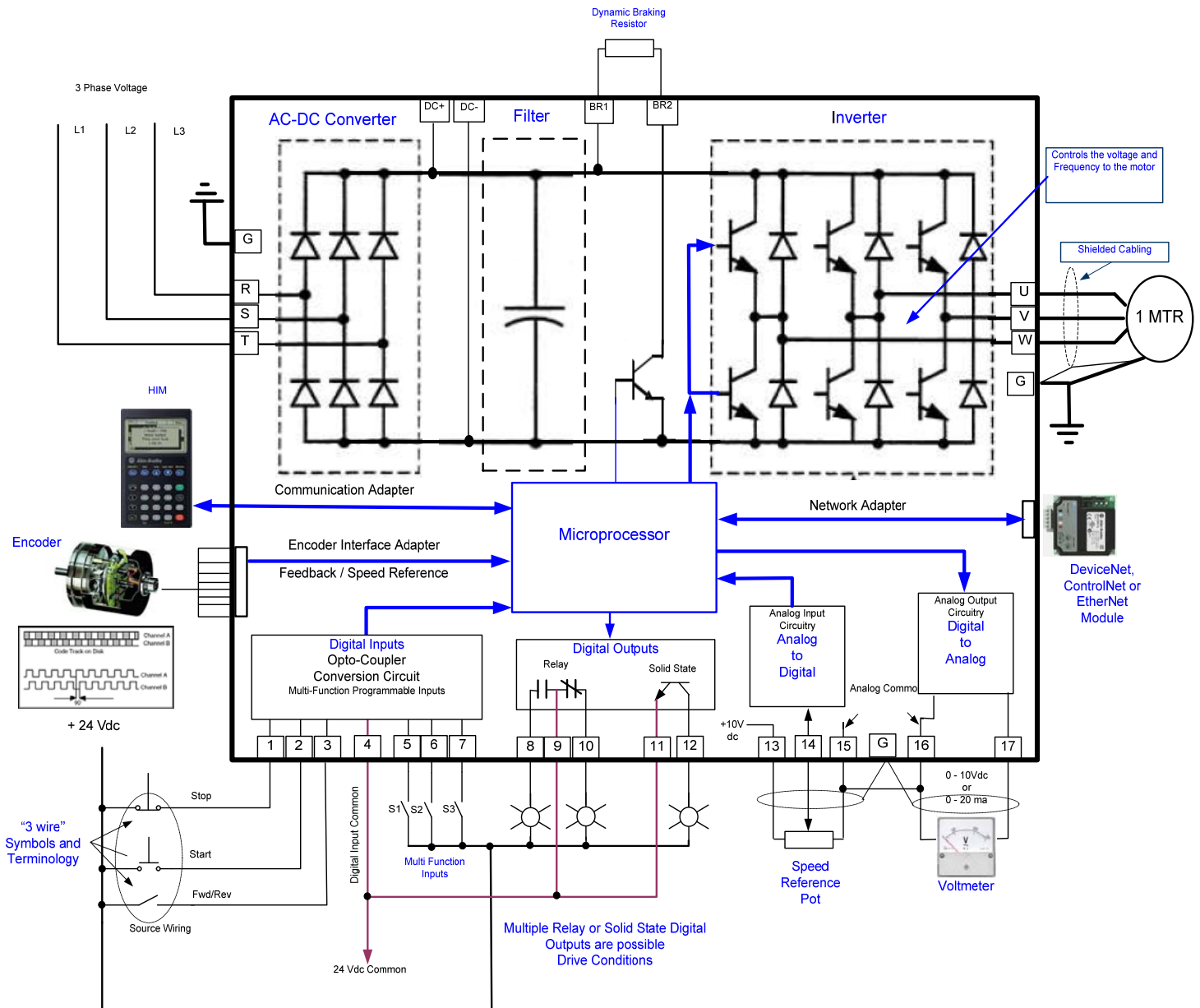
Block Diagram
Power Conversion Unit

The Power Conversion Unit has three major parts and performs these functions:

1. The **Converter** block rectifies the incoming fixed AC voltage (changes AC to DC).
2. The **Filter** block filters the DC voltage and creates a DC buss voltage approximately 1.35 times the line to line value of the AC supply voltage (650VDC for 480VAC).
3. The **Inverter** block then produces an AC current by creating the voltage output with the desired frequency to control the motor speed and torque.

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Generic AC Drive System



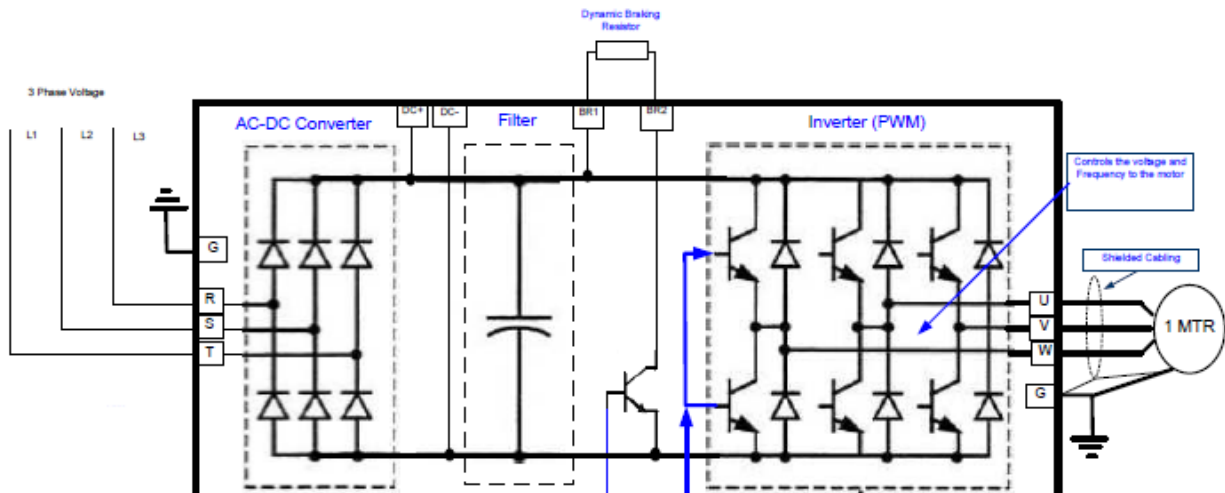
The generic drive system shown above illustrates all of the common components of any manufacturer's AC Drive controller. The terminology and parameters will vary between different manufacturers.

The Converter, Filter and Inverter sections are illustrated.

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Interface Terminals and Circuits

The next step in understanding an AC Drive system starts with the interface circuits.



Power Terminals

The **R, S, and T terminals** are usually connected to three phase but can be single phase. If connected to a single phase source, the drive capacity is generally derated about 65%.

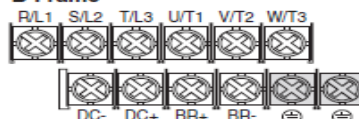
The **Motor connections** can be connected in a WYE or DELTA configuration. The motor connects to the U,V and W terminals. The motor may be wired as Delta or Wye.

The Braking Resistor is used to clip the excess voltage off the DC Buss when **Regeneration** (Deceleration) occurs. The external resistor connects between BR- and BR+ terminals.

Ground terminals connect Earth Ground to the drive and to the motor, along with all shields. Note, the motor cable **MUST** be shielded and connected to ground on both ends.

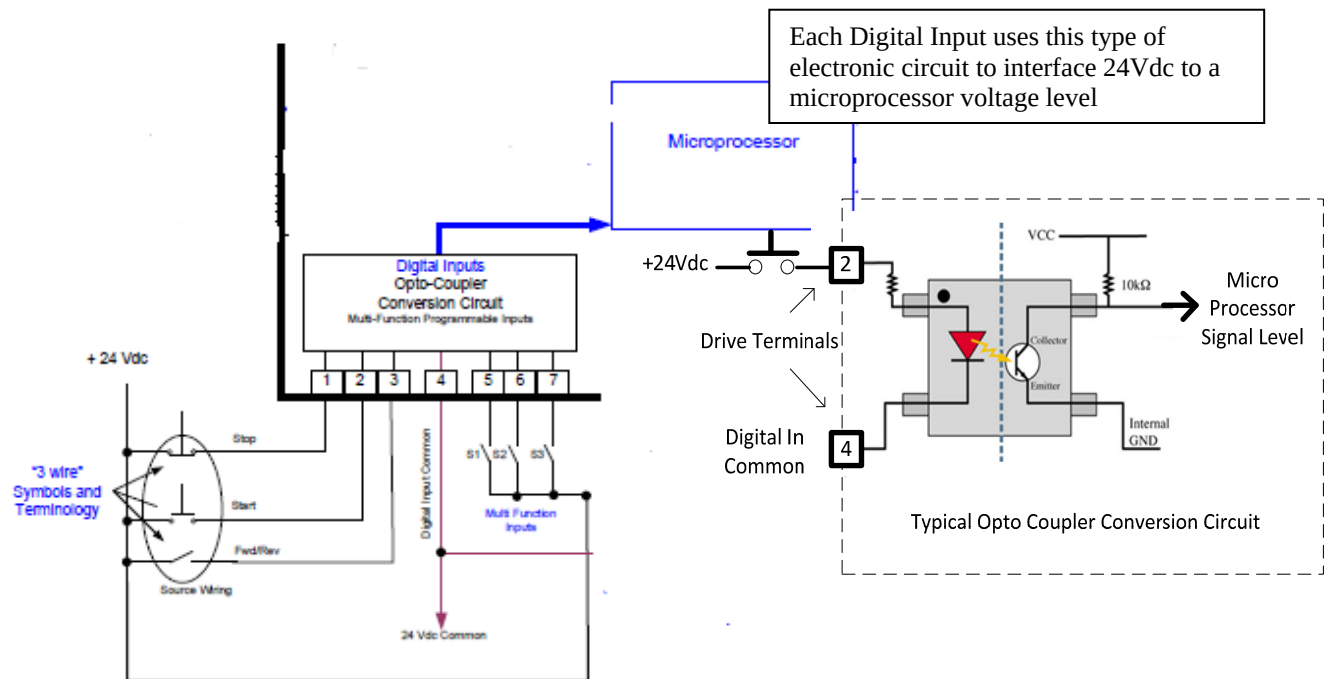
Power Terminal Block

B Frame



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Digital Inputs



Digital Inputs for the generic drive connect to terminals 1-3 and 5-7. These are called digital inputs because they are either **On** or **Off**. They are also programmable (multi-function) inputs on an AC Drive controller which means they can be configured to perform a variety of functions based on the user application (eg. Start, Stop, Jog, Preset Select, etc...).

The drive also has internal conversion circuitry using Opto-Isolators to interface +24Vdc to a 5 or 12Vdc used by the microprocessor. These signals are electrically isolated from each other through the Opto Coupler via light.

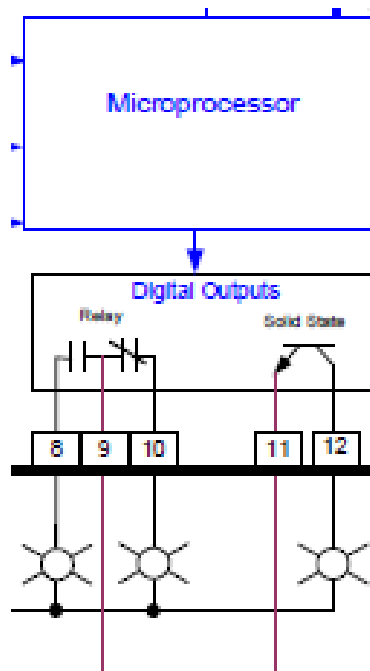
Digital Inputs may be powered through either an Internal (Drive supplied) or External DC Power Supply. An external supply is shown above. Most AC Drive controllers have an internal DC power supply for this purpose.

The number of digital inputs will vary between manufacturers.

The **Digital Inputs** of an AC Drive controller are wired to external circuits that may Enable the drive, Start/Stop the drive, Reverse the drive and/or have the ability to affect the output frequency of the drive.

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Digital Outputs

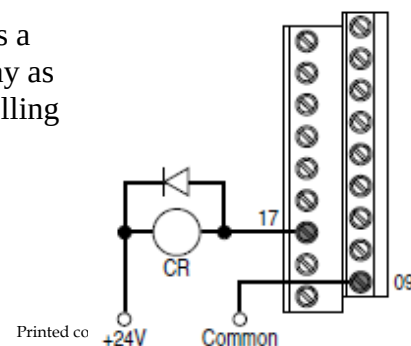


The Drive Conditions of a Powerflex AC Drive controller may vary from a simple LED, to programmable conditions that will cause **digital output(s)** to change state. These can be conditions such as "drive at speed" or "drive in current limit". ***When programming the digital relay output, the drive is configuring when the relay coil will energize.*** The normally open or closed contacts will follow the state of the relay coil.

The number of digital outputs will vary between manufacturers while some may be Relays, Solid State or a combination of both types of outputs. The example above shows a generic drive with two digital outputs.

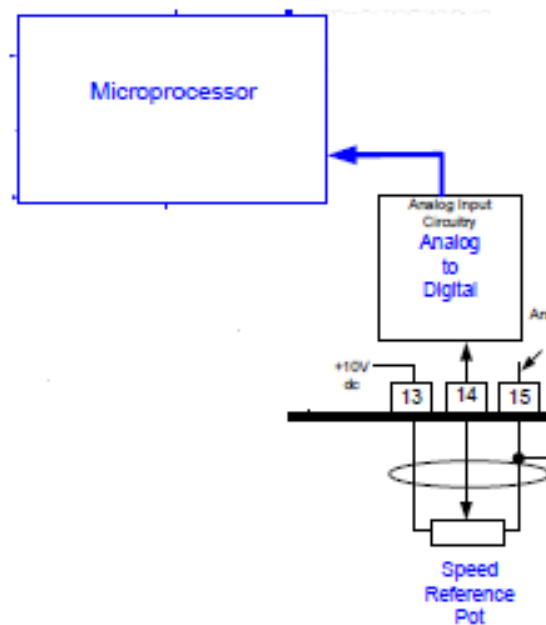
Digital outputs may be a Relay with Normally Open/Normally Closed contacts or a Solid State (Opto) Output. Relay outputs are more common and can handle more current. Solid State outputs are electronic outputs (Transistor). They are generally connected to PLC Inputs where current is small.

When using a solid state output with an inductive load such as a relay, install a recovery (Fly back) diode in parallel to the relay as shown, to prevent damage to the electronic component controlling the output.



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Analog Inputs



The Analog Input circuitry, inside the drive, performs an Analog to Digital conversion.

Analog Inputs may be used for obtaining a Speed Reference for a drive or as inputs to the drive under PID control.

They may be configured as 0-10Vdc or 4-20 ma signals. The inputs may also be configured for either a Unipolar or Bipolar input signal.

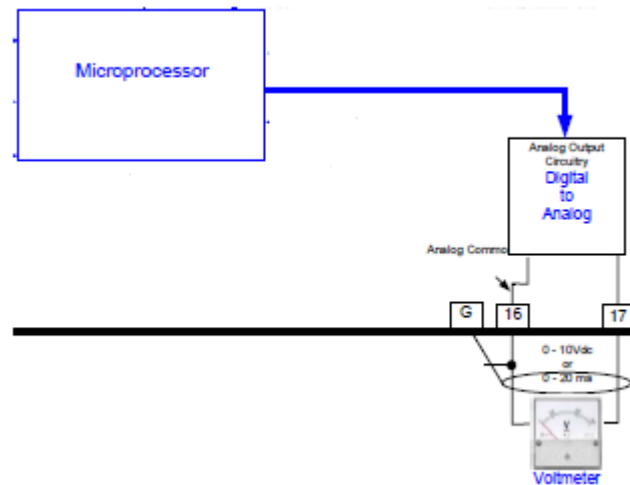
Most drives have at least one Analog Input and some may have more than one.

The Analog Input circuit shown above uses a potentiometer to obtain a **Unipolar Speed Reference**. The speed reference is based on the magnitude of the DC voltage. A digital input must be wired to provide a reverse capability for the drive.

Bipolar speed references require a bipolar DC power supply (- 10V and + 10V) which will provide a speed reference and a directional reference for the motor based on the magnitude and polarity of the signal. When using bi-polar speed references, the varying analog signal must be connected to a differential Analog input to the drive.

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Analog Outputs



The Analog Output circuitry, inside the drive, performs a Digital to Analog conversion.

Analog Outputs may be configured for one of several functions. They may be configured to provide an output voltage 0-10Vdc representing the output frequency of the controller between Minimum and Maximum frequencies.

They may be configured as 0-10Vdc or 4-20 ma signals. The outputs are configured for either a Unipolar or Bipolar input signal.

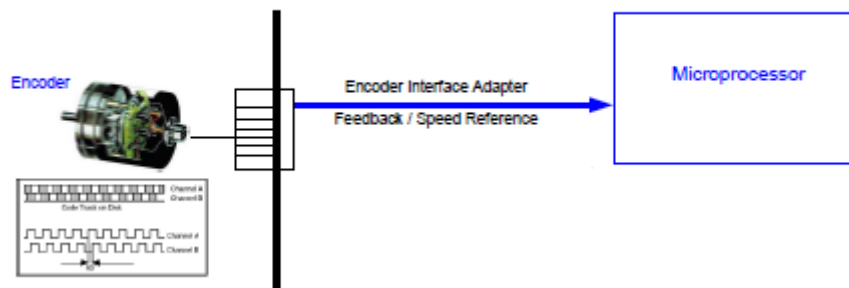
Most drives have at least one Analog Output and some may have more than one.

The Analog Output circuit shown above uses a voltmeter to display the output signal level.

The Analog Output from one drive can also be wired to another drive's Analog Input to provide a Speed Reference to the other drive (Master/Slave).

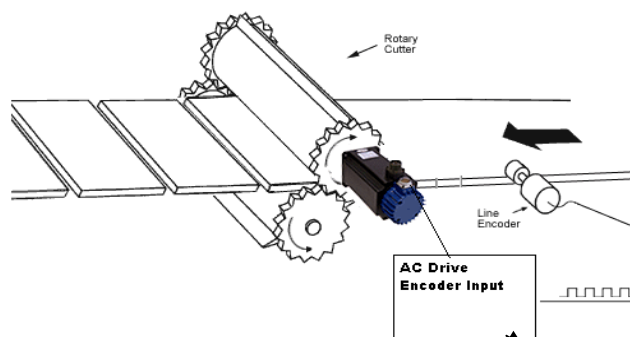
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Encoder Interface

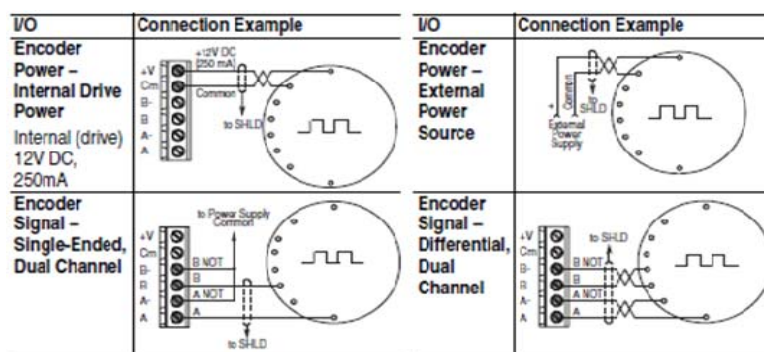


When an encoder interface used as a **Feedback device**, it is typically a quadrature incremental encoder. The encoder provides digital pulses to provide speed and direction signals to the drive. This would be normally be used in this configuration within a Closed Loop control system.

An Encoder may also be used as a Speed Reference for this drive (Example Below).

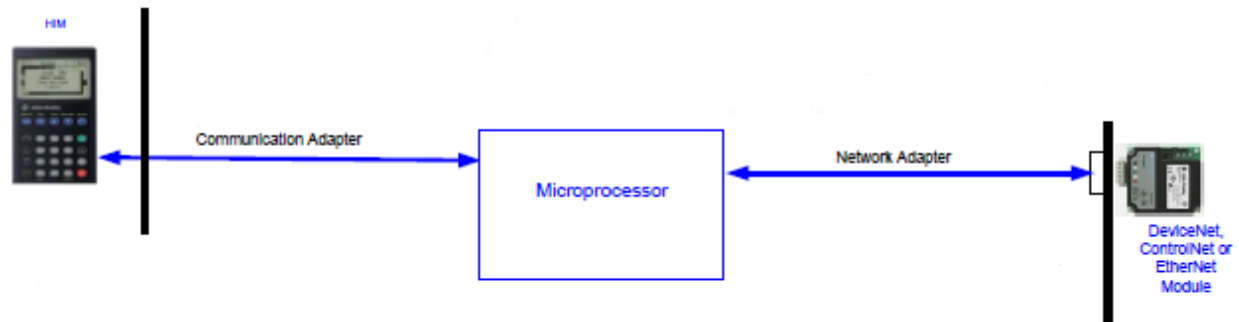


Types of encoders and connections are shown below. **The most common encoder used is the Differential Dual Channel encoder.** The twisted pair wiring provides better noise suppression for the small voltage level pulses.



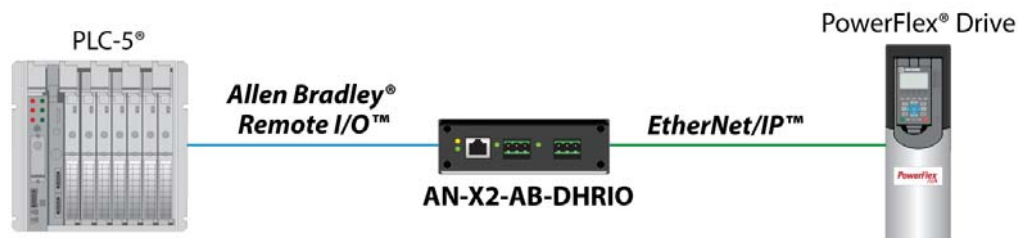
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Communications Adapters



Communications Adapters (Ports) allow connection of external devices such as the HIM or network communication modules (DeviceNet, ControlNet, Ethernet and Remote I/O communications).

There may be multiple communication adapters in a drive.



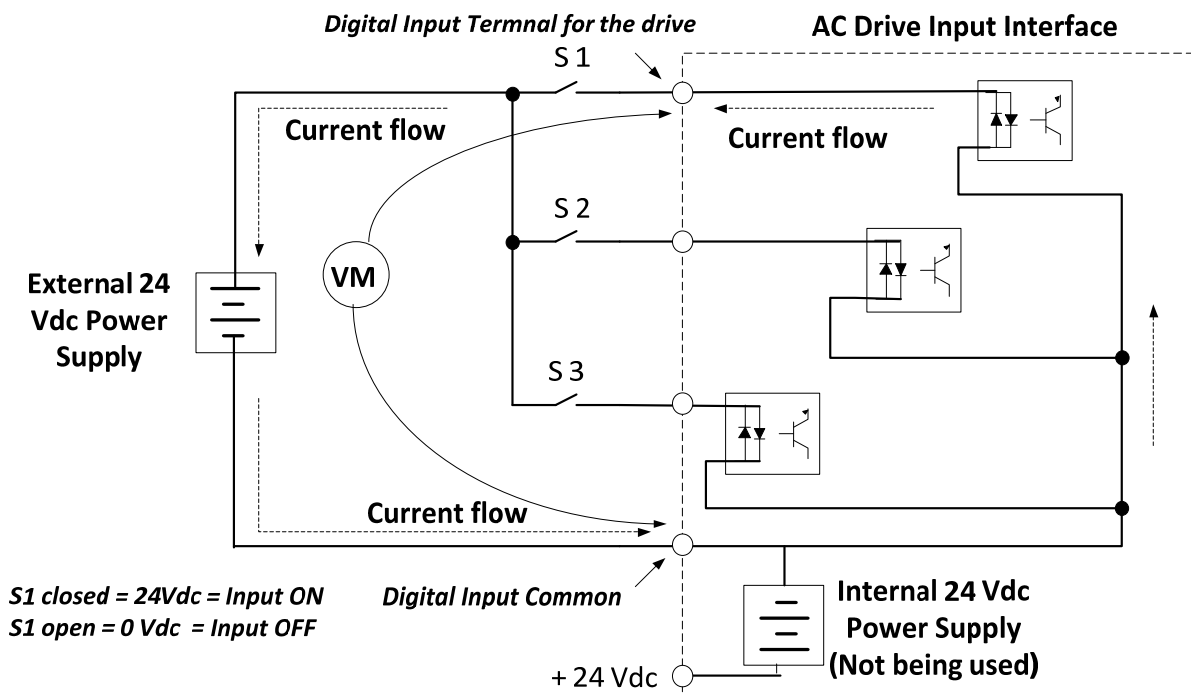
These basic interface circuits can be easily identified on almost all drives. Recognizing them and being able to verify their function is the **prime** responsibility of troubleshooters.

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I/O Wiring

AC Drive input interface circuits may be connected as either a **Source or Sink** wiring configuration on most manufacturer's drives. It is most important for the troubleshooter of drive systems to recognize the type of scheme being used. Otherwise the voltage measurement made at an input terminal may have the opposite meaning as what might be expected. Let's start with the Source wiring input scheme since they are more typical in Michelin applications with AC Drive interfacing. A source wiring input requires a **positive voltage** at the drive's input terminal with respect to common to activate (turn on) that input.

External DC Supply Source Wiring Example

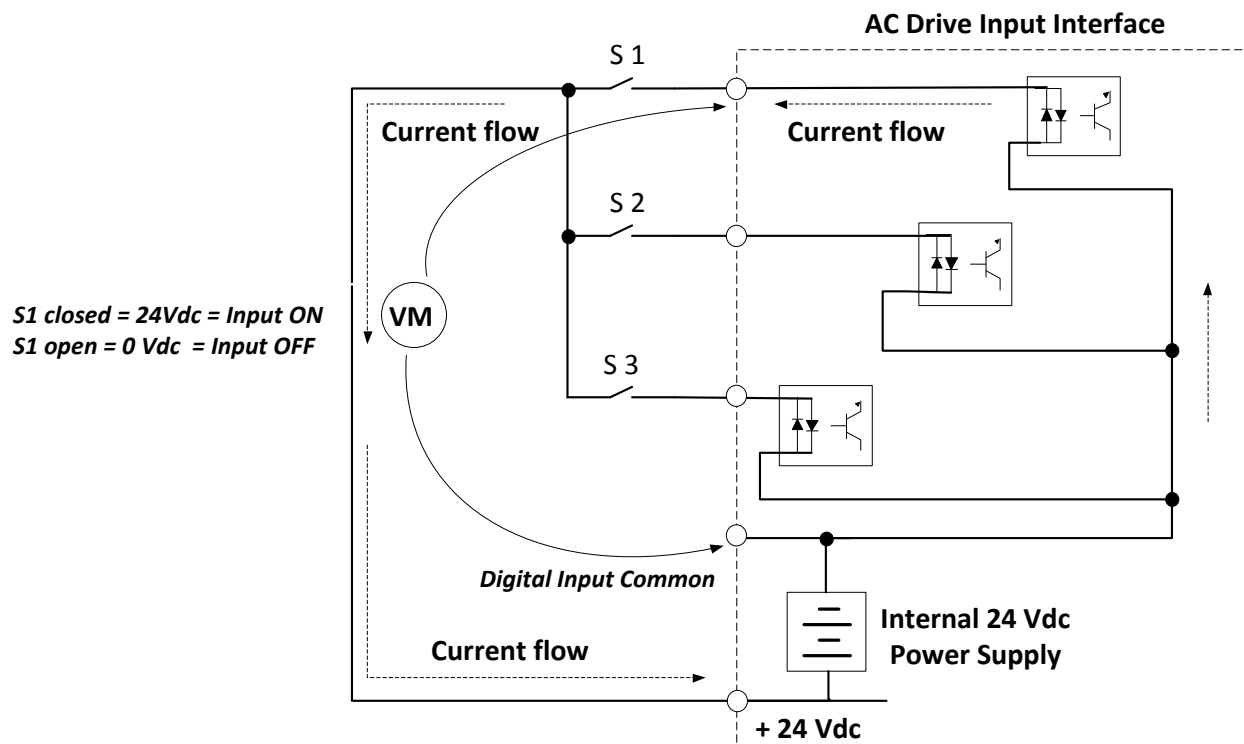


Most AC Drives have a built in DC 24 Volt Power Supply too. However, it is only designed to provide enough amperage to supply the drive's digital inputs. Rarely is it used within Michelin. The example shown above uses an external DC Power supply or User Supplied DC Voltage. This is most common within Michelin as this power supply may also feed other components not related to the drive such as PLC Inputs and safety relays, etc. In order to successfully connect an external supply, the 0V (common) from the external supply must also be connected to the Drive's digital input common.

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Next, we will examine the Source wiring input scheme which utilizes the drive's internal power supply. Either option produces the same result as far as the digital input required voltage. In other words this option still requires a **positive voltage** at the drive's input terminal with respect to common to activate that input.

Internal DC Supply Source Wiring Example



Notice the direction of current flowing in either of the Source wiring examples. The current is flowing **out** of the digital input terminal towards the positive side of the supply voltage. There must be a complete path for current to flow and this includes flowing through the drive's input interface circuitry.

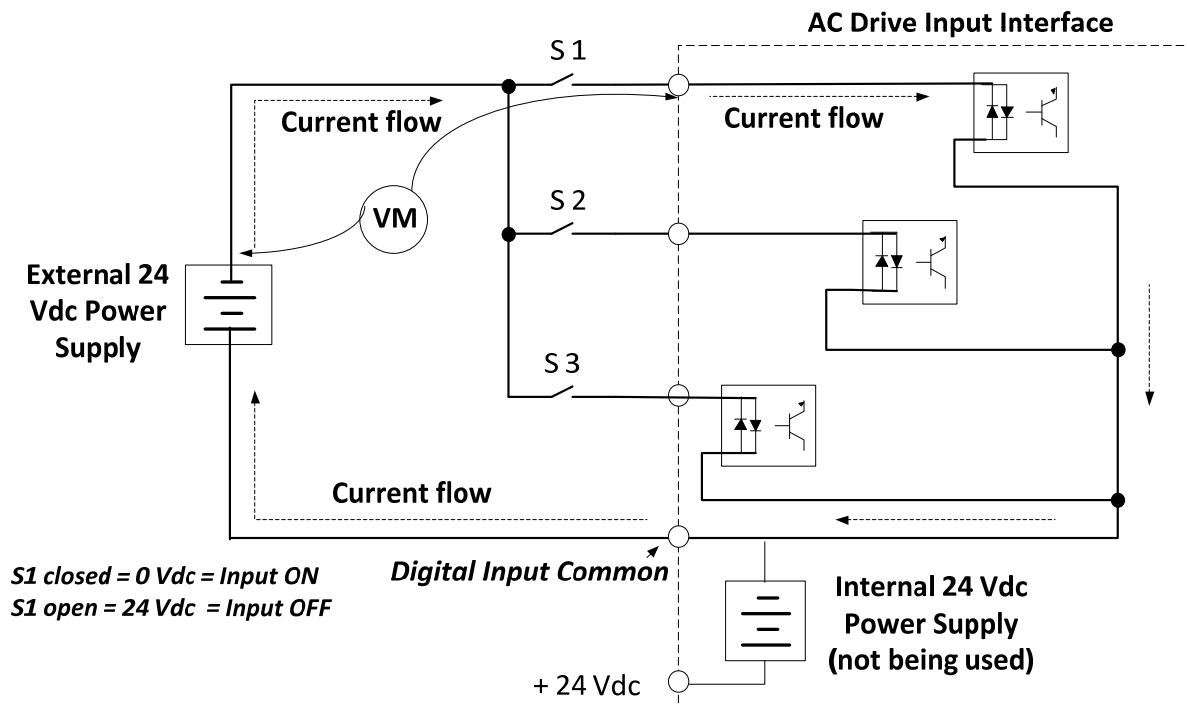
Inside the drive interface circuitry, there are typically Opto-Couplers which translate an external voltage to a digital voltage level (5Vdc or 12Vdc) which is the voltage level used inside the drive's microprocessor world. These components actually use light to make the signal translation which electrically isolates the outside world voltage from the drive's inside world voltage.

Note: *Whenever troubleshooting a digital input circuit, it is nice to have an idea what is inside the drive circuit. These examples show a simple, yet common Opto Coupler circuit used.*

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Now we will examine **Sink** wiring input configuration. Sometimes these may cause confusion by the troubleshooter and may cause incorrect interpretations of measurements. As stated earlier, Sink wiring inputs are rarely used within Michelin but *they do exist*.

External Supply Sink Wiring Example



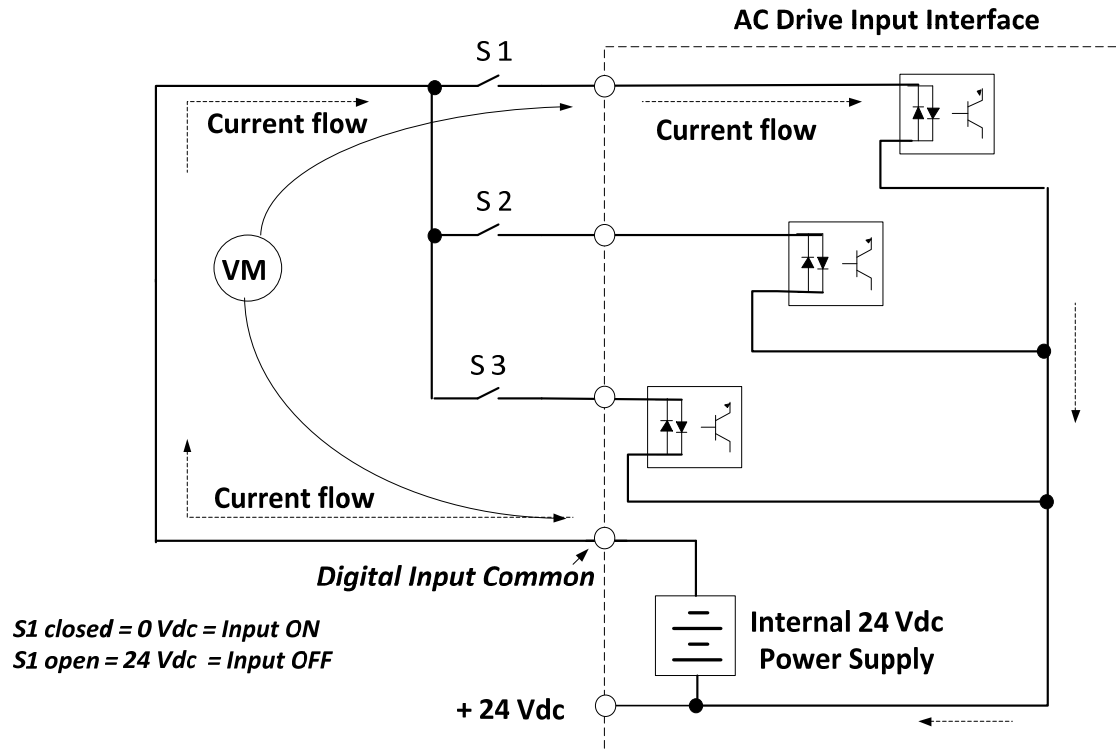
The voltages measured by the voltmeter have the **opposite** meaning as they did with the source wiring inputs we saw previously.

Notice that the external power supply's positive voltage is connected the drive's Digital Input Common. This may look or sound like a potential problem but the drive's internal power supply is not being used and this provides the complete path for current for the digital input circuit.

Now, notice the direction of current flowing in the Sink wiring example. The current is flowing **into** the digital input terminal towards the positive side of the supply voltage. Once again, there must be a complete path for current to flow and this includes flowing through the drive's input interface circuitry.

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Internal Supply Sink Wiring Example



Once again, notice the direction of current flowing in the above Sink wiring example. The current is flowing **into** the digital input terminal towards the positive side of the supply voltage. Once again, there must be a complete path for current to flow and this includes flowing through the drive's input interface circuitry.

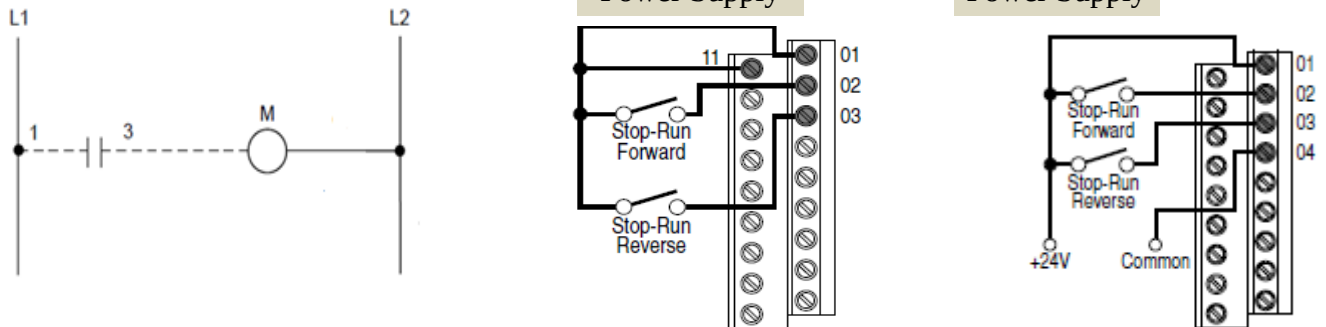
These basic interface circuits can be easily identified on all drive systems. Recognizing them and being able to verify their function is the prime responsibility of troubleshooters.

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AC Drives have the ability to be configured for different I/O Schemes.

- **2-Wire Control Scheme**

Below is a 2-wire control scheme, using a maintained contact pilot device in series with the starter coil.

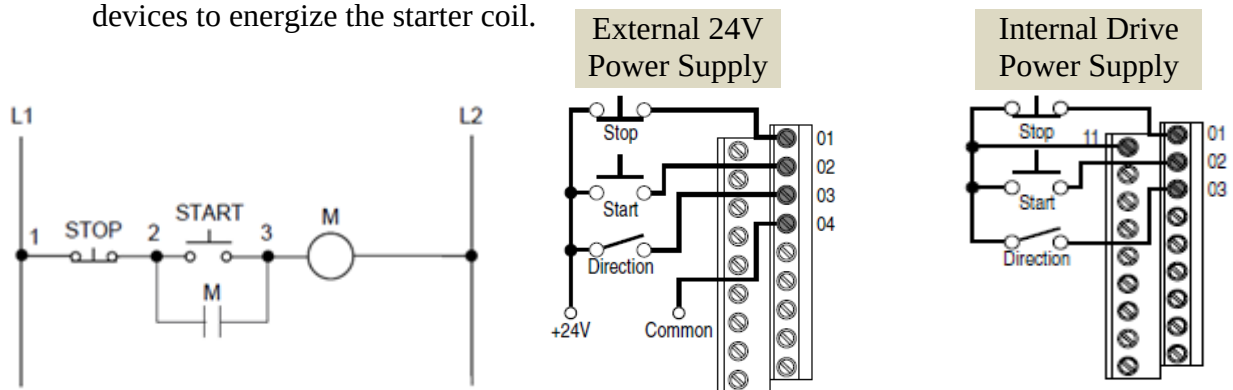


The term “2-wire” control is derived from the fact that in the basic circuit, only two wires are required to connect the pilot device to the motor starter (M).

The 2-Wire scheme requires maintained contacts in order for the drive to run and a separate Stop Input is not required. The terminology associated with 2-Wire is “Stop-Run”, “Stop-Run Forward” or “Stop-Run Reverse”.

- **3-Wire Control Scheme**

Below is a 3-wire control scheme, using momentary contact push buttons or similar pilot devices to energize the starter coil.



The term “3-wire” control is derived from the fact that in the basic ladder circuit, at least three wires are required to connect the pilot devices to the motor starter (M).

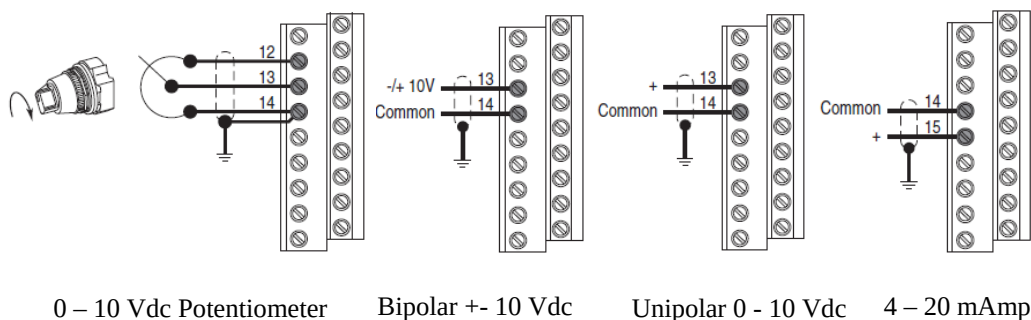
The 3-Wire Control Scheme utilizes momentary contacts to start the drive. A separate Stop Input must be configured with this scheme and the terminology used is “Start”, “Forward” or “Reverse”.

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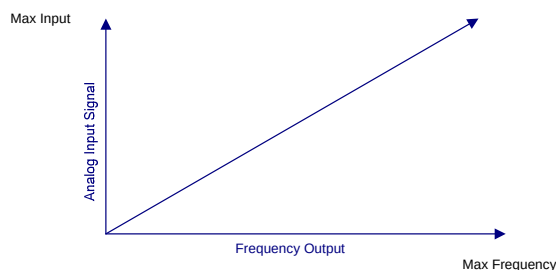
Speed References

AC Drives also have multiple capabilities of speed reference inputs. This allows flexibility in configuring a drive to a particular application. Below are some examples of the different methods of speed referencing:

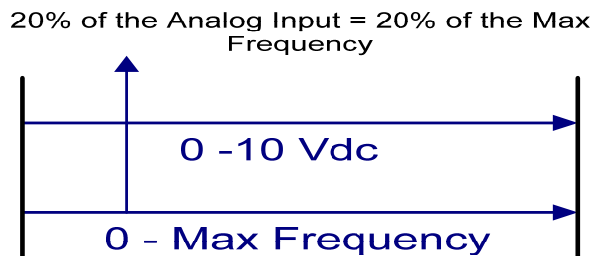
1. **Analog Input:** This method allows the user to configure an analog signal to provide the speed reference for the drive using many types of signals. Four unique methods are shown below.



The Analog Input allows for a linear range of speeds.

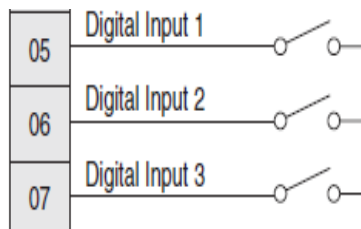


This linear range of frequencies is based on the drive's programmed minimum and maximum frequencies. Bipolar analog inputs allow for directional (Fwd/Rev) control of the drive based on the polarity of the analog input signal. Unipolar and 4-20 mA inputs control in one direction and will normally require a separate input to the drive for directional control.



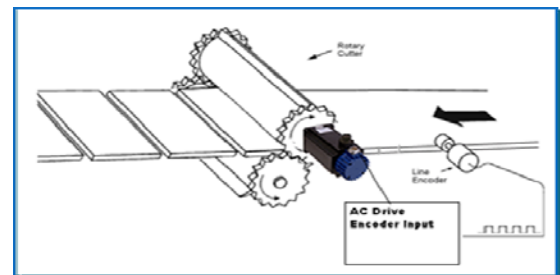
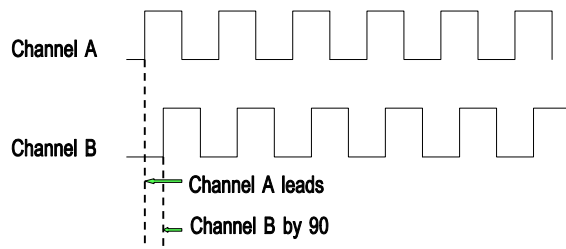
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2. **Preset frequency:** This method allows the user to program a block of frequencies into the drive that are selectable by digital inputs. Some drives may support up to eight (8) preset speeds and these speeds are selected by a binary number equal to the preset speed reference. Digital input #1 off, #2 on, and #3 off would represent 010_2 (binary) and the drive would select Preset Speed 2 (30 Hz) as the commanded frequency.



Preset #	Frequency	Input 3	Input 2	Input 1
Preset 0 Hz	10Hz	0	0	0
Preset 1 Hz	20 Hz	0	0	1
Preset 2 Hz	30 Hz	0	1	0
Preset 3 Hz	40 Hz	0	1	1
Preset 4 Hz	50 Hz	1	0	0
Preset 5 Hz	60 Hz	1	0	1
Preset 6 Hz	70 Hz	1	1	0
Preset 7 Hz	80 Hz	1	1	1

3. **Internal Frequency (MOP):** This method of speed reference utilizes a single programmed frequency in the drive stored as an internal value. The motor would run at this frequency at all times.
4. **Comm Port:** This method is used when control of the drive's speed reference is from an external communications port. This would be used for DeviceNet or EtherNet operation of the drive. The Speed Reference is issued from an external PLC using the particular network through the communications adapter.
5. **Encoder:** Some drives may obtain their speed reference from an encoder input. An incremental, quadrature encoder is typically used to provide a pulse train of inputs to the drive. The drive interprets this pulse train as a speed, direction or position reference.



6. **Jog Frequency:** The drive may be capable of executing user defined initiated Jog function which normally has a dedicated parameter called Jog Frequency used as the speed reference.

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Additional Drive Features

There are numerous features to understand with any AC drive controller. Many of the features of the PowerFlex drives will be studied in your labs. Typical features are:

- **Minimum Frequency**
The parameter setting on the drive which determines the lowest frequency the drive will output. Usually, this is set to 0Hz but, some applications may run at a setting other than 0Hz, such as an agitator or mixer.
- **Maximum Frequency**
The parameter setting on the drive which determines the highest frequency the drive will output.
- **Thermistor (Motor Thermistor)**
A thermistor is a non-linear resistance temperature device, made from semi-conductor material. Each specific thermistor has its own unique resistance vs. temperature characteristic.

There are two general types of Thermistors:

1. The positive temperature coefficient (PTC) type has a resistance that increases with increasing temperature. The PTC thermistor has a sharp rise in resistance when the temperature reaches the 'knee', or switching point.
 2. The negative temperature coefficient (NTC) type thermistor has a resistance that decreases with increasing temperature.
- **Jog**
Jogging is a means of accomplishing momentary motor movement by repetitive closure of a circuit using a single push-button or contact element (This is a manual function). Usually requires a parameter set for Jog Frequency.
 - **Factory Defaults:** This feature will automatically reset all of the software parameters back to a factory default condition. Sometimes, this is useful when many parameters have been changed and there is need to start over. Mainly used when replacing a drive in the machine before loading the unique parameter settings for that machine.

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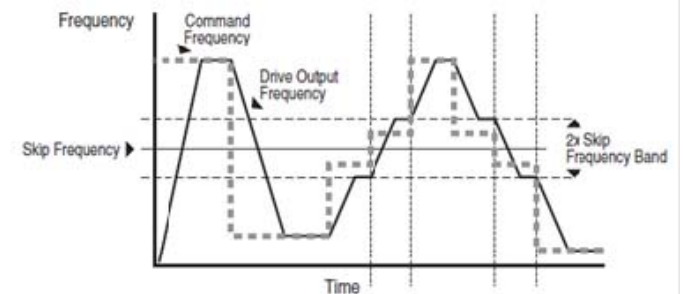
- **Autotune:**

Provides an automatic method for setting [IR Voltage Drop] and [Flux Current Ref], which greatly affect sensorless vector performance.

“Static Tune”: A temporary command that initiates a non-rotational motor stator resistance test for the best possible automatic setting of A128 [IR Voltage Drop]. A start command is required following initiation of this setting. The parameter returns to “Ready” (0) following the test, at which time another start transition is required operate the drive in normal mode. Used when motor cannot be uncoupled from the load.

“Rotate Tune”: A temporary command that initiates a “Static Tune” followed by a rotational test for the best possible automatic setting of A129 [Flux Current Ref]. A start command is required following initiation of this setting. The parameter returns to “Ready/Idle” (0) following the test, at which time another start transition is required to operate the drive in normal mode. ***Important: Used when motor is uncoupled from the load. Results may not be valid if a load is coupled to the motor during this procedure.***

- **Skip Frequency:** This feature is used when certain frequencies may have undesirable effects on the machine. It may allow the drive to accelerate from min to max speeds, skipping certain frequencies that may create excessive harmonics or vibrations which may be detrimental to the machine. Usually, the Skip Frequency is programmed in conjunction with a Skip Frequency Band to avoid a specific range of frequencies.



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DriveGuard (Safe Off)

The DriveGuard Safe Torque Off option provides the “Safe torque off (STO)” function defined in EN/IEC 61800-5-2. It blocks gate firing signals from reaching the IGBT output power devices of the drive. This prevents the IGBT’s from switching in the sequence necessary to generate torque in the connected motor.

Stop Modes

The Stop Mode Parameter sets the method of stopping the drive whenever a Stop command is asserted.

Stopping an AC motor can be accomplished in several ways with an AC Drive:

Coast to Stop --- This method simply disconnects the motor from the drive (electrically) which means the motor will simply coast to a stop. The time required to completely stop is based on the inertia of the drive system.

Ramp to Stop --- This method uses a Decel ramp to reduce the voltage and frequency proportionally. The user can adjust the Decel time and affect how long it will take to stop the motor. This is also referred to as Controlled Stop.

Motor Mounted or Separately Mounted Brake --- is a positive action, mechanical, friction device. Normal configuration is such that when the power is removed, the brake is set. This can be used as a holding brake. (Note: A Separately Mounted Brake is one which is located on some part of the mechanical drive train other than the motor.) This method would normally be used in conjunction with Coast to Stop.

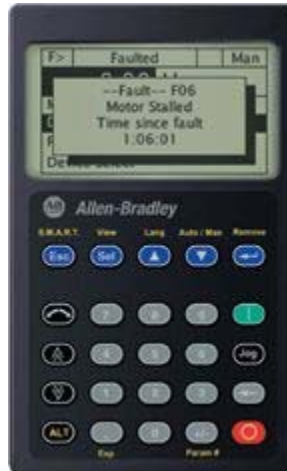
DC Injection --- This method of braking actually connects the motor to a DC Voltage/Current Source from the drive. This will immediately stop the motor.

Ramp to Hold --- With this method of braking the drive acknowledges the Stop command by decreasing or “ramping” the output voltage and frequency to zero in a programmed period (Decel Time), maintaining control of the motor until the drive output reaches zero.

Once the drive has reached zero output hertz on a Ramp-to-Stop and both parameters [DC Hold Time] and [DC Hold Level] are not zero, the drive applies DC to the motor producing current at the DC Hold Level for the DC Hold Time.

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- ***AC Drive Keypads (Human Interface Modules)***



Most AC Drives are equipped with a keypad (HIM), although, it may be listed as an option.

If a keypad is present, the troubleshooter can use it to:

- *Display certain read only parameters*
- *Program most basic parameters to get the motor going*
- *Program the Advanced set of parameters for the finer features of the Drive*
- *Display the Fault codes associated with a Drive Fault.*
- *Clear drive faults*
- *Start and Stop the drive*

The HIM module may be used as a storage device. This allows the user to Save the configured parameters to the module or transfer a parameter set from the HIM to the drive. When replacing a bad drive, these functions save enormous time in configuring the replacement drive.

Important: There is typically a procedure for removing the HIM when the drive is running. Removing the HIM while the drive is running may cause a drive fault and stop the drive.

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Parameters

AC drives use programming parameters to allow the user to configure the drive and motor to the application. Usually, the parameters are organized by Groups and Functions. They are also assigned a unique number for each parameter. All of this allows maximum flexibility in drive applications.

There are three types of software parameters in the Powerflex family of drives:

- ENUM**
 ENUM parameters allow a selection from 2 or more items. Each item is represented by a number.
- Numeric Parameters**
 Numeric parameters have a single numerical value (i.e. 0.1 Volts).
- Bit Parameters**
 Bit parameters have four individual bits associated with features or conditions. If the bit is 0, the feature is off or the condition is false. If the bit is 1, the feature is on or the condition is true.

Basic Parameters

Below are examples of Basic Programming Parameters an AC Drive typically uses. These parameters are essential in configuring the drive to the motor and the application.

 = Stop drive before changing this parameter.

No.	Parameter	Min/Max	Display/Options
P031 	[Motor NP Volts] Set to the motor nameplate rated volts.	20/Drive Rated Volts	1 VAC
P032 	[Motor NP Hertz] Set to the motor nameplate rated frequency.	15/400 Hz	1 Hz
P033	[Motor OL Current] Set to the maximum allowable motor current.	0.0/(Drive Rated Amps×2)	0.1 Amps
P034	[Minimum Freq] Sets the lowest frequency the drive will output continuously.	0.0/400.0 Hz	0.1 Hz
P035 	[Maximum Freq] Sets the highest frequency the drive will output.	0/400 Hz	1 Hz
P036 	[Start Source] Sets the control scheme used to start the drive. (1) When active, the Reverse key is also active unless disabled by A095 [Reverse Disable].	0/6	0 = "Keypad" ⁽¹⁾ 1 = "3-Wire" 2 = "2-Wire" 3 = "2-W Lvl Sens" 4 = "2-W Hi Speed" 5 = "Comm Port" 6 = "Momt FWD/REV"

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Display (Diagnostic) Parameters

Below are examples of the Display parameters, for the Read Only groupings, used for monitoring and/or troubleshooting. Display parameters may also be considered as diagnostic parameters and are very useful for obtaining information from the drive such as:

- Input Status
- Output Voltage
- Start Reference Source, etc...

These parameters may be used as troubleshooting tools.

No.	Parameter	Min/Max
d001	[Output Freq]	0.0/[Maximum Freq]
d002	[Commanded Freq]	0.0/[Maximum Freq]
d003	[Output Current]	0.00/(Drive Amps × 2)
d004	[Output Voltage]	0/Drive Rated Volts
d005	[DC Bus Voltage]	Based on Drive Rating
d006	[Drive Status]	0/1 (1 = Condition True)
d007-d009	[Fault x Code]	F2/F122

Advanced Parameters

Below, are some typical examples of the advanced parameters used to finely adjust and tune the drive/motor performance.

A078	[Jog Frequency]	0.0/[Maximum Freq]	0.1 Hz
A079	[Jog Accel/Decel]	0.1/600.0 Secs	0.1 Secs
A080	[DC Brake Time]	0.0/99.9 Secs	0.1 Secs
	A setting of 99.9 Secs = Continuous		
A081	[DC Brake Level]	0.0/(Drive Amps × 1.8)	0.1 Amps
A082	[DB Resistor Sel]	0/99	0 = "Disabled" 2 = "NoProtection"
		1 = "Normal RA Res"	3-99 = % of Duty Cycle

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Acceleration/Deceleration (Ramping)

Accel/Decel Ramping

AC drives can gradually increase or decrease the voltage and frequency supplied to a motor; therefore increasing or reducing these values automatically and in the correct proportions. The user can program or otherwise adjust how rapidly this reduction occurs. This process, called **ramping**, gradually accelerates/decelerates the motor in a controlled fashion.

The Accel/Decel Time parameters set the rate at which the drive ramps up/down its output frequency after a Start/Stop command or during an increase/decrease in command frequency (speed change). The acceleration and deceleration rates may be different values. The rate established is the result of the programmed Accel/Decel Time and the Maximum Frequency, as follows:

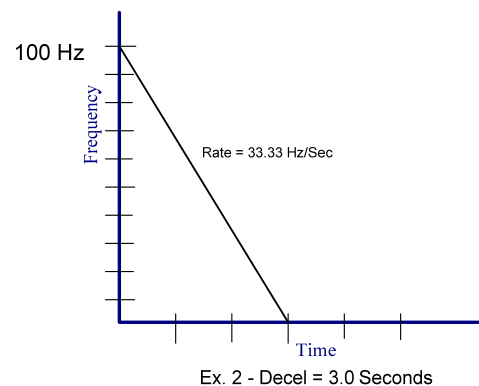
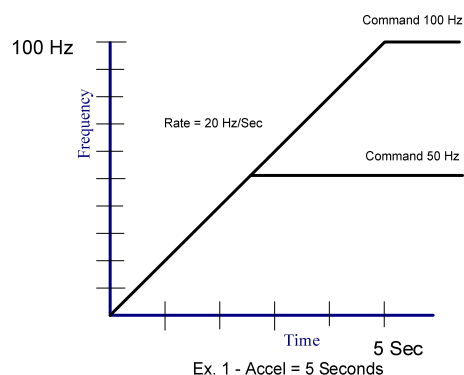
$$\text{Max. Frequency} / \text{Accel Time} = \text{Acceleration Rate (Hz/sec)}$$

$$\text{Max. Frequency} / \text{Decel Time} = \text{Deceleration Rate (Hz/sec)}$$

Example 1: A drive has a Maximum Frequency parameter set for 100 Hz and the Accel parameter set at 5 Seconds. The Accel Rate would equal **100 Hz / 5 Seconds = 20 Hz / Sec.** Using this rate the drive will perform the necessary calculations to ramp up the voltage, current and frequency output to the motor to match this rate.

Another way to look at this is to understand that if the Speed reference is 100 Hz and the Maximum Frequency is set to **100 Hz**, it would take **5 seconds** for the drive to ramp up to 100 Hz. Hence, the drive is accelerating at a rate of 20 Hz / second.

However, with the same parameter settings for Maximum Frequency and Accel time, if the drive is commanded to Start and run at 50Hz, it would ramp up to 50 Hz in 2.5 Seconds. In both cases, the drive is ramping up to speed at 20 Hz / second.



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Incremental Encoders

A **rotary encoder**, also called a **shaft encoder**, is an electro-mechanical device that converts the angular position or motion of a shaft or axle to a digital code or pulses.

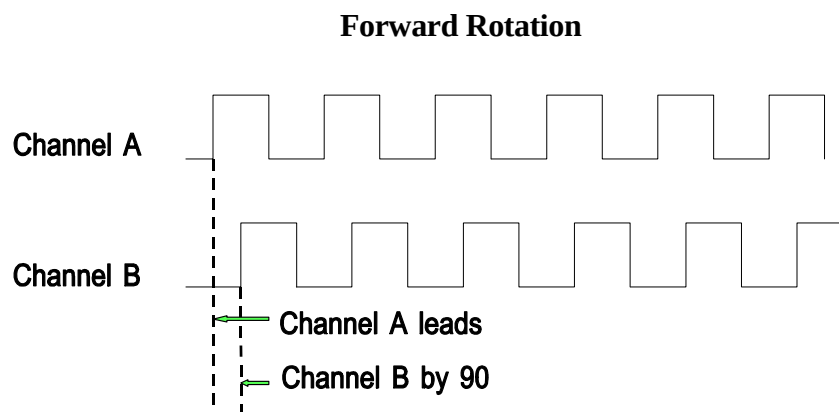
There are two main types: absolute and incremental.

- The output of absolute encoders indicates the current position of the shaft, making them angle transducers.
- The output of incremental encoders provides information about the *motion* of the shaft, which is typically further processed elsewhere into information such as speed, distance, and position.

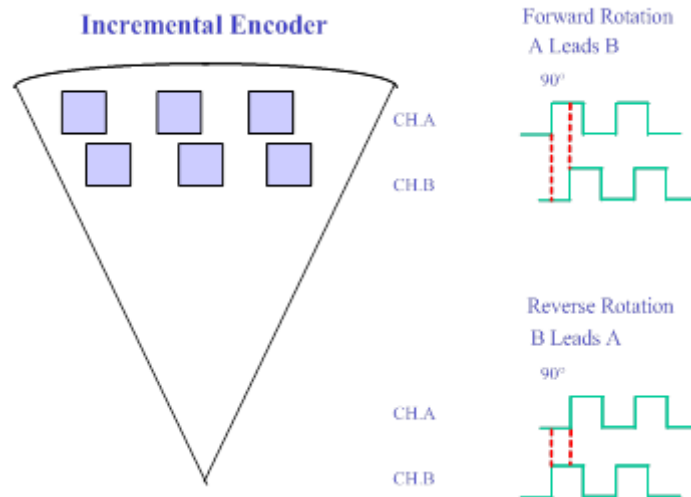
Rotary encoders are used in applications that require precise shaft feedback with unlimited rotation. They are used exclusively in industrial controls, robotics and computer input devices such as a mouse or a trackball.

An incremental encoder works operates by providing an A and a B pulse output that provide no usable count information in their own right. Rather, the counting is done in the external electronics. The point where the counting begins depends on the counter in the external electronics and not on the position of the encoder. To provide useful position information, the encoder position must be referenced to the device to which it is attached. The distinguishing feature of the incremental encoder is that it reports an incremental change in position of the encoder to the counting electronics.

The waveform shown below is considered standard for forward rotation where channel A leads channel B.

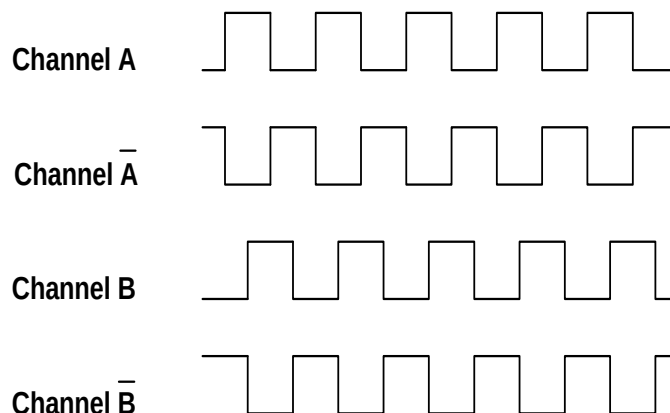


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The two output wave forms are 90 degrees out of phase, which is what quadrature means. These signals are decoded to produce a count-up pulse or a count-down pulse. For decoding in software, the A & B outputs are read by software, either via an interrupt on any edge or polling, and the above table is used to decode the direction. The use of complementary outputs will prevent false counting due to electrical noise developing stray electrical pulses. An additional benefit of a quadrature encoder is the ability to multiply the encoder disk resolution by four. By counting the leading and trailing edge of each window on both channels, each slot of window now gives four counts. Using this procedure, a 1000 window disk can have a resolution of 4000 counts per revolution.

Quadrature Encoder with Complementary Outputs



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Troubleshooting

The majority of troubleshooting revolves around the interface circuits. Some troubleshooting can be achieved by using the HIM/Keypad and/or software programming terminals.

Display parameters (keypad and/or computer) could be used to verify:

- Voltage and Frequency output to the motor
- Current output to the motor
- The logic status of the control inputs
- The value of an Analog input signal and its polarity
- The status of the digital inputs

However, a properly selected Digital Multi-meter can also be used to verify:

- Presence of input 3 phase voltage (L1/L2/L3)
- Voltage output voltage to the motor (T1/T2/T3)
- Analog speed reference coming from a 0v to +10volt potentiometer or Analog PLC Output
- Presence of Run Permissives

Typically, the digital Multi-meter will be the troubleshooter's first tool used in identifying problems encountered with an AC Drive.

Status LEDs provide a quick indication of drive status, changing colors from green to yellow to red to indicate various conditions. Typical status LEDs are Power, Ready, Drive, Communications Module, and one or more Network Connections. Normally, steady glowing green status lights mean everything is running properly. However, there are occasions when status lights are green, yet the drive is not operating properly. Flashing yellow or red lights indicate a problem with the drive that should be checked. Consult the drive manual for your specific drive to determine what a particular flashing light means and a detailed explanation of a particular fault or alarm that may be present.

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Basic Troubleshooting Procedure

1. Define The Problem

Clearly understand and define the problem you are having and keep the focus on this problem

2. Visual: Drive Powered Up, HIM display On, Faults or Alarms displayed?

HIM module on, Status LED conditions, Faults showing? Drive User Manual may be needed
Locate the drawings and be prepared to use them

3. Is the drive enabled? (Not Stopped)?

Verify presence of voltage at Stop and Enable Inputs (Source or Sink wiring?)
Locate the drawings and use them.

4. Is the drive receiving a Start Command (Input)? (2-wire/3-wire)?

Determine Start Source, Check for presence of voltage, Use Display Input Status Parameter, if needed. **Use the drawings.**

5. Determine and validate the Speed Reference Source.

Determine and validate if a speed reference is being sent to the drive. **Use the drawings.**

6. Is the drive outputting correct voltage and frequency based on the above?

Use the HIM module, Display Parameters and digital voltmeter to verify. Verify DriveGuard/Safe Off functioning. **Use the drawings.**

7. Verify Motor Circuit (Lockout/Tagout if needed) by using Ohm-meter

Verify ALL external contacts, disconnects, cables and terminals between drive and the motor.
Use the drawings.

While this procedure does not fully cover every scenario, these fundamental steps can be utilized every time.

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Introduction to the PowerFlex 40p

As part of the PowerFlex® family of AC drives, the PowerFlex 40P addresses the needs for closed loop control and category 3 Safe-off in its design. The PowerFlex 40P AC drive is based on the PowerFlex® 40 AC drive platform and shares common options and accessories. PowerFlex 40P AC drives are effective solutions for speed control of applications such as diverters, smart conveyors, packaging machines, palletizers, drafting machines, ring spinning machines and synthetic fiber spinning machines.

Features of the PowerFlex 40p

Start Up, Programming and Operation

- PowerFlex 40P drive programming is achieved by the use of an **external HIM** or via the resident DSI connection and the **PC programming tool**.
 - DriveTools™ SP
 - DriveExplorer™
 - Pocket DriveExplorer™
 -
- **StepLogic™ wizards** available for DriveTools SP and DriveExplorer (Lite or Full) assist in setup of position or velocity StepLogic programming.
- An **integral display and reset button** allows user to change display parameters and reset the drive if a fault occurs.
- **4 digit display** with 10 additional LED indicators provides a display of drive status and information.
- Integral **RS485 communications** can be used for programming from a PC. It can also be used in a multi-drop network configuration. A serial converter module provides connectivity to any controller with a DF1 port.
- The 10 most common application parameters are contained in the **Basic Program Group**. Common parameters shared with PowerFlex 40 are in the **Advanced Program Group** and additional position control parameters are located in the **Enhanced Program Group**.

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Sensorless Vector Control

- **Sensorless Vector Control** provides exceptional speed regulation and very high levels of torque across the entire speed range of the drive.
- The **Autotune feature** allows the PowerFlex 40P AC drive to adapt to individual motor characteristics.
- **Closed loop** with encoder feedback replaces slip compensation with trim for improved speed range and regulation.

Control Performance Features

- **V/Hz Motor Control**
- **Sensorless Vector Control** with Autotune feature develops high torque over a wide speed range and adapts to individual motor characteristics.
 - Slip compensation
 - Encoder trim
 -
- Integral **PID** functionality enhances application flexibility.
- **Position control** regulator mode.
- **Timer, Counter, Basic Logic and StepLogic™ functions** can reduce hardware design costs and simplify control schemes.
 - **Timer function:** Relay or opto outputs controlled by drive performing timer function. Timer is initiated by activating a digital input programmed as “Timer Start.”
 - **Counter function:** Relay or opto outputs controlled by drive performing counter function. Counter function is activated by a digital input programmed as “Counter Input.”
 - **Basic Logic:** Relay or opto outputs controlled by status of digital inputs programmed as “Logic Inputs.” Performs basic Boolean logic.
 - **Velocity and Position StepLogic™** function available providing repeatable speed or position changes based on StepLogic application profile.

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I/O Features

- Two (2) semi-programmable digital inputs.
 - Dedicated for start and stop
 - Simplifies wiring and troubleshooting
- Five (5) fully programmable digital inputs provide application versatility.
 - Standard 27 different settings, same as PowerFlex 40 AC drive providing application versatility
 - Additional 9 settings for Fibers and basic positioning enhancements
 - Reverse DI is now fully programmable via E202 [Digital Term 3]
- One (1) programmable relay output and two (2) opto-outputs can be used to indicate various drive or motor conditions.
- One (1) analog output is DIP Switch selectable for either voltage (0- 10V) or current (0-20 mA). This scalable, 10-bit output is suitable for metering or as a speed reference for another drive.
- One (1) analog input is DIP switch selectable for either voltage or current. Input is isolated from the rest of the drive I/O. Bipolar mode offers improved zero-cross performance.

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Catalog Number Explanation

It is very important to understand the PowerFlex 40p part numbering whenever you must replace a drive in the field. Below is the explanation of the 40p drives.

1-3	4	5	6-8	9	10	11	12	13-14
22D	-	B	2P3	N	1	0	4	AA
Drive	Dash	Voltage Rating	Rating	Enclosure	HIM	Emission Class	Version	Optional

Code	PowerFlex 40P
22D	

Code	Voltage	Ph.
B	240V AC	3
D	480V AC	3
E	600V AC	3

Code	EMC Filter
0	Not Filter

Code	HIM Version
1	Display/Fault Reset Only
2	Display/Fault Reset Only (Plate Drive)

Code	Purpose
AA	Reserved for
thru	custom firmware
ZZ	

Code	Enclosure
N	Panel Mount - IP 20 (NEMA Type Open)
F	Flange Mount - IP 20 (NEMA Type Open)
H	Plate Drive - IP 20 (NEMA Type Open)

Output Current @ 200-240V Input			Output Current @ 380-480V Input			Output Current @ 460-600V Input		
Code	Amps	kW (HP)	Code	Amps	kW (HP)	Code	Amps	kW (HP)
2P3	2.3	0.4 (0.5)	1P4	1.4	0.4 (0.5)	1P7	1.7	0.75 (1.0)
5P0	5.0	0.75 (1.0)	2P3	2.3	0.75 (1.0)	3P0	3.0	1.5 (2.0)
8P0	8.0	1.5 (2.0)	4P0	4.0	1.5 (2.0)	4P2	4.2	2.2 (3.0)
012	12	2.2 (3.0)	6P0	6.0	2.2 (3.0)	6P6	6.6	4.0 (5.0)
017	17.5	3.7 (5.0)	010	10.5	4.0 (5.0)	9P9	9.9	5.5 (7.5)
024	24	5.5 (7.5)	012	12	5.5 (7.5)	012	12	7.5 (10)
033	33	7.5 (10)	017	17	7.5 (10)	019	19	11 (15)
			024	24	11 (15)			

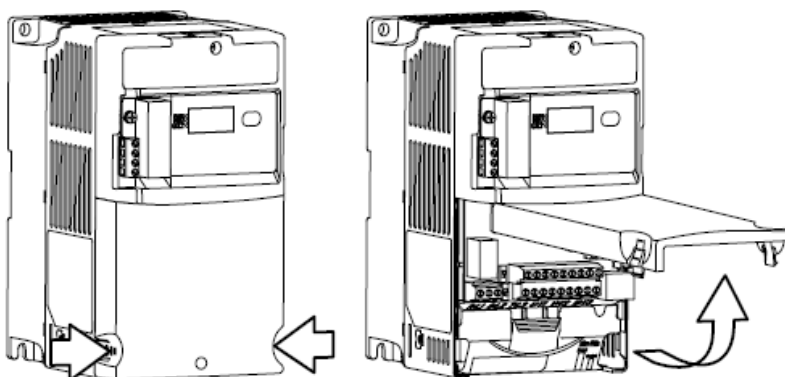
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Installation and Wiring

Opening the Cover:

1. Press and hold in the tabs on each side of the cover.
2. Pull the cover out and up to release.



Input Power Considerations

The drive is suitable for direct connection to input power within the rated voltage of the drive. Listed below are certain input power conditions which may cause component damage or reduction in product life. If any of the conditions exist, as described in below, install one of the devices listed under the heading *Corrective Action* on the line side of the drive.

Input Power Condition	Corrective Action
Low Line Impedance (less than 1% line reactance)	• Install Line Reactor⁽²⁾ • or Isolation Transformer • or Bus Inductor – 5.5 & 11 kW (7.5 & 15 HP) drives only
Greater than 120 kVA supply transformer	
Line has power factor correction capacitors	
Line has frequent power interruptions	• Install Line Reactor • or Isolation Transformer
Line has intermittent noise spikes in excess of 6000V (lightning)	
Phase to ground voltage exceeds 125% of normal line to line voltage	• Remove MOV jumper to ground. • or Install Isolation Transformer with grounded secondary if necessary.
Ungrounded distribution system	
240V open delta configuration (stinger leg) ⁽¹⁾	• Install Line Reactor

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Disconnecting the MOV's

To prevent drive damage, the MOVs connected to ground shall be disconnected if the drive is installed on an ungrounded distribution system where the line-to-ground voltages on any phase could exceed 125% of the nominal line-to-line voltage. To disconnect these devices, remove the jumper shown in the Figures 1.1 and 1.2.

1. Turn the screw counterclockwise to loosen.
2. Pull the jumper completely out of the drive chassis.
3. Tighten the screw to keep it in place.

Figure 1.1 Jumper Location (Typical)

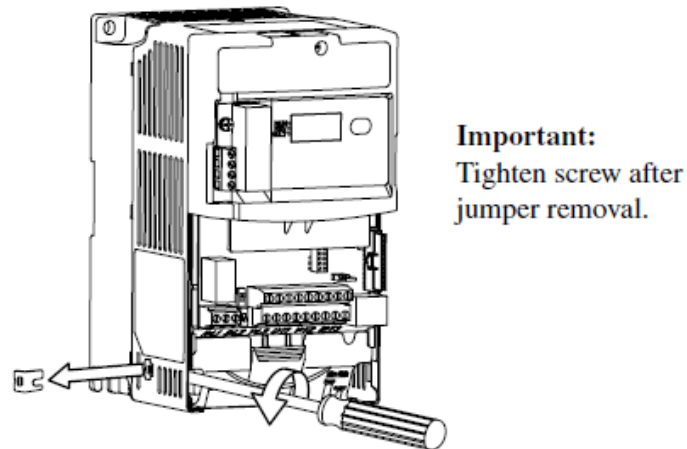
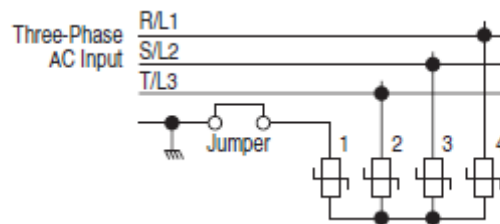


Figure 1.2 Phase to Ground MOV Removal



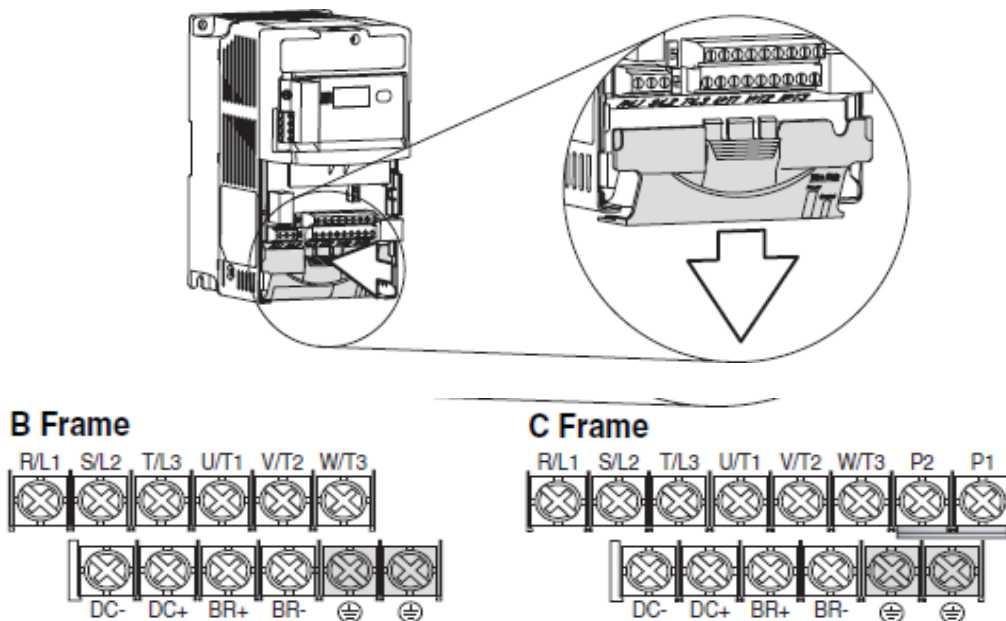
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PowerFlex 40p Power Diagram

The power terminal block is covered by a finger guard. To remove:

1. Press in and hold the locking tab.
2. Slide finger guard down and out.

Replace the finger guard when wiring is complete.



Terminal ⁽¹⁾	Description
R/L1, S/L2	1-Phase Input ⁽²⁾
R/L1, S/L2, T/L3	3-Phase Input
U/T1	To Motor U/T1
V/T2	To Motor V/T2
W/T3	To Motor W/T3
P2, P1	DC Bus Inductor Connection (C Frame drives only.) The C Frame drive is shipped with a jumper between Terminals P2 and P1. Remove this jumper only when a DC Bus Inductor will be connected. Drive will not power up without a jumper or inductor connected.
DC+, DC-	DC Bus Connection
BR+, BR-	Dynamic Brake Resistor Connection
⏏	Safety Ground - PE

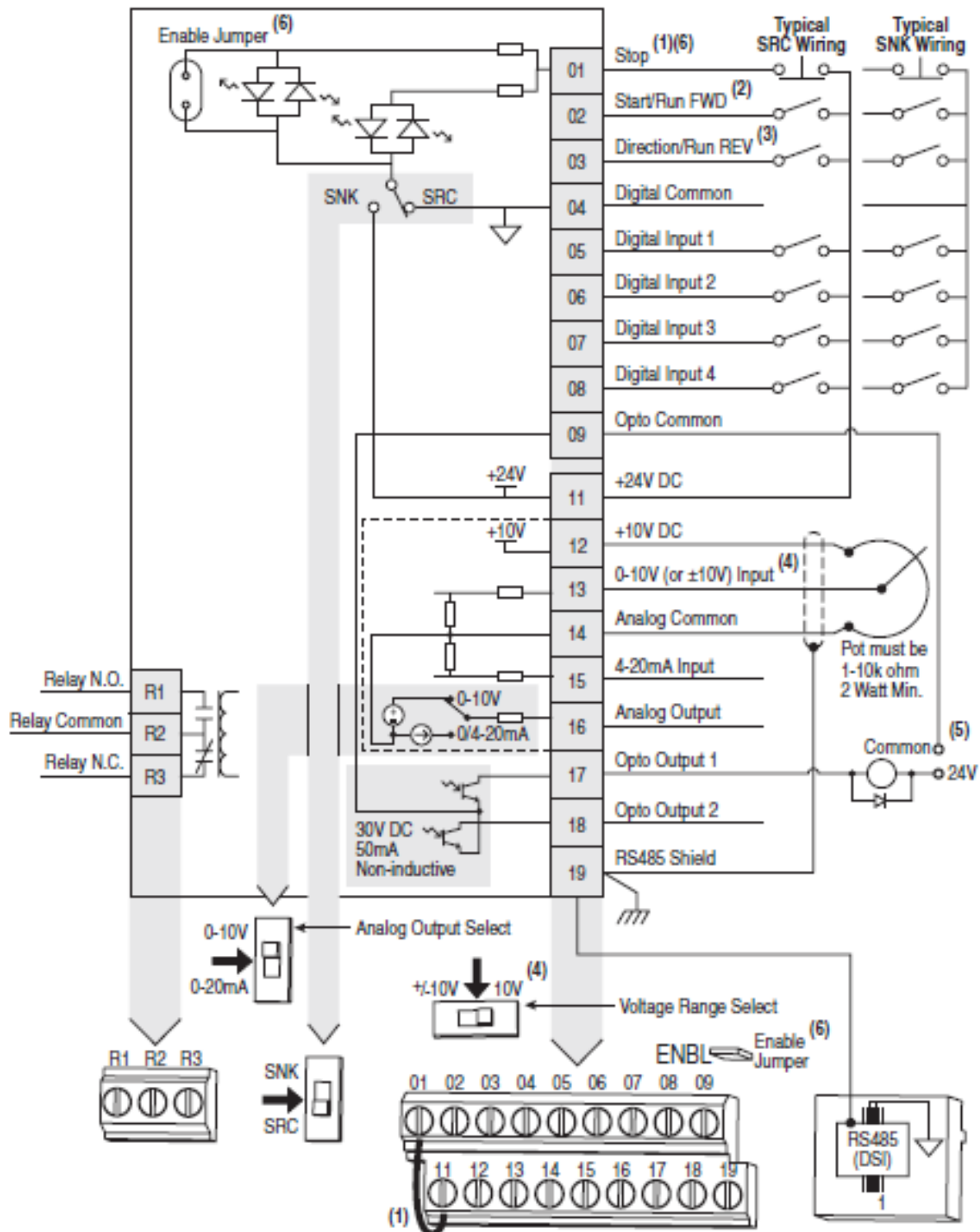


Switch any two motor leads to change forward direction.



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PowerFlex 40p I/O Control Diagram



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PowerFlex 40p Encoder Interface

The PowerFlex 40p supports the connection of a Quadrature Encoder. This provides a simple method of position control. The encoder provides rotor speed and direction. Therefore, the encoder must be wired such that the forward direction matches the motor forward direction. The PowerFlex 40P Encoder Interface can source 5 or 12 volt power and accept 5, 12 or 24 volt single ended or differential inputs.

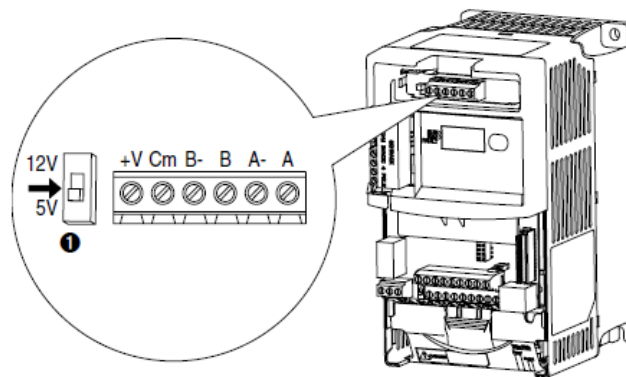


Table 1.J Terminal Description

No.	Signal	Description
+V	5V-12V Power ⁽¹⁾	Internal power source 250 mA (isolated).
Cm	Power Return	
B-	Encoder B (NOT)	Quadrature B input.
B	Encoder B	
A-	Encoder A (NOT)	Single channel, pulse train, or quadrature A input.
A	Encoder A	
1	Output	DIP switch selects 12 or 5 volt power supplied at terminals "+V" and "Cm" for the encoder.

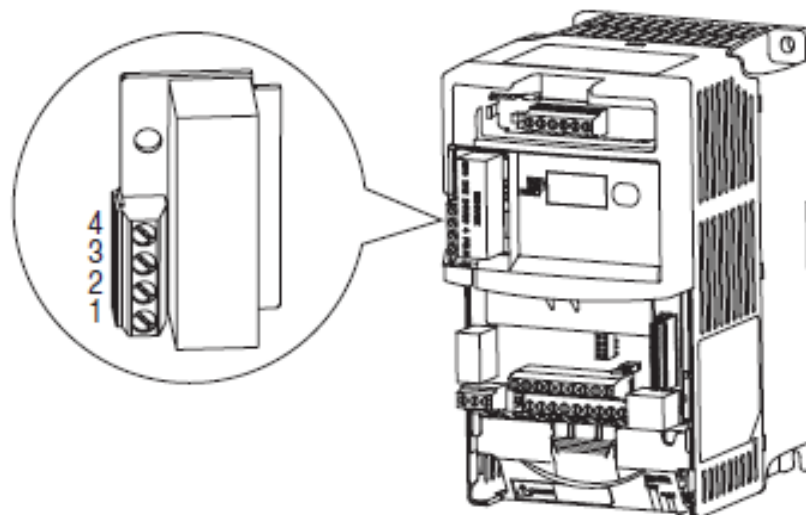
Examples of Quadrature Encoder Connections

I/O	Connection Example	I/O	Connection Example
Encoder Power – Internal Drive Power Internal (drive) 12V DC, 250mA		Encoder Power – External Power Source	
Encoder Signal – Single-Ended, Dual Channel		Encoder Signal – Differential, Dual Channel	

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DriveGuard Safe-Off Option (Series B)

The DriveGuard Safe-Off Option (Series B) board, when used with suitable safety components, provides protection according to EN 954-1:1997; Category 3 for safe off and protection against restart. The PowerFlex safe off option is just one safety control system.

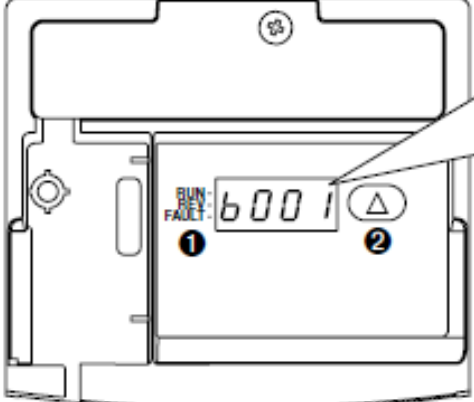


DriveGuard Safe Off Terminal Description

No.	Signal	Description
4	24V Common	Connections for user supplied power to energize coil.
3	+24V DC	
2	Common - N.C.	Normally closed contacts for monitoring relay status. Maximum Resistive Load: 250V AC / 30V DC / 50 VA / 60 Watts Maximum Inductive Load: 250V AC / 30V DC / 25 VA / 30 Watts
1	Monitor - N.C.	

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Display/Fault Reset



Menu	Description
b	Basic Display Group (View Only) Commonly viewed drive operating conditions.
d	Advanced Display Group (View Only) Consists of advanced drive operating conditions.
F	Fault Designator Consists of list of codes for specific fault conditions. Displayed only when fault is present.

No.	LED	LED State	Description
1	Run Status	Steady Red	Indicates drive is running.
		Flashing Red	Drive has been commanded to change direction.
	Direction Status	Steady Red	Indicates drive is running in reverse direction.
		Flashing Red	Drive has been commanded to change direction and motor is decelerating to zero.
	Fault Status	Flashing Red	Indicates drive is faulted.

No.	Key	Name	Description
2		Up Arrow	Scroll: Press and release to scroll through user-selectable display parameters. Refer to E201 [LED Display Opt] for options. Reset: Press and hold for three seconds to clear active fault.

Drive Programming Tools

For additional drive programming and control, a HIM or PC programming tools (DriveExplorer or DriveTools SP) should be used.

Description	Catalog Number
Serial Converter Module	22-SCM-232
DriveExplorer Software ⁽¹⁾	9306-4EXP02ENE
DriveTools SP Software ⁽¹⁾	9303-4DTS01ENE
Remote Panel Mount, LCD Display	22-HIM-C2S
Remote Handheld, LCD Display	22-HIM-A3

⁽¹⁾ Requires a Serial Converter Module.

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Programming and Parameters

This section provides a partial listing and brief description of the PowerFlex 40P parameters. Parameters are programmed, viewed and/or edited using either a HIM or PC programming tools (DriveExplorer™ or DriveTools SP™ software), a personal computer and a serial converter module.

- The 10 most common application parameters are contained in the **Basic Program Group**.
- Common parameters shared with PowerFlex 40 family are in the **Advanced Program Group**.
- Additional position control and fibers parameters are located in the **Enhanced Program Group**.

(Full parameter explanations are in the PowerFlex 40p Appendix, in this book)

To configure a drive to operate in a specific way, drive parameters may have to be set. Three types of parameters exist:

- **ENUM:** ENUM parameters allow a selection from 2 or more items. Each item is represented by a number.
- **Numeric Parameters:** These parameters have a single numerical value (i.e. 0.1 Volts).
- **Bit Parameters;** Bit parameters have four individual bits associated with features or conditions. If the bit is 0, the feature is off or the condition is false. If the bit is 1, the feature is on or the condition is true. Some parameters are marked as follows.



= Stop drive before changing this parameter.



= 32 bit parameter. Parameters marked 32 bit will have two parameter numbers when using RS485 communications and programming software.

(Full parameter explanations are in the PowerFlex 40p Appendix, in this book)

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Parameter Organization (Organized by Group)

Refer to the PowerFlex 40p Appendix in this book for a complete listing and description of each programming parameter.

Basic Display Group		Output Voltage	b004	Contrl In Status	b013	Analog In 4-20mA	b021
		DC Bus Voltage	b005	Dig In Status	b014	Output Power	b022
		Drive Status	b006	Comm Status	b015	Output Powr Fctr	b023
		Fault 1 Code	b007	Control SW Ver	b016	Drive Temp	b024
		Fault 2 Code	b008	Drive Type	b017	Counter Status	b025
		Fault 3 Code	b009	Elapsed Run Time	b018	Timer Status	b026
Output Freq	b001	Process Display	b010	Testpoint Data	b019	Stp Logic Status	b028
Commanded Freq	b002	Control Source	b012	Analog In 0-10V	b020	Torque Current	b029
Output Current	b003						
Basic Program Group		Motor NP Volts	P031	Minimum Freq	P034	Accel Time 1	P039
		Motor NP Hertz	P032	Maximum Freq	P035	Decel Time 1	P040
		Motor OL Current	P033	Start Source	P036	Reset To Defaults	P041
				Stop Mode	P037	Voltage Class	P042
				Speed Reference	P038	Motor OL Ret	P043
Advanced Program Group		Jog Frequency	A078	Anlg Out Setpt	A109	Stp Logic 0	A140
		Jog Accel/Decel	A079	Anlg In 0-10V Lo	A110	Stp Logic 1	A141
		DC Brake Time	A080	Anlg In 0-10V Hi	A111	Stp Logic 2	A142
		DC Brake Level	A081	Anlg In4-20mA Lo	A112	Stp Logic 3	A143
		DB Resistor Sel	A082	Anlg In4-20mA Hi	A113	Stp Logic 4	A144
		S Curve %	A083	Slip Hertz @ FLA	A114	Stp Logic 5	A145
Digital In1 Sel	A051	Boost Select	A084	Process Time Lo	A115	Stp Logic 6	A146
Digital In2 Sel	A052	Start Boost	A085	Process Time Hi	A116	Stp Logic 7	A147
Digital In3 Sel	A053	Break Voltage	A086	Bus Reg Mode	A117	Stp Logic Time 0	A150
Digital In4 Sel	A054	Break Frequency	A087	Current Limit 2	A118	Stp Logic Time 1	A151
Relay Out Sel	A055	Maximum Voltage	A088	Skip Frequency	A119	Stp Logic Time 2	A152
Relay Out Level	A056	Current Limit 1	A089	Skip Freq Band	A120	Stp Logic Time 3	A153
Opto Out1 Sel	A058	Motor OL Select	A090	Stall Fault Time	A121	Stp Logic Time 4	A154
Opto Out1 Level	A059	PWM Frequency	A091	Analog In Loss	A122	Stp Logic Time 5	A155
Opto Out2 Sel	A061	Auto Rstrt Tries	A092	10V Bipolar Enbl	A123	Stp Logic Time 6	A156
Opto Out2 Level	A062	Auto Rstrt Delay	A093	Var PWM Disable	A124	Stp Logic Time 7	A157
Opto Out Logic	A064	Start At PowerUp	A094	Torque Perf Mode	A125	EM Brk Off Delay	A160
Analog Out Sel	A065	Reverse Disable	A095	Motor NP FLA	A126	EM Brk On Delay	A161
Analog Out High	A066	Flying Start En	A096	Autotune	A127	MOP Reset Sel	A162
Accel Time 2	A067	Compensation	A097	IR Voltage Drop	A128	DB Threshold	A163
Decel Time 2	A068	SW Current Trip	A098	Flux Current Ref	A129		
Internal Freq	A069	Process Factor	A099	PID Trim Hi	A130		
Preset Freq 0	A070	Fault Clear	A100	PID Trim Lo	A131		
Preset Freq 1	A071	Program Lock	A101	PID Ref Sel	A132		
Preset Freq 2	A072	Testpoint Sel	A102	PID Feedback Sel	A133		
Preset Freq 3	A073	Comm Data Rate	A103	PID Prop Gain	A134		
Preset Freq 4	A074	Comm Node Addr	A104	PID Integ Time	A135		
Preset Freq 5	A075	Comm Loss Action	A105	PID Diff Rate	A136		
Preset Freq 6	A076	Comm Loss Time	A106	PID Setpoint	A137		
Preset Freq 7	A077	Comm Format	A107	PID Deadband	A138		
		Language	A108	PID Preload	A139		
Enhanced Program Group		Comm Write Mode	E207	Motor Fdbk Type	E216	Step Units 0	E230
		Power Loss Mode	E208	Motor NP Poles	E217	Step Units 1	E232
		Half Bus Enable	E209	Encoder PPR	E218	Step Units 2	E234
		Max Traverse	E210	Pulse In Scale	E219	Step Units 3	E236
		Traverse Inc	E211	Ki Speed Loop	E220	Step Units 4	E238
		Traverse Dec	E212	Kp Speed Loop	E221	Step Units 5	E240
LED Display Opt	E201	P Jump	E213	Positioning Mode	E222	Step Units 6	E242
Digital Term 3	E202	Find Home Freq	E214	Find Home Dir	E223	Step Units 7	E244
Accel Time 3	E203	Speed Ratio	E215	Encoder Pos Tol	E224	Pos Reg Filter	E246
Decel Time 3	E204			Counts Per Unit	E225	Pos Reg Gain	E247
Accel Time 4	E205				E226	Enh Control Word	E248
Decel Time 4	E206					Cmd Stat Select	E249
Advanced Display Group		Drive Status 2	d301	Slip Hz Meter	d303	Units Traveled H	d308
		Fibers Status	d302	Speed Feedback	d304	Units Traveled L	d309
				Encoder Speed	d306		

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Introduction to the PowerFlex 70

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Revision History		
Date	Revisor	Changes
7-15-2015	M. Chapman	Removed page 27 (Menu Flow Chart

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Introduction to the PowerFlex 70

The PowerFlex 70 can be programmed for either **Volts per Hertz** or **Sensorless Vector Control** to cover a wide range of applications from fans to extruders.

Optional internal communication modules provide fast and efficient control and/or data exchange with host controllers over popular interfaces.

These interfaces include:

- DeviceNet
- EtherNet
- ControlNet
- Remote I/O Serial Communications

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Catalog Number Explanation

1-3	4	5-7	8	9	10	11	12	13	14	15	16
20A	B	2P2	A	3	A	Y	Y	N	N	C	0
<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>l</i>

<i>a</i>		
Drive		
Code	Type	
20A	PowerFlex 70	

<i>b</i>		
Voltage Rating		
Code	Voltage	Ph.
B	240V AC	3
C	400V AC	3
D	480V AC	3
E	600V AC	3

<i>c1</i>			
ND Rating			
208V, 60 Hz Input			
Code	Amps	kW (Hp)	Frame
2P2	2.5	0.37 (0.5)	A
4P2	4.8	0.75 (1.0)	A
6P8	7.8	1.5 (2.0)	B
9P6	11	2.2 (3.0)	B
015	17.5	4.0 (5.0)	C
022	25.3	5.5 (7.5)	D
028	32.2	7.5 (10)	D
042	43	11 (15)	D
054	62.1	15 (20)	E
070	78.2	18.5 (25)	E

<i>c2</i>			
ND Rating			
240V, 60 Hz Input			
Code	Amps	kW (Hp)	Frame
2P2	2.2	0.37 (0.5)	A
4P2	4.2	0.75 (1.0)	A
6P8	6.8	1.5 (2.0)	B
9P6	9.6	2.2 (3.0)	B
015	15.3	4.0 (5.0)	C
022	22	5.5 (7.5)	D
028	28	7.5 (10)	D
042	42	11 (15)	D
054	54	15 (20)	E
070	70	18.5 (25)	E

<i>c3</i>			
ND Rating			
400V, 50 Hz Input			
Code	Amps	kW (Hp)	Frame
1P3	1.3	0.37 (0.5)	A
2P1	2.1	0.75 (1.0)	A
3P5	3.5	1.5 (2.0)	A
5P0	5.0	2.2 (3.0)	B
8P7	8.7	4.0 (5.0)	B
011	11.5	5.5 (7.5)	C
015	15.4	7.5 (10)	C
022	22	11 (15)	D
030	30	15 (20)	D
037	37	18.5 (25)	D
043	43	22 (30)	D
060	60	30 (40)	E
072	72	37 (50)	E

<i>c4</i>			
ND Rating			
480V, 60 Hz Input			
Code	Amps	kW (Hp)	Frame
1P1	1.1	0.37 (0.5)	A
2P1	2.1	0.75 (1.0)	A
3P4	3.4	1.5 (2.0)	A
5P0	5.0	2.2 (3.0)	B
8P0	8.0	3.7 (5.0)	B
011	11	5.5 (7.5)	C
014	14	7.5 (10)	C
022	22	11 (15)	D
027	27	15 (20)	D
034	34	18.5 (25)	D
040	40	22 (30)	D
052	52	30 (40)	E
065	65	37 (50)	E

<i>c5</i>			
ND Rating			
600V, 60 Hz Input *			
Code	Amps	kW (Hp)	Frame
0P9	0.9	0.37 (0.5)	A
1P7	1.7	0.75 (1.0)	A
2P7	2.7	1.5 (2.0)	A
3P9	3.9	2.2 (3.0)	B
6P1	6.1	4.0 (5.0)	B
9P0	9.0	5.5 (7.5)	C
011	11	7.5 (10)	C
017	17	11 (15)	D
022	22	15 (20)	D
027	27	18.5 (25)	D
032	32	22 (30)	D
041	41	30 (40)	E
052	52	37 (50)	E

* CE certification testing has not been performed on 600V class drives.

<i>d</i>	
Enclosure	
Code	Enclosure
A	Panel Mount - IP 20, NEMA/UL Type 1
C	Wall/Machine Mount = IP66, NEMA/UL Type 4X/12 for indoor use only
F	Flange Mount - Front Chassis = IP 20, NEMA/UL Type 1; Rear Heatsink = IP66, NEMA/UL Type 4X/12 for indoor/outdoor use
G	Wall/Machine Mount - IP54, NEMA/UL Type 12 *
L	Flange Mount with Conformal Coat
M	Panel Mount with Conformal Coat

* Only available on Frame E.

<i>e</i>	
HIM	
Code	Interface Module
0	Blank Cover
3	Full Numeric LCD
5	Prog. Only LCD
8	Wireless Interface Module - IP66, NEMA/UL Type 4X/12 Only

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Position Number											
1-3	4	5-7	8	9	10	11	12	13	14	15	16
20A	B	2P2	A	3	A	Y	Y	N	N	C	0
<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>l</i>

f

Documentation	
Code	Type
A	Manual
N	No Manual

g

Brake IGBT	
Code	w/Brake IGBT
Y	Yes

h

Internal Brake Resistor	
Code	w/Resistor
Y	Yes
N	No

i

Emission Class	
Code	Rating
A	Filtered* A* & B Frames (Optional) C, D, & E Frames (Standard)
N	Not Filtered* A & B Frames (Optional) C, D, & E Frames

* 600V Frames A through D available only without filter (Cat. Code N). 600V Frame E available only with filter (Cat. Code A).

* Increases size to Frame B.

j

Comm Slot	
Code	Network Type
B	BACnet
C	ControlNet (Coax)
D	DeviceNet
E	EtherNet/IP
R	Remote I/O
S	RS485 DF1
N	None

k

Control & I/O		
Code	Control	Safe-Off
N	Standard	N/A
C	Enhanced	No
G*	Enhanced	Yes

* Not available as factory installed option for 600V ratings.

l

Feedback	
Code	Feedback
N	NA - Standard Control
0	No Feedback - Enhanced Control
1	5V/12V Encoder w/Enhanced Control

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Power Wiring

The PowerFlex 70 has the built in protective features to help simplify installation:

- Ground fault protection during start-up and running helps ensure reliability
- Electronic motor overload protection increases motor life
- Removable MOV to ground and common mode capacitors to ground ensure compatibility with ungrounded systems.

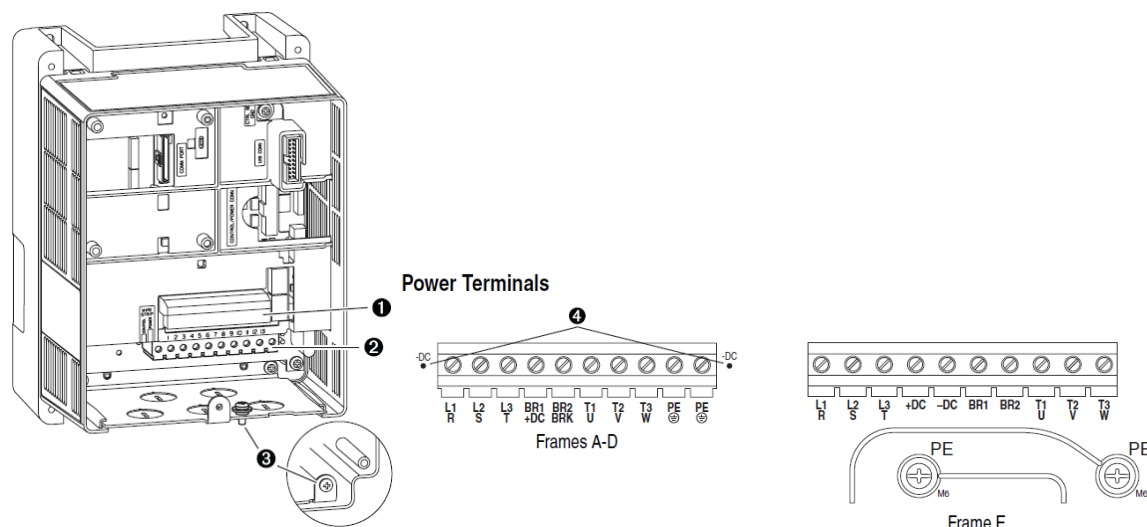
Component Locations and Power Terminals

Terminal Block Specifications

No.	Name	Description	Frame	Wire Size Range ⁽¹⁾		Torque	
				Maximum	Minimum	Maximum	Recommended
①	I/O Terminal Block	Signal and control connections	All	1.5 mm ² (16 AWG)	0.05 mm ² (30 AWG)	0.55 N-m (4.9 lb.-in.)	0.5 N-m (4.4 lb.-in.)
②	Power Terminal Block	Input power and motor connections	A, B, & C	3.5 mm ² (12 AWG)	0.3 mm ² (22 AWG)	0.66 N-m (5.5 lb.-in.)	0.6 N-m (5 lb.-in.)
			D	8.4 mm ² (8 AWG)	0.8 mm ² (18 AWG)	1.7 N-m (15 lb.-in.)	1.4 N-m (12 lb.-in.)
			E	25.0 mm ² (3 AWG)	2.5 mm ² (14 AWG)	2.71 N-m (24 lb.-in.)	2.71 N-m (24 lb.-in.)
③	SHLD terminal	Terminating point for wiring shields	All	—	—	1.6 N-m (14 lb.-in.)	1.6 N-m (14 lb.-in.)

⁽¹⁾ Maximum/minimum sizes that the terminal block will accept - these are not recommendations.

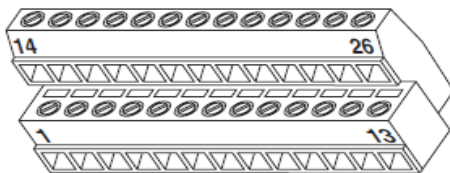
Terminal	Description	Notes
BR1	DC Brake (+)	DB Resistor Connection - Important: Do not connect both an internal and external DB resistor at the same time. This may violate the minimum allowed DB resistance and cause drive damage.
BR2	DC Brake (-)	
+DC	DC Bus (+)	④ Test point on Frames A-D located to the left or right of the Power Terminal Block. Frame E has a dedicated terminal.
-DC	DC Bus (-)	
PE	PE Ground	
U, V, W	U (T1), V (T2), W (T3)	To Motor
R, S, T	R (L1), S (L2), T (L3)	AC Line Input Power



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Control Wiring

Control Terminals

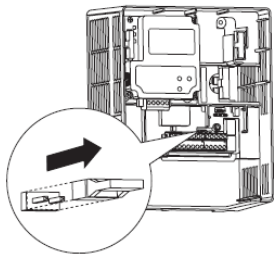


No.	Signal	Factory Default	Description	Related Param.
1	Digital In 1	Stop – CF (CF = Clear Fault)	11.2 mA @ 24V DC 19.2V minimum on state	361 - 366
2	Digital In 2	Start	3.2V maximum off state	
3	Digital In 3	Auto/Man	Important: Use only 24V DC, not suitable for 115V AC circuitry. ⁽³⁾ Inputs can be wired as sink or source.	
4	Digital In 4	Speed Sel 1		
5	Digital In 5	Speed Sel 2		
6	Digital In 6	Speed Sel 3		
7	24V Common	–	Drive supplied power for Digital In1-6 inputs only.	
8	Digital In Common	–	Not intended for use on circuits outside of the drive.	
9	+24V DC	–	See examples beginning on page 20 . 150mA maximum load.	
10	+10V Pot Reference	–	2 k ohm minimum load.	
11	Digital Out 1 – N.O. ⁽¹⁾	NOT Fault	<u>Max Resistive Load</u> 250V AC / 30V DC 50 VA / 60 Watts	380 - 387
12	Digital Out 1 Common		<u>Max Inductive Load</u> 250V AC / 30V DC 25 VA / 30 Watts	
13	Digital Out 1 – N.C. ⁽¹⁾	Fault	<u>Minimum DC Load</u> 10 µA, 10 mV DC	
14	Analog In 1 (– Volts)	⁽²⁾	Non-isolated, 0 to +10V, 10 bit, 100k ohm input impedance. ⁽⁴⁾	320 - 327
15	Analog In 1 (+ Volts)	Voltage – Reads value at 14 & 15	Non-isolated, 4-20mA, 10 bit, 100 ohm input impedance. ⁽⁴⁾	
16	Analog In 1 (– Current)			
17	Analog In 1 (+ Current)			
18	Analog In 2 (– Volts)	⁽²⁾	Isolated, bipolar, differential, 0 to +10V unipolar (10 bit) or ±10V bipolar (10 bit & sign), 100k ohm input impedance. ⁽⁵⁾	
19	Analog In 2 (+ Volts)	Voltage – Reads value at 18 & 19	Isolated, 4-20mA, 10 bit & sign, 100 ohm input impedance. ⁽⁵⁾	
20	Analog In 2 (– Current)			
21	Analog In 2 (+ Current)			
22	10V Pot Common Analog Out (– Volts) Analog Out (– Current)	⁽²⁾ Output Freq	0 to +10V, 10 bit, 10k ohm (2k ohm minimum) load. 0 to 20mA, 10 bit, 400 ohm maximum load. ⁽⁶⁾ Referenced to chassis ground.	340 - 344
23	Analog Out (+ Volts) Analog Out (+ Current)		Common if internal 10V supply (terminal 10) is used.	
24	Digital Out 2 – N.O. ⁽¹⁾	Run	See description at No.s 11-13.	380 - 387
25	Digital Out 2 Common			
26	Digital Out 2 – N.C. ⁽¹⁾	NOT Run		

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Hardware Enable Circuitry (Enhanced Control Only)

By default, the user can program a digital input as an Enable input. The status of this input is *interpreted by drive software*. If the application requires the drive to be disabled *without* software interpretation, a hardware enable configuration can be utilized. This is done by removing the enable jumper (ENBL JMP) and wiring the enable input to “Digital In 6”.



1. Remove drive cover.
2. Locate and remove the “Enable Jumper on the Main Control Board.
3. Wire Enable to “Digital In 6.”
4. Verify that 366 [Digital In6 Sel] is set to option 1 “Enable.”

Safe Off Board Terminal Block (Enhanced Control Only)

No.	Signal	Description
1	Monitor - N.C.	Normally closed contacts for monitoring relay status.
2	Common - N.C.	Maximum Resistive Load: 250V ac / 30V dc / 50 VA / 60 Watts Maximum Inductive Load: 250V ac / 30V dc / 25 VA / 30 Watts
3	+24V dc	Connections for user supplied power to energize coil.
4	24V Common	

Connection Examples

For detailed connection examples refer to the DriveGuard™ Safe-Off Option (Series B) for PowerFlex® 40P and PowerFlex® 70 AC Drives *User Manual*, publication PFLEX-UM003.

Encoder Interface Terminal Block (Enhanced Control Only)

No.	Signal	Description	Jumper Settings
1	5-12V Power ⁽¹⁾	Internal power source 250 mA (isolated)	Receive Voltage 12V = 5V 12V = 5V Send Voltage
2	Power Return		
3	Encoder B (NOT)	Single channel or quadrature B input.	
4	Encoder B		
5	Encoder A (NOT)	Single channel or quadrature A input.	
6	Encoder A		

⁽¹⁾ Jumper selectable +5/12V is available on 20A-ENC-1 Encoder Boards.

Sample Encoder Wiring

I/O	Connection Example	I/O	Connection Example
Encoder Power – Internal Drive Power Internal (drive) 12V DC, 250mA		Encoder Power – External Power Source	
Encoder Signal – Single-Ended, Dual Channel		Encoder Signal – Differential, Dual Channel	

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Speed Reference Control

Operation

The output frequency of the drive is controlled, in part, by the speed command or speed reference given to it. This reference can come from a variety of sources including:

- HIM (local or remote)
- Analog Input
- Preset Speed Parameter
- Jog Speed Parameter
- Communications Adapter
- Process PI Loop
- Digital Input MOP

Polarity

The reference can be selected as either unipolar or bipolar. Unipolar is limited to positive values and supplies only the speed reference. Bipolar supplies both the speed reference AND the direction command: + signals = forward direction and – signals = reverse direction.

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Speed Reference Selection Chart

PowerFlex 70 and 700 Default Auto Speed Reference Options								
Digital Inputs Status			Speed Reference Selections	Parameter	Default Setting	Default Selection	Default Setting	
Digital Input 6	Digital Input 5	Digital Input 4						
0	0	0	Speed Ref. A	90	2	Analog In "2"	0-10Vdc	
0	0	1	Speed Ref. B	93	11	Preset Speed 1	5 Hz	
0	1	0	Preset Speed 2	102	10	10 Hz		
0	1	1	Preset Speed 3	103	20	20 Hz		
1	0	0	Preset Speed 4	104	30	30 Hz		
1	0	1	Preset Speed 5	105	40	40 Hz		
1	1	0	Preset Speed 6	106	50	50 Hz		
1	1	1	Preset Speed 7	107	60	60 Hz		

Other Default Parameter Settings

Default Digital Inputs Selections				Default Digital Output Selections		
Dig In 1 Sel	361	4	Stop	Dig Out 1 Sel (380)	1	Fault
Dig In 2 Sel	362	5	Start	Dig Out 2 Sel (384)	4	Run
Dig In 3 Sel	363	18	Auto/Man	Dig Out 3 700 Only (388)	4	Run
Dig In 4 Sel	364	15	Speed Sel 1	Default Analog Output Selections		
Dig In 5 Sel	365	16	Speed Sel 2	Analog Out 1 Sel (342)	0	Output Frequency
Dig In 6 Sel	366	17	Speed Sel 3	Analog Out 2 Sel 700 Only (345)	0	Output Frequency

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Drive Status Indicators



#	Name	Color	State	Description
1	STS (Status)	Green	Flashing	Drive ready, but not running and no faults are present.
			Steady	Drive running, no faults are present.
		Yellow See page 4-8	Flashing, Drive Stopped	An inhibit condition exists, the drive cannot be started. Check parameter 214 [Start Inhibits].
			Flashing, Drive Running	An intermittent type 1 alarm condition is occurring. Check parameter 211 [Drive Alarm 1].
			Steady, Drive Running	A continuous type 1 alarm condition exists. Check parameter 211 [Drive Alarm 1].
		Red See page 4-3	Flashing	A fault has occurred. Check parameters 243, 245, 247 and 249
			Steady	A non-resetable fault has occurred.
2	PORT	Refer to the Communication Adapter User Manual.		Status of DPI port internal communications (if present).
	MOD			Status of communications module (when installed).
	NET A			Status of network (if connected).
	NET B			Status of secondary network (if connected).

(Refer to the User Manual for Detailed explanations)

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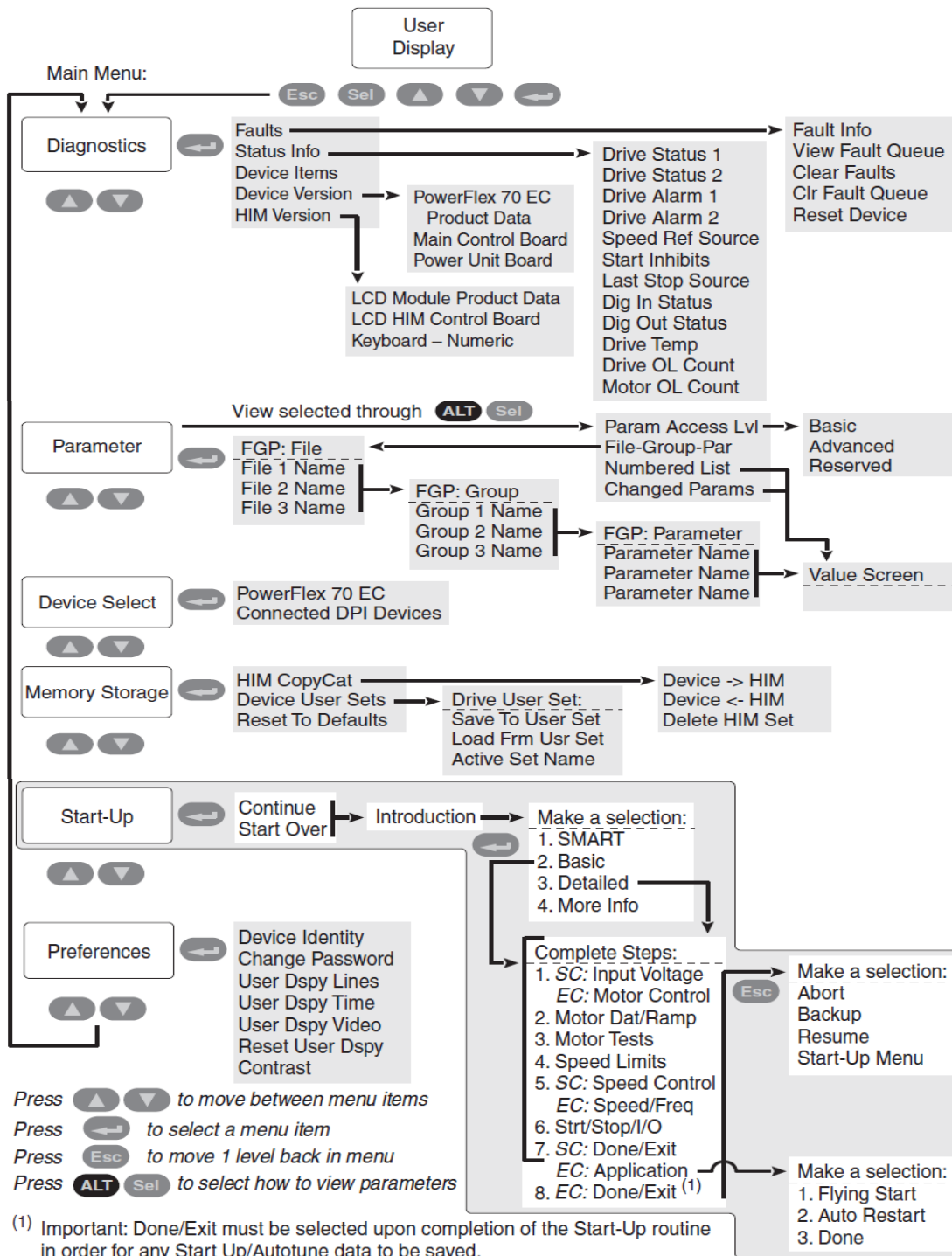
LCD Display Elements

Display	Description
F-> Power Loss Auto 0.0 Hz Main Menu: Diagnostics Parameter Device Select	Direction Drive Status Alarm Auto/Man Information Commanded or Output Frequency Programming / Monitoring / Troubleshooting

ALT Key and then ...	Performs this function ...	HIM Type
S.M.A.R.T.	Displays the S.M.A.R.T. screen.	LCD only
Log In/Out	Log in to change parameter settings. Log out to protect parameter settings. Change a password.	LED only
View	Allows the selection of how parameters will be viewed or detailed information about a parameter or component.	LCD only
Device	Select a connected adapter for editing.	LED only
ALT Lang	Displays the language selection screen.	LCD only
Auto / Man	Switches between Auto and Manual Modes.	LCD and LED
Remove	Allows HIM removal without causing a fault if the HIM is not the last controlling device and does not have Manual control of the drive.	LCD and LED
Exp	Allows value to be entered as an exponent. (Not available on PowerFlex 70.)	LCD only
Param #	Allows entry of a parameter number for viewing/editing.	LCD only

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Menu Structure



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Programming and Parameters

To configure a drive to operate in a specific way, drive parameters may have to be set.

Three types of parameters exist:

- **ENUM Parameters**

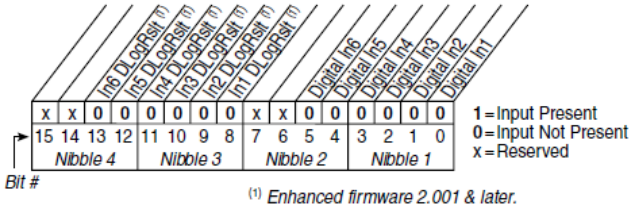
ENUM parameters allow a selection from 2 or more items. The LCD HIM will display a text message for each item. The LED HIM will display a number for each item.

- **Bit Parameters**

Bit parameters have individual bits associated with features or conditions. If the bit is 0, the feature is off or the condition is false. If the bit is 1, the feature is on or the condition is true.

- **Numeric Parameters**

These parameters have a single numerical value (i.e. 0.1 Volts). The example below and on the following page shows how each parameter type is presented in the drive user manual.

①	②	③	④	⑤	⑥
File	Group	No.	Parameter Name & Description	Values	Related
UTILITY (File E)	Drive ...	198	[Load Frm Usr Set]  Loads a previously saved set of parameter values from a selected user set location in drive nonvolatile memory to active drive memory.	Default: 0 "Ready" Options: 0 "Ready" 1 "User Set 1" 2 "User Set 2" 3 "User Set 3"	199 
	Diagnostics	216	[Dig In Status] Status of the digital inputs. 		
MOTOR ...	Torq ...	059	EC [SV Boost Filter] Sets the amount of filtering used to boost voltage during Sensorless Vector operation.	Default: 500 Min/Max: 0/32767 Units: 1	

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No.	Description																		
①	File – Lists the major parameter file category.																		
②	Group – Lists the parameter group within a file.																		
③	No. – Parameter number.  = Parameter value can not be changed until drive is stopped.  = 32 bit parameter.  = 32 bit parameter in Enhanced Control drives only.  = Parameter only displayed when [Motor Cntl Sel] is set to “4.”																		
④	Parameter Name & Description – Parameter name as it appears on an LCD HIM, with a brief description of the parameters function.  = This parameter is specific to Standard Control drives.  = This parameter will only be available with Enhanced Control drives.																		
⑤	Values – Defines the various operating characteristics of the parameter. Three types exist. <table border="1"> <tr> <td rowspan="2">ENUM</td><td>Default:</td><td>Lists the value assigned at the factory. “Read Only” = no default.</td></tr> <tr> <td>Options:</td><td>Displays the programming selections available.</td></tr> <tr> <td>Bit</td><td>Bit #:</td><td>Lists the bit place holder and definition for each bit.</td></tr> <tr> <td rowspan="4">Numeric</td><td>Default:</td><td>Lists the value assigned at the factory. “Read Only” = no default.</td></tr> <tr> <td>Min/Max:</td><td>The range (lowest and highest setting) possible for the parameter.</td></tr> <tr> <td>Units:</td><td>Unit of measure and resolution as shown on the LCD HIM.</td></tr> <tr> <td colspan="2"> Important: Some parameters will have two unit values: - Analog inputs can be set for current or voltage with 320 [Anlg In Config]. - Values that pertain to Enhanced Control drives only will be indicated by “. Important: When sending values through DPI ports, simply remove the decimal point to arrive at the correct value (i.e. to send “5.00 Hz,” use “500”). </td></tr> </table>		ENUM	Default:	Lists the value assigned at the factory. “Read Only” = no default.	Options:	Displays the programming selections available.	Bit	Bit #:	Lists the bit place holder and definition for each bit.	Numeric	Default:	Lists the value assigned at the factory. “Read Only” = no default.	Min/Max:	The range (lowest and highest setting) possible for the parameter.	Units:	Unit of measure and resolution as shown on the LCD HIM.	Important: Some parameters will have two unit values: - Analog inputs can be set for current or voltage with 320 [Anlg In Config]. - Values that pertain to Enhanced Control drives only will be indicated by “. Important: When sending values through DPI ports, simply remove the decimal point to arrive at the correct value (i.e. to send “5.00 Hz,” use “500”).	
ENUM	Default:	Lists the value assigned at the factory. “Read Only” = no default.																	
	Options:	Displays the programming selections available.																	
Bit	Bit #:	Lists the bit place holder and definition for each bit.																	
Numeric	Default:	Lists the value assigned at the factory. “Read Only” = no default.																	
	Min/Max:	The range (lowest and highest setting) possible for the parameter.																	
	Units:	Unit of measure and resolution as shown on the LCD HIM.																	
	Important: Some parameters will have two unit values: - Analog inputs can be set for current or voltage with 320 [Anlg In Config]. - Values that pertain to Enhanced Control drives only will be indicated by “. Important: When sending values through DPI ports, simply remove the decimal point to arrive at the correct value (i.e. to send “5.00 Hz,” use “500”).																		
⑥	Related – Lists parameters (if any) that interact with the selected parameter. The symbol “” indicates that additional parameter information is available in Appendix C.																		

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Parameter Organization

The LCD HIM displays parameters in a File-Group-Parameter or Numbered List view order. To switch display mode, access the Main Menu, press ALT, then Sel while cursor is on the parameter selection. In addition, using [Param Access Lvl], the user has the option to display the full parameter set (Advanced), commonly used parameters (Basic) or diagnostic/advanced tuning parameters (Reserved). To simplify programming, the displayed parameters will change according to the selection made with [Motor Cntl Sel]. For example, if “FVC Vector” is selected, the parameters associated solely with other operations such as Volts per Hertz or Sensorless Vector will be hidden.

File-Group-Parameter Order

This simplifies programming by grouping parameters that are used for similar functions. The parameters are organized into files. Each file is divided into groups, and each parameter is an element in a group. By default, the LCD HIM displays parameters by File-Group-Parameter view.

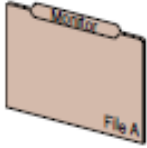
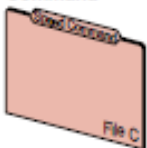
Numbered List View

All parameters are in numerical order.

Advanced Parameter View – Enhanced Control

Parameter 196 [Param Access Lvl] set to option 1 “Advanced.”

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File	Group	Parameters					
 File A	Metering	Output Freq	001	Output Powr Fctr	008	Torque Estimate	015 ^{3.x}
		Commanded Freq	002	Elapsed MWh	009	Analog In1 Value	016
		Output Current	003	Elapsed Run Time	010	Analog In2 Value	017
		Torque Current	004	MOP Frequency	011	Ramped Speed	022
		Flux Current	005	DC Bus Voltage	012	Speed Reference	023
		Output Voltage	006	DC Bus Memory	013	Commanded Torque**	024
		Output Power	007	Elapsed kWh	014	Speed Feedback	025
	Drive Data	Rated kW	026	Rated Amps	028		
		Rated Volts	027	Control SW Ver	029		
 File B	Motor Data	Motor Type	040	Motor NP RPM	044	Motor OL Factor	048
		Motor NP Volts	041	Motor NP Power	045	Motor Poles	049
		Motor NP FLA	042	Mtr NP Pwr Units	046	Motor OL Mode	050 ^{3.x}
		Motor NP Hertz	043	Motor OL Hertz	047		
	Torq Attributes	Motor Cntl Sel	053	Autotune	061	Torque Ref A Hi**	428
		Maximum Voltage	054	IR Voltage Drop	062	Torque Ref A Lo**	429
		Maximum Freq	055	Flux Current Ref	063	Torque Setpoint1**	435
		Compensation	056	Ixo Voltage Drop	064	Pos Torque Limit**	436
		Flux Up Mode	057	Autotune Torque**	066	Neg Torque Limit**	437
		Flux Up Time	058	Inertia Autotune**	067	Control Status**	440
		SV Boost Filter	059	Torque Ref A Sel**	427	Torq Current Ref**	441
	Volts per Hertz	StAcc Boost*	069	Break Voltage*	071		
		Run Boost*	070	Break Frequency*	072		
	Speed Feedback	Motor Fdbk Type	412	Enc Pos Feedback	414	Fdbk Filter Sel**	416
		Encoder PPR	413	Encoder Speed	415	Notch FilterFreq**	419
						Notch Filter K**	420
 File C	Spd Mode & Limits	Feedback Select	080	Skip Frequency 1	084	Skip Freq Band	087
		Minimum Speed	081	Skip Frequency 2	085	Speed/Torque Mod**	088
		Maximum Speed	082	Skip Frequency 3	086	Rev Speed Limit	454
		Overspeed Limit	083				
	Speed References	Speed Ref A Sel	090	Speed Ref B Sel	093	TB Man Ref Sel	096
		Speed Ref A Hi	091	Speed Ref B Hi	094	TB Man Ref Hi	097
		Speed Ref A Lo	092	Speed Ref B Lo	095	TB Man Ref Lo	098
	Discrete Speeds	Jog Speed 1	100	Preset Speed 1-7	101-107	Jog Speed 2	108
	Speed Trim	Trim % Setpoint	116	Trim In Select	117	Trim Hi	119
				Trim Out Select	118	Trim Lo	120
	Slip Comp	Slip RPM @ FLA	121	Slip Comp Gain*	122	Slip RPM Meter	123
	Process PI	PI Configuration	124	PI Lower Limit	131	PI Output Meter	138
		PI Control	125	PI Upper Limit	132	PI BW Filter	139
		PI Reference Sel	126	PI Preload	133	PI Deriv Time	459
		PI Setpoint	127	PI Status	134	PI Reference Hi	460
		PI Feedback Sel	128	PI Ref Meter	135	PI Reference Lo	461
		PI Integral Time	129	PI Fdbk Meter	136	PI Feedback Hi	462
		PI Prop Gain	130	PI Error Meter	137	PI Feedback Lo	463
	Speed Regulator	Ki Speed Loop**	445	Spd Err Filt BW	448 ^{3.x}	Total Inertia**	450
		Kp Speed Loop**	446	Speed Desired BW**	449	Speed Loop Meter**	451
		Kf Speed Loop**	447				
 File D	Restart Modes	Powerup Delay	167	Auto Rstrt Tries	174	Wake Level	180
		Start At PowerUp	168	Auto Rstrt Delay	175	Wake Time	181
		Flying Start En	169	Sleep Wake Mode	178	Sleep Level	182
		Flying StartGain	170	Sleep Wake Ref	179	Sleep Time	183
	Power Loss	Gnd Warn Level	177	Power Loss Time	185	Load loss Time	188
		Power Loss Mode	184	Load Loss Level	187		
	Ramp Rates	Accel Time 1	140	Decel Time 1	142	S Curve %	146
		Accel Time 2	141	Decel Time 2	143		
	Load Limits	Current Lmt Sel	147	Drive OL Mode	150	Regen Power Lim**	153
		Current Lmt Val	148	PWM Frequency	151	Current Rate Lim**	154
		Current Lmt Gain	149	Droop RPM@FLA	152	Shear Pin Time*	189

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File	Group	Parameters					
Dynamic Control <i>continued</i>	Stop/Brake Modes	DB While Stopped	145	DC Brake Time	159	Bus Reg Kp*	164
		Stop/Brk Mode A	155	Bus Reg Ki*	160	Bus Reg Kd*	165
		Stop/Brk Mode B	156	Bus Reg Mode A	161	Flux Braking	166
		DC Brake Lvl Sel	157	Bus Reg Mode B	162		
		DC Brake Level	158	DB Resistor Type	163		
Utility 	Direction Config	Direction Mode	190				
	HIM Ref Config	AutoMan Cnfg	192				
	MOP Config	Save MOP Ref	194	MOP Rate	195		
	Drive Memory	Param Access Lvl	196	Reset Meters	200	Dyn UserSet Cnfg	204
		Reset To Defaults	197	Language	201	Dyn UserSet Sel	205
		Load Frm User Set	198	Voltage Class	202	Dyn UserSet Actv	206
		Save To User Set	199	Drive Checksum	203		
	Diagnostics	Drive Status 1	209	Drive Temp	218	Status 1 @ Fault	227
		Drive Status 2	210	Drive OL Count	219	Status 2 @ Fault	228
		Drive Alarm 1	211	Motor OL Count	220	Alarm 1 @ Fault	229
		Drive Alarm 2	212	Mtr OL Trip Time	221 ^{3.x}	Alarm 2 @ Fault	230
		Speed Ref Source	213	Drive Status 3	222 ^{3.x}	Testpoint 1 Sel	234
		Start Inhibits	214	Status 3 @ Fault	223 ^{3.x}	Testpoint 1 Data	235
		Last Stop Source	215	Fault Frequency	224	Testpoint 2 Sel	236
		Dig In Status	216	Fault Amps	225	Testpoint 2 Data	237
		Dig Out Status	217	Fault Bus Volts	226		
	Faults	Fault Config 1	238	Fault Clear Mode	241	Fault 1-4 Code	243-249
		Fault Clear	240	Power Up Marker	242	Fault 1-4 Time	244-250
	Alarms	Alarm Config 1	259				
	Scaled Blocks	Scale1 In Value	476	Scale2 In Value	482		
		Scale1 In Hi	477	Scale2 In Hi	483		
		Scale1 In Lo	478	Scale2 In Lo	484		
Communication 	Comm Control	DPI Data Rate	270	Drive Ramp Rslt	273	DPI Ref Select	298
		Drive Logic Rslt	271	DPI Port Select	274		
		Drive Ref Rslt	272	DPI Port Value	275		
	Masks & Owners	Logic Mask	276	Fault Clr Mask	283	Reference Owner	292
		Start Mask	277	MOP Mask	284	Accel Owner	293
		Jog Mask	278	Local Mask	285	Decel Owner	294
		Direction Mask	279	Stop Owner	288	Fault Clr Owner	295
		Reference Mask	280	Start Owner	289	MOP Owner	296
		Accel Mask	281	Jog Owner	290	Local Owner	297
		Decel Mask	282	Direction Owner	291		
	Datalinks	Data In A1-D2	300-307	HighRes Ref	308	Data Out A1-D2	310-317
	Security	PortMask Act	595	Write Mask Act	597	Logic Mask Act	598
		Write Mask Cfg	596	Logic Mask	276		
	Inputs & Outputs 	Analog Inputs	Anlg In Config	320	Analog In 1 Lo	323	Analog In 2 Lo
Anlg In Sqr Root			321	Analog In 1 Loss	324	Analog In 2 Loss	327
Analog In 1 Hi			322	Analog In 2 Hi	325		
Analog Outputs		Anlg Out Config	340	Analog Out1 Hi	343	Anlg Out1 Setpt	377
		Anlg Out Absolut	341	Analog Out1 Lo	344		
		Analog Out1 Sel	342	Anlg Out Scale	354		
Digital Inputs		Digital In1-6 Sel	361-366	DigIn DataLogic	411		
Digital Outputs		Dig Out Setpt	379	Dig Out1 OnTime	382	Dig Out2 Level	385
		Digital Out1 Sel	380	Dig Out1 OffTime	383	Dig Out2 OnTime	386
		Dig Out1 Level	381	Digital Out2 Sel	384	Dig Out2 OffTime	387
Applications ^{3.x} 	Fiber Functions ^{3.x}	Fiber Control	620 ^{3.x}	Traverse Inc	623 ^{3.x}	P Jump	626 ^{3.x}
		Fiber Status	621 ^{3.x}	Traverse Dec	624 ^{3.x}		
		Sync Time	622 ^{3.x}	Max Traverse	625 ^{3.x}		

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Troubleshooting

Faults and alarms:

Faults and Alarms

A fault is a condition that stops the drive. There are three fault types.

Type	Fault Description	
①	Auto-Reset Run	When this type of fault occurs, and [Auto Rstrt Tries] (see page 3-35) is set to a value greater than "0," a user-configurable timer, [Auto Rstrt Delay] (see page 3-35) begins. When the timer reaches zero, the drive attempts to automatically reset the fault. If the condition that caused the fault is no longer present, the fault will be reset and the drive will be restarted. Drive must remain in Run state. If Stop is initiated, Restart function is aborted.
②	Non-Resettable	This type of fault normally requires drive or motor repair. The cause of the fault must be corrected before the fault can be cleared. The fault will be reset on power up after repair.
③	User Configurable	These faults can be enabled/disabled to annunciate or ignore a fault condition.

An alarm is a condition that, if left untreated, may stop the drive. There are two alarm types.

Type	Alarm Description	
①	User Configurable	These alarms can be enabled or disabled through [Alarm Config 1] on page 3-49 .
②	Non-Configurable	These alarms are always enabled.

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Manually Clearing Faults

Step	Key(s)
1. Press Esc to acknowledge the fault. The fault information will be removed so that you can use the HIM.	
2. Address the condition that caused the fault. The cause must be corrected before the fault can be cleared.	
3. After corrective action has been taken, clear the fault by one of these methods. • Press Stop • Cycle drive power • Set parameter 240 [Fault Clear] to "1." • "Clear Faults" on the HIM Diagnostic menu.	

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Introduction to the PowerFlex 700

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Revision History		
Date	Revisor	Changes
7-15-2015	M. Chapman	Added Opto-Coupler to Generic AC Drive Digital Input Picture
7-15-2015	M. Chapman	Removed Startup and Smart Start Information

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Introduction to the PowerFlex 700

The PowerFlex 700 AC drive offers high performance in an easy-to-use drive. The PowerFlex 700 AC drive is designed to control three-phase induction motors in applications with requirements ranging from the simplest speed control to demanding torque control. The drive has volts per hertz, sensorless vector and vector control.

Control and Performance

Multiple motor control algorithms allow performance matched to the application need:

- **Volts/Hertz** for simple Fan and Pump applications.
- **Sensorless Vector** for high torque production over a wide speed range.
- **Vector** for good torque regulation and excellent low speed/zero speed performance (w/Vector Control cassette).

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Catalog Number Explanation

¹⁻³ 20B	⁴ D	⁵⁻⁷ 2P1	⁸ A	⁹ 3	¹⁰ A	¹¹ Y	¹² N	¹³ A	¹⁴ R	¹⁵ C	¹⁶ 0	¹⁷⁻¹⁸ NN	¹⁹⁻²⁰ AD
<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>l</i>	<i>m</i>	<i>n</i>

<i>a</i> Drive	
Code	Type
20B	PowerFlex 700

<i>b</i> Voltage Rating				
Code	Voltage	Ph.	Prechg.	Frames
B	240V AC	3	-	0...6
C	400V AC	3	-	0...10
D	480V AC	3	-	0...10
E	600V AC	3	-	0...6
F	690V AC	3	-	5...6
H	540V DC	-	N	5...6, 10
J	650V DC	-	N	5...6, 10
N	325V DC	-	Y	5...6
P	540V DC	-	Y	5...9
R	650V DC	-	Y	5...9
T	810V DC	-	Y	5...6
W	932V DC	-	Y	5...6

<i>c1</i> ND Rating				
208/240V, 60 Hz Input				
Code	208V Amps	240V Amps	Hp	Frame
2P2	2.5	2.2	0.5	0
4P2	4.8	4.2	1.0	0
6P8	7.8	6.8	2.0	1
9P6	11	9.6	3.0	1
015	17.5	15.3	5.0	1
022	25.3	22	7.5	1
028	32.2	28	10	2
042	48.3	42	15	3
052	56	52	20	3
070	78.2	70	25	4
080	92	80	30	4
104	120	104	40	5
130	130	130	50	5
154	177	154	60	6
192	221	192	75	6
260	260	260	100	6

<i>c2</i> ND Rating			
400V, 50 Hz Input			
Code	Amps	kW	Frame
1P3	1.3	0.37	0
2P1	2.1	0.75	0
3P5	3.5	1.5	0
5P0	5.0	2.2	0
8P7	8.7	4.0	0
011	11.5	5.5	0
015	15.4	7.5	1
022	22	11	1
030	30	15	2
037	37	18.5	2
043	43	22	3
056	56	30	3
072	72	37	3
085	85	45	4
105	105	55	5
125	125	55	5
140	140	75	5
170	170	90	6
205	205	110	6
260	260	132	6
292	292	160	7
325	325	180	7
365	365	200	8
415	415	240	8
481	481	280	8
535	535	300	8
600	600	350	8
730	730	400	9
875	875	500	10

<i>c3</i> ND Rating			
480V, 60 Hz Input			
Code	Amps	Hp	Frame
1P1	1.1	0.5	0
2P1	2.1	1.0	0
3P4	3.4	2.0	0
5P0	5.0	3.0	0
8P0	8.0	5.0	0
011	11	7.5	0
014	14	10	1
022	22	15	1
027	27	20	2
034	34	25	2
040	40	30	3
052	52	40	3
065	65	50	3
077	77	60	4
096	96	75	5
125	125	100	5
156	156	125	6
180	180	150	6
248	248	200	6
292	292	250	7
325	325	250	7
365	365	300	8
415	415	350	8
481	481	400	8
535	535	450	8
600	600	500	8
730	730	600	9
875	875	700	10

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c4

ND Rating			
600V, 60 Hz Input			
Code	Amps	Hp	Frame
1P7	1.7	1.0	0
2P7	2.7	2.0	0
3P9	3.9	3.0	0
6P1	6.1	5.0	0
9P0	9.0	7.5	0
011	11	10	1
017	17	15	1
022	22	20	2
027	27	25	2
032	32	30	3
041	41	40	3
052	52	50	3
062	62	60	4
077	77	75	5
099	99	100	5
125	125	125	6
144	144	150	6

c5

ND Rating			
690V, 50 Hz Input			
Code	Amps	kW	Frame
052	52	45	5
060	60	55	5
082	82	75	5
098	98	90	6
119	119	110	6
142	142	132	6

d

Enclosure	
Code	Enclosure
A	IP20, NEMA/UL Type 1
F ✕	Open/Flange Mount Front: IP00, NEMA/UL Type Open Back/Heatsink: IP54, NEMA Type 12
N §	Open/Flange Mount Front: IP00, NEMA/UL Type Open Back/Heatsink: IP54, NEMA 12
G ✕	Stand-Alone/Wall Mount IP54, NEMA/UL Type 12
J	IP00, NEMA/UL Type Open with Conformal Coat
M ⇨	IP20, NEMA/UL Type 1 with Conformal Coat
U	Roll-In Front: IP00, NEMA/UL Type Open Back/Heatsink: IP54, NEMA 12 Frames 8 & 9 Only
V	Roll-In with Conformal Coat Front: IP00, NEMA/UL Type Open Back/Heatsink: IP54, NEMA 12 Frames 8 & 9 Only

e

HIM	
Code	Operator Interface
0	Blank Cover
3	Full Numeric LCD
5	Prog. Only LCD
J =	Remote (Panel Mount), IP66, NEMA/UL Type 12 Full Numeric LCD HIM
K =	Remote (Panel Mount), IP66, NEMA/UL Type 12 Prog. Only LCD HIM
= Available with Frames 5...6 Stand-Alone IP54 drives (Enclosure Code "G").	

f

Documentation	
Code	Type
A	Manual
N	No Manual
Q	No Shipping Package (Internal Use Only)

g

Brake	
Code	w/Brake IGBT ®
Y	Yes
N	No
® Brake IGBT is standard on Frames 0-3, optional on Frames 4-6 and not available on Frames 7...10.	

h

Internal Braking Resistor	
Code	w/Resistor
Y	Yes *
N	No
* Not available for Frame 3 drives or larger.	

i +

Emission		
Code	CE Filter ‡	CM Choke
A	Yes	Yes
B +	Yes	No
N	No	No

‡ Note: 600V class drives below 77 Amps (Frames 0-4) are declared to meet the Low Voltage Directive. It is the responsibility of the user to determine compliance to the EMC directive. Frames 7...10, 400/480V AC drives (Voltage Rating codes "C" and "D") meet CE certification requirements when installed per recommendations.

+ Refer to Internal EMC Filter for details on selecting this option for each frame size.

j

Comm Slot	
Code	Network Type
B	BACnet MS/TP
C	ControlNet (Coax)
D	DeviceNet
E	EtherNet/IP
R	Remote I/O
S	RS485 DF1
N	None

k

Control & I/O		
Code	Control	I/O Volts
A	Standard	24V DC/AC
B	Standard	115V AC
C	Vector ⚡	24V DC
D	Vector ⚡	115V AC
N	Standard	None
⚡ Vector Control Option utilizes DPI Only. Frame 7...10 drives only accept Vector Control.		

l

Feedback	
Code	Type
0	None
1	Encoder, 12V/5V

m

Future Use	
------------	--

n

Special Firmware (Frames 0...6 Only)	
Code	Type
AD =	60 Hz Maximum
AE =	Cascading Fan/Pump Control
AX =	82 Hz Maximum
BA =	Pump Off (for pump jack)
= Must be used with Vector Control option C or D (Position k). Positions m-n are only required when custom firmware is supplied.	

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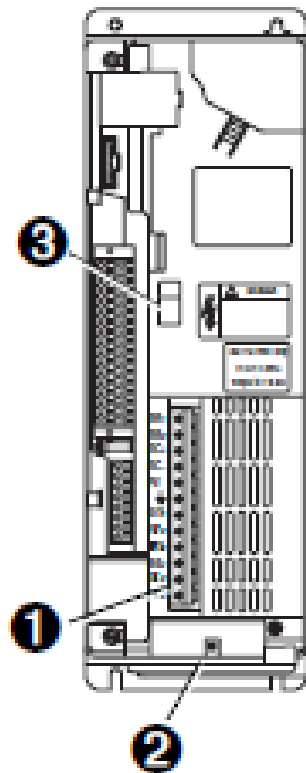
Opening the Cover

Locate the slot in the upper left corner. Slide the locking tab up and swing the cover open. Special hinges allow cover to move away from drive and lay on top of adjacent drive (if present).



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Component Locations and Power Terminals (Frame 0)



Frames 0 & 1

No.	Name	Frame	Description
①	Power Terminal Block	0 & 1	Input power and motor connections
		2	Input power and motor connections
		3	Input power and motor connections BR1, 2 terminals
		4	Input power and motor connections
		5 75 Hp, 480V 100Hp, 600V	Input power, DC+, DC-, BR1, 2, PE, motor connections
		5 100Hp	Input power, DC+, DC- and motor BR1, 2, PE terminals
		6	Input power, DC+, DC-, BR1, 2, PE, motor connections
		7	Input power, DC+, DC-, PE, motor connections
		8...10	Input power, DC+, DC-, PE, motor connections
②	SHLD Terminal	0...6	Terminating point for wiring shields
③	AUX Terminal Block ⁽¹⁾	0...4	Auxiliary Control Voltage PS+, PS- ⁽¹⁾
		5...6	
		7...10	

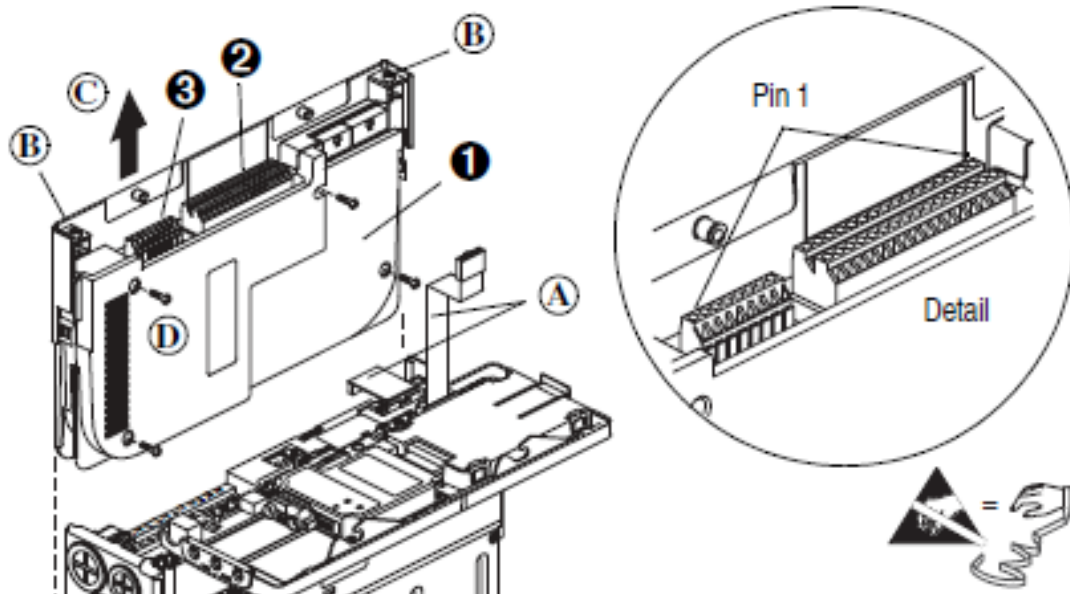
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I/O Control Cassette

The figure below shows the I/O Control Cassette and terminal block locations. The cassette provides a mounting point for the various PowerFlex 700 I/O options. To remove the cassette, follow the steps below. Cassette removal will be similar for all frames (Frame 0 drive shown).

Step	Description
A	Disconnect the two cable connectors shown below.
B	Loosen the two screw latches shown below.
C	Slide the cassette out.
D	Remove screws securing cassette cover to gain access to the boards.

Figure 1.7 PowerFlex 700 Cassette & I/O Terminal Blocks - Frames 0...6



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I/O Terminal Descriptions (Frame 0)

	No.	Signal	Factory Default	Description	Related Param.
	1	Analog In 1 (-) ⁽¹⁾	⁽²⁾	Isolated ⁽³⁾ , bipolar, differential, $\pm 10\text{V}/0\text{-}20\text{ mA}$, 11 bit & sign. For 0-20 mA, a jumper must be installed at terminals 17 & 18 (or 19 & 20). 88k ohm input impedance when configured for volt. & 95.3 ohm for current	320-327
	2	Analog In 1 (+) ⁽¹⁾			
	3	Analog In 2 (-) ⁽¹⁾			
	4	Analog In 2 (+) ⁽¹⁾			
	5	Pot Common	-	For (+) and (-) 10V pot references.	
	6	Analog Out 1 (-)	⁽²⁾	Single-ended bipolar (current output is not bipolar), $\pm 10\text{V}/0\text{-}20\text{ mA}$, 11 bit & sign, Voltage mode - limit current to 5 mA. Current mode - max. load is 400 ohms.	340-347
	7	Analog Out 1 (+)			
	8	Analog Out 2 (-)			
	9	Analog Out 2 (+)			
	10	HW PTC Input 1	-	1.8k ohm PTC, Internal 3.32k ohm pull-up resistor	238, 259
	11	Digital Out 1 - N.C. ⁽⁴⁾	Fault	Max. Resistive Load: 240V AC/30V DC - 1200VA, 150W Max. Current: 5A, Min. Load: 10 mA Max. Inductive Load: 240V AC/30V DC - 840VA, 105W Max. Current: 3.5A, Min. Load: 10 mA	380-391
	12	Digital Out 1 Common			
	13	Digital Out 1 - N.O. ⁽⁴⁾	NOT Fault		
	14	Digital Out 2 - N.C. ⁽⁴⁾	NOT Run		
	15	Digital Out 2/3 Com.			
	16	Digital Out 3 - N.O. ⁽⁴⁾	Run		
	17	Current In Jumper ⁽¹⁾ - Analog In 1		Placing a jumper across terminals 17 & 18 (or 19 & 20) will configure that analog input for current.	
	18				
	19	Current In Jumper ⁽¹⁾ - Analog In 2			
	20				
	21	-10V Pot Reference	-	2k ohm minimum load.	
	22	+10V Pot Reference	-		
	23	HW PTC Input 2	-	See above	
	24	+24VDC ⁽⁵⁾	-	Drive supplied logic input power. ⁽⁵⁾	
	25	Digital In Common	-		
	26	24V Common ⁽⁵⁾	-	Common for internal power supply.	
	27	Digital In 1 ⁽⁶⁾	Stop - CF	115V AC, 50/60 Hz - Opto isolated Low State: less than 30V AC High State: greater than 100V AC, 5.7 mA 24V DC - Opto isolated Low State: less than 5V DC High State: greater than 20V DC, 10 mA DC Digital Input Impedance: 21k ohm	361-366
	28	Digital In 2 ⁽⁶⁾	Start		
	29	Digital In 3 ⁽⁶⁾	Auto/Man.		
	30	Digital In 4 ⁽⁶⁾	Speed Sel 1		
	31	Digital In 5 ⁽⁶⁾	Speed Sel 2		
	32	Digital In 6/Hardware Enable ⁽⁶⁾ , see pg. 1-43	Speed Sel 3		

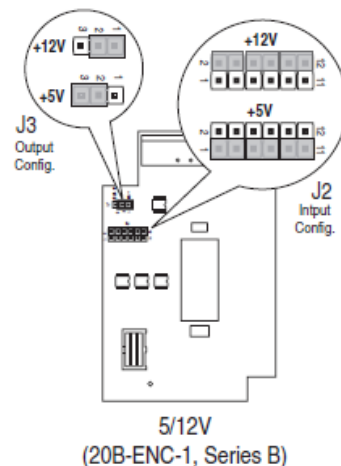
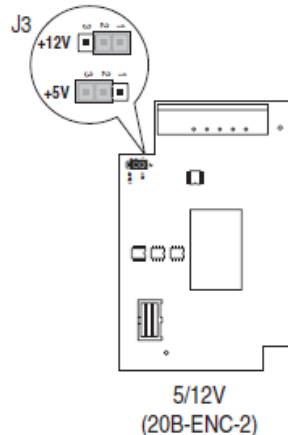
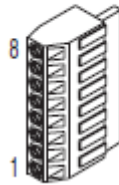
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Encoder Wiring Diagrams

Table 1.J Encoder Terminal Designations

No.	Description (refer to page A-4 for encoder specifications)
8	+12V ⁽¹⁾ DC Power
7	+12V ⁽¹⁾ DC Return (Common)
6	Encoder Z (NOT)
5	Encoder Z
4	Encoder B (NOT)
3	Encoder B
2	Encoder A (NOT)
1	Encoder A

See "Detail" on [page 1-39](#)



I/O	Connection Example	I/O	Connection Example
Encoder Power – (1) Internal Drive Power Internal (drive) 12V DC, 250mA		Encoder Power – External Power Source	
Encoder Signal – Single-Ended, Dual Channel		Encoder Signal – Differential, Dual Channel	

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Speed Reference Control

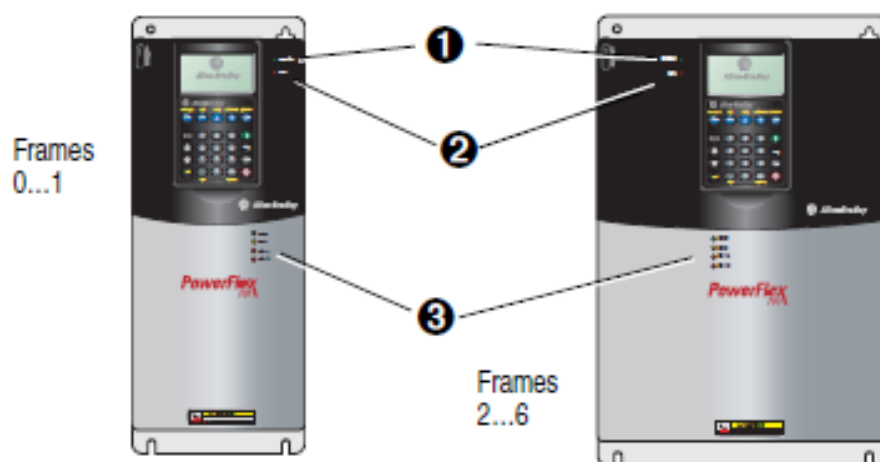
Speed Reference Selection Chart

PowerFlex 70 and 700 Default Auto Speed Reference Options								
Digital Inputs Status			Speed Reference Selections	Parameter	Default Setting	Default Selection	Default Setting	
Digital Input 6	Digital Input 5	Digital Input 4						
0	0	0	Speed Ref. A	90	2	Analog In "2"	0-10Vdc	
0	0	1	Speed Ref. B	93	11	Preset Speed 1	5 Hz	
0	1	0	Preset Speed 2	102	10	10 Hz		
0	1	1	Preset Speed 3	103	20	20 Hz		
1	0	0	Preset Speed 4	104	30	30 Hz		
1	0	1	Preset Speed 5	105	40	40 Hz		
1	1	0	Preset Speed 6	106	50	50 Hz		
1	1	1	Preset Speed 7	107	60	60 Hz		

Default Digital Inputs Selections				Default Digital Output Selections		
Dig In 1 Sel	361	4	Stop	Dig Out 1 Sel (380)	1	Fault
Dig In 2 Sel	362	5	Start	Dig Out 2 Sel (384)	4	Run
Dig In 3 Sel	363	18	Auto/Man	Dig Out 3 700 Only (385)	4	Run
Dig In 4 Sel	364	15	Speed Sel 1	Default Analog Output Selections		
Dig In 5 Sel	365	16	Speed Sel 2	Analog Out 1 Sel (342)	0	Output Frequency
Dig In 6 Sel	366	17	Speed Sel 3	Analog Out 2 Sel 700 Only (343)	0	Output Frequency

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Drive Status Indicators



#	Name	Color	State	Description
①	PWR (Power)	Green	Steady	Illuminates when power is applied to the drive.
②	STS (Status)	Green	Flashing	Drive ready, but not running and no faults are present.
			Steady	Drive running, no faults are present.
		Yellow See page 4-11	Flashing, Drive Stopped	A start inhibit condition exists, the drive cannot be started. Check parameter 214 [Start Inhibits].
			Flashing, Drive Running	An intermittent type 1 alarm condition is occurring. Check parameter 211 [Drive Alarm 1].
			Steady, Drive Running	A continuous type 1 alarm condition exists. Check parameter 211 [Drive Alarm 1].
		Red See page 4-5	Flashing	Fault has occurred. Check [Fault x Code] or Fault Queue.
			Steady	A non-resettable fault has occurred.
③	PORT	See the Communication Adapter User Manual.		Status of DPI port internal communications (if present).
	MOD			Status of communications module (when installed).
	NET A			Status of network (if connected).
	NET B			Status of secondary network (if connected).

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Programming and Parameters

To configure a drive to operate in a specific way, drive parameters may have to be set. Three types of parameters exist:

- **ENUM Parameters:** ENUM parameters allow a selection from 2 or more items. The LCD HIM will display a text message for each item.
- **Bit Parameters:** Bit parameters have individual bits associated with features or conditions. If the bit is 0, the feature is off or the condition is false. If the bit is 1, the feature is on or the condition is true.
- **Numeric Parameters:** These parameters have a single numerical value (that is, 0.1 Volts). The example on the following page shows how each parameter type is presented in this manual.

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①	②	③	④	⑤	⑥
File	Group	No.	Parameter Name & Description	Values	Related
UTILITY	Drive ...	198	[Load Frm Usr Set] Loads a previously saved set of parameter values from a selected user set location in drive nonvolatile memory to active drive memory.	Default: 0 "Ready" Options: 0 "Ready" 1 "User Set 1" 2 "User Set 2" 3 "User Set 3"	199
	Diagnostics	216	[Dig In Status] Status of the digital inputs. 1 = Input Present 0 = Input Not Present x = Reserved	Read Only	361 thru 366
MOTOR ...	Torq ...	434	[Torque Ref B Mult] Defines the value of the multiplier for the [Torque Ref B Sel] selection.	Default: 1.0 Min/Max: -/+32767.0 Units: 0.1	053

No.	Description
①	File – Lists the major parameter file category.
②	Group – Lists the parameter group within a file.
③	No. – Parameter number. Note that all parameters in the PowerFlex 700VC are 32 bit. = Parameter value cannot be changed until drive is stopped. = Parameter only displayed when [Motor Cntl Sel] is set to "4." = This parameter only available with firmware version 6.002 and later.
④	Parameter Name & Description – Parameter name as it appears on an LCD HIM, with a brief description of the parameter's function.
⑤	Values – Defines the various operating characteristics of the parameter. Three types exist.
	ENUM Default: Lists the value assigned at the factory. "Read Only" = no default. Options: Displays the programming selections available.
	Bit Bit: Lists the bit place holder and definition for each bit.
	Numeric Default: Lists the value assigned at the factory. "Read Only" = no default. Min/Max: The range (lowest and highest setting) possible for the parameter. Units: Unit of measure and resolution as shown on the LCD HIM.
	Important: Some parameters will have two unit values: • Analog inputs can be set for current or voltage with [Anlg In Config], param. 320. • Setting [Speed Units], parameter 79 selects Hz or RPM. Important: When sending values through DPI ports, simply remove the decimal point to arrive at the correct value (i.e. to send "5.00 Hz," use "500").
⑥	Related – Lists parameters (if any) that interact with the selected parameter. The symbol "" indicates that additional parameter information is available in Appendix C.

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Parameter Organization

The LCD HIM displays parameters in a **File-Group-Parameter** or **Numbered List** view order. To switch display mode, access the Main Menu, press ALT, then Sel while cursor is on the parameter selection. In addition, using [\[Param Access Lvl\]](#), the user has the option to display the full parameter set (Advanced), commonly used parameters (Basic) or diagnostic/advanced tuning parameters (Reserved). To simplify programming, the displayed parameters will change according to the selection made with [\[Motor Cntl Sel\]](#). For example, if “FVC Vector” is selected, the parameters associated solely with other operations such as Volts per Hertz or Sensorless Vector will be hidden.

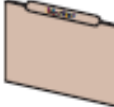
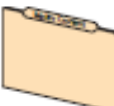
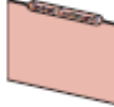
File-Group-Parameter Order

This simplifies programming by grouping parameters that are used for similar functions. The parameters are organized into files. Each file is divided into groups, and each parameter is an element in a group. By default, the LCD HIM displays parameters by File-Group-Parameter View.

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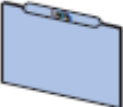
Advanced Parameter View

File	Group	Parameters								
	Monitor	Metering	Output Freq	001	Flux Current	005	DC Bus Memory	013		
			Commanded Speed	002	Output Voltage	006	Analog In1 Value	016		
			Ramped Speed	022	Output Power	007	Analog In2 Value	017		
			Speed Reference	023	Output Powr Fctr	008	Elapsed kWh	014		
			Commanded Torque**024		Elapsed MWh	009	PTC HW Value	018		
			Speed Feedback	025	Elapsed Run Time	010	Spd Fdbk No Filt	021		
			Output Current	003	MOP Reference	011				
			Torque Current	004	DC Bus Voltage	012				
	Drive Data	Rated kW	026	Rated Amps	028					
		Rated Volts	027	Control SW Ver	029					
	Motor Control	Motor Data	Motor Type	040	Motor NP RPM	044	Motor OL Factor	048		
			Motor NP Volts	041	Motor NP Power	045	Motor OL Mode ^{6,x}	050		
			Motor NP FLA	042	Mtr NP Pwr Units	046	Motor Poles	049		
			Motor NP Hertz	043	Motor OL Hertz	047				
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Troubleshooting (Faults and Alarms)

This chapter provides information to guide you in troubleshooting the PowerFlex 700. Included is a listing and description of drive faults (with possible solutions, when applicable) and alarms.

A fault is a condition that stops the drive. There are three fault types.

Type	Fault Description	
①	Auto-Reset Run	When this type of fault occurs, and [Auto Rstrt Tries] (see page 3-33) is set to a value greater than "0," a user-configurable timer, [Auto Rstrt Delay] (see page 3-33) begins. When the timer reaches zero, the drive attempts to automatically reset the fault. If the condition that caused the fault is no longer present, the fault will be reset and the drive will be restarted.
②	Non-Resettable	This type of fault normally requires drive or motor repair. The cause of the fault must be corrected before the fault can be cleared. The fault will be reset on power up after repair.
③	User Configurable	These faults can be enabled/disabled to annunciate or ignore a fault condition.

An alarm is a condition that, if left untreated, may stop the drive. There are two alarm types.

Type	Alarm Description	
①	User Configurable	These alarms can be enabled or disabled through [Alarm Config 1] on page 3-49 .
②	Non-Configurable	These alarms are always enabled.

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Manually Clearing Faults:

Step

1. Press the  key to acknowledge the fault. The fault information will be removed so that you can use the HIM.
 2. Address the condition that caused the fault.
The cause must be corrected before the fault can be cleared.
 3. After corrective action has been taken, clear the fault using one of these methods.
 - Press the  (Stop) key.
 - Cycle drive power.
 - Set parameter 240 [Fault Clear] to "1."
 - "Clear Faults" on the HIM Diagnostic menu.
-

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AC Drive Training Simulators

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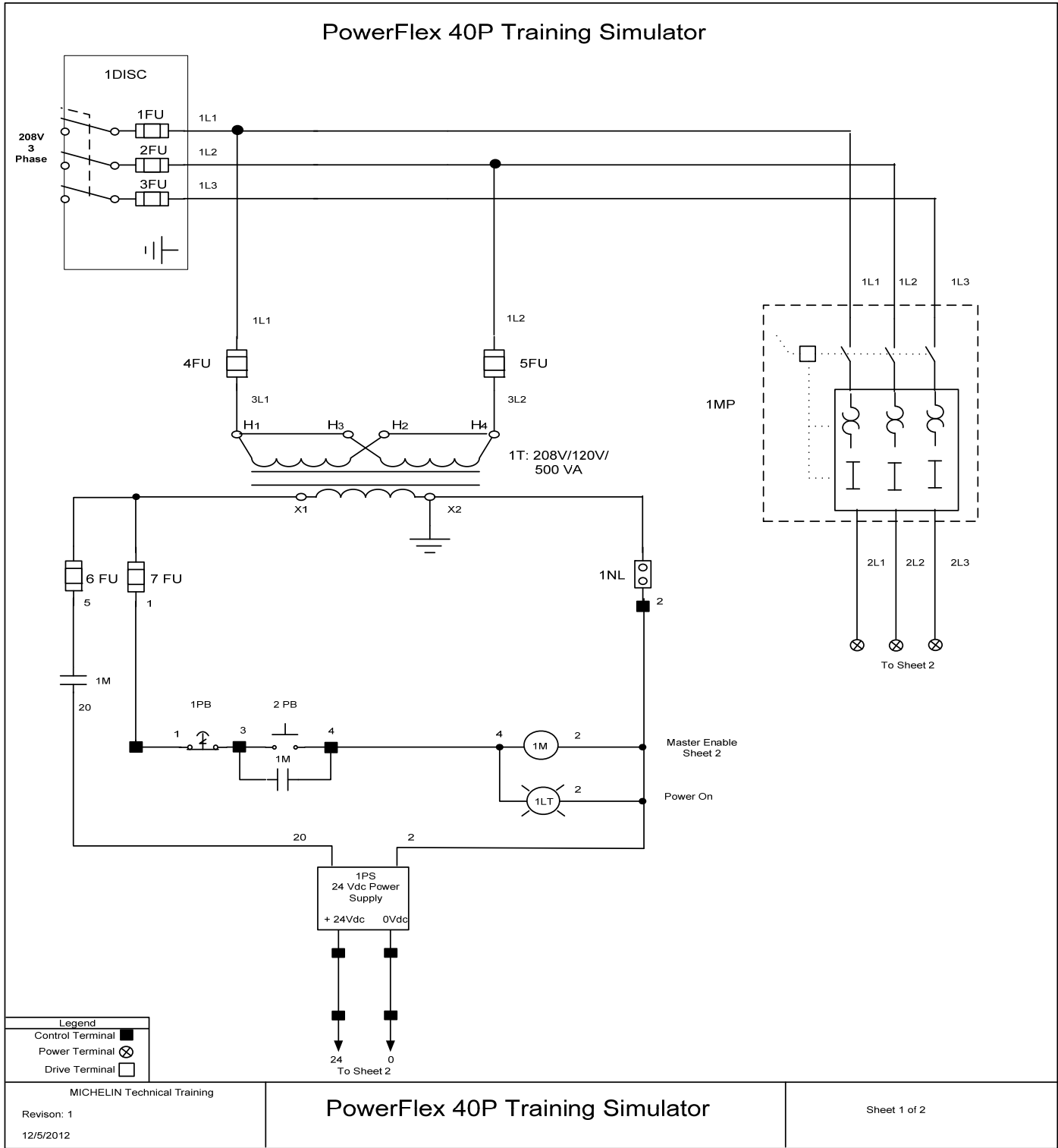
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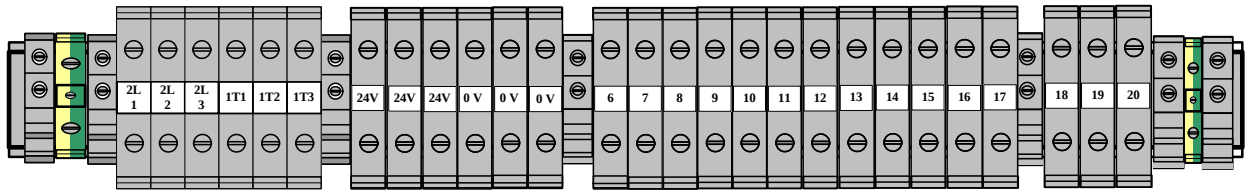
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PowerFlex 40p Simulator Wiring

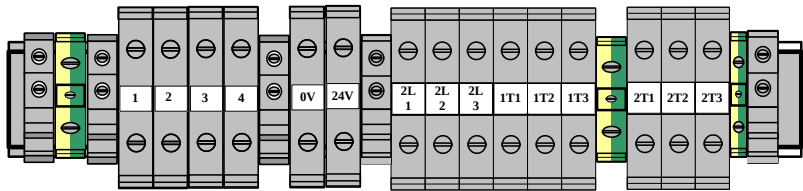


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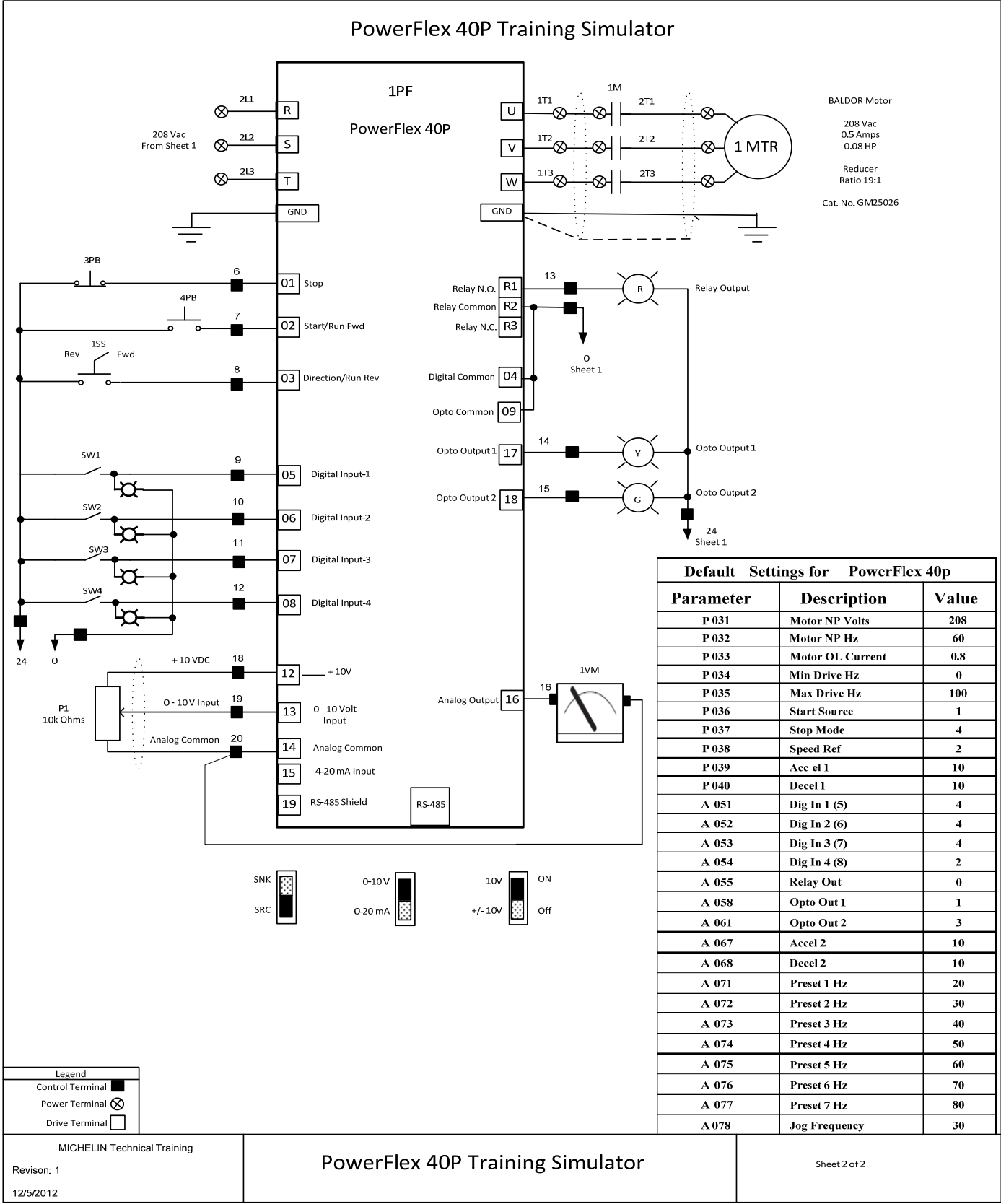
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Back Terminal Strip



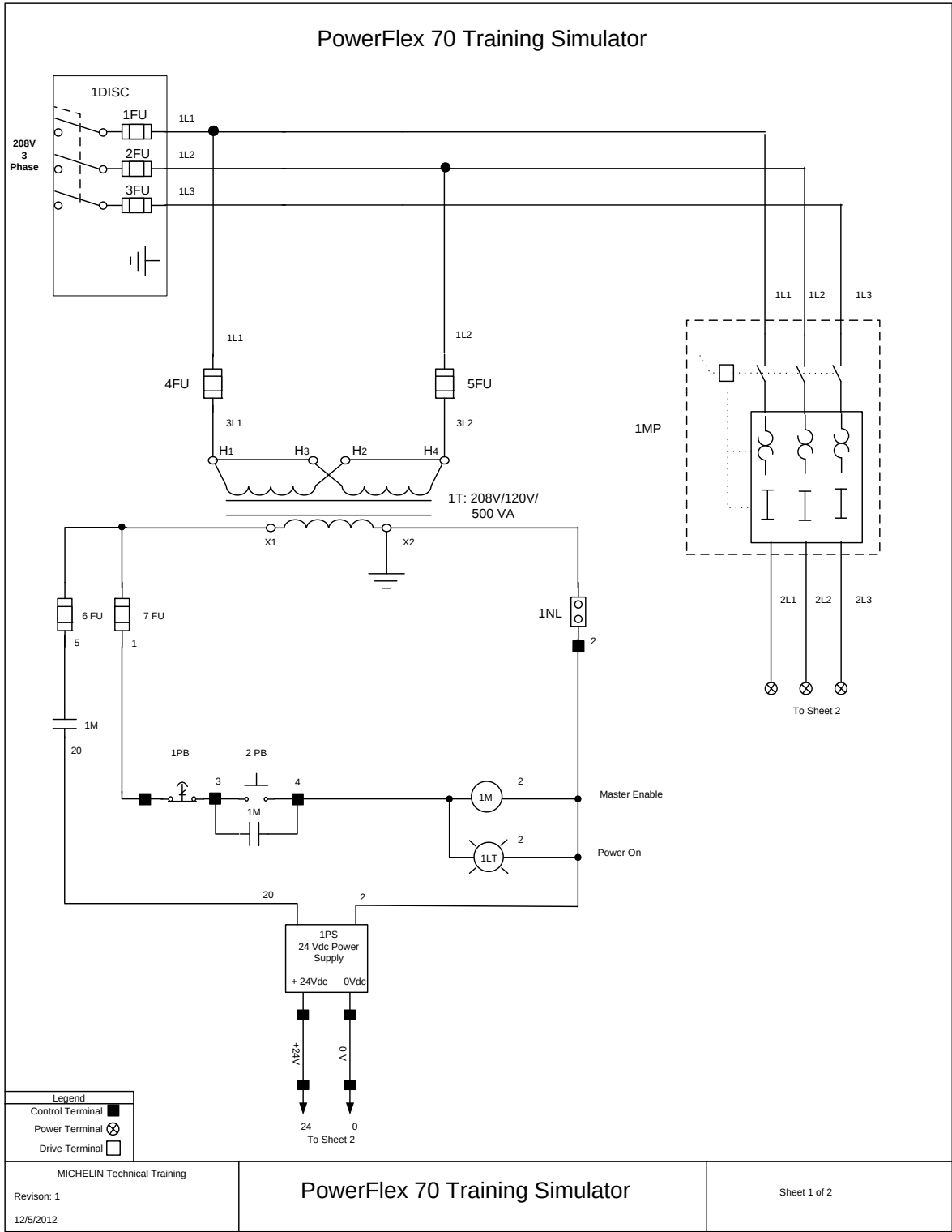
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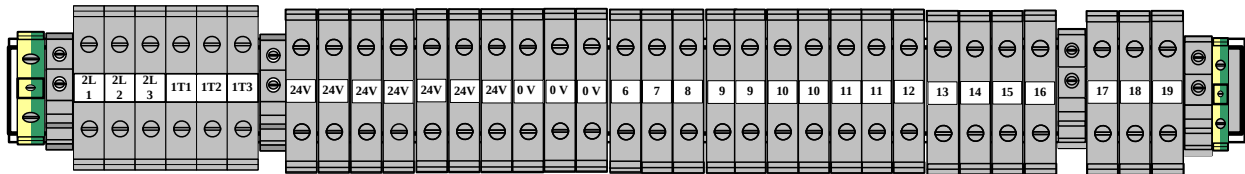
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PowerFlex 70 Simulator Wiring

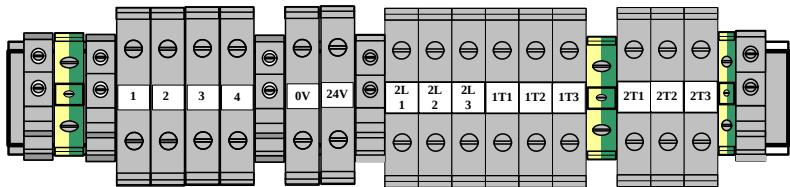


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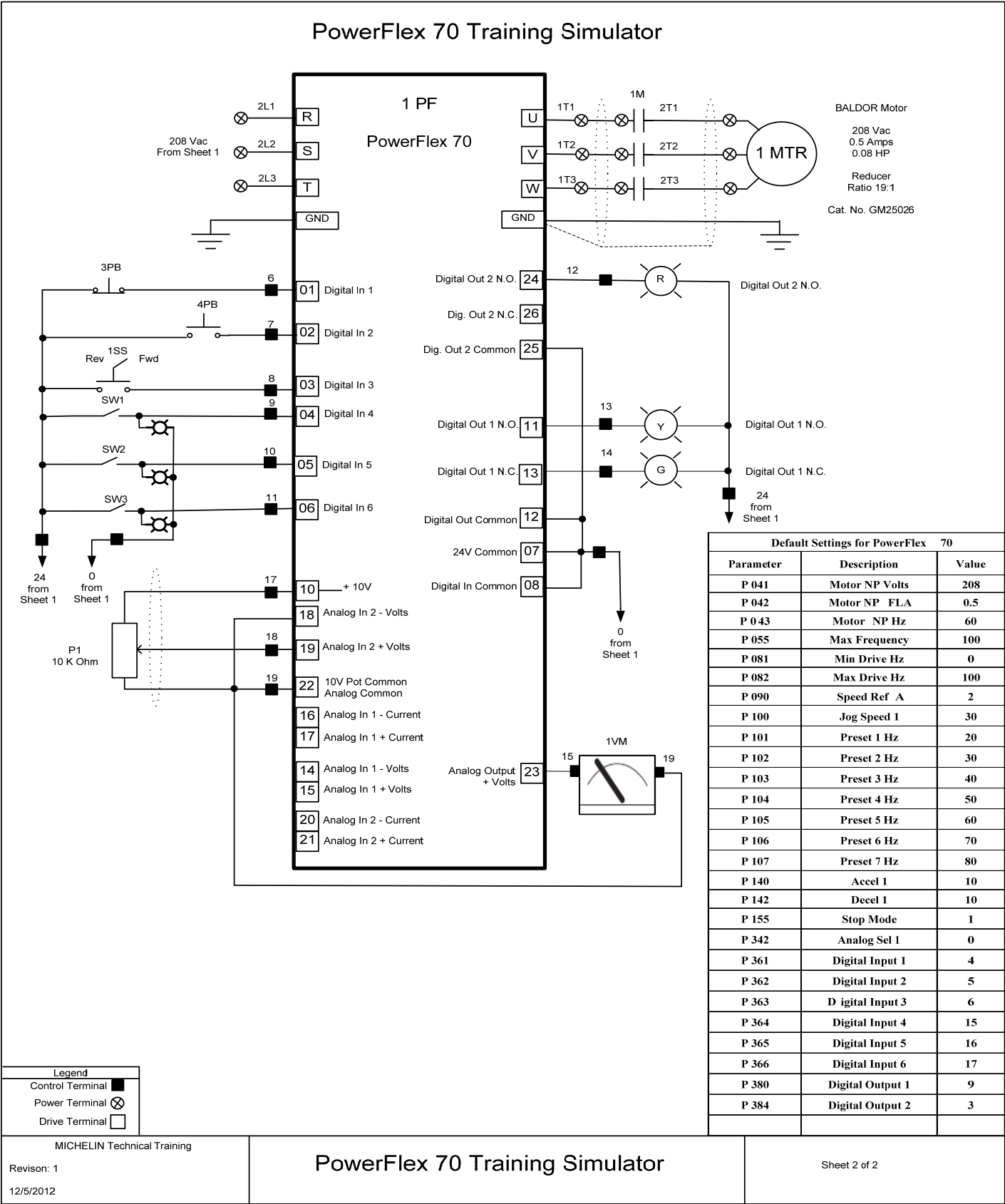
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Back Terminal Strip



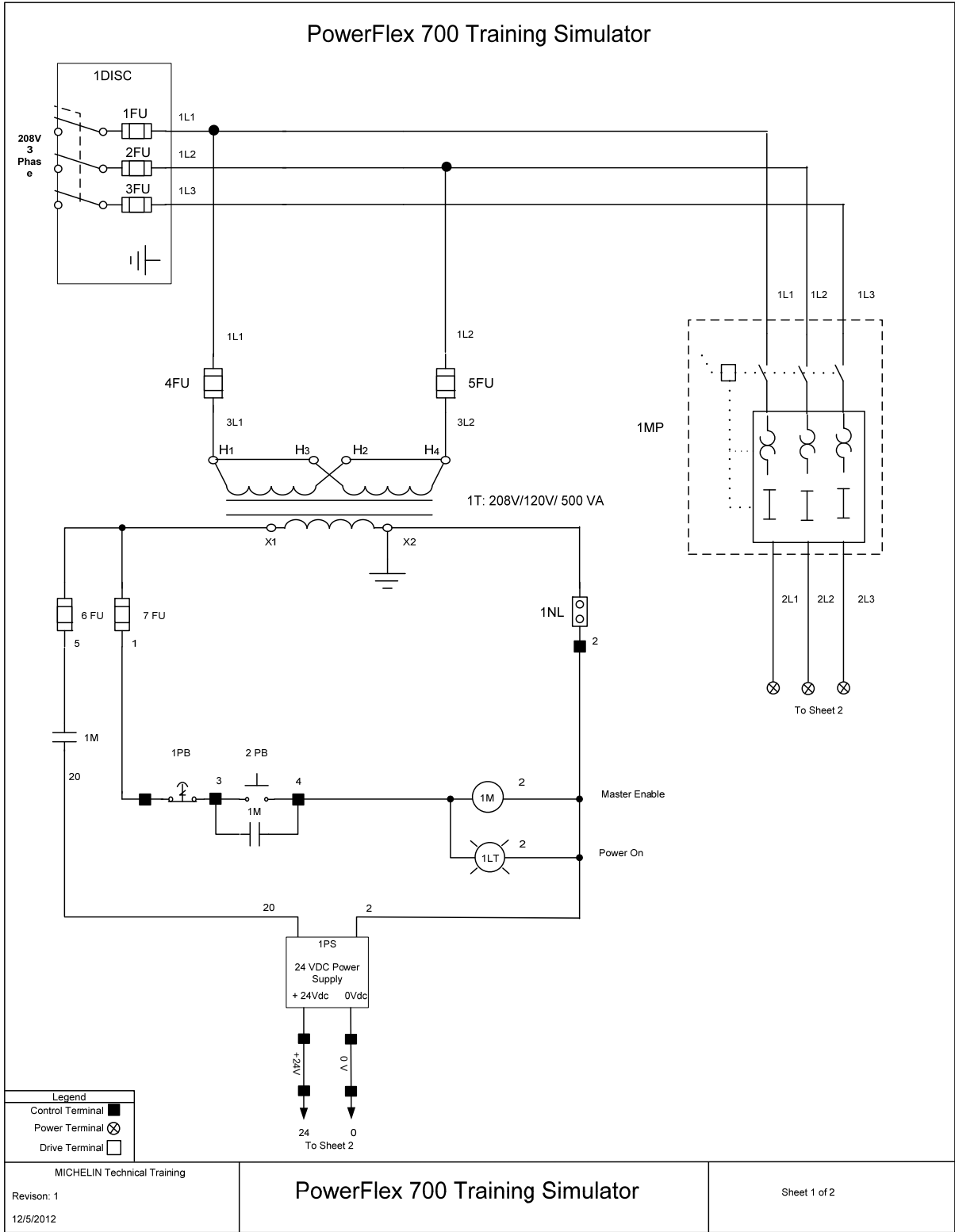
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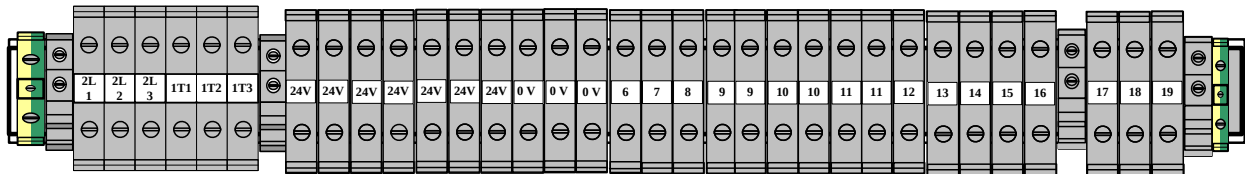
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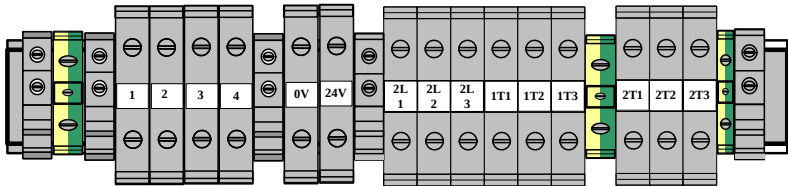


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Front Terminal Strip



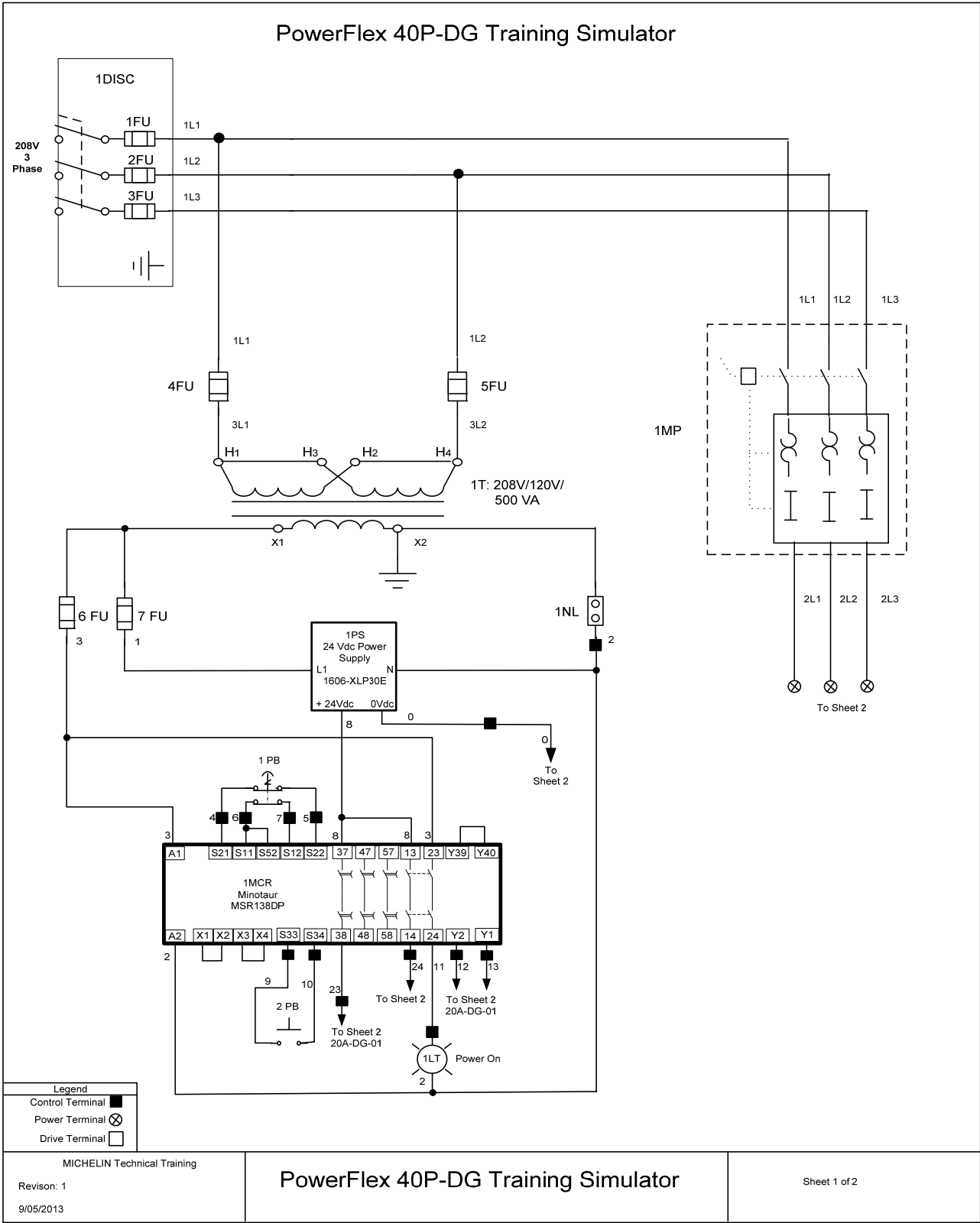
Back Terminal Strip



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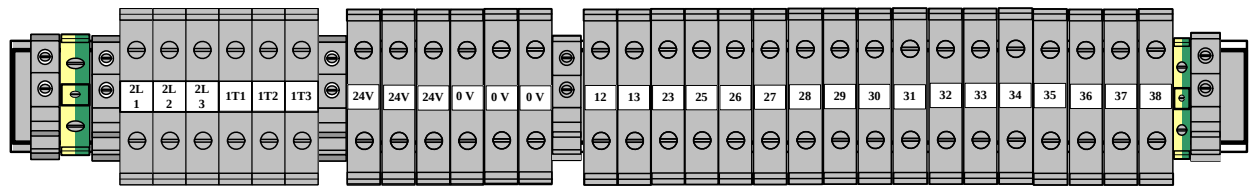
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PowerFlex 40p-DG Simulator Wiring

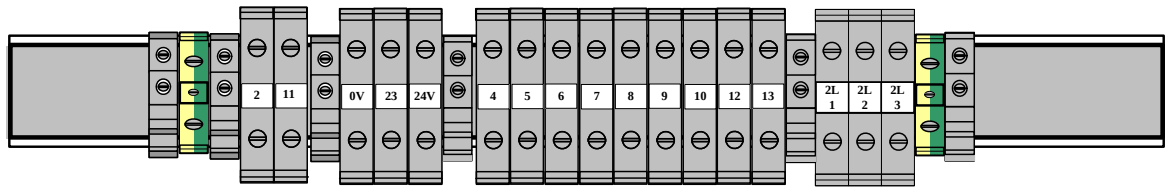


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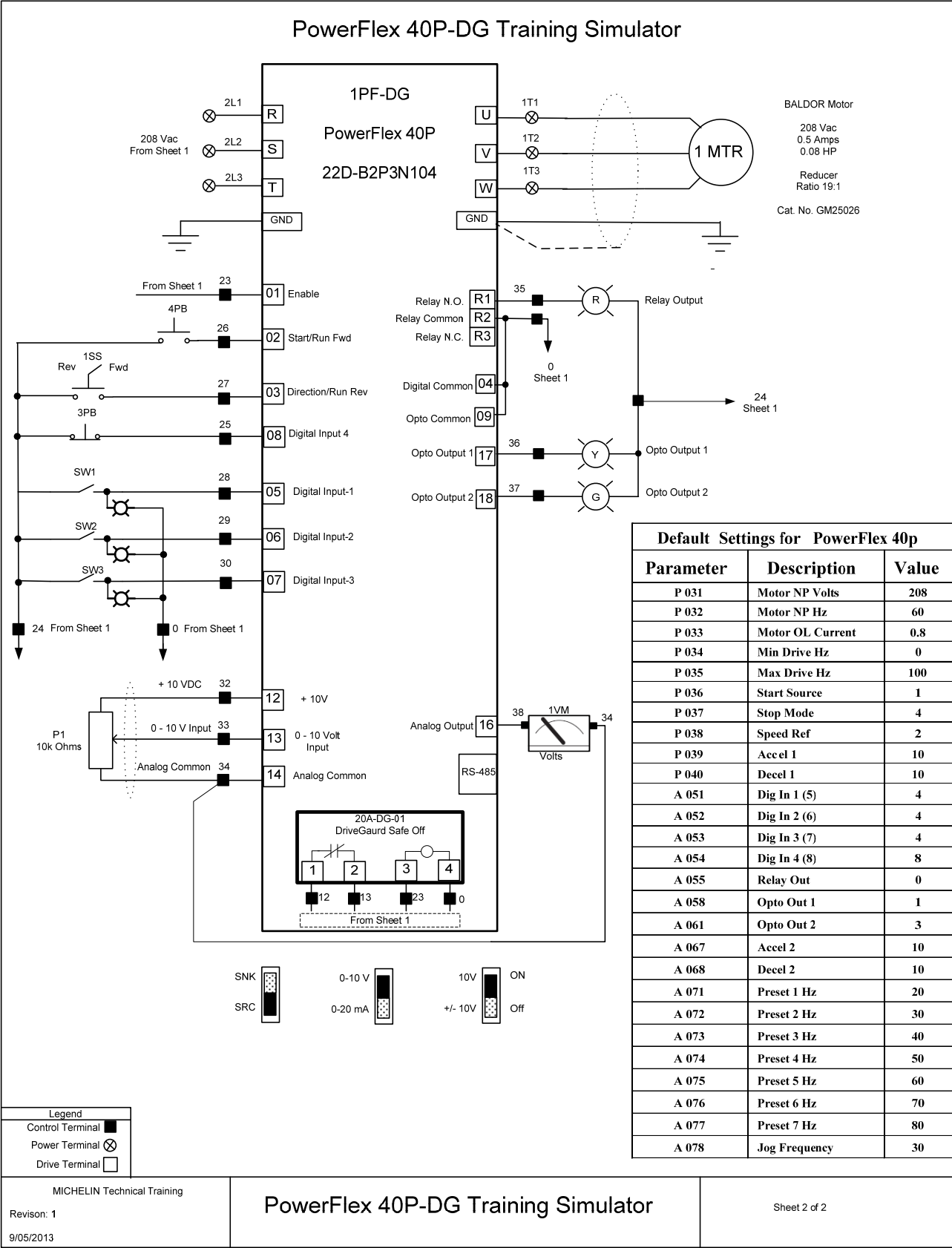
Front Terminal Strip



Back Terminal Strip



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PowerFlex Lockout Procedure, Drive Hardware, and Software Exercises

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Revision History		
Date	Reviser	Changes
7-15-2015	M. Chapman	Corrected PowerFlex 40p Exercise 4 error
7-17-2015	M. Chapman	Added Lockout/Tagout/Tryout Procedure
4-19-2019	M. Chapman	Revised lockout Tagout section

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AC Drive Trainer Lockout/Tagout/Tryout Procedure

PURPOSE AND SCOPE

The purpose of this procedure is to specifically outline the steps to be taken at the training center to fulfill the requirements outlined in "The Corporate Lockout/Tagout/Tryout Procedure.

As in The Corporate Lockout/Tagout/Tryout Procedure, "lockout" shall mean - the placement of a lockout device on an energy isolating device, in accordance with an established procedure, ensuring that the energy isolating device and the equipment being controlled cannot be operated until the lockout device is removed.

This procedure is designed for the AP School training centers only and is not intended to be interpreted as the official procedure of any other department in Michelin.

RESPONSIBILITY

It is the responsibility of the instructors to train all trainees on the lockout procedures. It is the responsibility of the trainees to comply with this procedure when doing any work that involves electrical or fluid circuits training or otherwise.

LOCKOUT TAGS

Lockout tags and locks shall be stored in the individual tool drawers and a responsibility of the trainees.

EQUIPMENT

A Multi-meter will be required for this task, which is located in the tool box assigned to the student. **Before each use of the multi-meter**, the student shall thoroughly inspect the meter case and display for damage. Also, both meter leads shall be inspected for cracked or chaffed insulation as well as insulation hardening or stiffness on the leads. The leads should be flexible and easy to bend. The meter lead probe tips should also be inspected for damage as well.

PROCEDURE

The following procedure shall apply in cases where equipment or systems are to be serviced or repaired which will expose personnel to the hazards of moving machinery, energized electrical circuits, or hydraulic/pneumatic circuits.

This Lockout/Tagout/Tryout procedure applies to the AC Drive Trainers in the AP school training centers.

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I. **Lockout/Tagout/Tryout Procedure For The AC Drive Trainers.**

- A. **Preparation for Shutdown.** Prior to shutting down any equipment, inform the person responsible for that particular area.
- B. **Equipment Shutdown.** Shut the power off by switching the main disconnect switch to the “off” position for the Drive Trainer.
- C. **Applying LOCKOUT/TAGOUT Devices.** On the AC Drive trainer, turn the disconnect switch to the OFF position. Lock out and tag out the disconnect switch. After locking and tagging out the main electrical disconnect, it shall be tested to ensure that it cannot be moved from the locked position.
- D. **Verifying Isolation of Equipment.** Verify the Multi-meter functions correctly by measuring a known voltage at the Line side (top) of the disconnect switch, which should be approximately 208 VAC. Afterwards, take measurements from phase to phase (3 phase to phase measurements) on the **load** side (Bottom) of the disconnect switch. All of which must indicate zero volts. Also, on load side (bottom) of the disconnect switch, measure each phase with respect to ground (3 phase to ground measurements). All of which must indicate zero volts. The green and yellow terminal strip (adjacent to the disconnect) which is Ground can be used for these measurements. After verifying that the power to the load side of the disconnect switch is zero volts, re-check your meter at the line side (top) of the disconnect switch again (208VAC) to confirm that the multi meter is still functioning and measuring the appropriate voltage. **Because this is an AC Drive, also verify the DC Buss on the drive with the multi-meter set to DC Volts. The DC Buss can be found on the power terminal block at the bottom of the Drive and its labeled +DC and –DC. A decaying DC voltage value will generally be seen, which indicates the meter is functioning on the DC setting. Wait until the DC voltage approaches zero volts.** Additionally, the controls for the equipment shall be operated in all modes to ensure that the electrical circuit has been broken. After this test, the controls shall be returned to the "off" position.

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E. **Perform the work safely** – ensure that the work performed is done safely. (For Arc/Flash safety: adhere to the Personnel Protective Equipment required for the drive trainers.
- Look ahead, and avoid doing anything that could re-activate the equipment.
- Don't bypass the lockout at any time.

F. **Removal of Lockout/Tagout** - In general, the removal of a lockout can only be done by the person who has installed the lock.
- Make sure the equipment is safe to operate.
- Remove all tools from the work area.
- Be sure the system is fully assembled. Wire and components in place etc...
- Notify everyone who works in the area that LOCKOUT/TAGOUT is being removed.
- Remove the lock and tag and safely re-energized the panel.
PowerFlex 40p Drive Hardware Exercises

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PowerFlex 40p Hardware Exercise 1

Hardware Exercise Objective:

To become familiar with the drive's interfaces, locations, and configuration parameters.

Equipment Needed:

Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedure:

Power up the simulator and set the drive up to the factory default parameter settings.

Which parameter did you use?

Use course manual, User Manual and simulator drawing package to answer the following questions

1) Locate and record the PowerFlex 40p Part number from the simulator

- .Describe (in detail) the catalog explanation of the drive.

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2) Using the User Manual and/or Schematic drawing, locate each of the following interface terminals to answer the following questions.

- **AC Input Terminals:**
What terminals does the AC Line connect to?
- **Drive Output Terminals:**
What terminals does the motor connect on the drive?
- **Drive Guard / Safe Off**
Is the drive configured for Drive Guard / Safe Off?
- **Digital Inputs:**
What are the terminal numbers of the Digital Inputs?

Are the simulator's Digital Inputs (as drawn on the schematic) setup for sinking or sourcing?

Which parameters are used to configure each of the Digital Inputs?

Digital Input 1 Parameter # _____ Value/Description _____

Digital Input 2 Parameter # _____ Value/Description _____

Digital Input 3 Parameter # _____ Value/Description _____

Digital Input 4 Parameter # _____ Value/Description _____

What is the type of parameter (ENUM,Numeric or Bit)? _____

Which parameter will allow you to view the status of the Digital Inputs?

What is the type of parameter (ENUM,Numeric or Bit)? _____

Turn on SW-1 and SW-3 and use this parameter to display the Digital Input Status.

Digital Input	4	3	2	1
Status (1 or 0)				

Which parameter will allow you to view the Control Input Status?

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What is the type of parameter (ENUM,Numeric or Bit)? _____

Press the Start or Stop buttons or Fwd/Rev Switch and use this parameter to display the Digital Input Status.

Control Status Input	Dynamic Brake Transistor On	I/O Terminal 1 Stop Input	I/O Terminal 3 Fwd/Rev Input	I/O Terminal 2 Start/Run Input
Status (1 or 0)				
Bit Number				

- **Analog Input(s):**

How many Analog Inputs does the drive have?

How many are used (wired) on the simulator and what is connected to them?

What are the drive's actual analog input terminals and wire numbers (from the schematic drawing)?

Can the Analog voltage be unipolar or bipolar?

How should the drive react differently between unipolar and bipolar Speed References?

Which parameter(s) will allow you to view the Analog Input values?

- **Analog Output(s):**

How many Analog Outputs does the drive have?

Which parameter(s) is used to configure the Analog Output(s)?

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- **Digital Outputs:**

How many Digital Outputs does the drive have?

List each Digital Output and whether it is a Relay or Solid State Output.

List each parameter associated with configuring each Digital Output.

- **Feedback:**

Does the drive have a terminal strip for connection to an encoder?

What type(s) of encoders can be used?

What terminals could an encoder be connected?

- **Motor:**

How can you determine the resistance of the Stator windings?

What is the resistance of the stator?

Can the motor be connected as Delta or Wye?

Which parameters are used to determine the Volts/Hz ratio for this drive?

Calculate the Volts/Hz ratio. _____ V/Hz

- **Dynamic Braking Resistor:**

Can a dynamic braking resistor be used on the drive?

If so, what are the braking resistor terminal numbers?

What is purpose of the dynamic braking resistor?

- **DC Bus:**

What are the DC bus voltage terminal numbers?

Using the voltmeter, what does the DC Buss measure?

While measuring the bus voltage, turn off the main disconnect and measure the time required for the voltage level to drop below 20 Vdc.

Why did the bus not go to zero volts instantaneously?

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PowerFlex 40p Hardware Exercise 2

Hardware Exercise Objective:

To become familiar with Configuring and Verifying operations of the Allen-Bradley Power Flex 40p AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

- 1) **Power up and Configure the drive with the HIM using the parameters shown on the simulator drawing package.**

After loading all of the parameters, start with step #2 below :

- 2) Turn off SW-1, SW-2, SW-3 and SW-4 on the simulator.
- 3) Set the Direction Selector Switch to Forward.
- 4) Set the Speed Potentiometer to the minimum setting.
- 5) Does the Status Line on the HIM display show '**Stopped**' ?
- 6) Why does the Status line show '**Stopped**' ?
- 7) Which Light(s) connected to the digital Outputs are '**On**' ?
- 8) What parameter and conditions will turn on these lights?
Red Light : Parameter # _____ Condition : _____
Yellow Light : Parameter # _____ Condition : _____
Green Light : Parameter # _____ Condition : _____
- 9) Press the Start Button and Observe the Status Line on the HIM.
- 10) Does the motor rotate?

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- 11) Does the Status Line on the HIM Display show '**AT Reference**' ?
- 12) Why does the drive think it is '**At Reference**' when the motor is not rotating?
- 13) Slowly increase the Speed Pot until reaching the maximum setting. Does the drive output the Maximum frequency, as programmed?
- 14) Slowly reduce the setting of the potentiometer until the drive is running at 50 Hz. Then, change the direction of the drive. Does the drive change direction instantly?
- 15) If it doesn't explain why.
- 16) Set the Selector switch back to Forward and set the potentiometer to about 10%.
- 17) Cycle the SW-1, SW-2 and SW-3 switches through each of the Preset Frequency combinations and complete the table for each of the 8 combinations.

SW-3	SW-2	SW-1	Output Frequency	Parameter #	Preset Number
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

- 18) Explain why the drive doesn't operate preset frequency #0 (Refer to Parameter P 038 in the user Manual).
- 19) Stop the drive using the Stop button.
- 20) Which parameter configures the SW-4 Digital Input and what function does SW-4 perform when operated.

Parameter # _____ Function _____

- 21) Turn on SW-4 and observe what happens.

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22) Release SW-4 and press the Start button on the simulator.

23) Turn on SW-4 again.

24) Did SW-4 perform the same function as before?

25) If not, what is the reason for this?

26. Which parameter configures the Analog Output and what is it configured for??

Parameter # _____ Configured as _____

27) Set the drive to run at 68Hz using the 0 - 10V input as a speed reference.

28) Calculate what the Analog Input Voltage should be to run at this frequency.

Calculated Voltage: _____ Vdc *Analog Input Vdc/10V = 68Hz/Max Freq.*

29) Measure the Analog Input Voltage at the drive terminals using a Voltmeter.

Measured Voltage _____ Vdc

30) Which Basic Display Parameter will display the Analog Input Value on the HIM?

Parameter # _____ Verify the value using this parameter.

31) Calculate what should the Analog Output voltage equal and then verify with your voltmeter.

Analog Output Vdc/10 Vdc = Output Frequency / Max Frequency

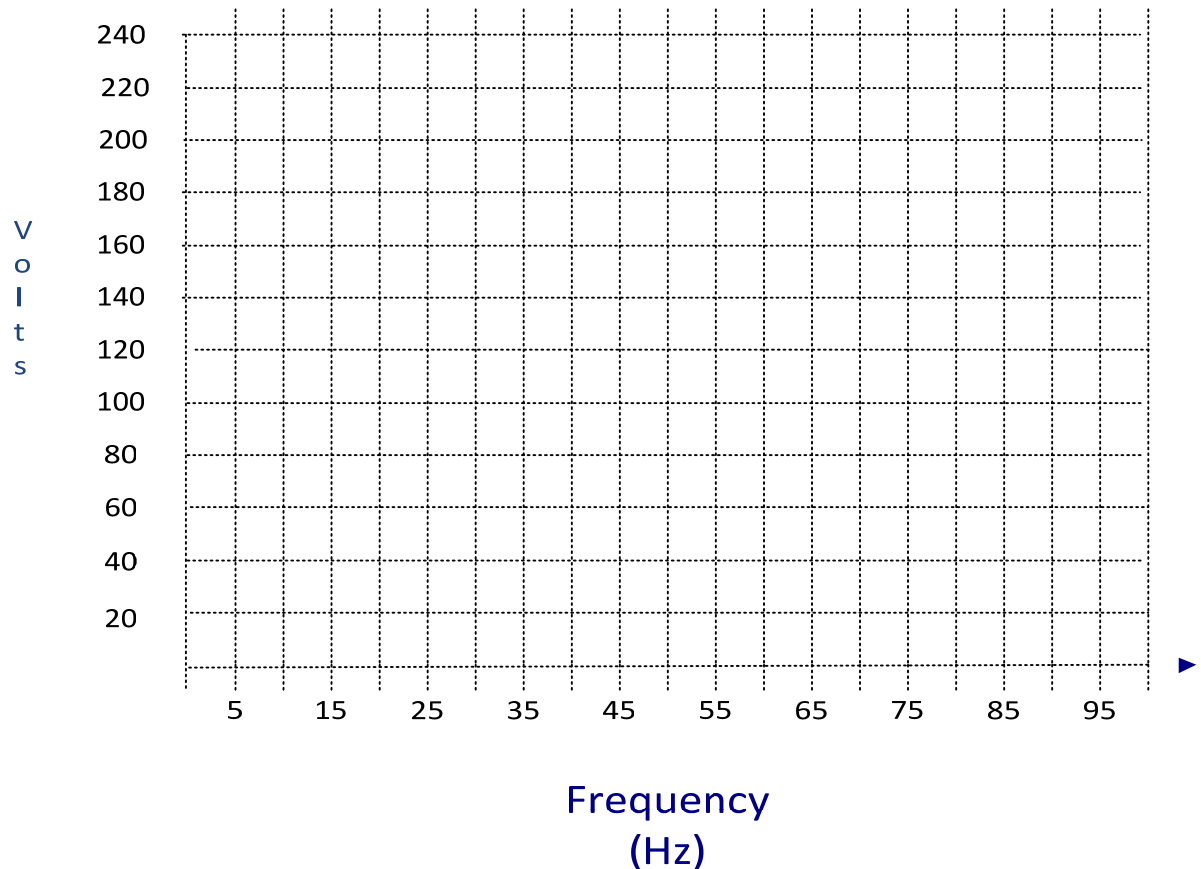
Calculated Voltage _____ Vdc Measured Value _____ Vdc

On the following page, you will plot the Volts/Hz curve for this drive.
Before proceeding, change the Accel 1 parameter (P 039) to a value of 0.2 seconds and
the Torque Performance Mode parameter (A 125) to 0 (Volts/Hz).

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Volts/Hertz Curve (Plotted)



Step 1: Calculate the Volts/Hertz ratio for your drive

(V/Hz = Motor Nameplate Voltage / Motor Nameplate Frequency)

V/Hz = _____ Plot this curve for each 5Hz change.

Step 2: Using the HIM Display to view the Output Voltage and the Potentiometer as your Speed Reference, slowly increase the speed from 0Hz in 5 Hz increments and plot the Output Voltage on the chart.

Step 3: Compare the theoretical graph to the actual graph

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PowerFlex 40p Hardware Exercise 3

To become familiar with the Accel/Decel functions and using the Internal Frequency to control the speed reference of the Allen-Bradley Power Flex 40 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedures:

The drive should be configured for the parameters associated with the drive simulator.

Hardware Exercise Procedure:

- 1) Verify operation of the drive.
- 2) Program Accel Time 1 for 5 Seconds and Decel Time 1 for 15 seconds.
- 3) Reset all SW-1 – Sw-4 switches.
- 4) Turn the Potentiometer to maximum (CW).
- 5) Start the drive and record the actual Accel time to reach 100 Hz. ____Seconds
- 6) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 7) Calculate the Acceleration Rate. _____ Hz/Sec
- 8) Calculate the Deceleration Rate. _____ Hz/Sec
- 9) *Start the drive and set the Potentiometer to output 50 Hz to the motor and then leave it at this setting. Then, Stop the drive and proceed to the next step.*
- 10) Start the drive and record the actual Accel time to reach 50 Hz. ____Seconds
- 11) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 12) Explain why the times are different between Step 6 and Step 11?

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- 13) Configure Accel Time 2 (Parameter # ____) for 10 seconds.
- 14) Configure Decel Time 2 (Parameter # ____) for 10 seconds.

- 15) Configure Accel Time 3 (Parameter # ____) for 15 seconds.
- 16) Configure Decel Time 3 (Parameter # ____) for 15 seconds.

- 17) Configure Accel Time 4 (Parameter # ____) for 20 seconds.
- 18) Configure Decel Time 4 (Parameter # ____) for 20 seconds.

- 19) Configure Digital Input 1 (Parameter A051) for a Value = 1.
- 20) Configure Digital Input 2 (Parameter A052) for a value of 28.

- 21) Turn off SW-1, SW-2, SW-3 and SW-4
- 22) Turn the Potentiometer to maximum (CW).
- 23) Start the drive and record the Accel time in the following table and then stop the drive and record the Decel time, repeating for each condition in the table (Cycling SW-1 and SW-2 for each step).

SW-2	SW-1	Accel Time (Seconds)	Decel Time (Seconds)
0	0		
0	1		
1	0		
1	1		

- 24) Stop the drive.
- 25) Configure the drive so that the Speed Reference is “**Internal Frequency**”.

Parameter # ____ = _____. What is the setting of the Internal Frequency
Parameter? ____ Hz

- 26) Configure SW-1 to cause the drive to “**MOP Up**”. Parameter # ____ = ____
- 27) Configure SW-2 to cause the drive to “**MOP Dn**”. Parameter # ____ = ____
- 28) Start the drive and document the output frequency. _____ Hz
- 29) Toggle SW-1 and observe the output frequency. Explain why the output frequency changed and in which direction.

- 30) Toggle SW-2 and observe the output frequency. Explain why the output frequency changed and in which direction.

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PowerFlex 40p Hardware Exercise 4

Hardware Exercise Objective :

To become familiar with Constant Torque and Constant Horsepower control using the Allen-Bradley Power Flex 40 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

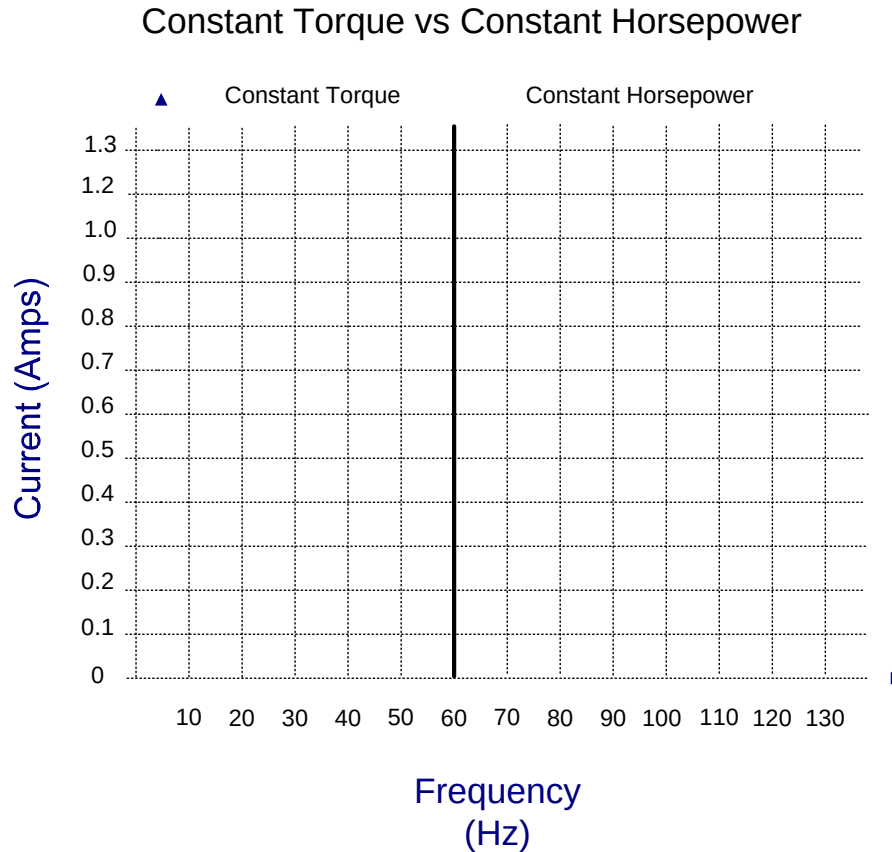
Hardware Exercise Procedure:

- 1) Configure the Speed Reference to use the 0-10Volt input and set the Potentiometer to the minimum setting.
- 2) Configure the Accel 1 Parameter setting to 0.2 Seconds.
- 3) Configure Max Frequency to 130 Hz.
- 4) Use the HIM to display the Output Current and Output Frequency.
- 5) Use the Potentiometer to control the output frequency for each of the settings shown below and document the motor current (Output Current).

10 Hz	Output Current = _____ amps
20 Hz	Output Current = _____ amps
30 Hz	Output Current = _____ amps
40 Hz	Output Current = _____ amps
50 Hz	Output Current = _____ amps
60 Hz	Output Current = _____ amps
70 Hz	Output Current = _____ amps
80 Hz	Output Current = _____ amps
90 Hz	Output Current = _____ amps
100 Hz	Output Current = _____ amps
110 Hz	Output Current = _____ amps
120 Hz	Output Current = _____ amps
130 Hz	Output Current = _____ amps

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6) Use the following chart to graph the motor current for each frequency above.



7) Based on the findings in the step 4 and step 5, and the basic motor horsepower formula **Motor HP = Torque * Speed** and the assumption that motor current is directly related to Motor Torque, we can make a determination that the drive controls in Constant Torque from _____ Hz up to _____ Hz.

It controls in Constant Horsepower from _____ Hz to _____ Hz.

Summary

Constant Torque Area – The AC Drive supplies rated motor nameplate voltage and motor develops full horsepower at 60 hertz base frequency.

Constant Horsepower Area – The AC Drive delivers motor nameplate rated voltage from 60 Hertz to 130 hertz (or drive maximum). Motor horsepower is constant in this range but motor torque is reduced as frequency increases.

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PowerFlex 40p Hardware Exercise 5

Hardware Exercise Objective :

To become familiar with programming Skip Frequencies and various other features of the Allen-Bradley Power Flex 40 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

- 1) Configure the Relay Output light to turn on when your trainer is at speed. Start the drive and verify operation.

- 2) Configure Opto -1 Output to turn on when the motor is running. Start the drive and verify the operation.

- 3) Configure the **Max Frequency** to equal **100 Hz**.

- 4) Configure the analog output to vary its voltage based on drive output current. Start the drive and verify the operation by varying the speed from max to half speed.

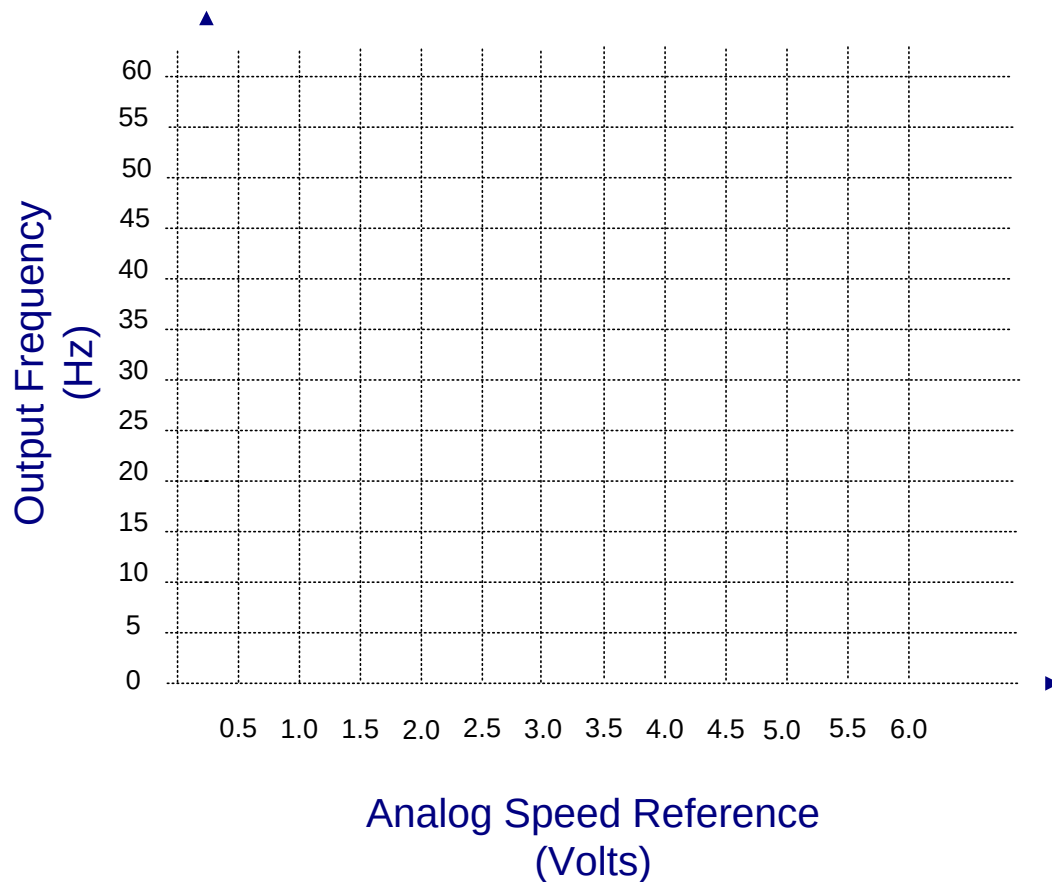
Analog Output Voltage / 10 Vdc = Motor Current / 200% Drive Rated Output Current

Do not change the analog output to current. Example: 0 – 20 ma or 4 – 20 ma.

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- 5) Set a skip frequency of 40 hertz and a bandwidth of 10 hertz. Reduce Accel1 time to 0.2 Seconds. Using the Analog input as your speed reference, start the drive at minimum speed and slowly increase the speed reference in 5 Hz increments and graph the output frequency below.

Skip Frequency Graphing Chart



- 6) What range of frequencies were actually skipped?

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- 8) What parameter(s) stores the drive's fault code?
- 9) How do you determine what your most recent fault is, and how do you determine what is your least recent fault?
- 10) Which parameter allows you to set the drive up to run as a Volts/Hz drive or a Sensorless Vector Control drive?
- 11) What is the purpose of parameter A114?
- 12) What is the difference between volts per hertz and sensorless vector control?
- 13) What is the purpose of parameter A126?
- 14) What is the purpose of parameter A127?
- 15) What is the difference between "Static Tune" and "Rotate Tune" when performing an Autotune?

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PowerFlex 70 Drive Hardware Exercises

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PowerFlex 70 Hardware Exercise 1

Hardware Exercise Objective:

To become familiar with the drive's interfaces, locations, and configuration parameters.

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
PowerFlex 70 User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedure:

Power up the simulator and set the drive up to the factory default parameter settings.

Which parameter did you use?

Hardware Exercise Procedure:

- 1) Locate and record the PowerFlex 70 Part number from the simulator
 - .Describe (in detail) the catalog number explanation of the drive.

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2) Using the User Manual and/or Schematic drawing, locate each of the following interface terminals to answer the following questions.

- **AC Input Terminals:**
What terminals does the AC Line connect to?
- **Drive Output Terminals:**
What terminals does the motor connect on the drive?
- **Drive Guard / Safe Off**
Is the drive configured for Drive Guard / Safe Off?
- **Digital Inputs:**
What are the terminal numbers of the Digital Inputs?

Are the simulator's Digital Inputs (as drawn on the schematic) setup for sinking or sourcing?

Which parameters are used to configure each of the Digital Inputs?

Digital Input 1: _____ Value/Description _____

Digital Input 2: _____ Value/Description _____

Digital Input 3: _____ Value/Description _____

Digital Input 4: _____ Value/Description _____

Digital Input 5: _____ Value/Description _____

Digital Input 6: _____ Value/Description _____

Which parameter(s) will allow you to view the status of the Digital Inputs?

Parameter # _____ Value _____

What type of parameter is this (ENUM, Numeric, or Bit)? _____

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Analog Input(s):

How many Analog Inputs does the drive have?

How many are used (wired) on the simulator and what is connected to them?

What are the drive's actual analog input terminals and wire numbers (from the schematic drawing)?

Can the Analog voltage be unipolar or bipolar?

How does the drive react differently between unipolar and bipolar control?

Which parameters will allow you to view the Analog Input 1 and Analog Input 2 values?

- **Analog Output(s):**

How many Analog Outputs does the drive have?

Which parameter(s) is used to configure the Analog Output(s)?

- **Digital Outputs:**

How many Digital Outputs does the drive have?

List each Digital Output and whether it is a Relay or Solid State Output.

- **Feedback:**

Does the drive have a terminal strip for connection for an encoder?

What type(s) of encoders can be used?

What terminals could an encoder be connected?

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- **Motor:**
How can you determine the resistance of the Stator windings?

What is the resistance of the stator?

Can the motor be connected as Delta or Wye?

- **Dynamic Braking Resistor:**
Can a dynamic braking resistor be used on the drive?

If so, what are the braking resistor terminal numbers?

What is purpose of the dynamic braking resistor?

- **DC Bus:**
What are the DC bus voltage terminal numbers?

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PowerFlex 70 Hardware Exercise 2

Hardware Exercise Objective:

To become familiar with Configuring and Verifying the Allen-Bradley Power Flex 70 AC Drive. To understand the Volts/Hertz ratio as it applies to the drive.

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
PowerFlex 70 User Manual
Digital Multi-meter
Simulator Drawing Package

- 1) **Power up and Configure the drive with the HIM using the parameters shown on the simulator drawing package.**

After completing Step 1, verify the following parameters:

Parameter 83 is set for 0 (Overspeed Limit)

Parameter 91 is set for 100Hz (Max. Frequency).

Refer to the following table as a reference for verifying the operation of the drive

PowerFlex 70 and 700 Default Auto Speed Reference Options							
Digital Inputs Status			Speed Reference Selections	Parameter	Default Setting	Default Selection	Default Setting
Digital Input 6	Digital Input 5	Digital Input 4					
0	0	0	Speed Ref. A	90	2	Analog In "2"	0-10Vdc
0	0	1	Speed Ref. B	93	11	Preset Speed 1	20 Hz
0	1	0	Preset Speed 2	102	10	30 Hz	
0	1	1	Preset Speed 3	103	20	40 Hz	
1	0	0	Preset Speed 4	104	30	50 Hz	
1	0	1	Preset Speed 5	105	40	60 Hz	
1	1	0	Preset Speed 6	106	50	70 Hz	
1	1	1	Preset Speed 7	107	60	80 Hz	

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- 2) Turn off SW-1, SW-2 and SW-3 on the simulator.
- 3) Set the Direction Selector Switch to Forward.
- 4) Set the Speed Potentiometer to the minimum setting.
- 5) Does the Status Line on the HIM display show '**Stopped**' ?
- 6) Why does the Status line show '**Stopped**' ?
- 7) Which Light(s) connected to the digital Outputs are '**On**' ?
- 8) What conditions will turn on these lights?
- 9) Press the Start Button.
- 10) Does the motor rotate?
- 11) If not, why does it not rotate?
- 12) Does the Status Line on the HIM Display shows '**AT Speed**' ?
- 13) Why does the drive think it is '**At Speed**' when the motor is not rotating?
- 14) Slowly increase the Speed Pot until reaching the maximum setting. Does the drive output the Maximum frequency, as programmed?
- 15) Which light(s) are on and what drive conditions cause them to be on?

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16) Slowly reduce the setting of the potentiometer until the drive is outputting 50 Hz. Then, change the direction of the drive. Does the drive change direction instantly?

17) If it doesn't explain why.

18) Set the Selector switch back to Forward.

19) Using the reference table from above, cycle the SW-1, SW-2 and SW-3 switches through each of the Preset Frequency selections and verify the correct output frequency for each of the 8 combinations.

20) Does the drive operate all of the different preset frequencies as shown in the table?

21) Stop the drive using the Stop button.

22) What is the Analog Output configured as on the simulator?

23) If the drive is running at 68Hz, what should the Analog Output voltage equal?

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PowerFlex 70 Hardware Exercise 3

Hardware Exercise Objective:

To become familiar with the Accel/Decel functions and using the MOP to control the speed reference of the Allen-Bradley Power Flex 70 AC Drive. To become familiar with using the HIM as a diagnostic tool during troubleshooting.

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
PowerFlex 70 User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedures:

The drive should be configured for the parameters associated with the drive simulator.

Hardware Exercise Procedure:

- 1) Verify operation of the drive.
- 2) Program Accel Time 1 for 5 Seconds and Decel Time 1 for 15 seconds.
- 3) Reset all SW-1 – Sw-4 switches.
- 4) Turn the Potentiometer to maximum (CW).
- 5) Start the drive and record the actual Accel time to reach 100 Hz. ____Seconds
- 6) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 7) Calculate the Acceleration Rate. _____ Hz/Sec
- 8) Calculate the Deceleration Rate. _____ Hz/Sec
- 9) Set the Potentiometer to 50% (Outputs 50 Hz).
- 10) Start the drive and record the actual Accel time to reach 50 Hz. ____Seconds
- 11) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 12) Explain why the times are different between Step 5 and Step 6?

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- 13) Stop the drive and reset all switches.
14) Configure the drive so that the Speed Reference is “**MOP Level**”.

Parameter # ____ = ____.

- 15) Configure SW-1 to cause the drive to “MOP Inc”. Parameter # ____ = ____
16) Configure SW-2 to cause the drive to “MOP Dec”. Parameter # ____ = ____
17) Start the drive and document the output frequency. _____ Hz
18) Toggle SW-1 and observe the output frequency. Explain why the output frequency changed and in which direction.

- 19) Toggle SW-2 and observe the output frequency. Explain why the output frequency changed and in which direction.

The following steps involve using the HIM to obtain diagnostic information from the drive during troubleshooting.

- 20) Follow the proper procedure for removing the keypad while the drive is running.
21) When viewing a parameter what purpose does **alt/view** serve?
22) What is the purpose of parameter 212?
23) What is the purpose of parameter 213? Verify operation.
24) What is the purpose of parameter 214? Verify operation.
25) What is the purpose of parameter 215? Verify operation.
26) What is the purpose of parameter 216? Verify operation.
27) What is the purpose of parameter 217? Verify operation.

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28) Write the purpose of the following parameters:

Parameter 224:

Parameter 225:

Parameter 226:

Parameter 227:

Parameter 228:

Parameter 229:

Parameter 230:

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PowerFlex 70 Hardware Exercise 4

Hardware Exercise Objective:

To become familiar with Alarms, Conflicts and Faults in the Allen-Bradley Power Flex 70 AC Drive as well as its applications.

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive
ACD Course Manual
PowerFlex 70 User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

Note: Use the user manual as a reference to complete this Hardware Exercise

- 1) Configure Digital Input #1 as “**Not Used**” (Parameter 361 = 0). What type of conflict did this create? How is this conflict indicated?
- 2) Re-configure Digital Input # 1 back to a “**Stop**” input. Then, configure Digital Input #2 as a “Run” input (Parameter 362 = 8). What type of Conflict did this create.
- 3) Re-configure Digital Input # 2 back to a “**Start**” input.
- 4) What is Alarm 27?
- 5) What is Fault # 38? How would you confirm this fault?
- 6) What is Fault 41? How could you confirm this fault?
- 7) Refer to Parameters 238 and 259. Can certain alarms and faults be enabled/disabled? If so, how? Can these two faults be enabled or disabled?

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8) A Fault is a condition that will stop the drive. What are the three types of faults associated with this drive?

Type (Number)	Fault	Description

9) An alarm is a condition that, if left untreated, may stop the drive. What are the two alarm types.

Type (Number)	Alarm	Description

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PowerFlex 70 Hardware Exercise 5

Hardware Exercise Objective:

To become familiar with the Diagnostics Parameters of the Allen-Bradley Power Flex 70 AC Drive during troubleshooting

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive
ACD Course Manual
PowerFlex User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

- 1) What parameter(s) display the drive's status?
- 2) What parameter displays the drive's DC BUS Voltage?
- 3) What parameter displays the drive's Output Frequency?
- 4) What parameter displays the drive's Output Current?
- 5) What parameter displays the drive's Output Voltage?
- 6) How do you determine the value of the commanded frequency?
- 7) How do you change the drive from volts per hertz to sensorless vector control?
- 8) What is the difference between volts per hertz and sensorless vector control?
- 9) What is the purpose of parameter 042?

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Note: Motor commissioning parameters are from parameter 040 – 046.

10) What parameter initiates the auto-tune?

11) Name the three types of auto-tune available, and give a brief description of each.

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PowerFlex 70 Hardware Exercise 6

Hardware Exercise Objective:

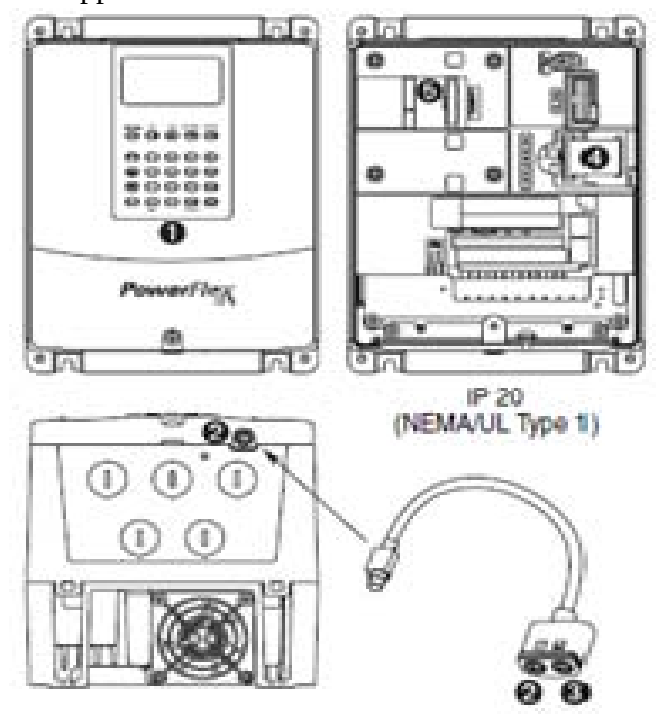
To become familiar with the operations of the communications ports and HIM used with the Allen-Bradley Power Flex 70 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 70 AC Drive
ACD Course Manual
PowerFlex 70 User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

Refer to Appendix B in the User Manual to answer the following:



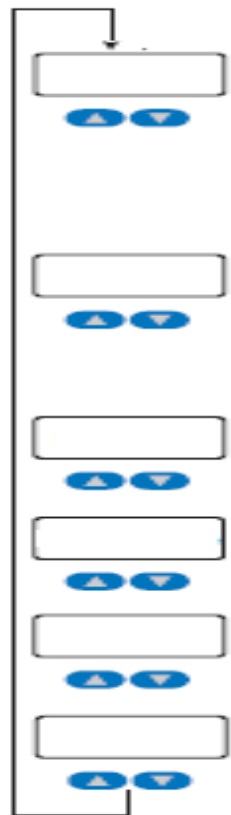
1) Complete the table below for each of the items shown:

Number	Connector	Description
1		
2		
3		
4		
5		

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2) Explain how to safely disconnect the HIM from the drive.

3) Fill in the structure of the Main Menu accessible from the HIM?



4) After selecting the Diagnostics Menu, fill in the description information for the following table:

Option	Description
Faults	
Status Info	
Device Items	
Device Version	
HIM Version	

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5) What functions can be performed from the Device Select Menu?

6) What functions can be performed from the Memory Storage Menu?

Option	Description
HIM Copycat Device -> HIM Device <- HIM	
Device User Sets	
Reset To Defaults	

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PowerFlex 700 Drive Hardware Exercises

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PowerFlex 700 Hardware Exercise 1

Hardware Exercise Objective:

To become familiar with the drive's interfaces, locations, and configuration parameters.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedure:

Power up the simulator and set the drive up to the factory default parameter settings.

Which parameter did you use?

Hardware Exercise Procedure:

1) Locate and record the PowerFlex 700 Part number from the simulator

- .Describe (in detail) the catalog explanation of the drive.

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2) Using the User Manual and/or Schematic drawing, locate each of the following interface terminals to answer the following questions.

- **AC Input Terminals:**
What terminals does the AC Line connect to?
- **Drive Output Terminals:**
What terminals does the motor connect on the drive?
- **Drive Guard / Safe Off**
Is the drive configured for Drive Guard / Safe Off?
- **Digital Inputs:**
What are the terminal numbers of the Digital Inputs?

Are the simulator's Digital Inputs (as drawn on the schematic) setup for sinking or sourcing?

Which parameters are used to configure each of the Digital Inputs?

Digital Input 1: _____ Value/Description _____

Digital Input 2: _____ Value/Description _____

Digital Input 3: _____ Value/Description _____

Digital Input 4: _____ Value/Description _____

Digital Input 5: _____ Value/Description _____

Digital Input 6: _____ Value/Description _____

Which parameter(s) will allow you to view the status of the Digital Inputs?

Parameter # _____ Value _____

What type of parameter is this (ENUM, Numeric, or Bit)? _____

Turn on SW-1 and SW-3 and use this parameter to display the Digital Input Status.

Digital Input	6	5	4	3	2	1
Status (1 or 0)						
Bit Number						

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Analog Input(s):

How many Analog Inputs does the drive have?

How many are used (wired) on the simulator and what is connected to them?

What are the drive's actual analog input terminals and wire numbers (from the schematic drawing)?

Can the Analog voltage be unipolar or bipolar?

How does the drive react differently between unipolar and bipolar control?

Which parameters will allow you to view the Analog Input 1 and Analog Input 2 values?

- **Analog Output(s):**

How many Analog Outputs does the drive have?

Which parameter(s) is used to configure the Analog Output(s)?

- **Digital Outputs:**

How many Digital Outputs does the drive have?

List each Digital Output and whether it is a Relay or Solid State Output.

- **Feedback:**

Does the drive have a terminal strip for connecting an encoder?

What type(s) of encoders can be used?

What terminals could an encoder be connected?

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- **Motor:**
How can you determine the resistance of the Stator windings?

What is the resistance of the stator?

Can the motor be connected as Delta or Wye?

- **Dynamic Braking Resistor:**
Can a dynamic braking resistor be used on the drive?

If so, what are the braking resistor terminal numbers?

What is purpose of the dynamic braking resistor?

- **DC Bus:**
What are the DC bus voltage terminal numbers?

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PowerFlex 700 Hardware Exercise 2

Hardware Exercise Objective:

To become familiar with Configuring and Verifying the Allen-Bradley Power Flex 700 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

- 1) **Power up and Configure the drive with the HIM using the parameters shown on the simulator drawing package.**

After completing Step 1, verify the following parameters:

Parameter 83 is set for 0

Parameter 91 is set for 100Hz (Max. Frequency).

Refer to the following table as a reference for verifying the operation of the drive

PowerFlex 70 and 700 Default Auto Speed Reference Options								
Digital Inputs Status			Speed Reference Selections	Parameter	Default Setting	Default Selection	Default Setting	
Digital Input 6	Digital Input 5	Digital Input 4						
0	0	0	Speed Ref. A	90	2	Analog In "2"	0-10Vdc	
0	0	1	Speed Ref. B	93	11	Preset Speed 1	20 Hz	
0	1	0	Preset Speed 2	102	10	30 Hz		
0	1	1	Preset Speed 3	103	20	40 Hz		
1	0	0	Preset Speed 4	104	30	50 Hz		
1	0	1	Preset Speed 5	105	40	60 Hz		
1	1	0	Preset Speed 6	106	50	70 Hz		
1	1	1	Preset Speed 7	107	60	80 Hz		

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- 2) Turn off SW-1, SW-2 and SW-3 on the simulator.
- 3) Set the Direction Selector Switch to Forward.
- 4) Set the Speed Potentiometer to the minimum setting.
- 5) Does the Status Line on the HIM display show '**Stopped**' ?

- 6) Why does the Status line show '**Stopped**' ?

- 7) Which Light(s) connected to the digital Outputs are '**On**' ?

- 8) What conditions will turn on these lights?

- 9) Press the Start Button.
- 10) Does the motor rotate?

- 11) If not, why does it not rotate?

- 12) Does the Status Line on the HIM Display shows '**AT Speed**' ?

- 13) Why does the drive think it is '**At Speed**' when the motor is not rotating?

- 14) Slowly increase the Speed Pot until reaching the maximum setting. Does the drive output the Maximum frequency, as programmed?

- 15) Which light(s) are on and what drive conditions cause them to be on?

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16) Slowly reduce the setting of the potentiometer until the drive is outputting 50 Hz. Then, change the direction of the drive. Does the drive change direction instantly?

17) If it doesn't explain why.

18) Set the Selector switch back to Forward.

19) Using the reference table from above, cycle the SW-1, SW-2 and SW-3 switches through each of the Preset Frequency selections and verify the correct output frequency for each of the 8 combinations.

20) Does the drive operate all of the different preset frequencies as shown in the table?

21) Stop the drive using the Stop button.

22) What is the Analog Output configured as on the simulator?

23) If the drive is running at 68Hz, what should the Analog Output voltage equal?

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PowerFlex 700 Hardware Exercise 3

Hardware Exercise Objective:

To become familiar with the Accel/Decel functions and using the MOP to control the speed reference of the Allen-Bradley Power Flex 700 AC Drive. To become familiar with using the HIM as a diagnostic tool during troubleshooting.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedures:

The drive should be configured for the parameters associated with the drive simulator.

Hardware Exercise Procedure:

- 1) Verify operation of the drive.
- 2) Program Accel Time 1 for 5 Seconds and Decel Time 1 for 15 seconds.
- 3) Reset all SW-1 – SW-3 switches.
- 4) Turn the Potentiometer to maximum (CW).
- 5) Start the drive and record the actual Accel time to reach 100 Hz. ____Seconds
- 6) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 7) Calculate the Acceleration Rate. ____ Hz/Sec
- 8) Calculate the Deceleration Rate. ____ Hz/Sec
- 9) Set the Potentiometer to 50% (Outputs 50 Hz).
- 10) Start the drive and record the actual Accel time to reach 50 Hz. ____Seconds
- 11) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 12) Explain why the times are different between Step 5 and Step 6?

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- 13) Stop the drive and reset all SW switches.
- 14) Configure the drive so that the Speed Reference is “MOP Level”.
Parameter # _____ = _____.
- 15) Configure SW-1 to cause the drive to “MOP Inc”. Parameter # _____ = _____
- 16) Configure SW-2 to cause the drive to “MOP Dec”. Parameter # _____ = _____
- 17) Start the drive and document the output frequency. _____ Hz
- 18) Toggle SW-1 and observe the output frequency. Explain why the output frequency changed and in which direction.
- 19) Toggle SW-2 and observe the output frequency. Explain why the output frequency changed and in which direction.

The following steps involve using the HIM to obtain diagnostic information from the drive during troubleshooting.

- 20) Follow the proper procedure for removing the keypad while the drive is running.
- 21) When viewing a parameter what purpose does ***alt/view*** serve?
- 22) What is the purpose of parameter 212?
- 23) What is the purpose of parameter 213? Verify operation.
- 24) What is the purpose of parameter 214? Verify operation.
- 25) What is the purpose of parameter 215? Verify operation.
- 26) What is the purpose of parameter 216? Verify operation.
- 27) What is the purpose of parameter 217? Verify operation.

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28) Write the purpose of parameters: 224 – 230?

224:

225:

226:

227:

228:

229:

230:

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PowerFlex 700 Hardware Exercise 4

Hardware Exercise Objective:

To become familiar with Alarms and Conflicts in the Allen-Bradley Power Flex 700 AC Drive as well as its applications.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

Note: Use the user manual as a reference to complete this Hardware Exercise

- 1) Configure Digital Input #1 as “**Not Used**” (Parameter 361 = 0). What type of conflict did this create? How is this conflict indicated?
- 2) Re-configure Digital Input # 1 back to a “**Stop**” input. Then, configure Digital Input #2 as a “Run” input (Parameter 362 = 8). What type of Conflict did this create.
- 3) Re-configure Digital Input # 2 back to a “**Start**” input.
- 4) What is Alarm 27?
- 5) What is Fault # 38? How would you confirm this fault?
- 6) What is Fault 41? How could you confirm this fault?
- 7) Refer to Parameters 238 and 259. Can certain alarms and faults be enabled/disabled? If so, how? Can these two faults be enabled or disabled?

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- 8) A Fault is a condition that will stop the drive. What are the three types of faults associated with this drive?

Type (Number)	Fault	Description

- 9) An alarm is a condition that, if left untreated, may stop the drive. What are the two alarm types.

Type (Number)	Alarm	Description

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PowerFlex 700 Hardware Exercise 5

Hardware Exercise Objective:

To become familiar with the Diagnostics Parameters of the Allen-Bradley Power Flex 700 AC Drive during troubleshooting.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

- 1) What parameter(s) display the drive's status?
- 2) What parameter displays the drive's DC BUS Voltage?
- 3) What parameter displays the drive's Output Frequency?
- 4) What parameter displays the drive's Output Current?
- 5) What parameter displays the drive's Output Voltage?
- 6) How do you determine the value of the commanded frequency?
- 7) How do you change the drive from volts per hertz to sensorless vector control?

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8) What is the difference between volts per hertz and sensorless vector control?

9) What is the purpose of parameter 042?

Note: Motor commissioning parameters are from parameter 040 – 046.

12) What parameter initiates the auto-tune?

13) Name the three types of auto-tune available, and give a brief description of each.

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PowerFlex 700 Hardware Exercise 6

Hardware Exercise Objective:

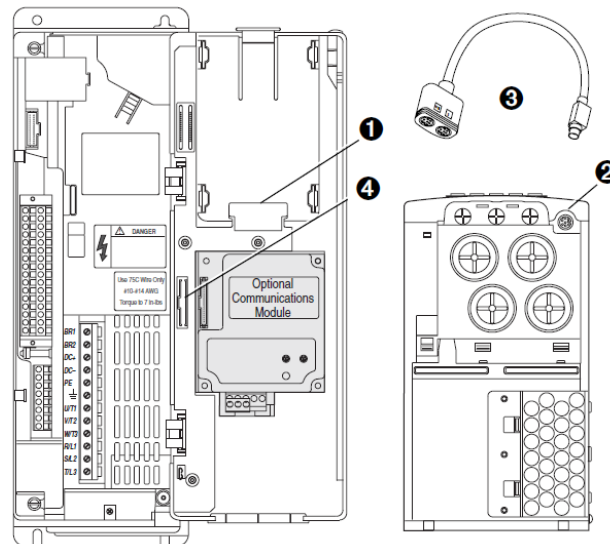
To become familiar with the operations of the communications ports and HIM used with the Allen-Bradley Power Flex 700 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive
ACD Course Manual
PowerFlex 700 User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

Refer to Appendix B in the User Manual to answer the following:



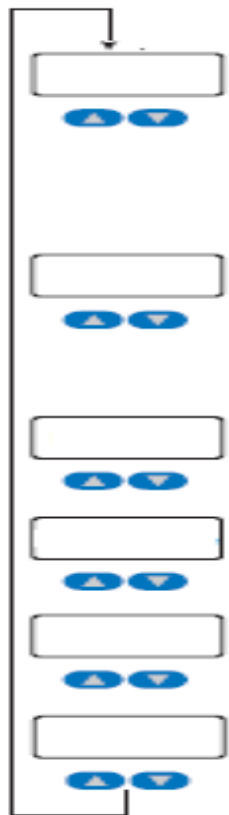
7) Complete the table below for each of the items shown:

Number	Connector	Description
1		
2		
3		
4		

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8) Explain how to safely disconnect the HIM from the drive.

9) Fill in the structure of the Main Menu accessible from the HIM?



10) After selecting the Diagnostics Menu, fill in the description information for the following table:

Option	Description
Alarms	
Faults	
Status Information	
Device Version	
HIM Version	

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11) What functions can be performed from the Device Select Menu?

12) What functions can be performed from the Memory Storage Menu?

Option	Description
HIM Copycat Device -> HIM Device <- HIM	
Device User Sets	
Reset To Defaults	

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PowerFlex 40P-DG Drive Hardware Exercises

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PowerFlex 40p (DG) Hardware Exercise 1

Hardware Exercise Objective:

To become familiar with the drive's interfaces, locations, and configuration parameters.

Equipment Needed:

Allen-Bradley Power Flex 40p (DG) AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedure:

Power up the simulator and set the drive up to the factory default parameter settings.

Which parameter did you use?

Hardware Exercise Procedure:

Use the course manual, User Manual and simulator drawing package to answer the following questions

1) Locate and record the PowerFlex 40p Part number from the simulator

- .Describe (in detail) the catalog explanation of the drive.

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2) Using the User Manual and/or Schematic drawing, locate each of the following interface terminals to answer the following questions.

- **AC Input Terminals:**
What terminals does the AC Line connect to?
- **Drive Output Terminals:**
What terminals does the motor connect on the drive?
- **Drive Guard / Safe Off**
Is the drive configured for Drive Guard / Safe Off?
- **Digital Inputs:**
What are the terminal numbers of the Digital Inputs?

Are the simulator's Digital Inputs (as drawn on the schematic) setup for sinking or sourcing?

Which parameters are used to configure each of the Digital Inputs?

Digital Input 1 Parameter: _____ Value/Description _____

Digital Input 2 Parameter: _____ Value/Description _____

Digital Input 3 Parameter: _____ Value/Description _____

Digital Input 4 Parameter: _____ Value/Description _____

What is the type of parameter (ENUM,Numeric or Bit)? _____

Which parameter(s) will allow you to view the status of the Digital Inputs?

What is the type of parameter (ENUM,Numeric or Bit)? _____

Turn on SW-1 and SW-3 and use this parameter to display the Digital Input Status.

Digital Input	4	3	2	1
Status (1 or 0)				
Bit Number				

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Which parameter will allow you to view the Control Input Status?

What is the type of parameter (ENUM,Numeric or Bit)? _____

Press the Start or Stop buttons or Fwd/Rev Switch and use this parameter to display the Digital Input Status.

Control Status Input	Dynamic Brake Transister On	I/O Terminal 1 Enable Input	I/O Terminal 3 Fwd/Rev Input	I/O Terminal 2 Start/Run Input
Status (1 or 0)				

- **Analog Input(s):**

How many Analog Inputs does the drive have?

How many are used (wired) on the simulator and what is connected to them?

What are the drive's actual analog input terminals and wire numbers (from the schematic drawing)?

Can the Analog Input voltage be unipolar or bipolar?

How should the drive react differently between unipolar and bipolar Speed References?

Which parameter will allow you to view the Analog Input values?

- **Analog Output(s):**

How many Analog Outputs does the drive have?

Which parameter is used to configure the Analog Output(s)?

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- **Digital Outputs:**

How many Digital Outputs does the drive have?

List each Digital Output and whether it is a Relay or Solid State Output.

List each parameter associated with configuring each Digital Output.

- **Feedback:**

Does the drive have a terminal strip for connection to an encoder?

What type(s) of encoders can be used?

What terminals could an encoder be connected?

- **Motor:**

How can you determine the resistance of the Stator windings?

What is the resistance of the stator?

Can the motor be connected as Delta or Wye?

Which parameters can be used to determine the Volts/Hz ratio for this drive?

Calculate the Volts/Hz ratio. _____ V/Hz

- **Dynamic Braking Resistor:**

Can a dynamic braking resistor be used on the drive?

If so, what are the braking resistor terminal numbers?

What is purpose of the dynamic braking resistor?

- **DC Bus:**

What are the DC bus voltage terminal numbers?

Using the voltmeter, what does the DC Buss measure?

While measuring the bus voltage, turn off the main disconnect and measure the time required for the voltage level to drop below 20 Vdc.

Why did the bus not go to zero volts instantaneously?

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PowerFlex 40p (DG) Hardware Exercise 2

Hardware Exercise Objective:

To become familiar with configuring and verifying operations of the Allen-Bradley Power Flex 40p AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p (DG) AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

Power up and Configure the drive with the HIM using the parameters shown on the simulator drawing package.

After loading all of the parameters, start with step #1 below :

- 1) Turn off SW-1, SW-2, SW-3 and SW-4 on the simulator.
- 2) Set the Direction Selector Switch to Forward.
- 3) Set the Speed Potentiometer to the minimum setting.
- 4) Does the Status Line on the HIM display show '**Stopped**' ?
- 5) Why does the Status line show '**Stopped**' ?
- 6) Which Light(s) connected to the digital Outputs are '**On**' ?
- 7) What parameter and conditions will turn on these lights?
Red Light : Parameter # _____ Condition : _____
Yellow Light : Parameter # _____ Condition : _____
Green Light : Parameter # _____ Condition : _____
- 8) Press the Start Button and Observe the Status Line on the HIM.
- 9) Does the motor rotate?

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- 10) Does the Status Line on the HIM Display show '**AT Reference**' ?
- 11) Why does the drive think it is '**AT Reference**' when the motor is not rotating?
- 12) Slowly increase the Speed Pot until reaching the maximum setting. Does the drive output the Maximum frequency, as programmed?
- 13) Which light(s) are on and what drive conditions cause them to be on?
- 14) Slowly reduce the setting of the potentiometer until the drive is running at 50 Hz. Then, change the direction of the drive. Does the drive change direction instantly?
- 15) If it doesn't explain why.
- 16) Set the Selector switch back to Forward and set the potentiometer to about 10%.
- 17) Cycle the SW-1, SW-2 and SW-3 switches through each of the Preset Frequency combinations and complete the table for each of the 8 combinations.

SW-3	SW-2	SW-1	Output Frequency	Parameter #	Preset #
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

- 18) Explain why the drive doesn't operate preset frequency #0 (Refer to Parameter P 038 in the user Manual).

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19) Stop the drive using the Stop button.

20) Which parameter configures the Analog Output and what is it configured for??

Parameter # _____ Configured as _____

21) Set the drive to run at 68Hz using the 0 - 10V input as a speed reference.

22) Calculate what the Analog Input Voltage should be to run at this frequency.

Calculated Voltage: _____ Vdc *Analog Input Vdc/10V = 68Hz/Max Freq.*

23) Measure the Analog Input Voltage at the drive terminals using a Voltmeter.

Measured Voltage _____ Vdc

24) Which Basic Display Parameter will display the Analog Input Value on the HIM?

Parameter # _____ Verify the value using this parameter.

25) Calculate what should the Analog Output voltage equal and then verify with your voltmeter.

Analog Output Vdc/10 Vdc = Output Frequency / Max Frequency

Calculated Voltage _____ Vdc Measured Value _____ Vdc

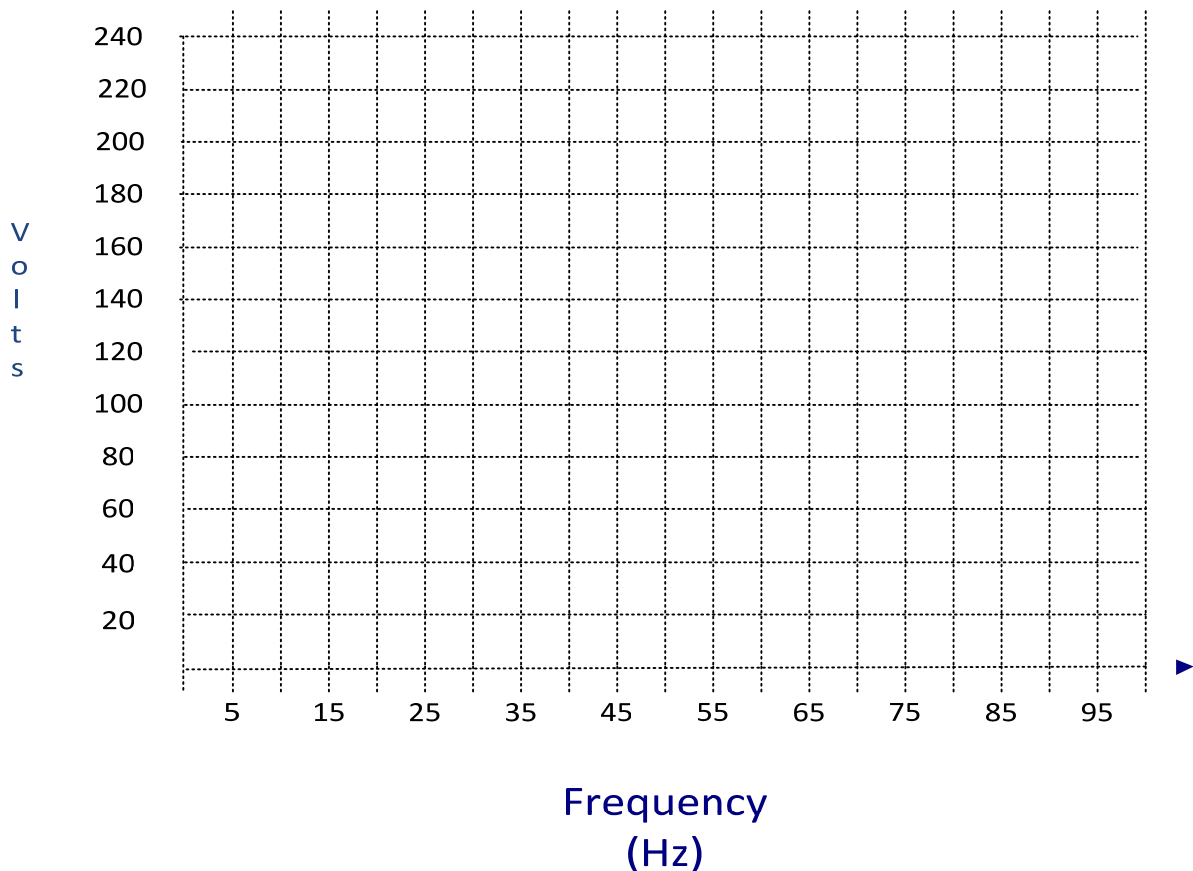
Note : If you have already plotted the Volts/Hz ratio on another Lab Exercise with another drive, you may proceed to the next exercise.

On the following page, you will plot the Volts/Hz curve for this drive. Before proceeding, change the Accel 1 parameter (P 039) to a value of 0.2 seconds and the Torque Performance Mode parameter (A 125) to 0 (Volts/Hz).

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Volts/Hertz Curve (Plotted)



Step 1: Calculate the Volts/Hertz ration for your drive

(V/Hz = Motor Nameplate Voltage / Motor Nameplate Frequency)

V/Hz = _____ Plot this curve for each 5Hz change.

Step 2: Using the HIM Display to view the Output Voltage and the Potentiometer as your Speed Reference, slowly increase the speed from 0Hz in 5 Hz increments and plot the Output Voltage on the chart.

Step 3: Compare the theoretical graph to the actual graph

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PowerFlex 40p (DG) Hardware Exercise 3

To become familiar with the Accel/Decel functions and using the Internal Frequency to control the speed reference of the Allen-Bradley Power Flex 40 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p (DG) AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Preliminary Procedures:

The drive should be configured for the parameters associated with the drive simulator.

Hardware Exercise Procedure:

- 1) Verify operation of the drive.
- 2) Program Accel Time 1 for 5 Seconds and Decel Time 1 for 5 seconds.
- 3) Reset all SW-1 – Sw-4 switches.
- 4) Turn the Potentiometer to maximum (CW).
- 5) Start the drive and record the actual Accel time to reach 100 Hz. ____Seconds
- 6) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 7) Calculate the Acceleration Rate. _____ Hz/Sec
- 8) Calculate the Deceleration Rate. _____ Hz/Sec
- 9) *Start the drive and set the Potentiometer to output 50 Hz to the motor and then leave it at this setting. Then, Stop the drive and proceed to the next step.*
- 10) Start the drive and record the actual Accel time to reach 50 Hz. ____Seconds
- 11) Stop the drive and record the actual Decel time to reach 0 Hz. ____Seconds
- 12) Explain why the times are different between Step 6 and Step 11?

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- 13) Configure Accel Time 2 (Parameter # _____) for 10 seconds.
- 14) Configure Decel Time 2 (Parameter # _____) for 10 seconds.

- 15) Configure Accel Time 3 (Parameter # _____) for 15 seconds.
- 16) Configure Decel Time 3 (Parameter # _____) for 15 seconds.

- 17) Configure Accel Time 4 (Parameter # _____) for 20 seconds.
- 18) Configure Decel Time 4 (Parameter # _____) for 20 seconds.

- 19) Configure Digital Input 1 (Parameter A051) for a Value = 1.
- 20) Configure Digital Input 2 (Parameter A052) for a value of 28.

- 21) Turn off SW-1, SW-2, SW-3 and SW-4
- 22) Turn the Potentiometer to maximum (CW).
- 23) Start the drive and record the Accel time in the following table and then stop the drive and record the Decel time for each condition in the table (Cycling SW-1 and SW-2 for each step).

SW-2	SW-1	Accel Time (Seconds)	Decel Time (Seconds)
0	0		
0	1		
1	0		
1	1		

- 24) Stop the drive.
- 25) Configure the drive so that the Speed Reference is “**Internal Frequency**”.
Parameter # _____ = _____.

- 26) Configure SW-1 to cause the drive to “**MOP Up**”. Parameter # _____ = _____
- 27) Configure SW-2 to cause the drive to “**MOP Down**”. Parameter # _____ = _____
- 28) Start the drive and document the output frequency. _____ Hz
- 29) Toggle SW-1 and observe the output frequency. Explain why the output frequency changed and in which direction.

- 30) Toggle SW-2 and observe the output frequency. Explain why the output frequency changed and in which direction.

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PowerFlex 40p (DG) Hardware Exercise 4

Hardware Exercise Objective :

To become familiar with Constant Torque and Constant Horsepower control using the Allen-Bradley Power Flex 40p AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p (DG) AC Drive Simulator
ACD Course Manual
PowerFlex User Manual
Digital Multi-meter
Simulator Drawing Package

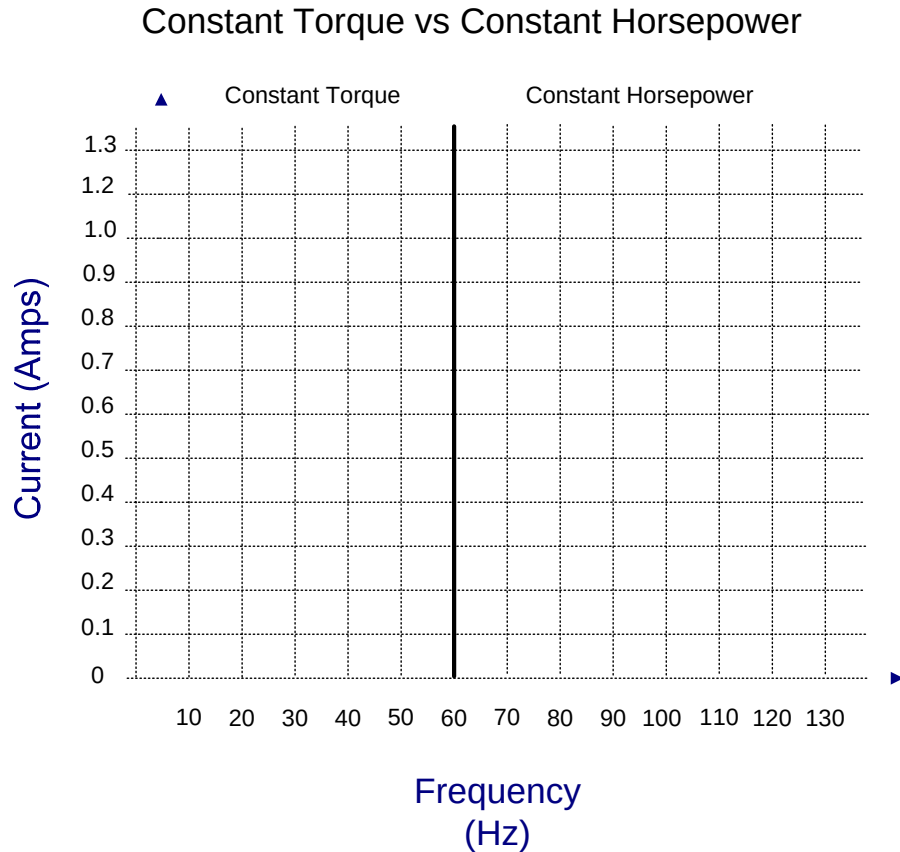
Hardware Exercise Procedure:

- 1) Configure the Speed Reference to use the 0-10Volt input and set the Potentiometer to the minimum setting.
- 2) Configure the Accel 1 Parameter setting to 0.2 Seconds.
- 3) Configure Max Frequency to 130 Hz.
- 4) Use the HIM to display the Output Current and Output Frequency.
- 5) Use the Potentiometer to control the output frequency for each of the settings shown below and document the motor current (Output Current).

10 Hz	Output Current = _____ amps
20 Hz	Output Current = _____ amps
30 Hz	Output Current = _____ amps
40 Hz	Output Current = _____ amps
50 Hz	Output Current = _____ amps
60 Hz	Output Current = _____ amps
70 Hz	Output Current = _____ amps
80 Hz	Output Current = _____ amps
90 Hz	Output Current = _____ amps
100 Hz	Output Current = _____ amps
110 Hz	Output Current = _____ amps
120 Hz	Output Current = _____ amps
130 Hz	Output Current = _____ amps

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6) Use the following chart to graph the motor current for each frequency above.



7) Based on the findings in the step 4 and step 5, and the basic motor horsepower formula **Motor HP = Torque * Speed** and the assumption that motor current is directly related to Motor Torque, we can make a determination that the drive controls in Constant Torque from _____ Hz up to _____ Hz.

It controls in Constant Horsepower from _____ Hz to _____ Hz.

Summary

Constant Torque Area – The AC Drive supplies rated motor nameplate voltage and motor develops full horsepower at 60 hertz base frequency.

Constant Horsepower Area – The AC Drive delivers motor nameplate rated voltage from 60 Hertz to 130 hertz (or drive maximum). Motor horsepower is constant in this range but motor torque is reduced as frequency increases.

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PowerFlex 40p (DG) Hardware Exercise 5

Hardware Exercise Objective :

To become familiar with programming Skip Frequencies and various other features of the Allen-Bradley Power Flex 40 AC Drive.

Equipment Needed:

Allen-Bradley Power Flex 40p (DG) AC Drive Simulator
ACD Course Manual
PowerFlex 40P User Manual
Digital Multi-meter
Simulator Drawing Package

Hardware Exercise Procedure:

- 1) Configure the Relay Output light to turn on when your trainer is at speed. Start the drive and verify operation.
- 2) Configure Opto -1 Output to turn on when the motor is running. Start the drive and verify the operation.
- 3) Configure the **Max Frequency** to equal **100 Hz**.
- 4) Configure the analog output to vary its voltage based on drive output current. Start the drive and verify the operation by varying the speed from max to half speed.

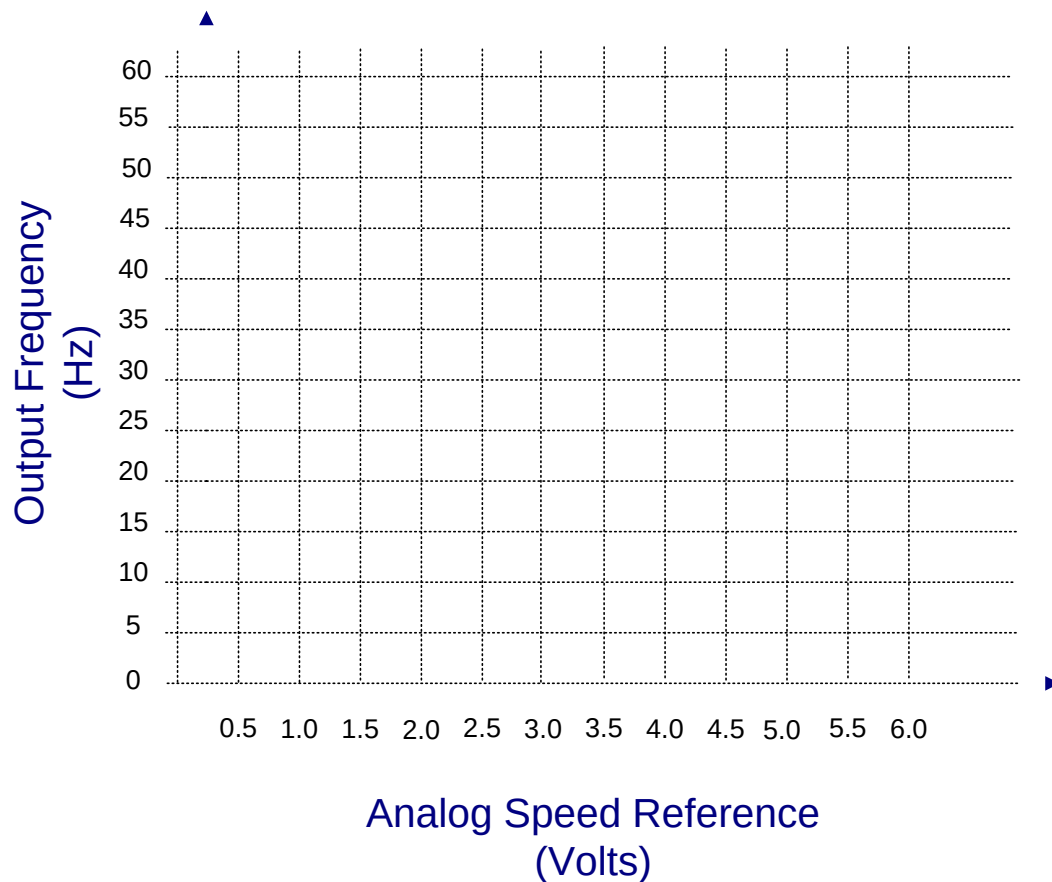
Analog Output Voltage / 10 Vdc = Motor Current / 200% Drive Rated Output Current

Do not change the analog output to current. Example: 0 – 20 ma or 4 – 20 ma.

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- 5) Set a skip frequency of 40 hertz and a bandwidth of 10 hertz. Reduce Accel1 time to 0.2 Seconds. Using the Analog input as your speed reference, start the drive at minimum speed and slowly increase the speed reference in 5 Hz increments and graph the output frequency below.

Skip Frequency Graphing Chart



- 6) What range of frequencies were actually skipped?

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- 7) What parameter(s) stores the drive's fault code?
- 8) How do you determine what your most recent fault is, and how do you determine what is your least recent fault?
- 9) Which parameter allows you to set the drive up to run as a Volts/Hz drive or a Sensorless Vector Control drive?
- 10) What is the purpose of parameter A114?
- 11) What is the difference between volts per hertz and sensorless vector control?
- 12) What is the purpose of parameter A126?
- 13) What is the purpose of parameter A127?
- 14) What is the difference between "Static Tune" and "Rotate Tune" when performing an Autotune?

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PowerFlex Drive Software Exercises

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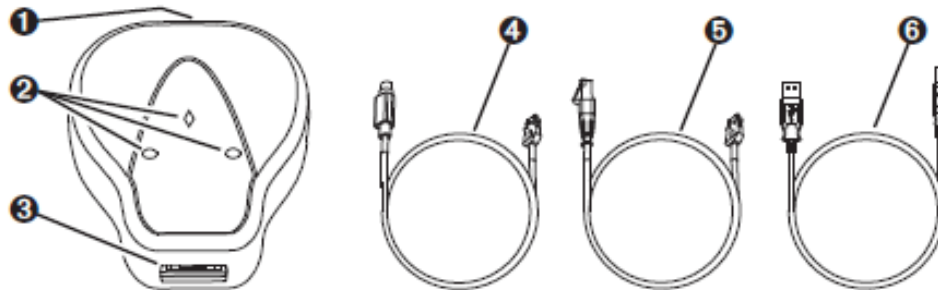
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Establishing Communications

The 1203-USB converter provides a communications interface between a computer and any Allen-Bradley product supporting these drive communication interfaces:

- DPI (Drive Peripheral Interface) for PowerFlex® 7-Class drives and PowerFlex 750-Series drives
- DSI (Drive Serial Interface) for PowerFlex 4-Class drives and PowerFlex 520-Series drives

The 1203-USB converter uses the full-duplex, RS-232 DF1 protocol..



Item	Part	Description
①	Computer Cable Port	The USB cable is plugged into this connector.
②	Status Indicators	LEDs that indicate converter operation, data being received from the computer, and data being sent to the computer. See Chapter 4, Troubleshooting , for more information.
③	Drive Cable Port	• DPI Drives or SCANport Legacy Drives: Plug the 20-HIM-H10 cable into this port. • DSI Drives: Plug the 22-HIM-H10 cable into this port.
④	20-HIM-H10 Cable	DPI or SCANport drive cable (1 m) with male 26-pin-to-male 8-pin circular mini-DIN connectors.
⑤	22-HIM-H10 Cable	DSI drive cable (1 m) with male 26-pin-to-male RJ45 connectors.
⑥	USB Cable	USB cable (2 m) with a Type B connector on one end to connect to the 1203-USB converter, and a Type A connector on the other end of the cable to connect to a computer.

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The converter is compatible with the following software packages:

Configuration Tool	1203-USB	DPI Drive	DSI Drive	SCANport Drive
Connected Components Workbench Software, version 1.02 or later	Yes	Yes	Yes	No
DriveExplorer Lite or Full software, version 4.04 or later	Yes	Yes	Yes ⁽¹⁾	Yes
DriveExecutive software, version 1.01 or later	Yes	Yes	Yes ⁽¹⁾	Yes
Terminal emulation software, such as HyperTerminal	Yes	not recommended		

The software used in this course will be Drive Executive. It is compatible with the following Allen Bradley Products:

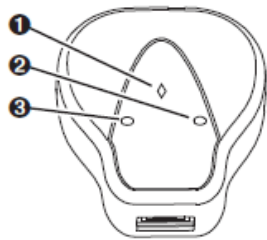
DPI Products

- PowerFlex 70/70EC drives
- PowerFlex 700/700VC drives
- PowerFlex 700 Series B drives
- PowerFlex 700H drives
- PowerFlex 700S drives
- PowerFlex 700L
- PowerFlex 753 drives
- PowerFlex 755 drives
- PowerFlex Digital DC drives
- PowerFlex 7000 drives
- SMC™ Flex
- SMC-50 Smart Motor Controller
- MSR57 Speed Monitoring Safety Relay

DSI Products

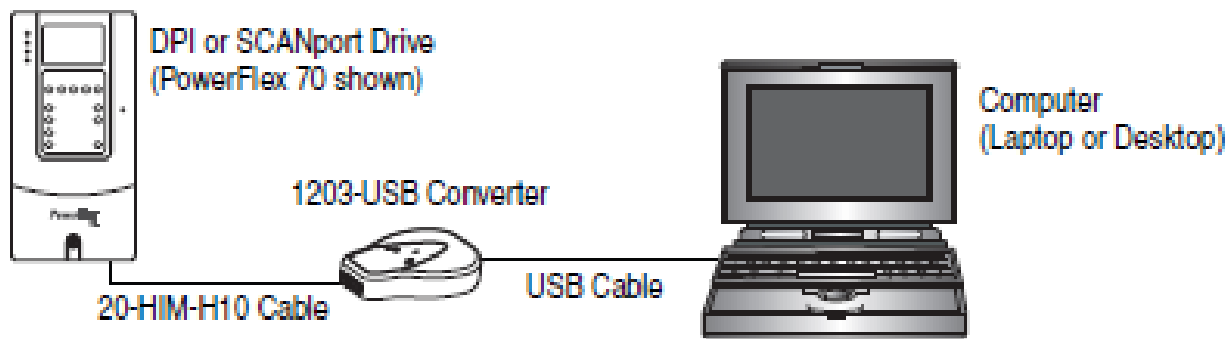
- PowerFlex 4 drives
- PowerFlex 4M drives
- PowerFlex 40 drives
- PowerFlex 40P drives
- PowerFlex 400 drives
- PowerFlex 520-Series drives (Some limitations of use)

The converter has three status indicators to report its operating status.

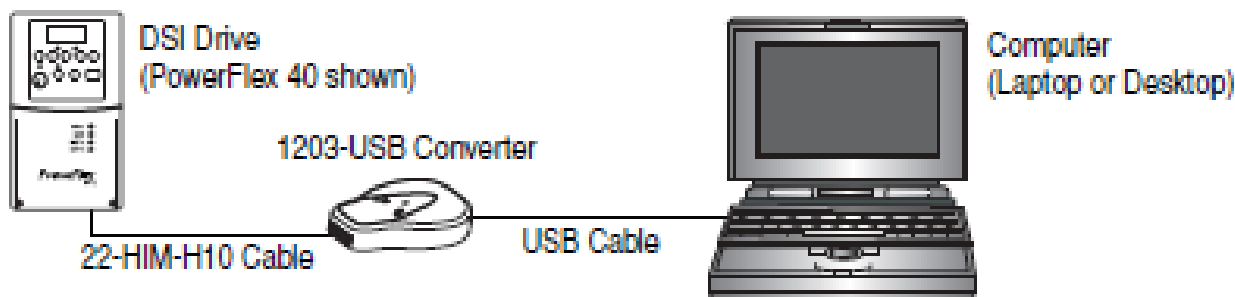


Item	Status Indicator	Description
❶	Diamond	Status of converter.
❷	TX	Converter is transmitting data.
❸	RX	Converter is receiving data.

Connecting to a DPI (PowerFlex 70/700) device with a computer:



Connecting to a DSI (PowerFlex 40p) device with a computer:



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PowerFlex 40p Software Exercise 1

Exercise Objective:

Become familiar with establishing communications between the PowerFlex 40p Drive and a computer.

Equipment Needed:

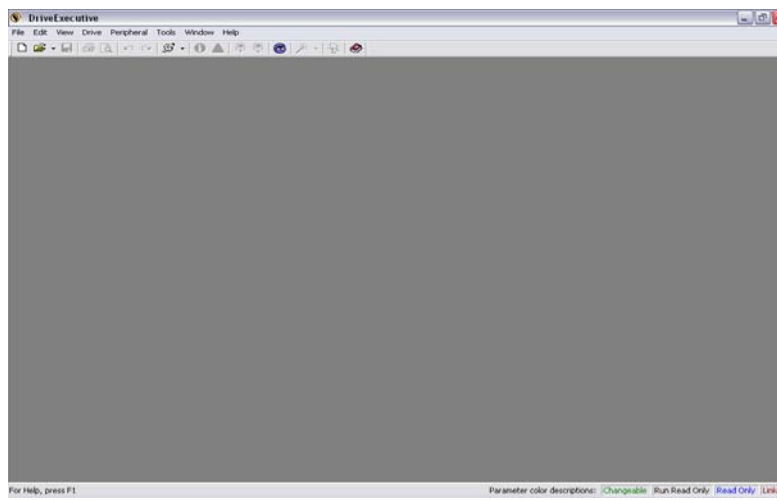
Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications).

Exercise Procedure:

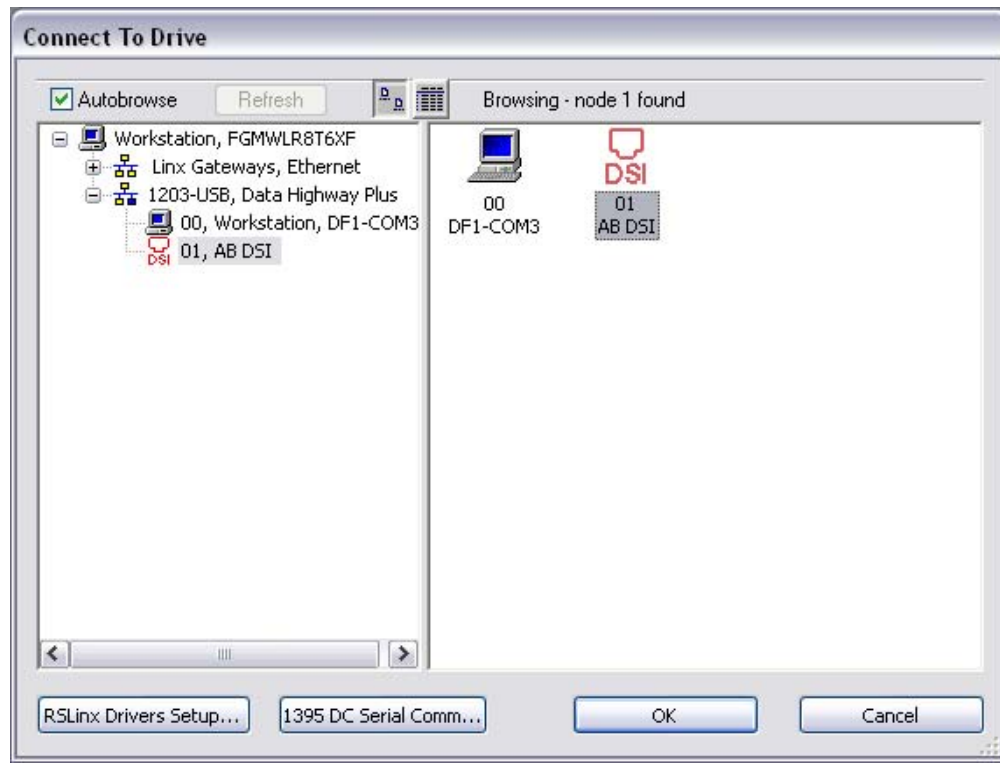
1. Start the “Drive Executive Software.



2. On the main Tool Bar, Click **Drive > Connect to Drive**

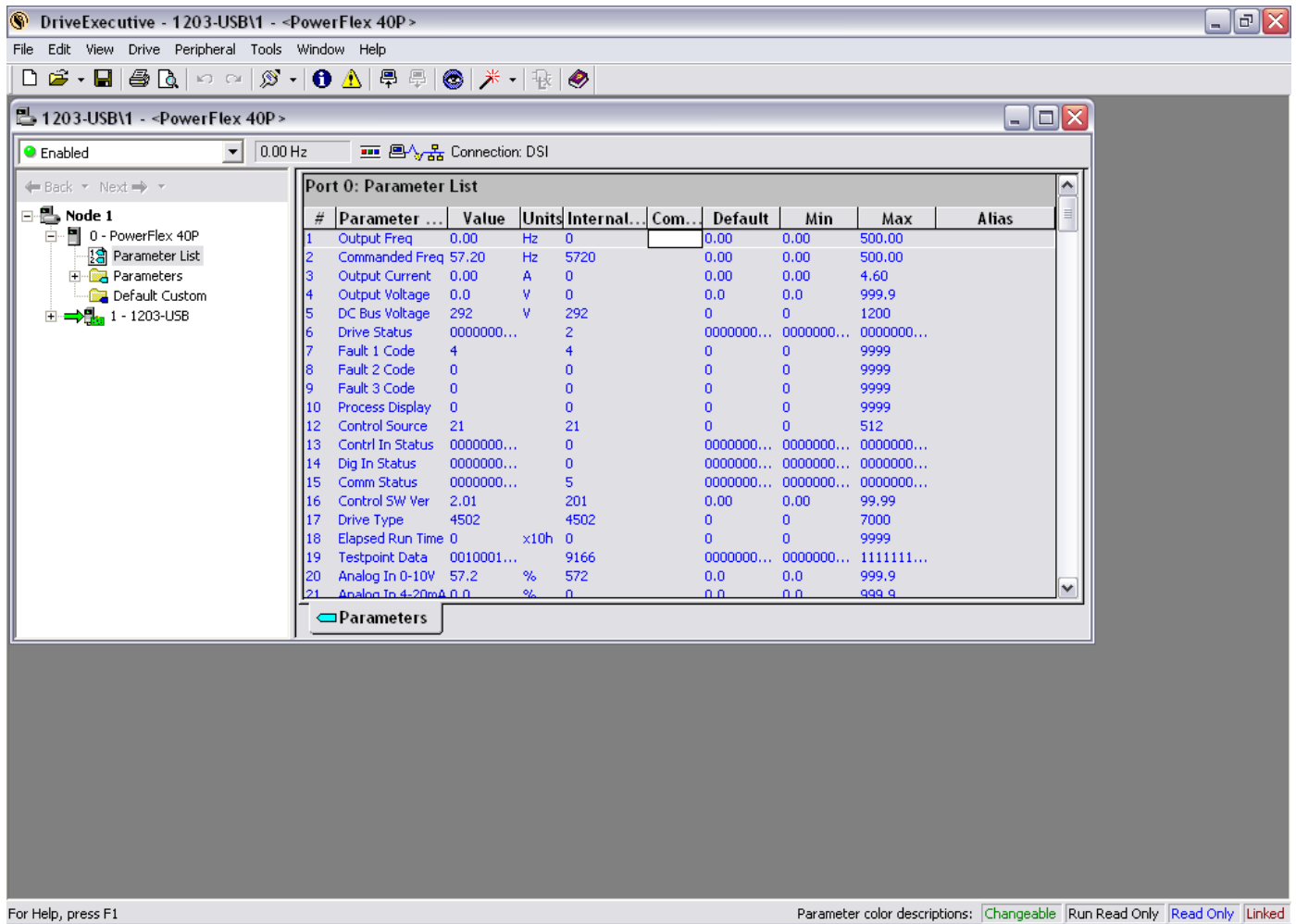
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3. You will be presented with this dialog screen:



4. Select the station “**01 AB DSI**” and click **OK** and you should connect to the Drive On-Line and see the following screen:

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5. At this point you are connected directly on-line with the drive and there are many options available for you to explore and become familiar with:
 - a. Monitor or change parameters
 - b. Create Custom Parameter sets for monitoring
 - c. Create Chart Recordings of specified parameters or drive conditions

Notice the arrangement of the folders in the Tree. These are arranged by Parameter Groups that we have discussed in the class. The first view allows you to scroll the entire parameter list in numerical order. You can view specific parameter groups and you can create your own Custom Parameter List.

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PowerFlex 40p Software Exercise 2

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 40p drive.

Equipment Needed:

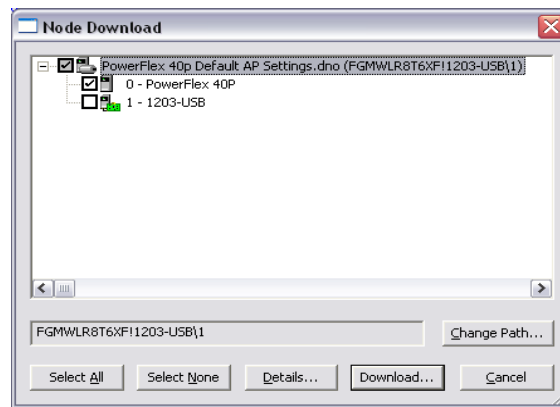
Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

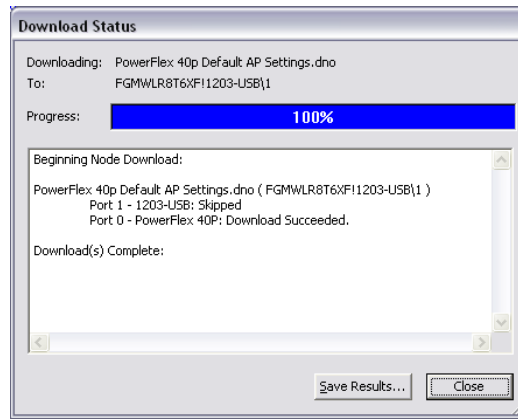
Exercise Procedure:

1. From the Main Tool Bar, open a file named **PowerFlex 40P Node 1_Sim_Default.dno**
2. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:

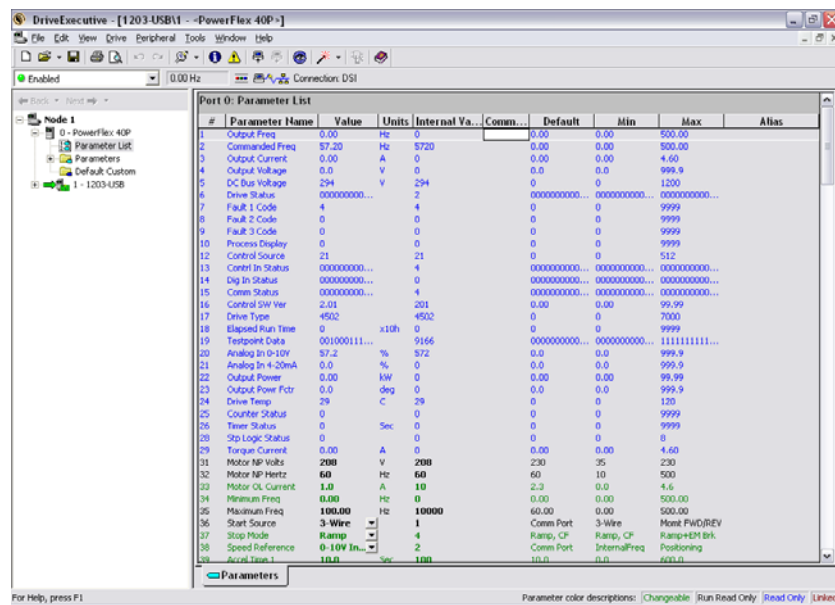


3. Select **PowerFlex 40p** and click **Download**. Once the download has completed, you should see the following screen:

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- Click **Close**.
- You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.
- After a few seconds, you should be connected to the drive and viewing the following screen:



- Verify the drive operation. It should operate as per the default AP settings for this drive.
- Locate the parameter for **Preset Speed 1** (Parameter 71) using the parameter view list functions. Change this parameter setting to 60 Hz and verify the setting by starting the drive using Preset Speed 1.

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9. Next, click on **File -> Save As** and change the file name to **PowerFlex 40p Exercise 2.dno** and then click **Save**.

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PowerFlex 40p Software Exercise 3

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 40p drive.

Equipment Needed:

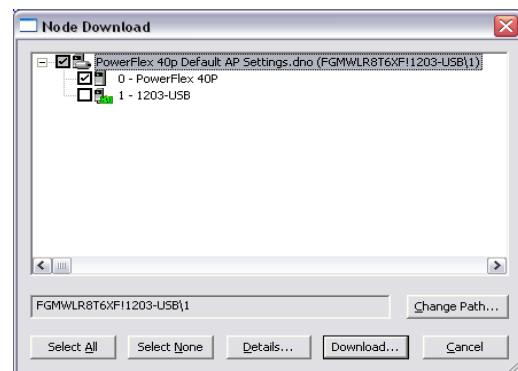
Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

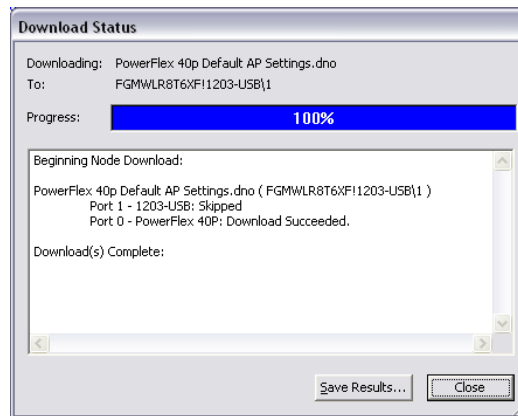
Exercise Procedure:

1. If the **PowerFlex 40P Node 1_Sim_Default.dno** has been loaded connect to the drive with Drive Executive, skip the following steps and proceed to step 8.
2. From the Main Tool Bar, open a file named **PowerFlex 40P Node 1_Sim_Default.dno**.
3. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:
4. Select **PowerFlex 40p** and click **Download**.



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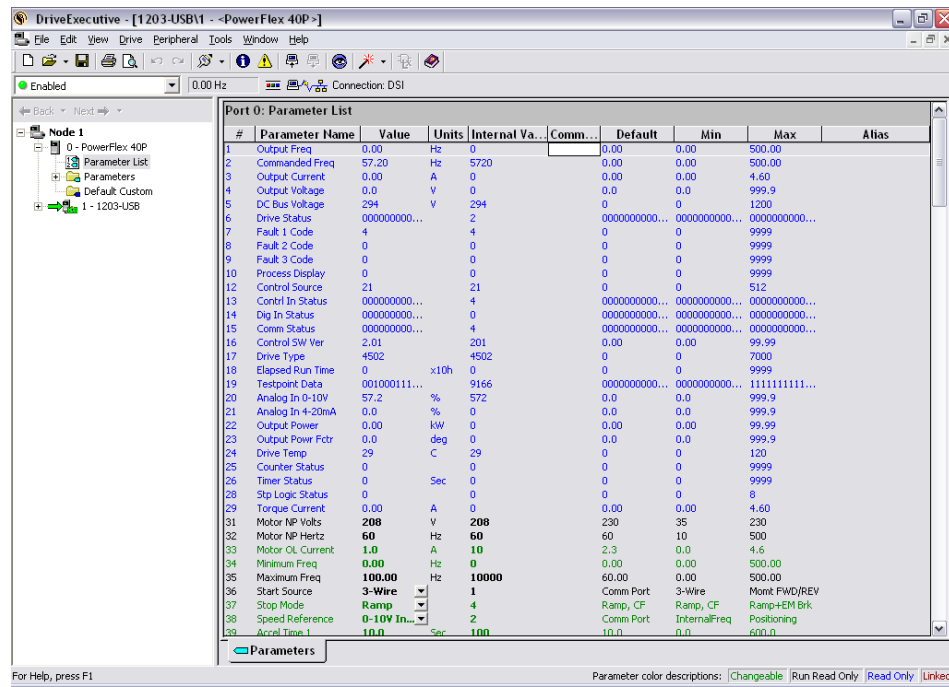
5. Once the download has completed, you should see the following screen:



6. Click **Close**.

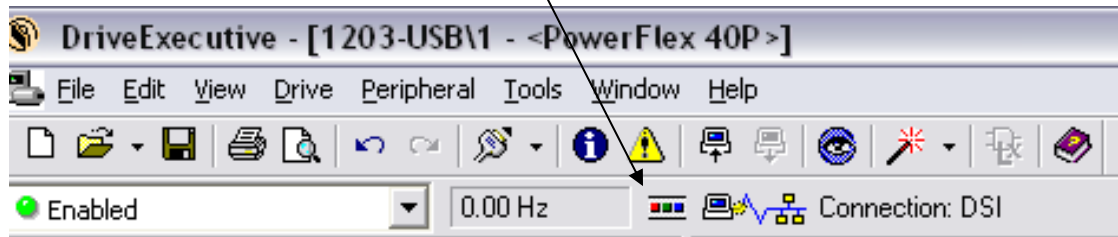
7. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.

8. After a few seconds, you should be connected to the drive and viewing the following screen:



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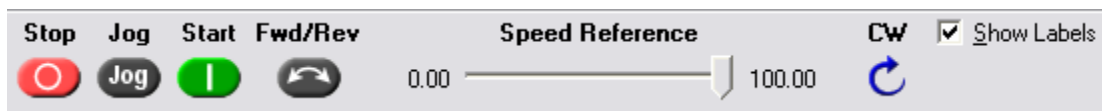
9. From the following screen, click on this ICON (Control Bar).



10. Change parameter # 36 (Start Source) to Comm Port.

11. Change parameter # 38 (Speed Reference) to Comm Port

12. You will now be able to control the drive from your computer using the following screen functions (Stop, Start, Jog, Fwd/Rev, Speed Reference).



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PowerFlex 70 Software Exercise 1

Exercise Objective:

Become familiar with establishing communications between the PowerFlex 70 Drive and a computer.

Equipment Needed:

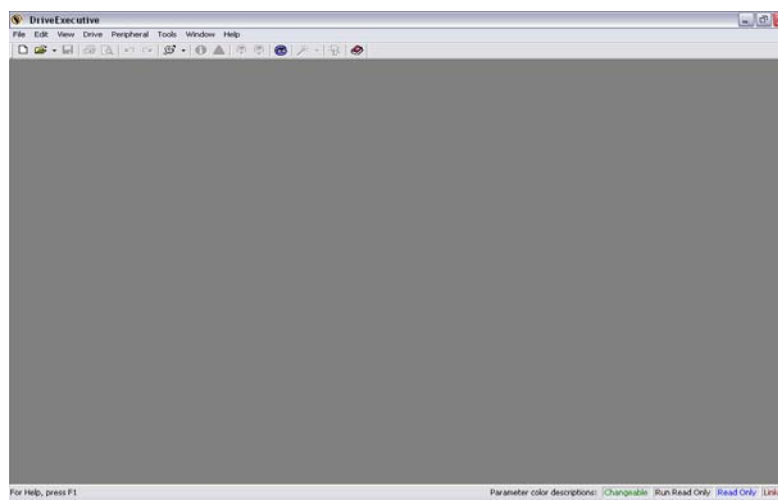
Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 1203-SSS Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications).

Exercise Procedure:

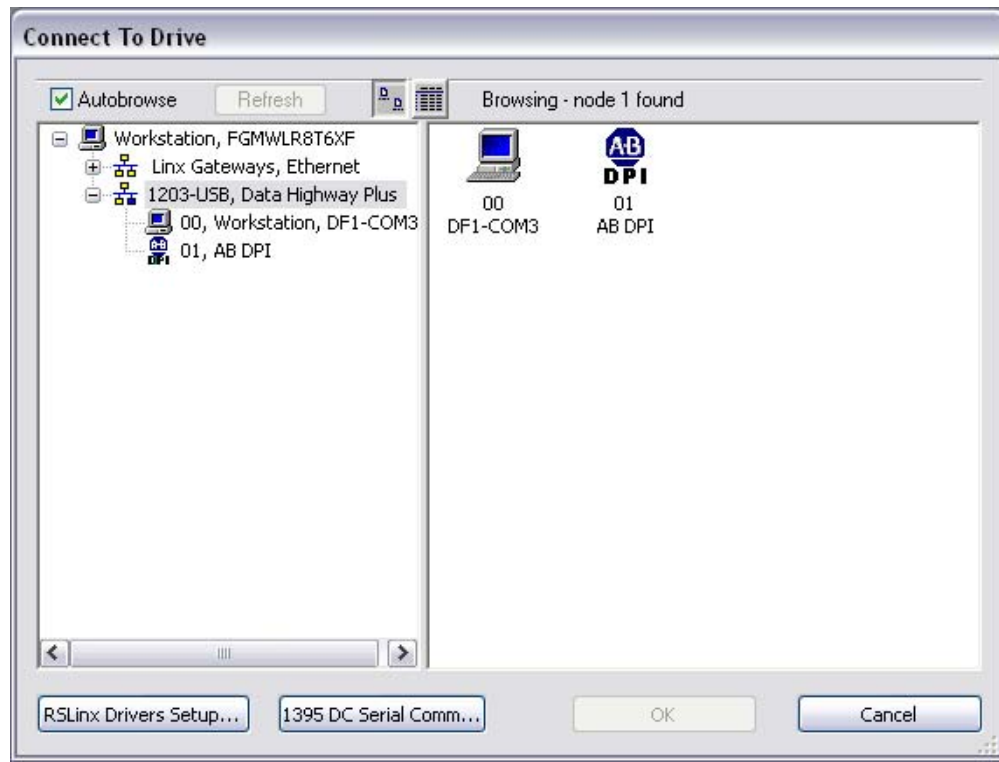
1. Start the “Drive Executive Software.



2. On the main Tool Bar, Click **Drive > Connect to Drive**

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4. You will be presented with this dialog screen:



5. Select the station **“01 AB DPI”** and click **OK** and you should connect to the Drive On-Line and see the following screen:

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DriveExecutive - 1203-USB1 - <PowerFlex 70>

File Edit View Drive Peripheral Tools Window Help

1203-USB1 - <PowerFlex 70>

Enabled 0.00 Hz Connection: DPI

Node 1

- 0 - PowerFlex 70
 - Parameter List
 - Monitor
 - Motor Control
 - Speed Command
 - Dynamic Control
 - Utility
 - Communication
 - Inputs & Outputs
 - Default: Custom
 - 1 - LCD Module
 - 2 - 1203-USB
 - 5 - 20-COMM-D

Port 0: Parameter List

#	Parameter ...	Value	Units	Internal...	Com...	Default	Min	Max	Alias
1	Output Freq	0.0	Hz	0		-400.0	-400.0	400.0	
2	Commanded Freq	40.0	Hz	400		-400.0	-400.0	400.0	
3	Output Current	0.0	Amps	0		0.0	0.0	3276.7	
4	Torque Current	0.0	Amps	0		-819.2	-819.2	819.2	
5	Flux Current	0.0	Amps	0		-819.2	-819.2	819.2	
6	Output Voltage	0.0	VAC	0		0.0	0.0	600.0	
7	Output Power	0.0	kW	0		0.0	0.0	2400.0	
8	Output Powr Fctr	0.00		0		0.00	0.00	1.00	
9	Elapsed MWh	0.0	MWh	0		0.0	0.0	4294967...	
10	Elapsed Run Time	69.0	Hrs	690		0.0	0.0	4294967...	
11	MOP Frequency	0.0	Hz	0		-400.0	-400.0	400.0	
12	DC Bus Voltage	292.2	VDC	2922		0.0	0.0	3276.7	
13	DC Bus Memory	292.1	VDC	2921		0.0	0.0	819.2	
16	Analog In1 Value	0.000		0		-32.767	-32.767	32.767	
17	Analog In2 Value	0.000		0		-32.767	-32.767	32.767	
26	Rated kW	0.75	kW	75		0.00	0.00	3000.00	
27	Rated Volts	240.0	VAC	2400		0.0	0.0	6553.5	
28	Rated Amps	4.2	Amps	42		0.0	0.0	6553.5	
29	Control SW Ver	2.009		2009		0.000	0.000	65.535	
40	Motor Type	Induc		0		Induction	Induction	Synchr PM	

Parameters

For Help, press F1

Parameter color descriptions: Changeable Run Read Only Read Only Linked

6. At this point you are connected directly on-line with the drive and there are many options available for you to explore and become familiar with:
 - a. Monitor or change parameters
 - b. Create Custom Parameter sets for monitoring
 - c. Create Chart Recordings of specified parameters or drive conditions

Notice the arrangement of the folders in the Tree. These are arranged by Parameter Groups that we have discussed in the class. The first view allows you to scroll the entire parameter list in numerical order. You can view specific parameter groups and you can create your own Custom Parameter List.

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PowerFlex 70 Software Exercise 2

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 70 drive.

Equipment Needed:

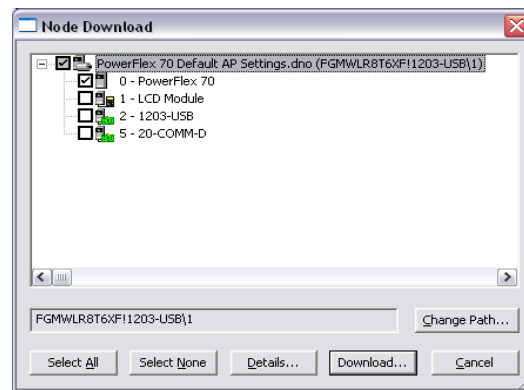
Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 1203-SSS Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

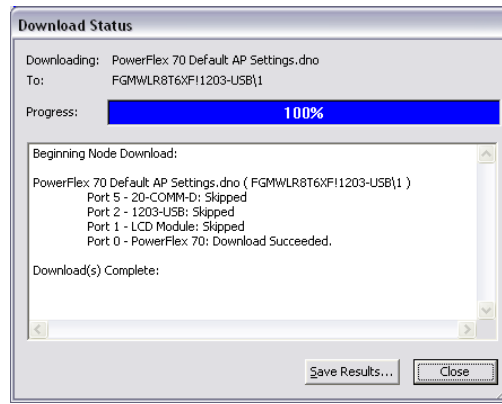
Exercise Procedure:

1. From the Main Tool Bar, open a file named **PowerFlex 70 Default AP Settings.dno**.
2. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:

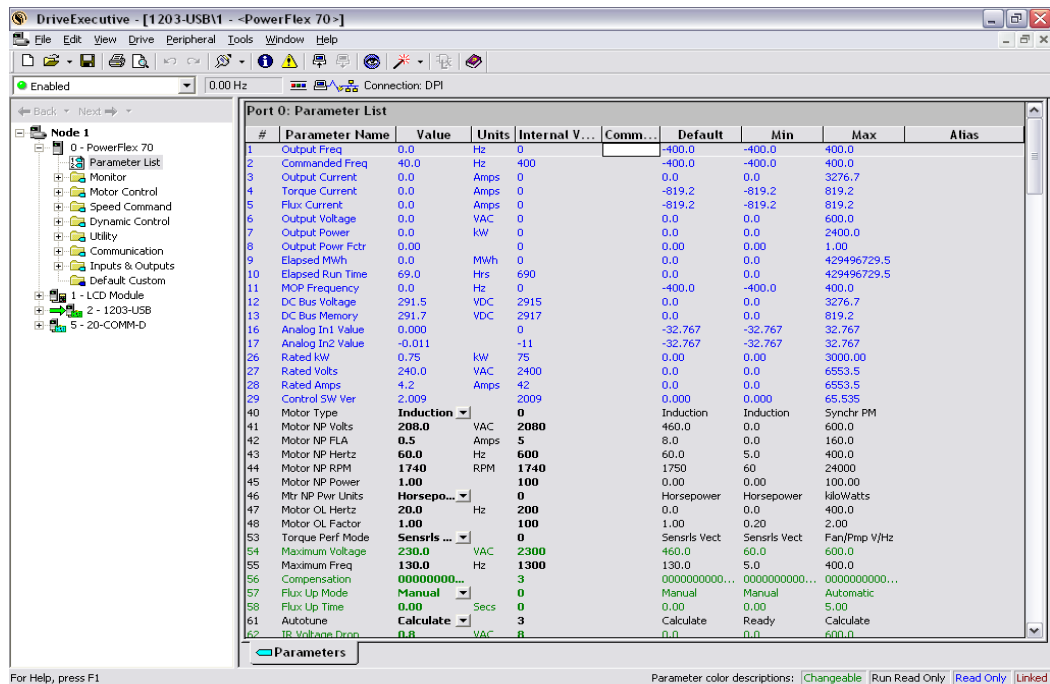


3. Select **PowerFlex 70** and click **Download**. Once the download has completed, you should see the following screen:

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4. Click **Close**.
5. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.
6. After a few seconds, you should be connected to the drive and viewing the following screen:



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7. Verify the drive operation. It should operate as per the default AP settings for this drive.
8. Locate the parameter for **Preset Speed 1** (Parameter 101) using the parameter view list functions. Change this parameter setting to 60 Hz and verify the setting by starting the drive using Preset Speed 1.
9. Next, click on **File -> Save As** and change the file name to **PowerFlex 70 Exercise 2.dno** and then click **Save**.

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PowerFlex 70 Software Exercise 3

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 70 drive.

Equipment Needed:

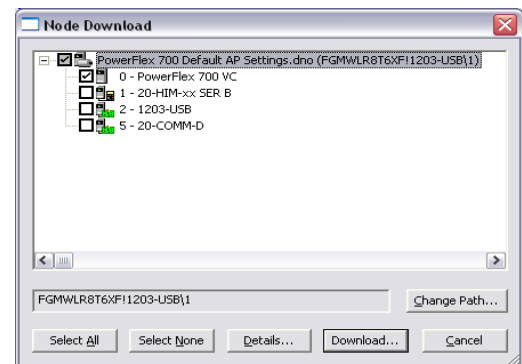
Allen-Bradley Power Flex 70 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

Exercise Procedure:

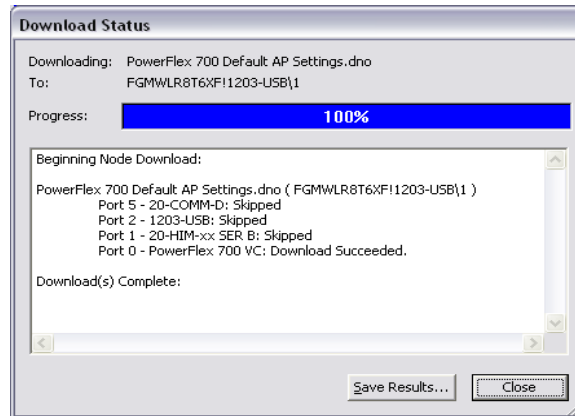
1. If the **PowerFlex 70 VC Node 1_Sim Default.dno** has been loaded connect to the drive with Drive Executive, skip the following steps and proceed to step 8.
2. From the Main Tool Bar, open a file named **PowerFlex 70 VC Node 1_Sim Default.dno**
3. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:
4. Select **PowerFlex 70** and click **Download**.



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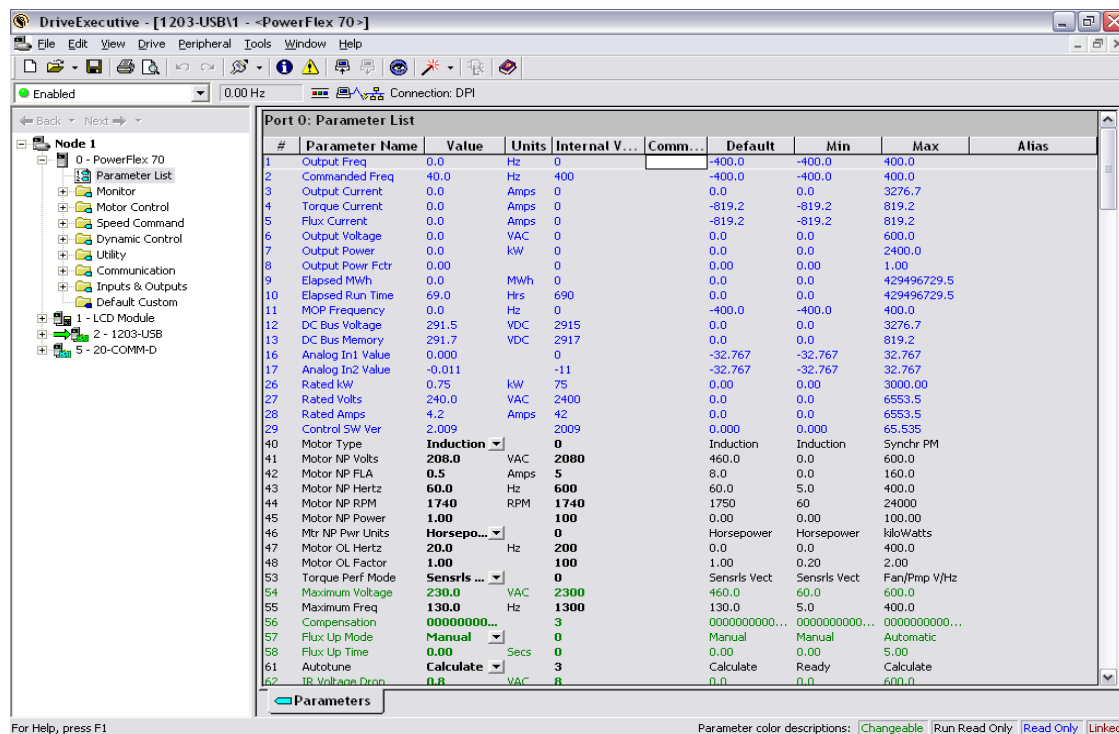
5. Once the download has completed, you should see the following screen:

6. Click **Close**.



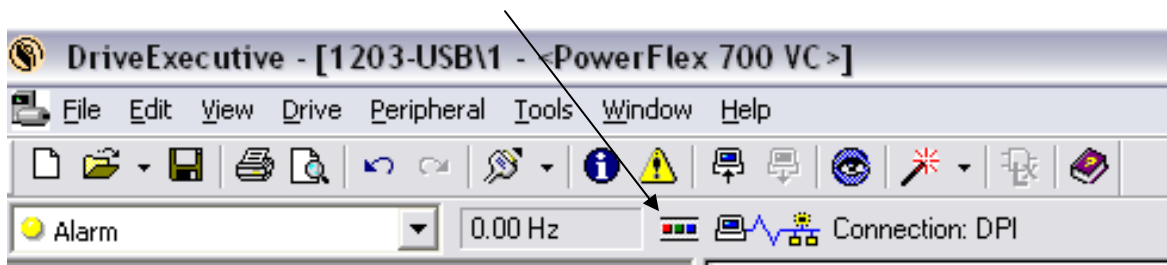
7. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.

8. After a few seconds, you should be connected to the drive and viewing the following screen:



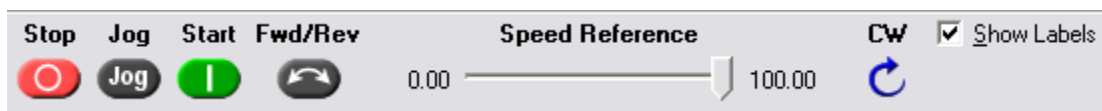
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9. From the following screen, click on this ICON (Control Bar).



10. Change parameter # 90 (Speed Reference A) to DPI Port-2.

11. You will now be able to control the drive from your computer using the following screen functions (Stop, Start, Jog, Fwd/Rev, Speed Reference).



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PowerFlex 700 Software Exercise 1

Exercise Objective:

Become familiar with establishing communications between the PowerFlex 700 Drive and a computer.

Equipment Needed:

Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 1203-SSS Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications).

Exercise Procedure:

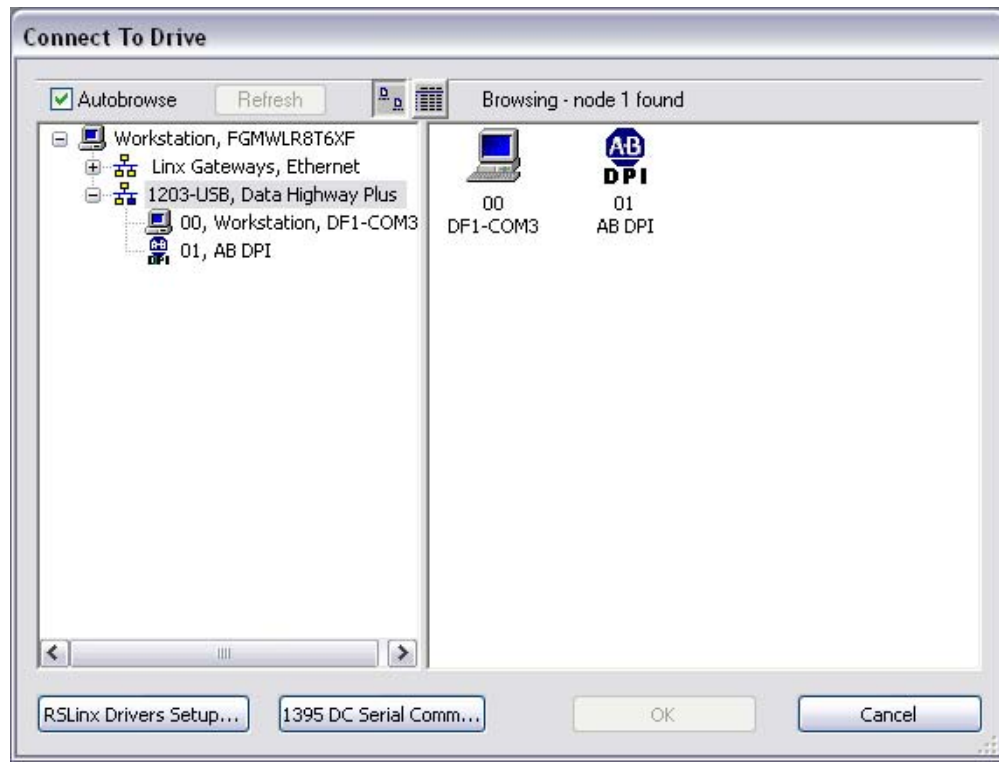
1. Start the “Drive Executive Software.



2. On the main Tool Bar, Click **Drive** > **Connect to Drive**

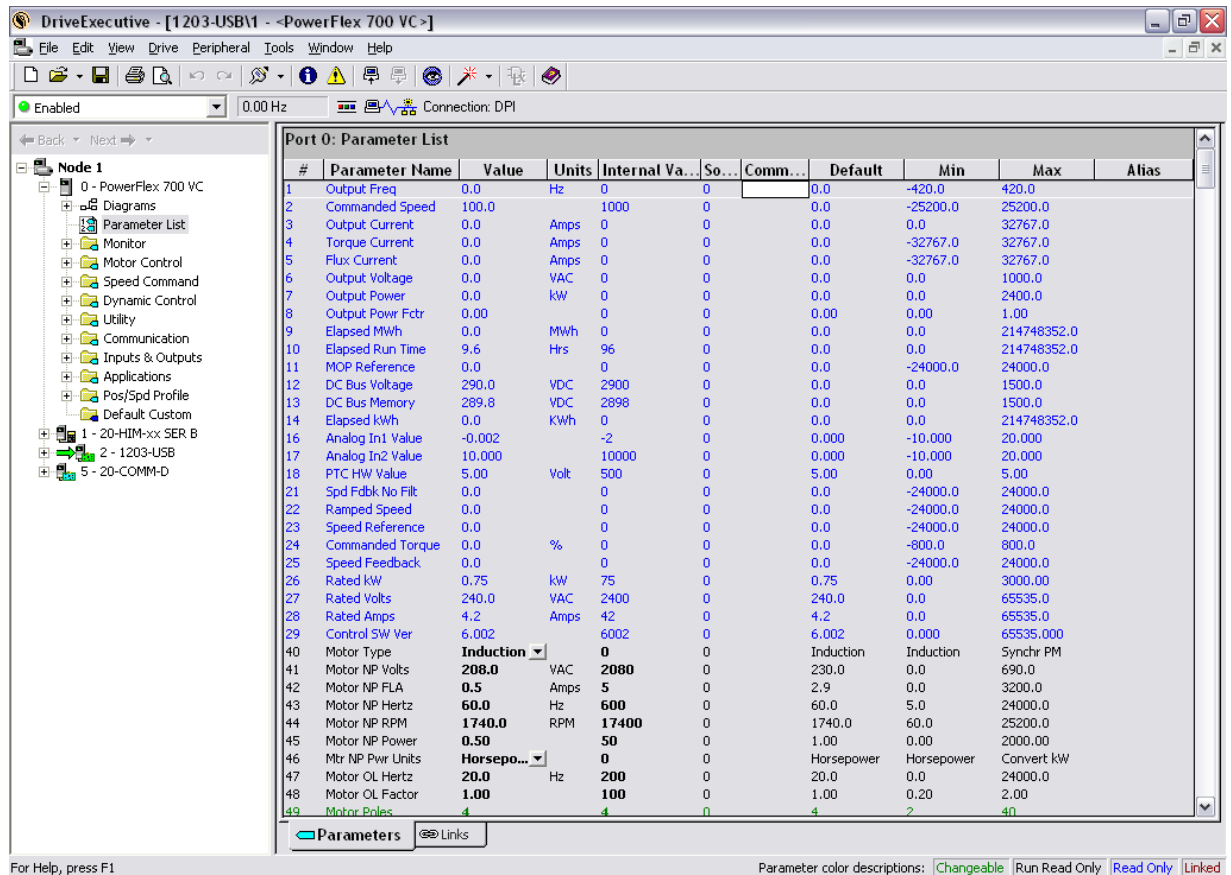
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3. You will be presented with this dialog screen:



4. Select the station **“01 AB DPI”** and click **OK** and you should connect to the Drive On-Line and see the following screen:

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5. At this point you are connected directly on-line with the drive and there are many options available for you to explore and become familiar with:
 - a. Monitor or change parameters
 - b. Create Custom Parameter sets for monitoring
 - c. Create Chart Recordings of specified parameters or drive conditions

Notice the arrangement of the folders in the Tree. These are arranged by Parameter Groups that we have discussed in the class. The first view allows you to scroll the entire parameter list in numerical order. You can view specific parameter groups and you can create your own Custom Parameter List.

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PowerFlex 700 Software Exercise 2

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 700 drive.

Equipment Needed:

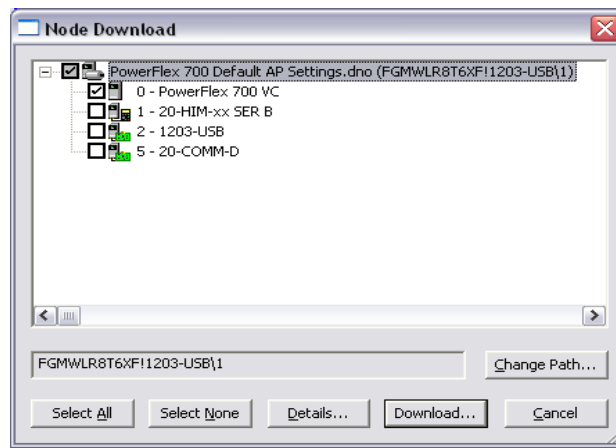
Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 1203-SSS Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

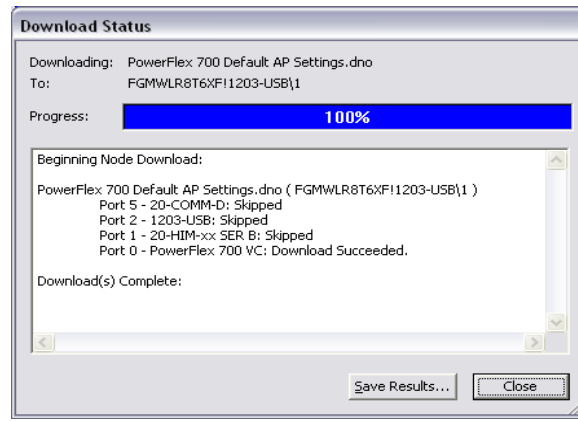
Exercise Procedure:

1. From the Main Tool Bar, open a file named **PowerFlex 700 VC Node 1_Sim Default.dno**
2. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:

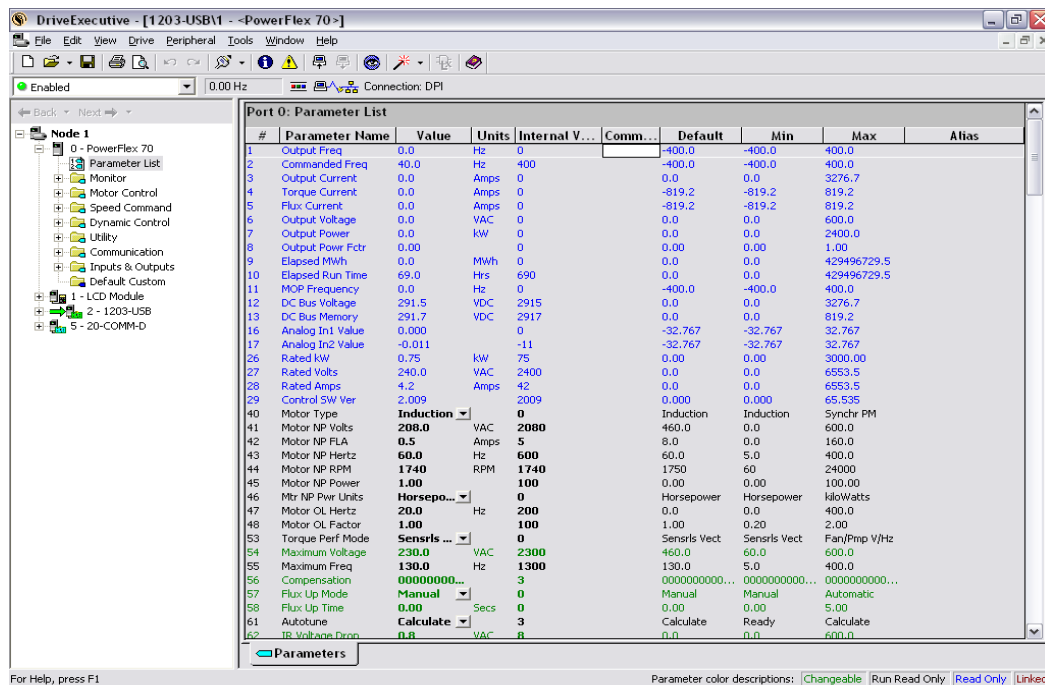


3. Select **PowerFlex 700** and click **Download**. Once the download has completed, you should see the following screen:

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4. Click **Close**.
5. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.
6. After a few seconds, you should be connected to the drive and viewing the following screen:



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7. Verify the drive operation. It should operate as per the default AP settings for this drive.
8. Locate the parameter for **Preset Speed 1** (Parameter 101) using the parameter view list functions. Change this parameter setting to 60 Hz and verify the setting by starting the drive using Preset Speed 1.
9. Next, click on **File -> Save As** and change the file name to **PowerFlex 700 Exercise 2.dno** and then click **Save**.

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PowerFlex 700 Software Exercise 3

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 700 drive.

Equipment Needed:

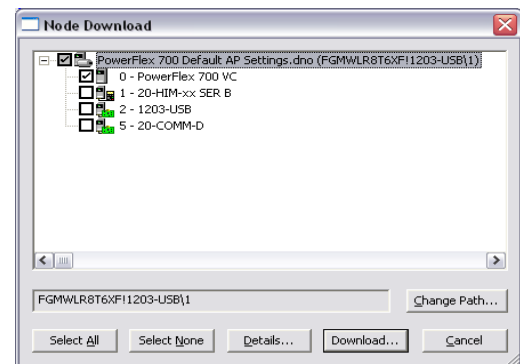
Allen-Bradley Power Flex 700 AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

Exercise Procedure:

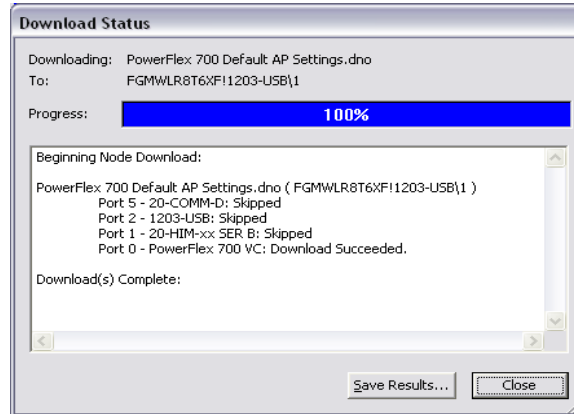
1. If the **PowerFlex 700 VC Node 1_Sim Default.dno** has been loaded connect to the drive with Drive Executive, skip the following steps and proceed to step 8.
2. From the Main Tool Bar, open a file named **PowerFlex 700 VC Node 1_Sim Default.dno**
3. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:
4. Select **PowerFlex 700** and click **Download**.



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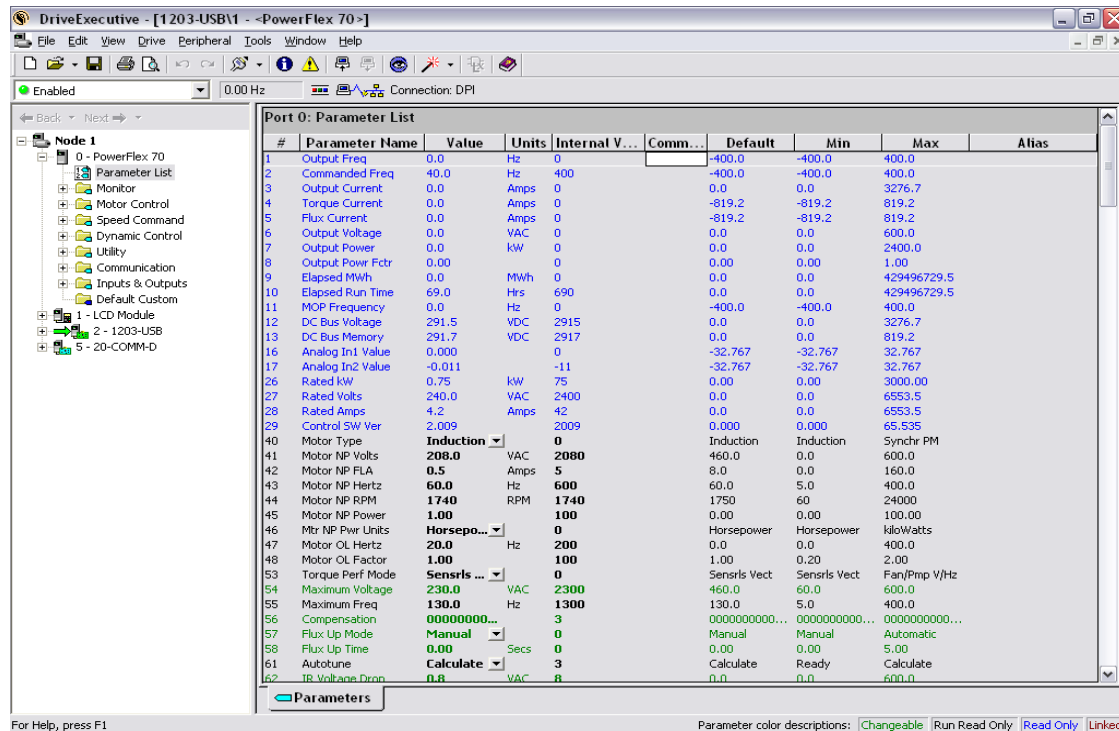
5. Once the download has completed, you should see the following screen:

6. Click **Close**.



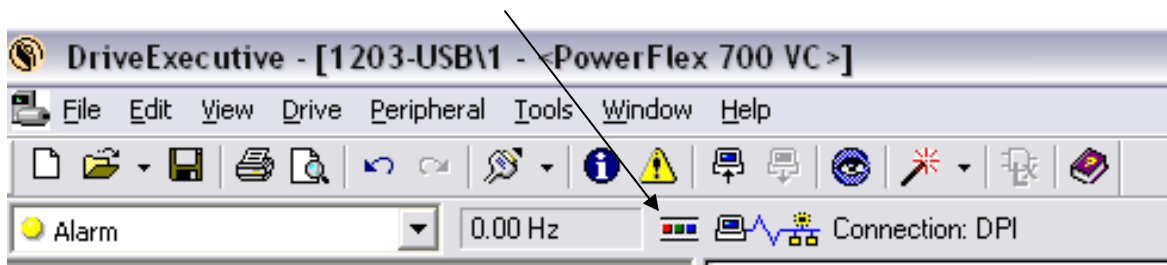
7. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.

8. After a few seconds, you should be connected to the drive and viewing the following screen:



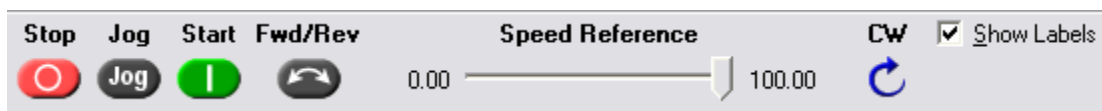
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9. From the following screen, click on this ICON (Control Bar).



10. Change parameter # 90 (Speed Reference A) to DPI Port-2.

11. You will now be able to control the drive from your computer using the following screen functions (Stop, Start, Jog, Fwd/Rev, Speed Reference).



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PowerFlex 40p DG Software Exercise 1

Exercise Objective:

Become familiar with establishing communications between the PowerFlex 40p-DG Drive and a computer.

Equipment Needed:

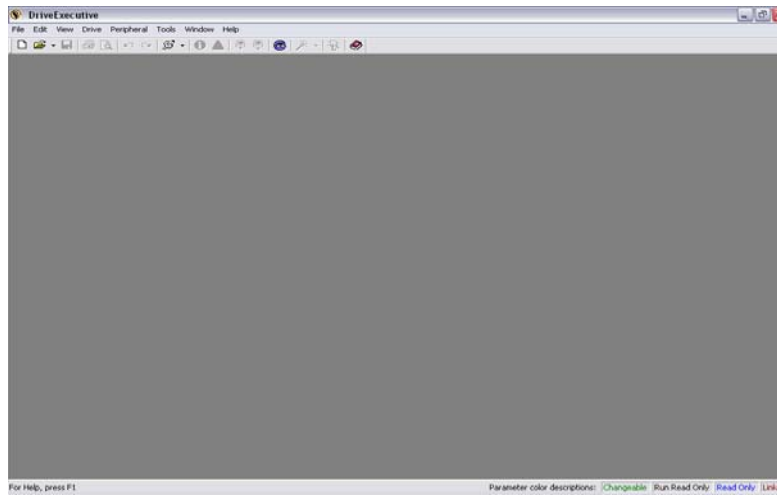
Allen-Bradley Power Flex 40p-DG AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications).

Exercise Procedure:

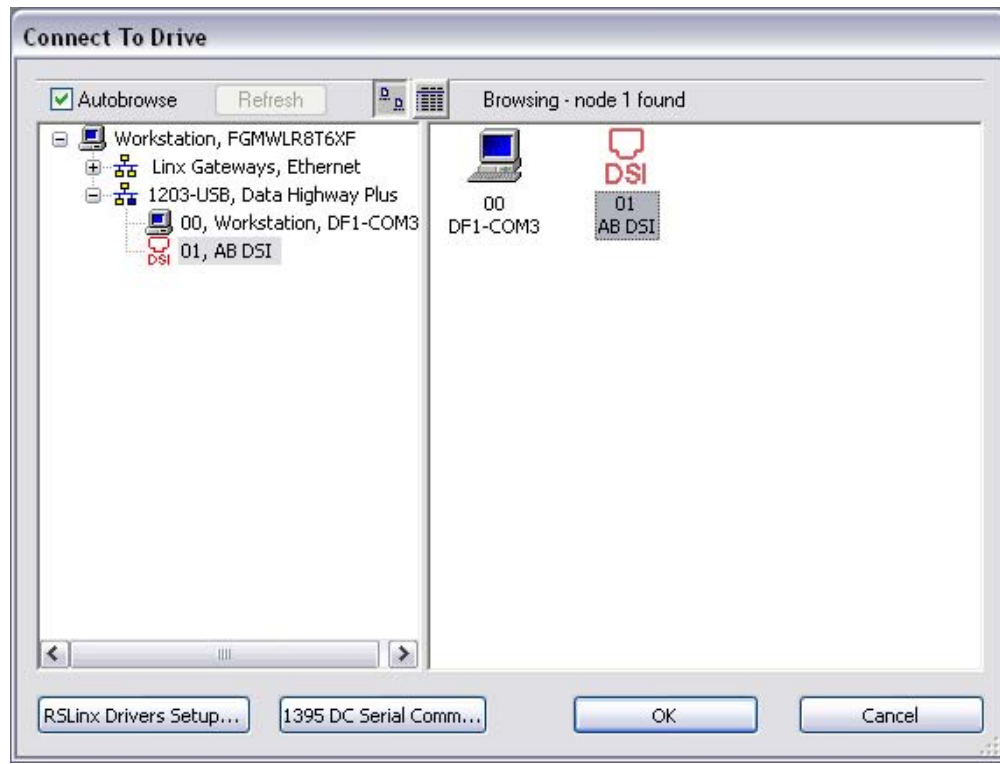
3. Start the “Drive Executive Software.



4. On the main Tool Bar, Click **Drive > Connect to Drive**

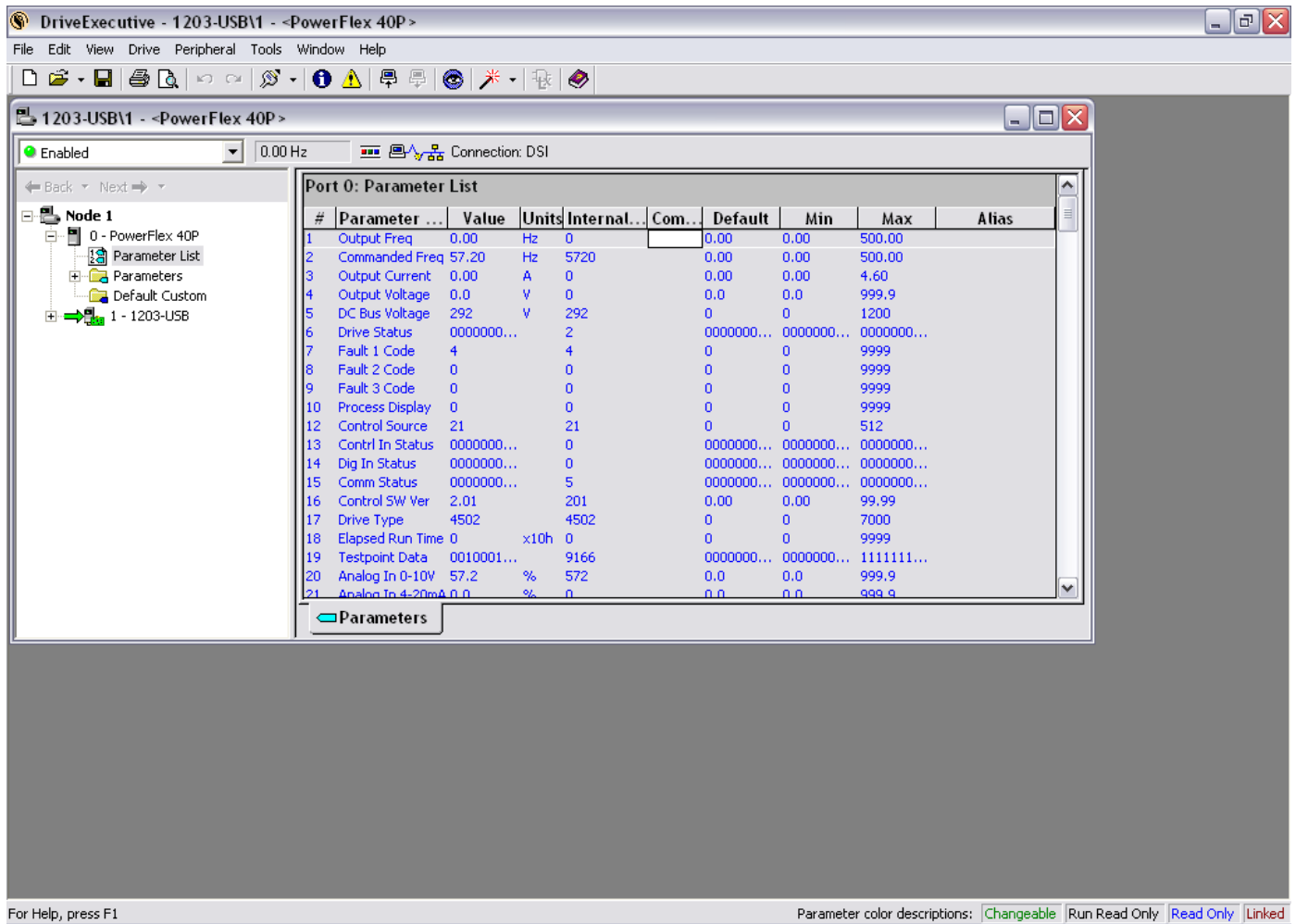
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5. You will be presented with this dialog screen:



6. Select the station “**01 AB DSI**” and click **OK** and you should connect to the Drive On-Line and see the following screen:

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7. At this point you are connected directly on-line with the drive and there are many options available for you to explore and become familiar with:
 - d. Monitor or change parameters
 - e. Create Custom Parameter sets for monitoring
 - f. Create Chart Recordings of specified parameters or drive conditions

Notice the arrangement of the folders in the Tree. These are arranged by Parameter Groups that we have discussed in the class. The first view allows you to scroll the entire parameter list in numerical order. You can view specific parameter groups and you can create your own Custom Parameter List.

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PowerFlex 40p-DG Software Exercise 2

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 40p-DG drive.

Equipment Needed:

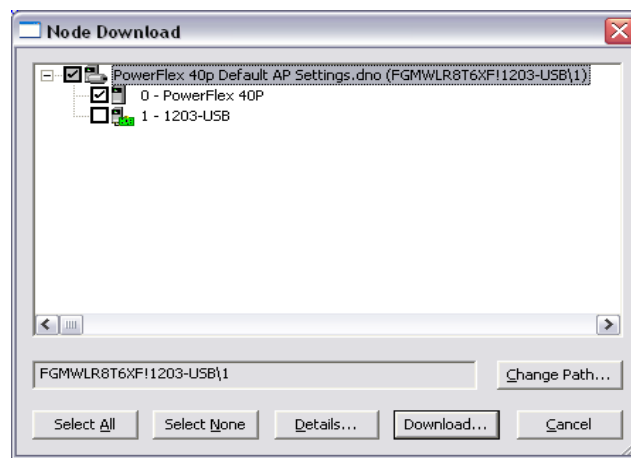
Allen-Bradley Power Flex 40p AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

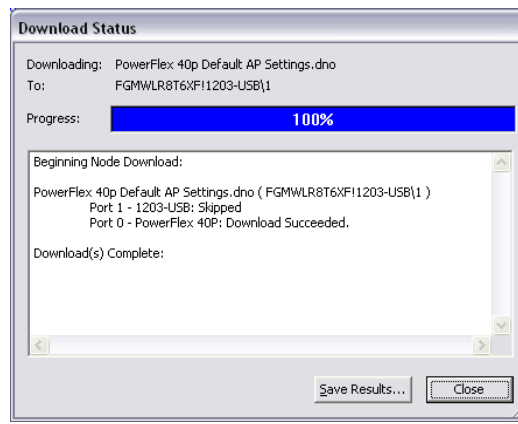
Exercise Procedure:

1. From the Main Tool Bar, open a file named **PowerFlex 40P-DG_Sim_Default.dno**
2. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:

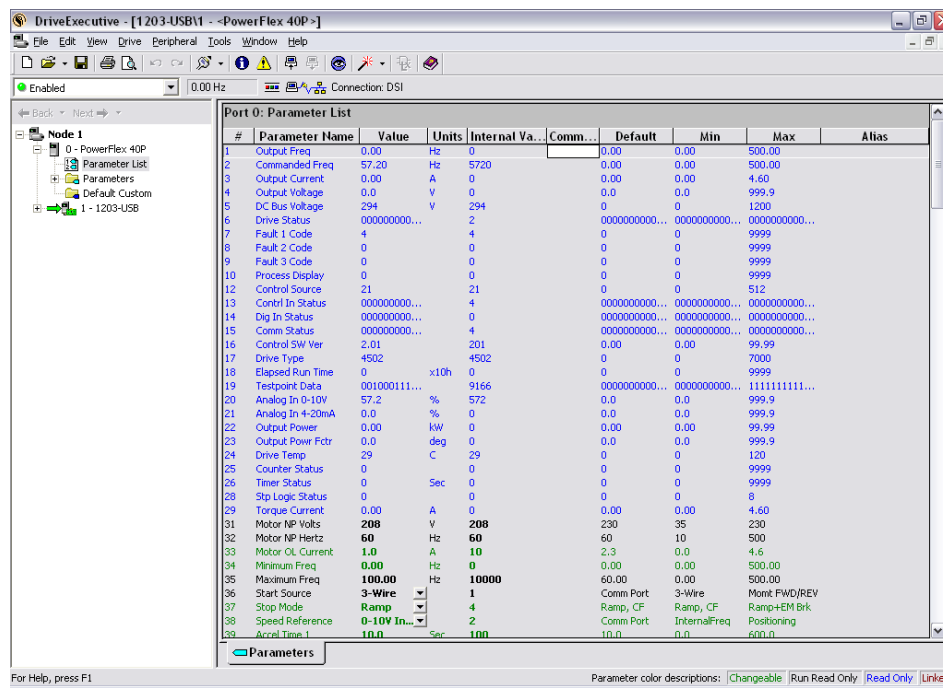


3. Select **PowerFlex 40p** and click **Download**. Once the download has completed, you should see the following screen:

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4. Click **Close**.
5. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.
6. After a few seconds, you should be connected to the drive and viewing the following screen:



7. Verify the drive operation. It should operate as per the default AP settings for this drive.

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8. Locate the parameter for **Preset Speed 1** (Parameter 71) using the parameter view list functions. Change this parameter setting to 60 Hz and verify the setting by starting the drive using Preset Speed 1.
9. Next, click on **File -> Save As** and change the file name to **PowerFlex 40p-DG Exercise 2.dno** and then click **Save**.

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PowerFlex 40p-DG Software Exercise 3

Exercise Objective:

Become familiar with Downloading/Modifying/Saving a configuration to the PowerFlex 40p drive.

Equipment Needed:

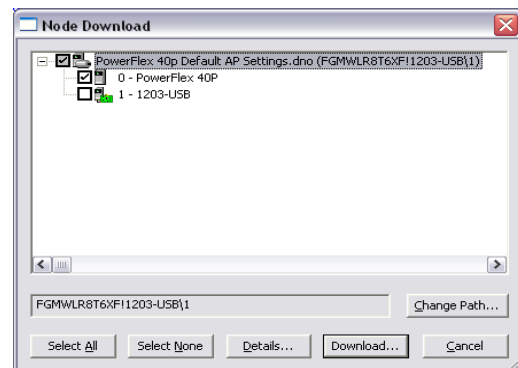
Allen-Bradley Power Flex 40p-DG AC Drive Simulator
ACD Course Manual
Simulator Drawing Package
1203-USB Converter or 22-SCM-232 Converter
Dell Computer

Preliminary Procedure:

Power-up the simulator and the computer. Connect the computer to the drive using the appropriate converter and cabling (See Establishing Communications). Start **Drive Executive** on your computer but **DO NOT** connect to the drive from the program, yet.

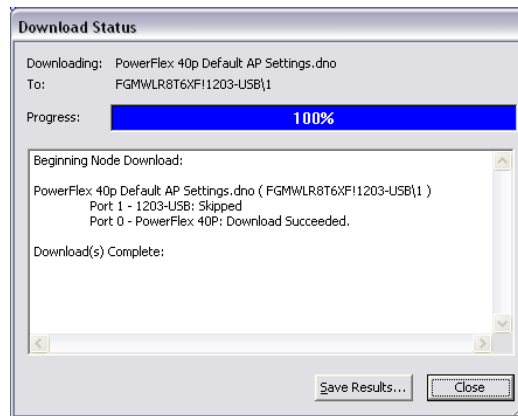
Exercise Procedure:

1. If the **PowerFlex 40P-DG_Sim_Default.dno** has been loaded connect to the drive with Drive Executive, skip the following steps and proceed to step 8.
2. **From the Main Tool Bar, open a file named *PowerFlex 40P-DG_Sim_Default.dno*.**
3. From the Main Tool Bar select **Drive -> Download To 0**. You will be presented the following screen:
4. Select **PowerFlex 40p** and click **Download**.



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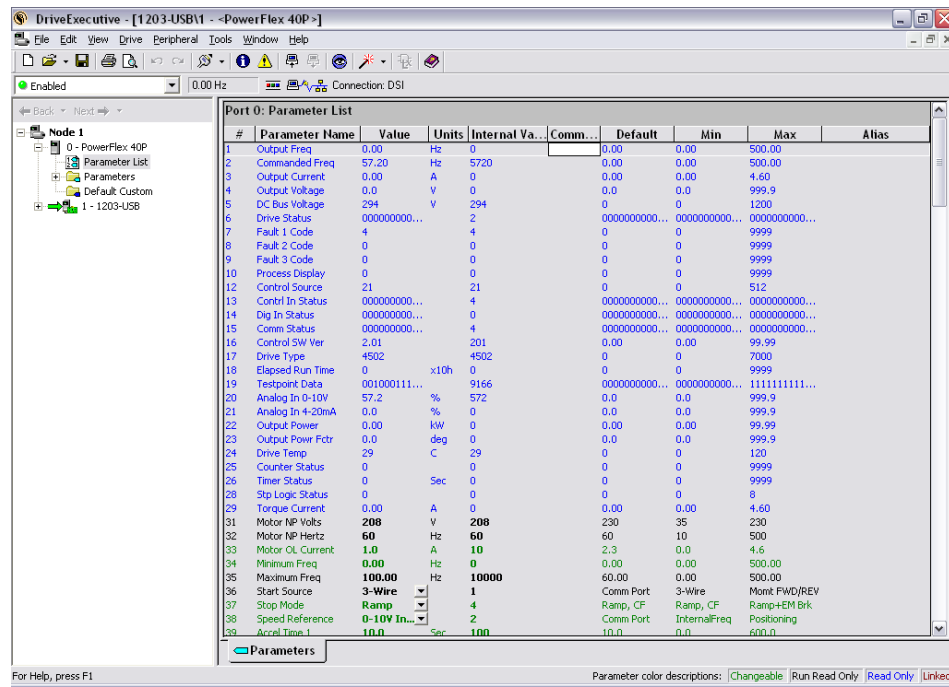
5. Once the download has completed, you should see the following screen:



6. Click **Close**.

7. You will then be asked if you would like to connect to the drive that you just downloaded to. Enter **Yes**.

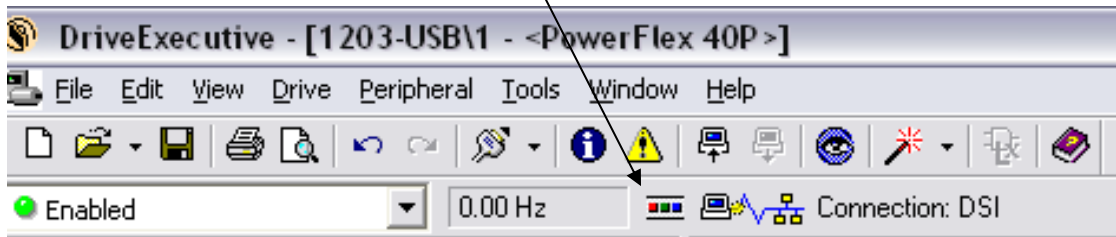
8. After a few seconds, you should be connected to the drive and viewing the following screen:



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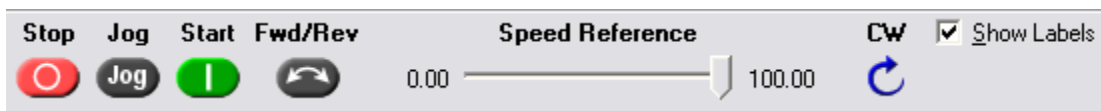
9. From the following screen, click on this ICON (Control Bar).



10. Change parameter # 36 (Start Source) to Comm Port.

11. Change parameter # 38 (Speed Reference) to Comm Port

12. You will now be able to control the drive from your computer using the following screen functions (Stop, Start, Jog, Fwd/Rev, Speed Reference).



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Introduction to Safe-Off Circuits

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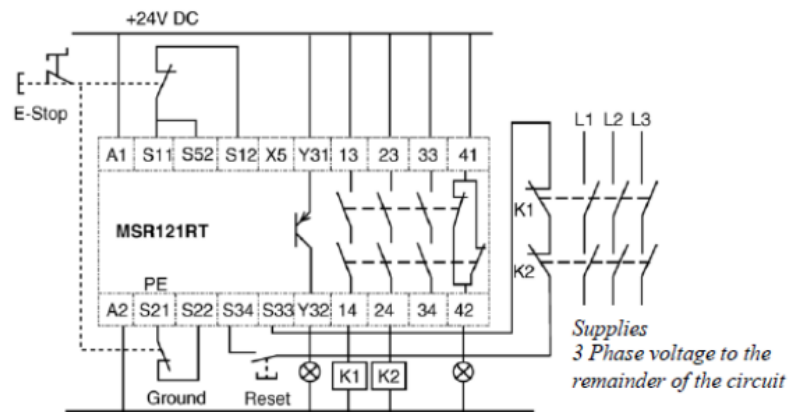
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The circuit shown on the previous page may be replaced by the following Safety Relay in new Michelin installations that may require Stop category 0 safety requirements. The circuit shown below is configured as a Dual Channel E. Stop circuit.

The Allen-Bradley Guard Master Minotaur MSR121RT is a safety monitoring relay that provides three normally open safety contacts, a normally open auxiliary contact and a solid state auxiliary output for use with interfacing to PLCs.



The Allen-Bradley GuardMaster Minotaur MSR121RT is a safety monitoring relay that provides three normally open safety contacts, a normally open auxiliary contact and a solid state auxiliary output for use with interfacing to PLCs. The basic tasks of safety relays are:

1. To detect faults on safety-related electrical control circuits, e.g. faults in sensors, wiring, contactors, etc.
2. To provide an ensured switching action, e.g. to act as an intermediate relay to amplify a signal or distribute it to multiple devices.
3. To provide a manual reset facility. They achieve their function by using internal redundancy (e.g. duplication) and monitoring.

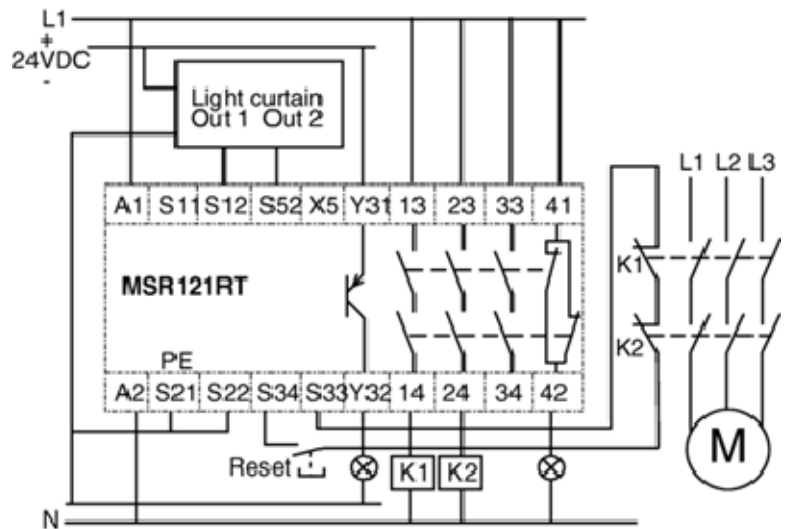
Reset and output monitoring is determined by the wiring configuration. Automatic/manual reset can use a jumper or can be used to check operation of external contactors. Monitored manual requires the use of a manually operated normally open momentary switch.

The safety outputs have independent and redundant internal contacts to help ensure the safety function. It has one normally closed auxiliary output as well as a solid state PNP auxiliary output to indicate the status of the MSR121RT to an indicator lamp or a PLC.

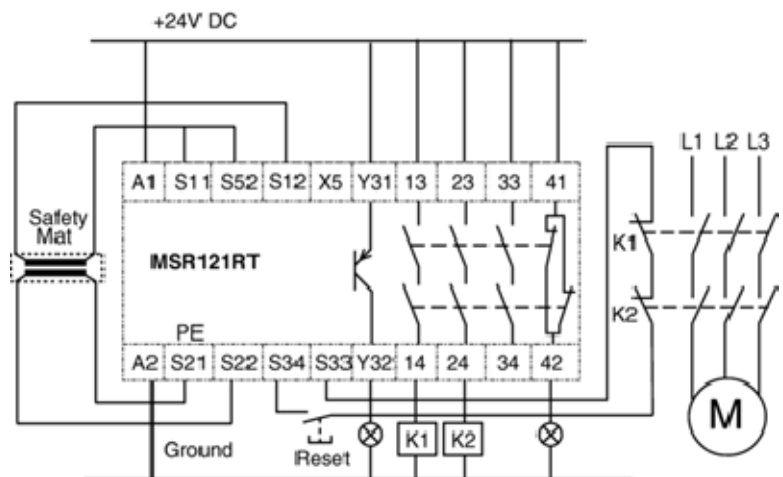
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Typical wiring configurations for the MSR121 Safety Relay

24V DC Light Curtain 115/230V AC
Safety Relay, Monitored Manual Reset,
Monitored Output

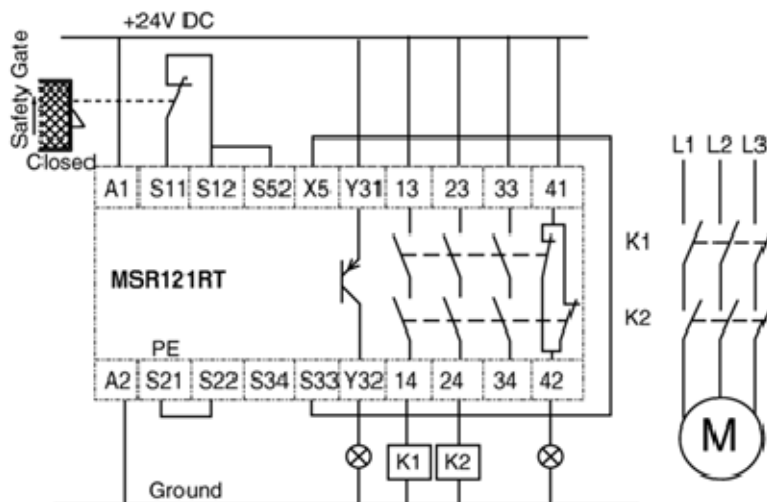


Safety Mat Monitored Manual Reset,
Monitored Output

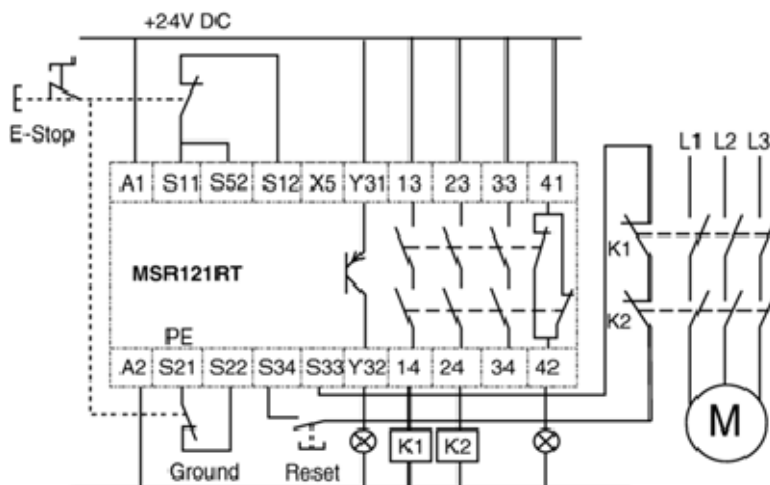


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Single Channel Safety Gate, Automatic Reset, No Output Monitoring



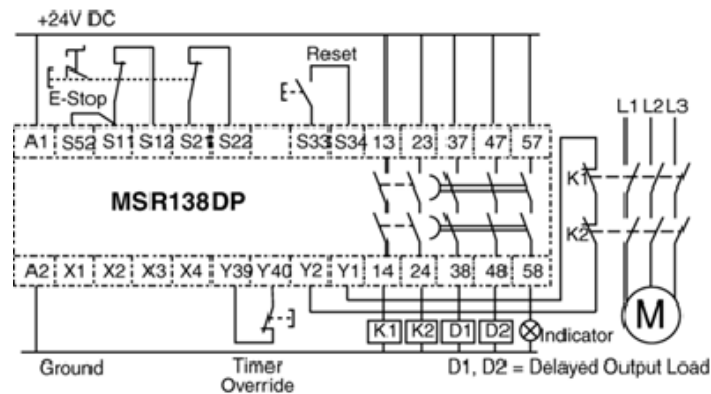
Dual Channel E-Stop, Monitored Manual Reset, Monitored Output



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The heart of this SRCS is the Emergency Relay (MSR138). We can also see that the circuit has at least two Emergency Stop Push Buttons connected in a series configuration. We can also see that there is an expansion module (Safety Expander) used. A reset circuit (not shown) is wired to the Emergency Relay and connects to terminals S33 and S34. Four control relays are connected to the output of the Emergency Relay. Two of the relays are connected to the delayed outputs of the Emergency Relay.

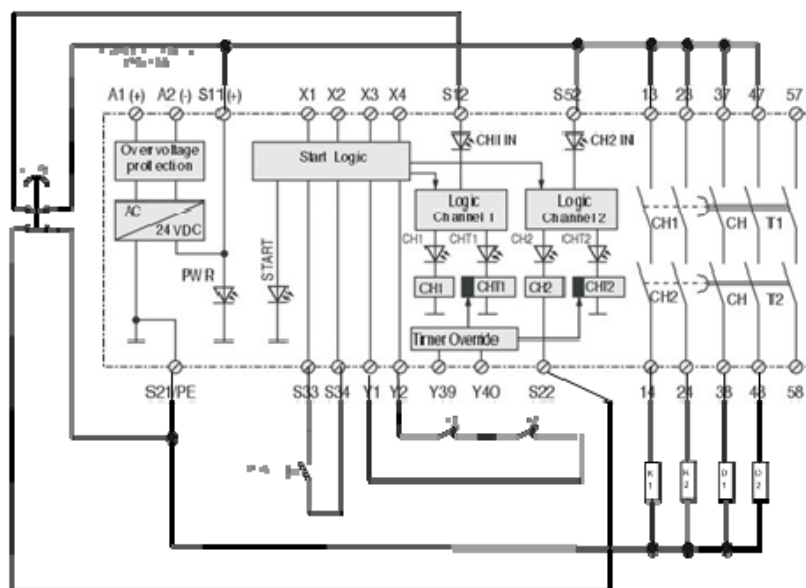
A simplified version of the SRCS is shown below.



The following illustration shows the same Emergency Relay in a block diagram.

Description:

The MSR138DP can be connected in 3 different input wiring configurations: 1 normally closed, 2 normally closed, or 2 OSSD. When connected in the two normally closed fashion, the MSR138DP checks for cross faults across the two inputs.



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The MSR138DP has output monitoring that can accommodate either automatic/manual reset or a monitored manual reset. When configured with automatic/manual reset (jumpers on X1-X2 and X3-X4), the MSR138DP can have the reset terminals S33-S34 jumped or can be converted to an unmonitored manual reset by adding a normally open switch in the monitoring loop (S33-S34). When configured to monitored manual reset, the MSR138DP checks the output monitoring circuit through the manual application of the reset switch. The unit cannot be reset until the timing function has completed.

The outputs of the MSR138DP include two normally open immediate safety outputs and three normally open delayed safety outputs. The safety outputs have independent and redundant internal contacts to support the safety function. If a reset request is made during the time cycle, it will cause a lockout condition. Cycle inputs after timing has completed and reset after the delay time has expired to clear lockout.

A normally closed timer reset switch can be added to force the delayed contacts open prior to the completion of the timing cycle.

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Troubleshooting

The first step in troubleshooting this circuit and its components is a visual step. First look, at the three green LED's (Lights on the front of the Safety Relay). Under normal conditions, they should all be illuminated. Below is troubleshooting guide for possible conditions that may arise. In most cases, you may use an ohmmeter to perform the corrective actions. Before proceeding with an ohmmeter, you must Lock Out, Tag Out, and Verify absence of voltage at the machine. While this guide may not identify every situation that occurs, it is a useful starting point.



	LED Condition	Possible cause(s)	Corrective Action
1.	All three Lights on but circuit not operating properly	- Reset Pushbutton may be malfunctioning. - Incorrect voltage applied to the A₁ and A₂ terminals. - Absence of voltage at terminals 13, 23, 37, and/or 47. - K₁ or K₂ Contactors may be welded closed. - N.C. contacts on K₁ or K₂ may be bad in feedback circuit between Y₁ and Y₂ terminals. - Faulty K₁ or K₂ contactor operation. - Safety Relay may be malfunctioning.	- Verify the operation of the Reset Pushbutton and its N.O. contacts. - Verify the correct voltage is applied to A₁ and A₂ terminals. - Verify correct voltage is present at these terminals. - Verify condition of K₁ and K₂ contactors (Not Stuck) - Using an ohmmeter, verify continuity between Y₁ and Y₂ terminals; if open, K₁ or K₂ N.C. contacts are the problem; Isolate which contacts are bad using an ohmmeter. - Verify operation of K₁ and K₂ contactors. - Replace the Safety Relay.
2.	Power ON CH-1 Off CH-2 Off Circuit not operating properly	- The E. Stop pushbutton may be pressed. - No Continuity between S₂₁ and S₂₂ or between S₁₁ and S₁₂ terminals. - Safety Relay may be malfunctioning.	- Verify the Estop is not pressed; Verify each set of the N.C. contacts of the E. Stop Pushbutton operate correctly. - Verify continuity loops between Terminals S₂₁ and S₂₂ along with S₁₁ and S₁₂. - Replace the Safety Relay.
3.	Power ON CH-1 On CH-2 Off Circuit not operating properly	- No Continuity between S₂₁ and S₂₂. - Safety Relay may be malfunctioning.	- Using ohmmeter, verify continuity loop between Terminals S₂₁ and S₂₂; Faulty set of N.C. Contacts on E. Stop button between terminals S₂₁ and S₂₂. - Replace the Safety Relay.
4.	Power ON CH-1 Off CH-2 On Circuit not operating properly	- No Continuity between S₁₁ and S₁₂. - Safety Relay may be malfunctioning.	- Using ohmmeter, verify continuity loop between Terminals S₁₁ and S₁₂; Faulty set of N.C. Contacts on E. Stop button between terminals S₁₁ and S₁₂. - Replace the Safety Relay.

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Interfacing AC Drives with Safety Relays

This chapter shows specific examples of how to interface AC Drives with safety relays. This document is intended to provide support for typical applications and only covers a sub task of a typical customer application involving safety technology. The examples shown do not claim to be complete and do not represent customer specific solutions.

Because there are many application-specific variables, users **MUST** ensure that products are correctly used and the appropriate architecture is deployed.

IMPORTANT: *A risk assessment should be performed to ensure that all task-hazard combinations have been identified and addressed. The risk assessment may require additional circuitry to reduce the risk to a tolerable level. Safety circuits must take into consideration safety distance calculations which are not part of the scope of this document. This application note is targeted for advanced users and assumes knowledge of drive and safety systems.*

Stop Categories

- Stop Category 0 – Coast, Immediate removal of power to the motor. With immediate removal of power to the motor, the motor will coast to stop. The time required to stop motion is dependent on the load inertia and speed as well as the friction in the mechanical power transmission equipment used in the system.
- Stop Category 1 - Controlled. Controlled Stop, then removal of power to the motor. A ramp to stop will be used to control the mechanical power transmission to rest then power is removed from the motor. The time required to bring the mechanical system to rest is dependent upon load inertia and speed as well as the regenerative dissipation capacity of the drive.

References:

- NFPA79, Electrical Standards for Industrial Machinery, 2007 Edition
- EN954-1, Safety of Machinery
 - o Safety related parts of a control system
 - o Part 1: General Principles for Design
- EN60204-1 : 2006 Safety of Machinery
 - o Electrical Equipment of Machines
 - o Part 1: General Requirements

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Safety Categories

Safety Categories per EN954-1

SUMMARY OF REQUIREMENTS:	SYSTEM BEHAVIOR:
Category B (see note 1) - Safety related parts of machine control systems and/or their protective equipment, as well as their components, shall be designed, constructed, selected, assembled and combined in accordance with relevant standards so that they can withstand the expected influence.	When a fault occurs it can lead to a loss of the safety function.
Category 1 - The requirements of B apply together with the use of well tried safety components and the safety principles.	As described for category B but with higher safety related reliability of the safety related function. (The higher the reliability, the less the likelihood of a fault.)
Category 2- The requirements of B and the use of well tried safety principles apply. The safety function(s) shall be checked at machine start-up and periodically by the machine control system. If a fault is detected a safe state shall be initiated or if this is not possible a warning shall be given.	The loss of safety function is detected by the check. The occurrence of a fault can lead to the loss of safety function between the checking intervals.
Category 3 (see notes 2 & 3) - The requirements of B and the use of well tried safety principles apply. The system shall be designed so that a single fault in any of its parts does not lead to the loss of safety function.	When the single fault occurs the safety function is always preformed. Some but not all faults will be detected. An accumulation of undetected faults can lead to the loss of safety function.
Category 4 (see notes 2 & 3) - The requirements of B and the use of well tried safety principles apply. The system shall be designed so that a single fault in any of its parts does not lead to the loss of safety function. The single fault is detected at or before the next demand on the safety function. If this detection is not possible then an accumulation of faults shall not lead to a loss of safety function.	When the faults occur the safety function is always preformed. The faults will be detected in time to prevent the loss of safety functions.

Note 1: Category B in itself has no special measures for safety but it forms the base for the other categories.

Note 2: Multiple faults caused by a common cause or as inevitable consequences of the first fault shall be counted as a single fault.

Note 3: The fault review may be limited to two faults in combination if it can be justified but complex circuits may require more faults in combination to be considered.

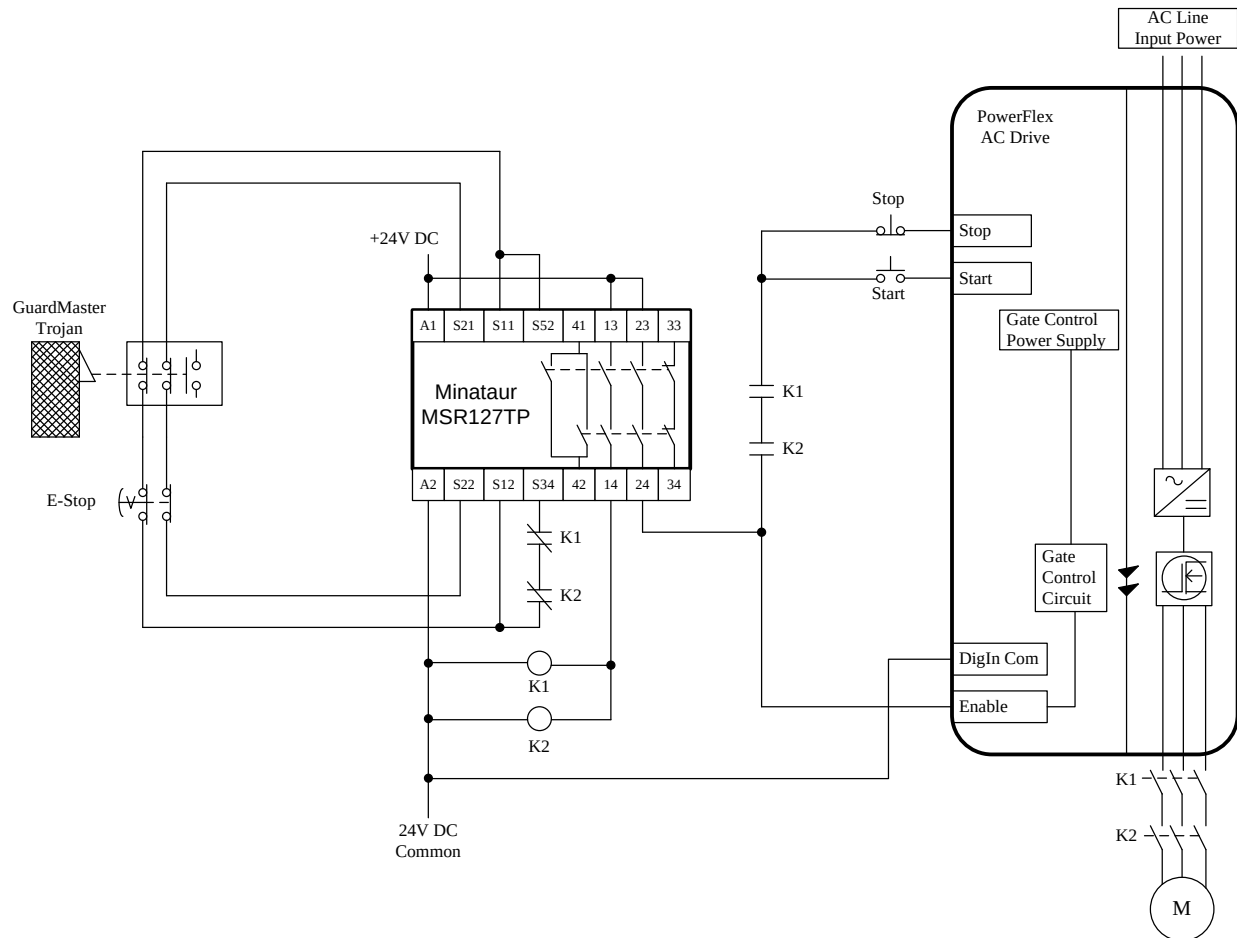
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Component Descriptions

- Gate Interlock - The Trojan gate interlock uses direct opening contacts. When the gate is opened, the contacts in the interlock are forced open by non-resilient components (e.g., not by springs). A redundant (two) set of double break contacts are designed to ensure that at least one signal is sent to the safety relay, when the gate is opened.
- E-Stop - When the E-Stop button is pressed the safety function is activated (Drive goes to safe state). When a contactor is used it is opened by the E-Stop circuit. The E-stop button must utilize direct-opening contacts. The button must latch to an open state when the contacts open (i.e., you must not be able to tickle the contacts without latching the button). The button must be a red with a yellow background.
- Contactor - Provides electromechanical disconnection of the motor from the drive. Its normally-closed mechanically linked contacts are monitored by the safety relay to help ensure that the contactor has dropped out before the next resetting of the safety relay.
- Start / Stop Pushbuttons - The Start and Stop pushbuttons are used to turn the drive on and off. They are also symbolic of the non-safety related parts of the machine control system. These buttons can be replaced by a programmable control system.
- The PowerFlex® Safe-off Option Board:
 - Is designed to safely remove power from the gate firing circuits of the drives output power devices (IGBT's). This prevents them from switching in the pattern necessary to generate AC power to the motor.
 - Can be used in combination with other safety devices to meet the Stop and protection against restart requirements of EN954-1.
- Safety Relay Reset - The examples in this document use an automatic reset scheme for the safety relay. The use of a manual or automatic reset should be dictated by the application.

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Drive Without Safe-off, Coast Stop

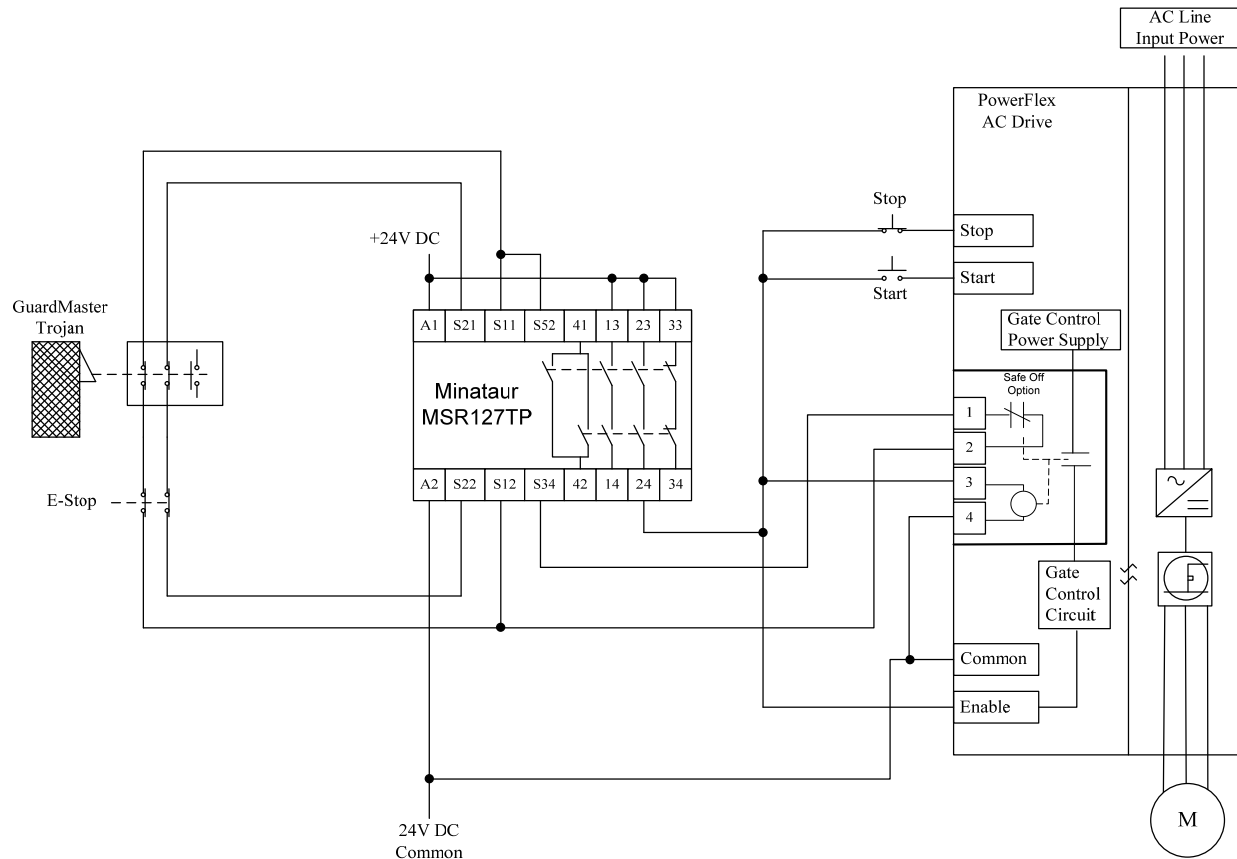


Circuit Operation: The gate will trigger the safety function. When triggered the safety relay outputs (Terminals 13 to 14 and 23 to 24) open. This immediately removes +24vdc from the drive Start/Stop input. The result is the drive stop signal is asserted and a ramp to stop is issued. Once the time delay in the safety relay expires, the safety relay time delay contacts (Terminals 37 to 38, 47 to 48, and 57 to 58) open. The +24vdc is then removed from the drive enable as well as both contactor coils K1 & K2. The N.O. auxiliary contact of K1 and K2 open to prevent a drive Start before the contactors close. The N.C. auxiliary contacts are monitored by the safety relay at terminals Y1 & Y2.

Note: The time delay of the safety relay must be greater than the deceleration ramp time of the drive otherwise a coast stop will occur when the timer expires. When the gate is restored and satisfies the safety relay inputs, the safety relay is reset (terminals Y39 and Y40 are N.C.). The safety relay outputs are energized restoring +24vdc to the drive Enable input, the coils of the contactors and the Drive Start/Stop. A start command must be issued to the drive to restart.

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Drive with Safe Off, Coast Stop

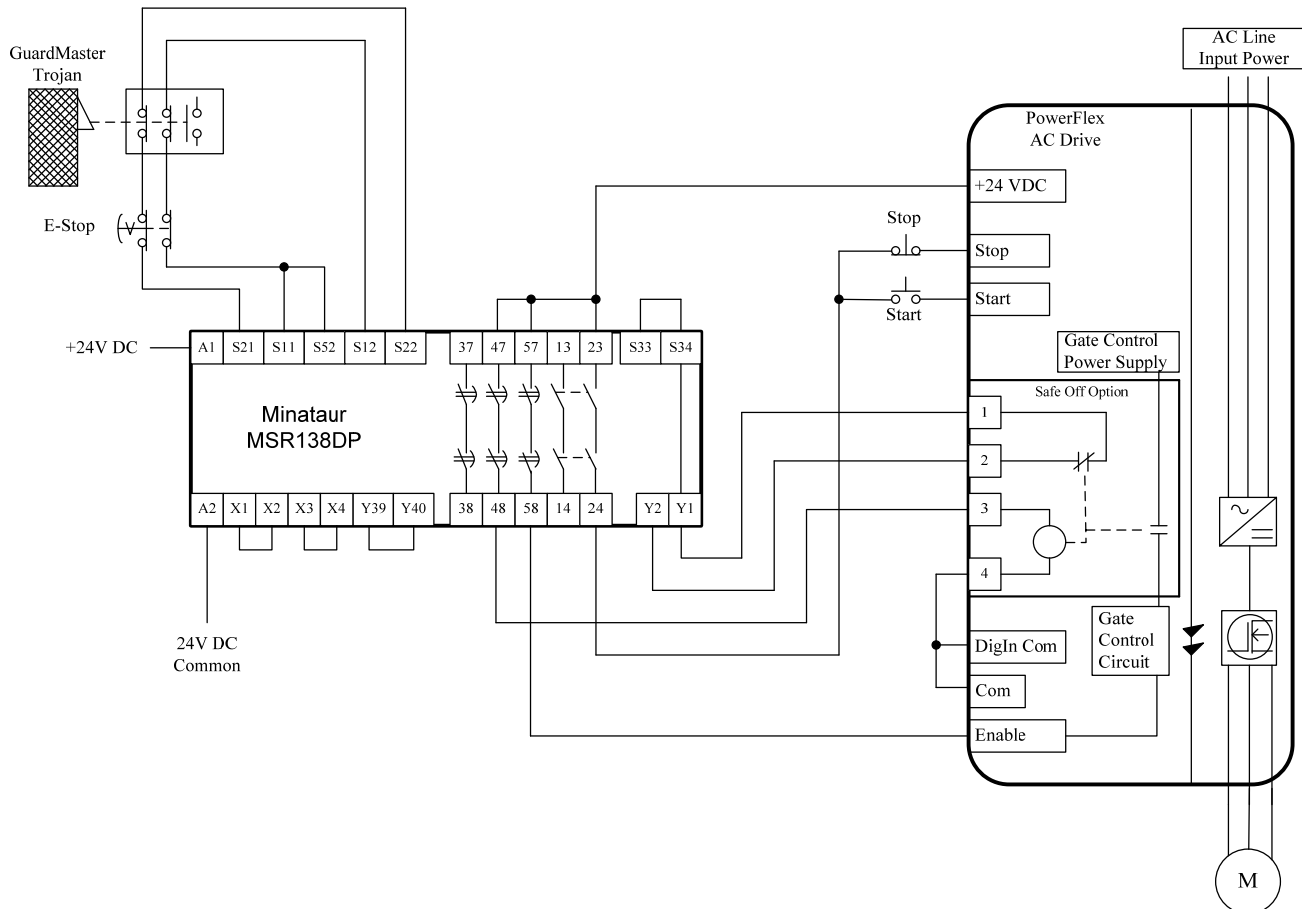


Circuit Operation: The gate will trigger the safety function. When triggered the safety relay outputs (Terminals 13 to 14, 23 to 24, and 33 to 34) open. This immediately removes +24vdc from the drives enable and Start/Stop inputs, and the Safe-off Option relay. The result is the drive is disabled, a stop signal is asserted, the Safe-Off Option goes to a safe state (off), and the motor will coast to stop. The Safe-off Option N.C. auxiliary contacts are monitored by the safety relay at terminals S52 & S34.

When the gate is restored and satisfies the safety relay inputs, the safety relay is reset. The safety relay outputs are energized restoring +24vdc to the drive Enable input and Drive Not Stop, and the Safe-off Option relay. A start command must be issued to the drive to restart. If the Safe-off monitor contact is in an abnormal state, as monitored by the safety relay (S52 & S34) the safety relay will prevent operation at the next cycling of the “safety” circuit. When the E-Stop is activated, the safety function is triggered and performs the same as the gate triggered function. Restoration of the E-Stop also recovers in the same manner as the gate restoration.

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Safety Drive, Controlled Stop



Circuit Operation: The gate will trigger the safety function. When triggered the safety relay outputs (Terminals 13 to 14 and 23 to 24) open. This immediately removes +24vdc from the drive Start/Stop input. The result is the drive stop signal is asserted and a ramp to stop is issued.

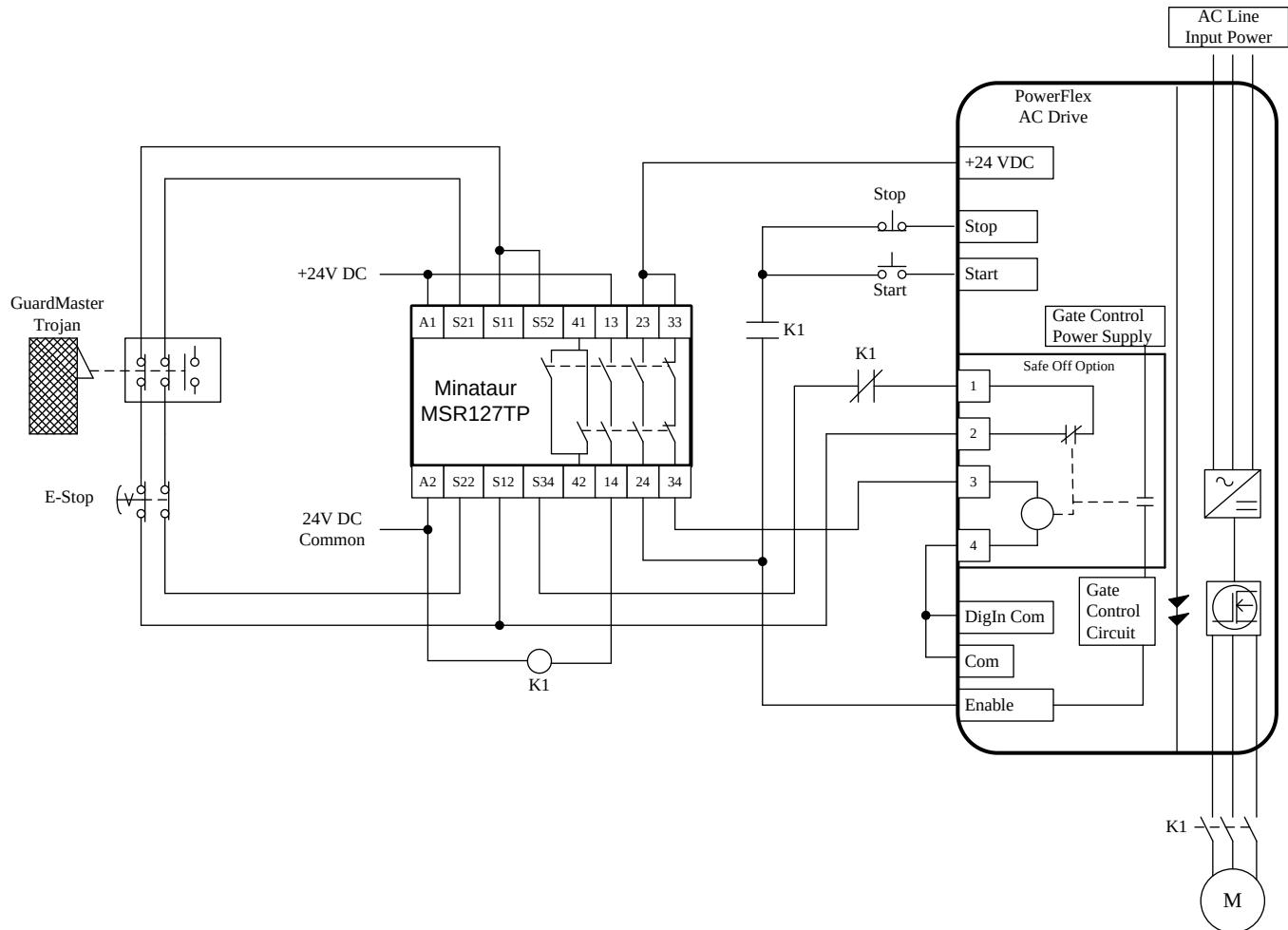
Once the time delay in the safety relay expires, the safety relay time delayed contacts (Terminals 37 to 38, 47 to 48, and 57 to 58) open. The +24vdc is then removed from the drive Enable input as well as the Safe-off Option. The Safe-off Option goes to a safe state (off).

Note: The time delay of the safety relay must be greater than the deceleration ramp time of the drive otherwise a coast stop will occur when the timer expires.

When the gate is restored and satisfies the safety relay inputs, the safety relay is reset (terminals Y39 and Y40 are N.C.). The safety relay outputs are energized restoring +24vdc to the drive Enable input, the Safe-off Option and the Drive Start/Stop. A start command must be issued to the drive to restart. If the Safe-off monitor contact is in an abnormal state, as monitored by the safety relay (Y1 & Y2) the safety relay will prevent operation at the next cycling of the “safety” circuit.

MICHELIN NORTH AMERICA	TITLE: Ch7 Introduction to Safe Off Circuits.doc	Edition Date: 2 May 2019	PAGE 20 OF 20
Entity: NA/TTG	Author: Marty Chapman	Validator: N. Dean Campbell NA/TTG	Classification: D3

Safety Drive, Coast Stop



Circuit Operation: The gate will trigger the safety function. When triggered the safety relay outputs (Terminals 13 to 14, 23 to 24, and 33 to 34) open. This immediately removes +24vdc from the drives enable and Start/Stop inputs, K1's contactor coil, and the Safe-off Option relay. The result is the drive is disabled, a stop signal is asserted, the Safe-off Option goes to a safe state (off), K1 contactor opens, and the motor will coast to stop. The N.O. auxiliary contact of K1 opens to prevent a drive Start before the contactor closes. The N.C. auxiliary contacts are monitored by the safety relay at terminals S52 & S34.

When the gate is restored and satisfies the safety relay inputs, the safety relay is reset. The safety relay outputs are energized restoring +24vdc to the drive Enable input and Drive Not Stop, the coil of the K1 contactor, and the Safe-off Option relay. A start command must be issued to the drive to restart. If Safe-off monitor contact and/or K1's contactor N.C. auxiliary contact are in an abnormal state, as monitored by the safety relay (S52 & S34) the safety relay will prevent operation at the next cycling of the "safety" circuit.

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Supplemental Exercises

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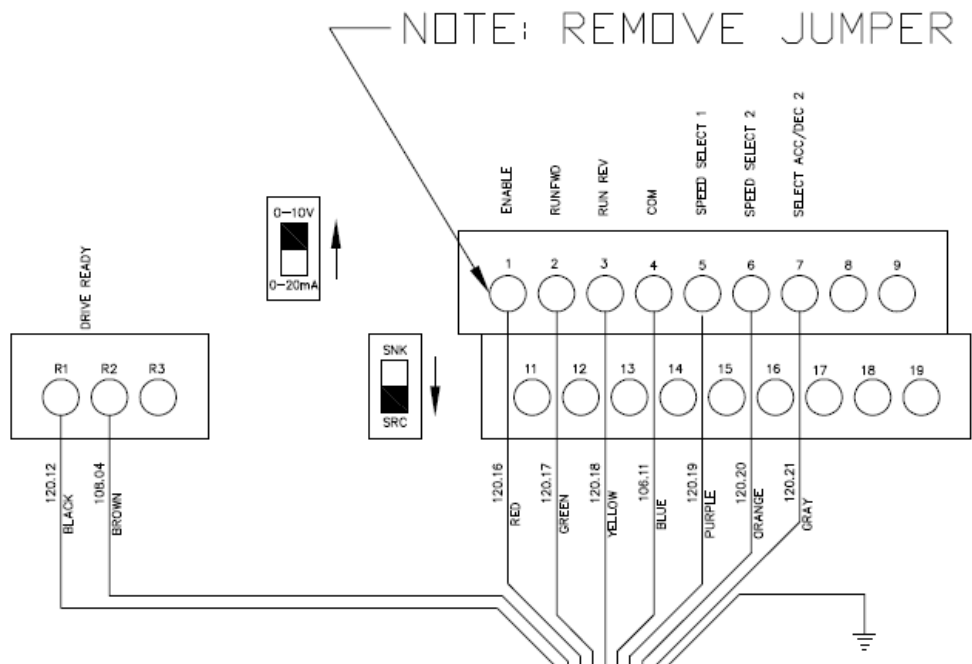
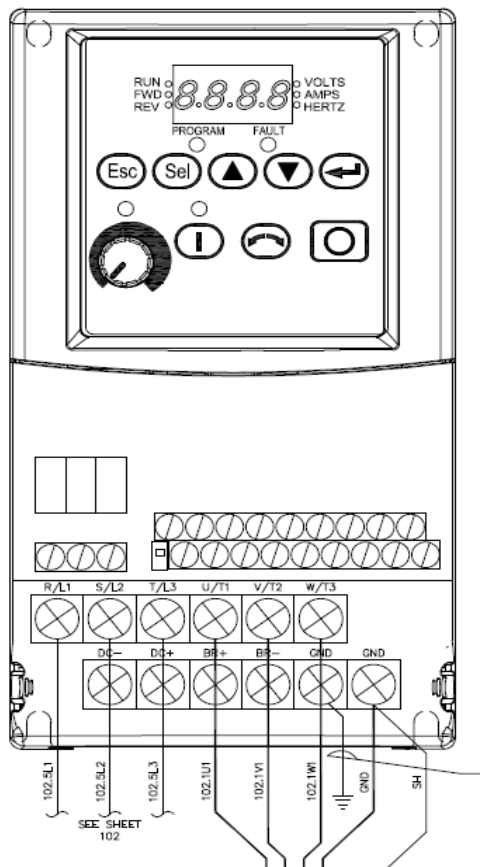
Challenge Exercise 1

Use the Challenge Circuit Drawings (Pages 6-10), to answer the following questions:

1. What is the motor FLA connected to 102drv22 ?
2. Does the motor have an external brake?
3. Does the VFD (102drv31) have any line conditioning in the circuit?
4. What is the purpose of 102MCP31 ?
5. Where is shielded cable used in the circuit for the Bed Conveyor?
6. 109MCR2 is used for what purpose?
7. The line side wires for VFD 102dvr22) connects to which terminals on the Drive?
8. What type of inputs are the drives configured for (Source or Sink)?
9. What does the R1/R2 terminals connect to for the VFD (102drv22) ?
10. Where could you measure the DC Bus voltage on VFD (102drv31) ?

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VFD 102drv22

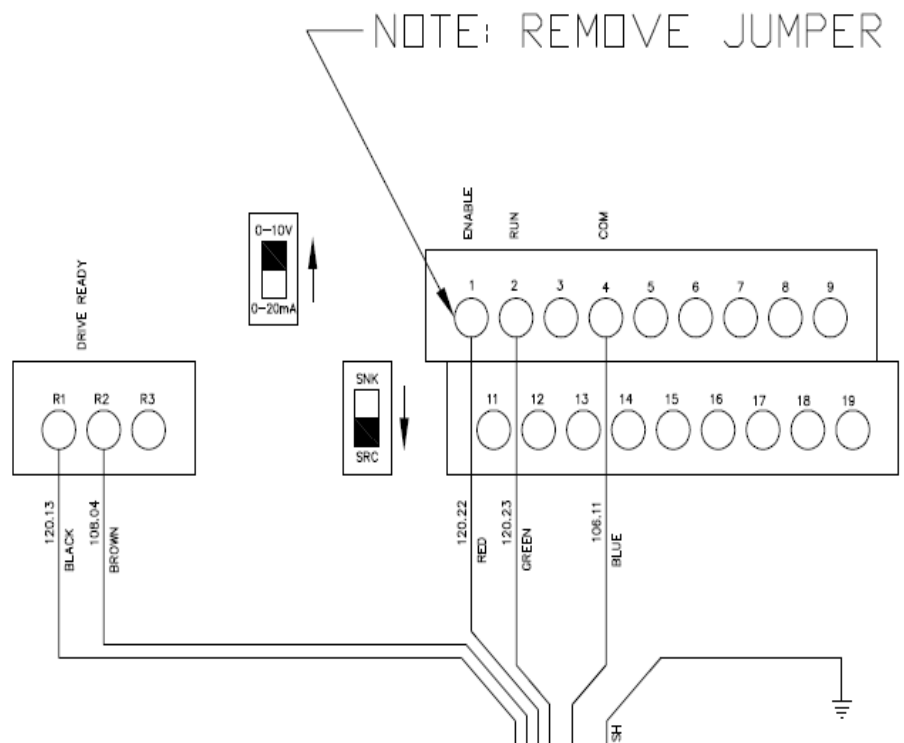
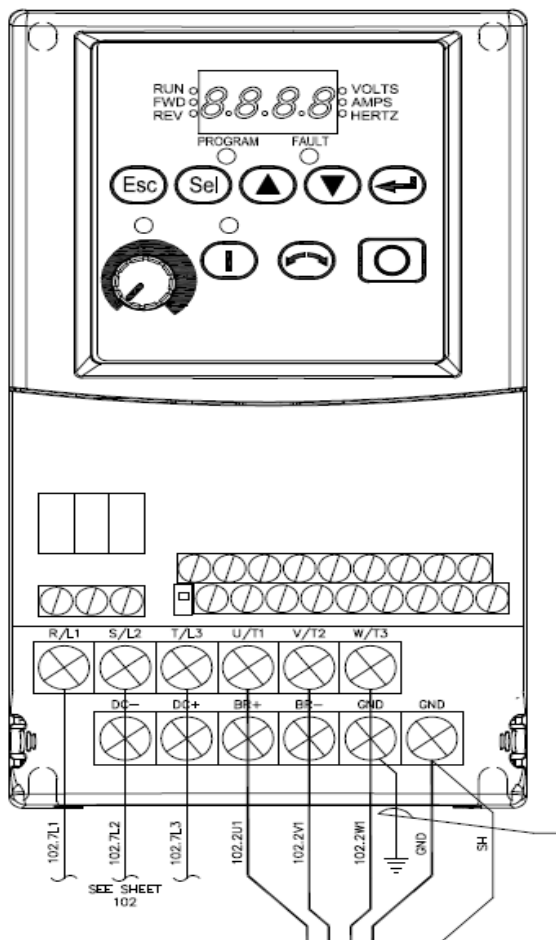


Settings for 102drv22

Parameter	Description	Value	Parameter	Description	Value
P 031	Motor NP Volts	460	A 067	Accel 2	1.5
P 032	Motor NP Hz	60	A 068	Decel 2	0.6
P 033	Motor OL Current	1.7	A 070	Preset 0 Hz	72
P 034	Min Drive Hz	10.0	A 071	Preset 1 Hz	24
P 035	Max Drive Hz	90.0	A 072	Preset 2 Hz	45
P 036	Start Source	2	A 073	Preset 3 Hz	60
P 037	Stop Mode	4	A 074	Preset 4 Hz	30
P 038	Speed Ref	4	A 075	Preset 5 Hz	40
P 039	Accel 1	0.5	A 076	Preset 6 Hz	50
P 040	Decel 1	0.3	A 077	Preset 7 Hz	60
A 051	Dig In 1 (5)	4			
A 052	Dig In 2 (6)	4			
A 053	Dig In 3 (7)	1			
A 054	Dig In 4 (8)	0			
A 055	Relay Out	0			

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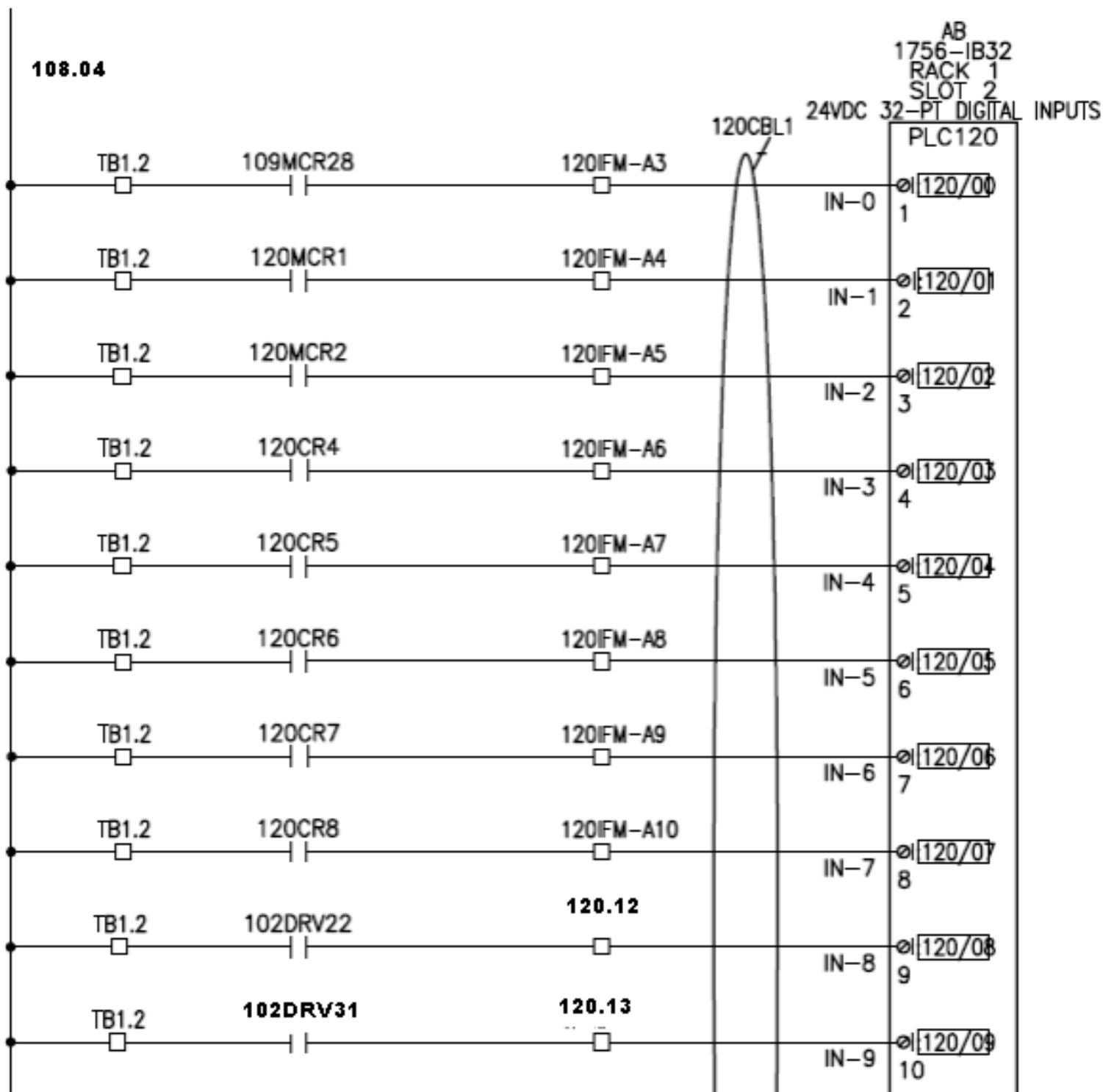
VFD 102drv31



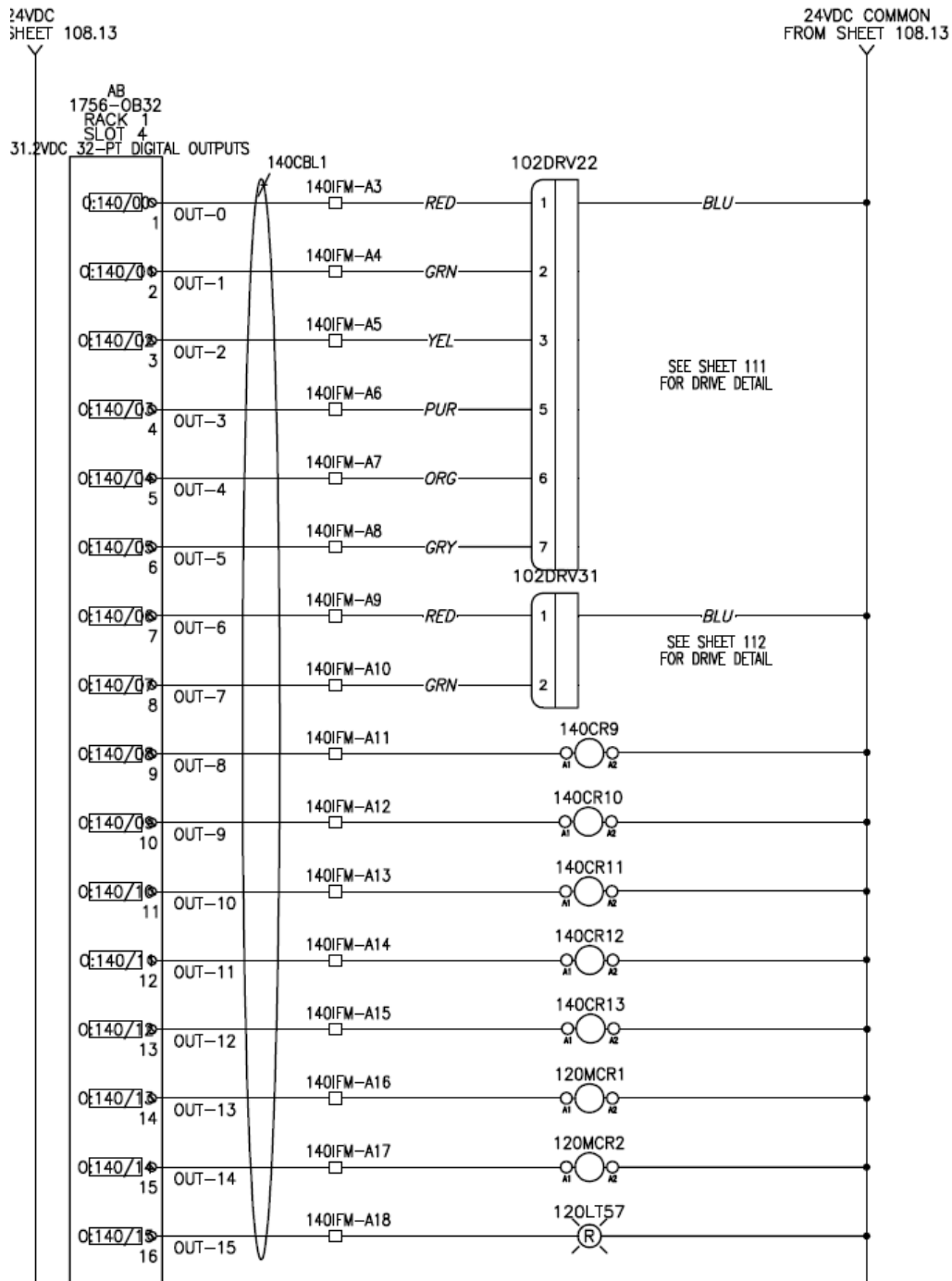
Settings for 102drv31

Parameter	Description	Value	Parameter	Description	Value
P 031	Motor NP Volts	460	A 067	Accel 2	0.5
P 032	Motor NP Hz	60	A 068	Decel 2	0.3
P 033	Motor OL Current	1.7	A 070	Preset 0 Hz	72
P 034	Min Drive Hz	10.0	A 071	Preset 1 Hz	24
P 035	Max Drive Hz	90.0	A 072	Preset 2 Hz	45
P 036	Start Source	2	A 073	Preset 3 Hz	60
P 037	Stop Mode	4	A074	Preset 4 Hz	30
P 038	Speed Ref	1	A 075	Preset 5 Hz	40
P 039	Accel 1	0.5	A 076	Preset 6 Hz	50
P 040	Decel 1	0.3	A 077	Preset 7 Hz	60
A 051	Dig In 1 (5)	0			
A 052	Dig In 2 (6)	0			
A 053	Dig In 3 (7)	4			
A 054	Dig In 4 (8)	0			
A 055	Relay Out	0			

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Challenge Exercise 2

Use the Challenge Circuit Drawings (Pages 6-10), to answer the following questions:

1. The Start Source for VFD (102drv31) is configured as:
2. The Stop Mode for VFD (102drv31) is configured as:
3. The Digital Input #1 for VFD (102drv31) is configured as:
4. What condition will cause the Relay output for either drive to energize?
5. What Frequency should VFD (102dvr31) output to the motor when running (assuming the drive is enabled and commanded to run)?
6.
 - a. What Frequency should VFD (102drv22) output to the motor if terminal 5 measures 0Vdc and terminals 6 and 7 measure 24Vdc (assuming the drive is enabled and commanded to run)?
 - b. What should the three phase voltage to the motor be?
7. If we want to output 60 Hz to the motor connected to VFD (102dvr22), what should be the voltage measured between the following terminals and common? The motor should run Forward using Accel 2.

1 _____

2 _____

3 _____

5 _____

6 _____

7 _____

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8. What is the Acceleration rate of either of the VFD's, using Accel1?
9. What is the Deceleration Rate of either of the drives, using Decel1?
10. If we want to output 60 Hz to the motor connected to VFD (102dvr22), using Accel 1, how long would it take for the drive to accelerate to this frequency upon receiving a run command?
11. What is the Analog Output used for in 102dvr22?
12. If you want to reset all parameters to Factory Default, which parameter allows you do this?
13. What is the Speed Reference for 102drv31 configured as?
14. What parameters will determine the Volts/Hertz ratio for 102drv22?

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Parameter Configuration

The following PowerFlex 40P parameters are used to configure the drive to operate on a network.

Parameter	Details	Reference
P036 [Start Source]	Set to 5 "Comm Port" if Start is controlled from the network.	Page 3-10
P038 [Speed Reference]	Set to 5 "Comm Port" if the Speed Reference is controlled from the network.	Page 3-12
A103 [Comm Data Rate]	Sets the data rate for the RS485 (DSI) Port. All nodes on the network must be set to the same data rate.	Page 3-34
A104 [Comm Node Addr]	Sets the node address for the drive on the network. Each device on the network requires a unique node address.	Page 3-34
A105 [Comm Loss Action]	Selects the drive's response to communication problems.	Page 3-34
A106 [Comm Loss Time]	Sets the time that the drive will remain in communication loss before the drive implements A105 [Comm Loss Action].	Page 3-34
A107 [Comm Format]	Sets the transmission mode, data bits, parity and stop bits for the RS485 (DSI) Port. All nodes on the network must be set to the same setting.	Page 3-35
E207 [Comm Write Mode]	Set to 0 "EEPROM" when programming drive. Set to 1 "RAM only" to only write to volatile memory.	Page 3-52

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Writing (06) Logic Command Data

The PowerFlex 40P drive can be controlled via the network by sending Function Code 06 writes to register address 8192 (Logic Command). [P036](#) [Start Source] must be set to 5 “Comm Port” in order to accept the commands. Parameter [E249](#) [Cmd Stat Select] is used to select Velocity or Position Bit definitions.

Velocity Bit Definitions

Logic Command - E249 = 0 “Velocity”		
Address (Decimal)	Bit(s)	Description
8192	0	1 = Stop, 0 = Not Stop
	1	1 = Start, 0 = Not Start
	2	1 = Jog, 0 = No Jog
	3	1 = Clear Faults, 0 = Not Clear Faults
	5,4	00 = No Command
		01 = Forward Command
		10 = Reverse Command
		11 = No Command
	6	1 = Logic In 1
	7	1 = Logic In 2
	9,8	00 = No Command
		01 = Accel Rate 1
		10 = Accel Rate 2
		11 = Hold Accel Rate Selected
	11,10	00 = No Command
		01 = Decel Rate 1
		10 = Decel Rate 2
		11 = Hold Decel Rate Selected
	14,13,12	000 = No Command
		001 = Freq. Source = P036 [Start Source]
		010 = Freq. Source = A069 [Internal Freq]
		011 = Freq. Source = Comms (Addr 8193)
		100 = A070 [Preset Freq 0]
		101 = A071 [Preset Freq 1]
		110 = A072 [Preset Freq 2]
		111 = A073 [Preset Freq 3]
	15	Not Used

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Writing (06) Reference

The Speed Reference to a PowerFlex 40P drive can be controlled via the network by sending Function Code 06 writes to register address 8193 (Reference). [P038](#) [Speed Reference] must be set to 5 “Comm Port” in order to accept the Speed Reference.

Reference	
Address (Decimal)	Description
8193	A decimal value entered as xxx.xx where the decimal point is fixed. For example, a decimal “100” equals 1.00 Hz and “543” equals 5.43 Hz.

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Reading (03) Logic Status Data

The PowerFlex 40P Logic Status data can be read via the network by sending Function Code 03 reads to register address 8448 (Logic Status). Parameter [E249](#) [Cmd Stat Select] is used to select Velocity or Position Bit definitions.

Velocity Bit Definitions

Logic Status - E249 = 0 "Velocity"		
Address (Decimal)	Bit(s)	Description
8448	0	1 = Ready, 0 = Not Ready
	1	1 = Active (Running), 0 = Not Active
	2	1 = Cmd Forward, 0 = Cmd Reverse
	3	1 = Rotating Forward, 0 = Rotating Reverse
	4	1 = Accelerating, 0 = Not Accelerating
	5	1 = Decelerating, 0 = Not Decelerating
	6	Not Used
	7	1 = Faulted, 0 = Not Faulted
	8	1 = At Reference, 0 = Not At Reference
	9	1 = Reference Controlled by Comm
	10	1 = Operation Cmd Controlled by Comm
	11	1 = Parameters have been locked
	12	Digital Input 1 Status
	13	Digital Input 2 Status
	14	Digital Input 3 Status
	15	Digital Input 4 Status

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Reading (03) Feedback

The Feedback (Output Frequency) from the PowerFlex 40P drive can be read via the network by sending Function Code 03 reads to register address 8451 (Feedback).

Feedback ⁽¹⁾	
Address (Decimal)	Description
8451	A xxx.x decimal value where the decimal point is fixed. For example, a decimal "123" equals 12.3 Hz and "300" equals 30.0 Hz.

⁽¹⁾ Returns the same data as Reading (03) Parameter d001 [Output Freq].

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Reading (03) Drive Error Codes

The PowerFlex 40P Error Code data can be read via the network by sending Function Code 03 reads to register address 8449 (Drive Error Codes).

Logic Status		
Address (Decimal)	Value (Decimal)	Description
8449	0	No Fault
	2	Auxiliary Input
	3	Power Loss
	4	Undervoltage
	5	Overvoltage
	6	Motor Stalled
	7	Motor Overload
	8	Heatsink Overtemperature
	12	HW Overcurrent (300%)
	13	Ground Fault
	29	Analog Input Loss
	33	Auto Restart Tries
	38	Phase U to Ground Short
	39	Phase V to Ground Short
	40	Phase W to Ground Short
	41	Phase UV Short
	42	Phase UW Short
	43	Phase VW Short
	48	Parameters Defaulted
	63	Software Overcurrent
	64	Drive Overload
	70	Power Unit Fail
	71	Network Loss Fault
	80	AutoTune Fail
	81	Communication Loss
	94	Encoder Loss Fault
	100	Parameter Checksum Error
	111	Hardware Enable Fault
	122	I/O Board Fail

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Chariot Transfer Confection OR Finishing Configuration Work Line

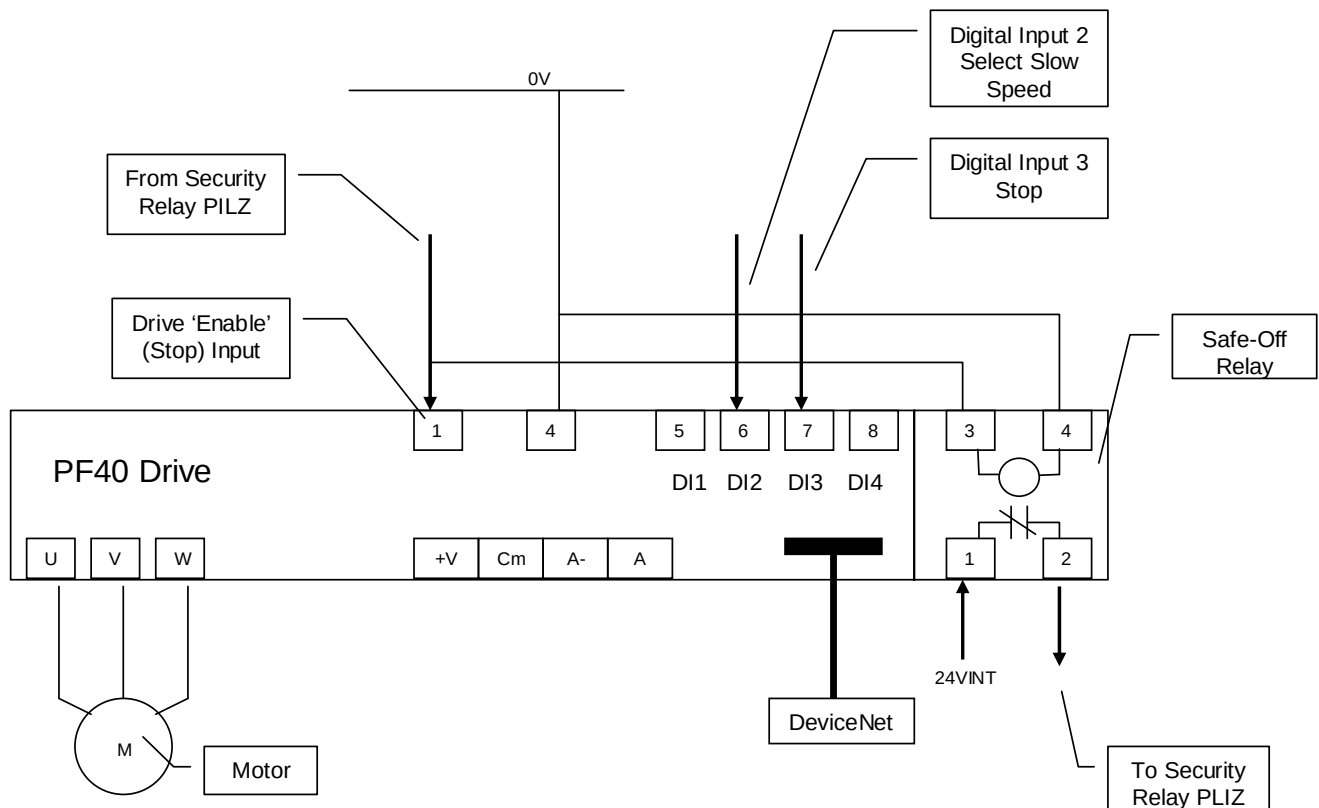
The 'chariot transfer' drive is configured as below. The speed reference is controlled by the 'Command Word' or by two external digital inputs connected to terminals 6 and 7.

Digital input 2 is connected to the 'transfer slowdown' proximity detector which is used to switch from a transfer speed to a final positioning speed.

Digital input 3 is connected to the 'transfer stop' proximity detector which is used to switch from a slow speed to a zero stop reference.

The 'state' of the digital inputs is also transferred from the drive via the DeviceNet network 'status word' (bits 13 & 14) to the ControlLogix to enable certain grafacs to 'step-on', etc..

The 'stop' and 'safe-off' inputs are controlled from the 'Transfer Securities' which are monitored via the PILZ security automate..



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Configuration of digital inputs 2 & 3 within PF40P drive

A52	Digital In 2 Sel (term 6)	4 – Preset Freq	Refer to A70 to A77
A53	Digital In 3 Sel (term 7)	4 – Preset Freq	Refer to A70 to A77
A38	Speed Reference	5 – Comm Port	From ‘Command Word’ but overridden if A52 or A53 input active

A52 (Slow)	A53 (Stop)	Frequency Source	Speed Applied
0	0	A70 (Preset Freq 0)	Reference (Command Word)
1	0	A72 (Preset Freq 2)	PV – Slow Speed
0	1	A74 (Preset Freq 4)	GV – Fast Speed (n/u)
1	1	A76 (Preset Freq 6)	0 – Zero Speed

From the above parameter setting the chariot can be transferred at the following speeds:

- Fast speed controlled by the ‘command Word’
- When A52 = 1 (A72), speed ramps from fast to slow.
- When A52 & A53 = 1 (A76), speed ramps from slow to zero.

At certain times it is necessary to perform a ‘Jog’ cycle and ignore the status of the external inputs. This is achieved by changing the status of parameters A52 & A53, setting both there values to zero.

This is done by ‘write’ message instructions executed by the Contrologix. When both A52 & A53 equal zero, the drive speed reference is controlled by the ‘Command Word’ whose value (Jog Ref) is determined by the program. When this operation is complete another ‘write’ message instruction is executed to change A52 & A53 back to external input selection.

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DeviceNet

The PF40P drives and associated motors will be controlled from the ControlLogix via the DeviceNet network. The PF40P drive has two input words, 'logic command' and 'reference' and two output words, 'logic status' and 'feedback'.

These input and output words allow the ControlLogic to 'command' the drive and also to monitor its 'status'. See below for 'logic command word' and 'logic status word'.

Parameters within the PF40P drive can be 'written to' or 'read' by the use of 'explicit message' instruction. Message instruction can be continuously written to the drive without any reduction in its memory 'life cycle' if parameter 207 is set to RAM only.

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Logic Command / Status Words

Logic Command Word

Logic Bits																Command	Description
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
															X	Stop	0 = Not Stop 1 = Stop
														X		Start	0 = Not Start 1 = Start
													X			Jog	0 = Not Jog 1 = Jog
												X				Clear Faults	0 = Not Clear Faults 1 = Clear Faults
										X	X					Direction	00 = No Command 01 = Forward Command 10 = Reverse Command 11 = Change Direction (toggle)
									X							Not Used	
								X								MOP Increment	0 = Not Increment 1 = Increment
						X	X									Accel Rate	00 = No Command 01 = Accel Rate 1 Command 10 = Accel Rate 2 Command 11 = Hold Accel Rate
				X	X											Decel Rate	00 = No Command 01 = Decel Rate 1 Command 10 = Decel Rate 2 Command 11 = Hold Decel Rate
	X	X	X													Reference Select	000 = No Command 001 = Freq Source = Select 010 = Freq Source = Int. Freq 011 = Freq Source = Comm 100 = Preset Freq 1 101 = Preset Freq 2 110 = Preset Freq 3 111 = Preset Freq 4
X																MOP Decrement	0 = Not Decrement 1 = Decrement

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Logic Status Word

Logic Bits																Status	Description
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
															X	Ready	0 = Not Ready 1 = Ready
														X		Active	0 = Not Active 1 = Active
													X			Command Direction	0 = Reverse 1 = Forward
												X				Actual Direction	0 = Reverse 1 = Forward
											X					Accel	0 = No Accelerating 1 = Accelerating
										X						Decel	0 = Not Decelerating 1 = Decelerating
									X							Alarm	0 = No Alarm 1 = Alarm
								X								Fault	0 = No Fault 1 = Fault
							X									At Speed	0 = Not At Reference 1 = At Reference
						X										Main Freq	0 = Not Controlled By Comm 1 = Controlled By Comm
					X											Operation Command	0 = Not Controlled By Comm 1 = Controlled By Comm
				X												Parmeters	0 = not locked 1 = Locked
			X													Digital Input 1 Status	
		X														Digital Input 2 Status	
	X															Digital Input 3 Status	
X																Digital Input 4 Status	

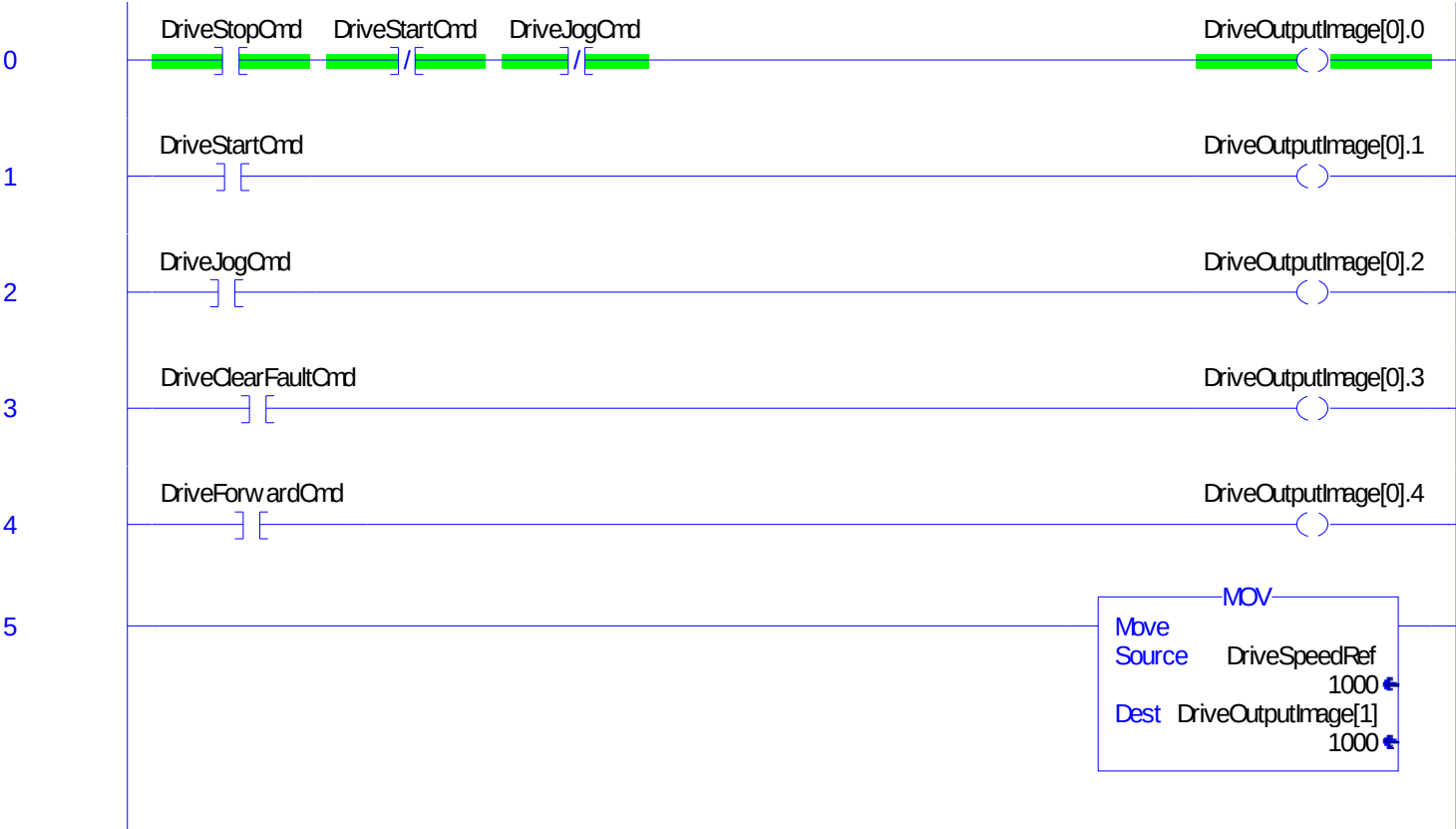
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ControlLogix Ladder Logic Program

Information received from Status and Feedback Words

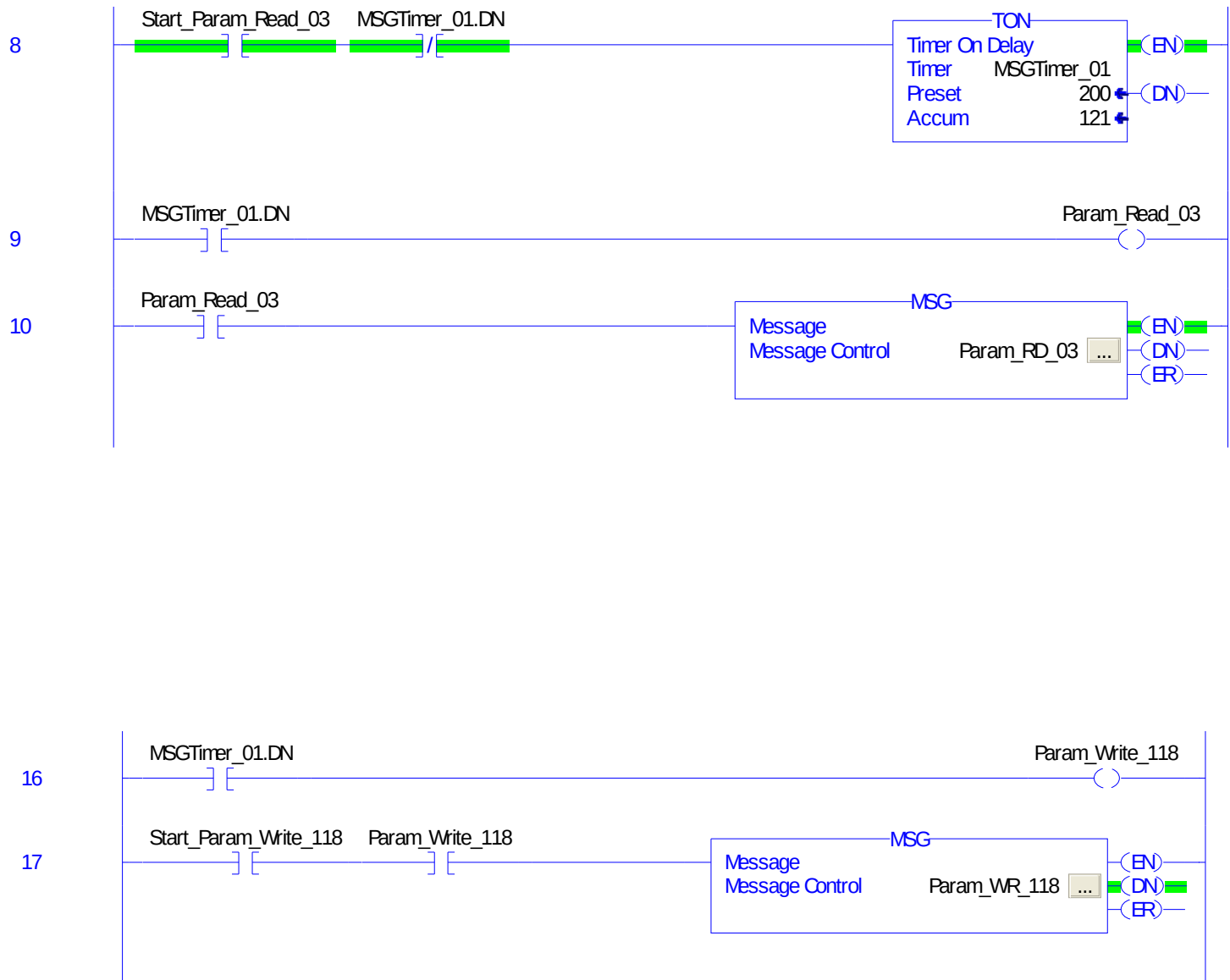


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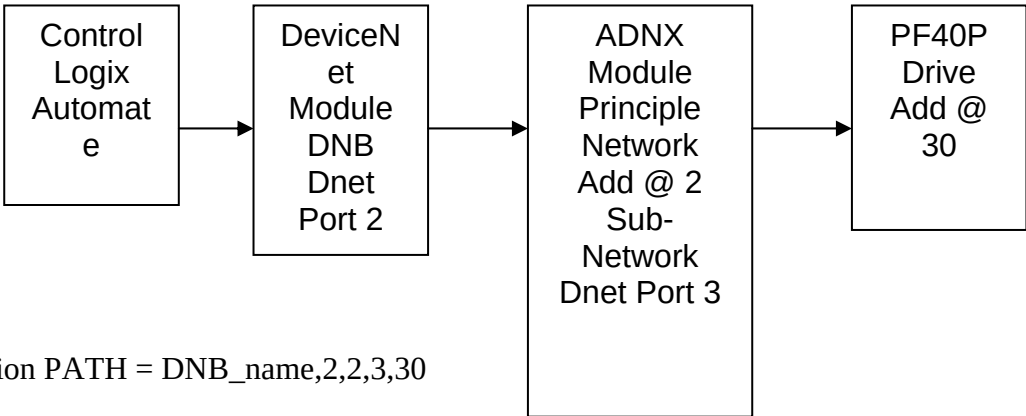
Logic commands and Reference words transmitted to the drive



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Communication Path

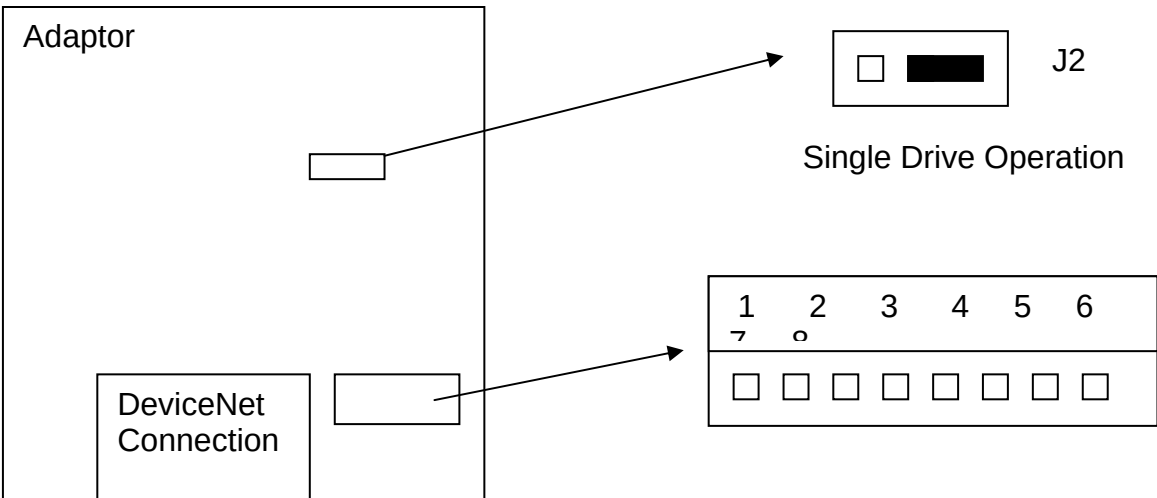
This is required to allow the message to reach its final destination, a PF40P drive connected to the devicenet network.



Communication PATH = DNB_name,2,2,3,30

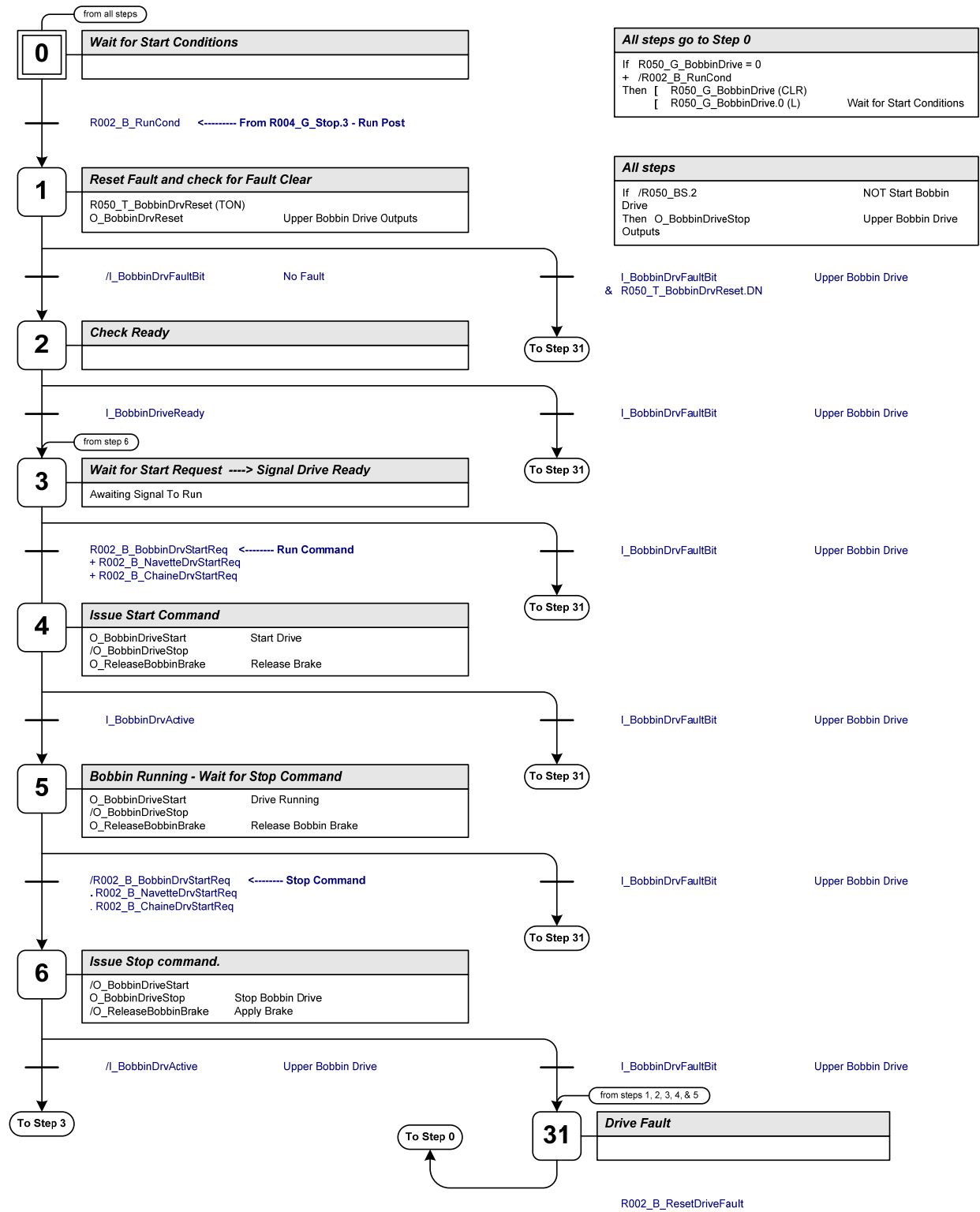
Setting DeviceNet Address

Communication to the drive via DevieNet is achieved by an adaptor interface (22-COMM-D) installed inside the front cover of the drive. To enable the adaptor / drive to communicate with the DeviceNet the ‘node address’ and ‘data rate’ must be set. This is achieved by setting the switches as displayed below.



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Drive Start-Up Grafcet

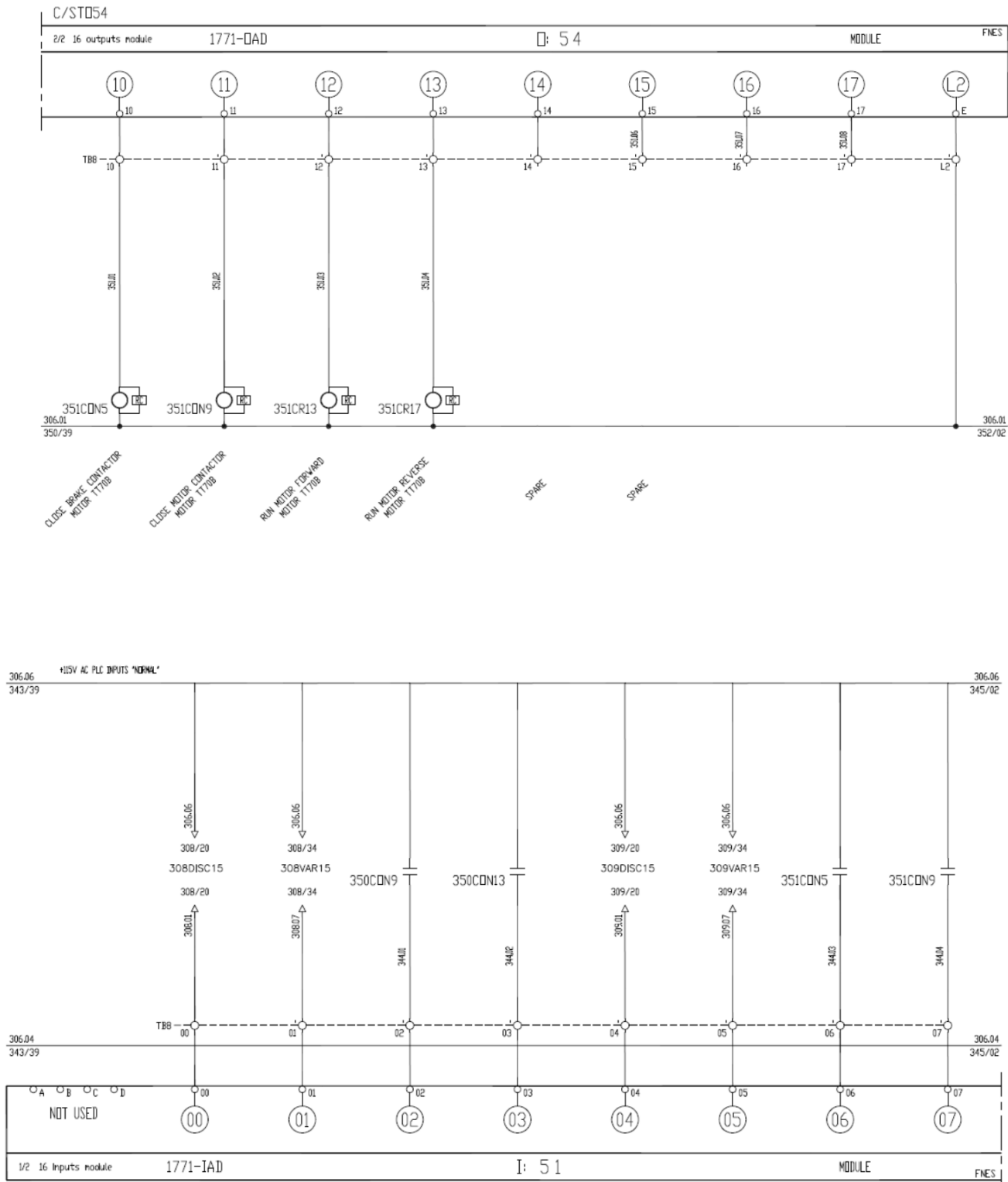


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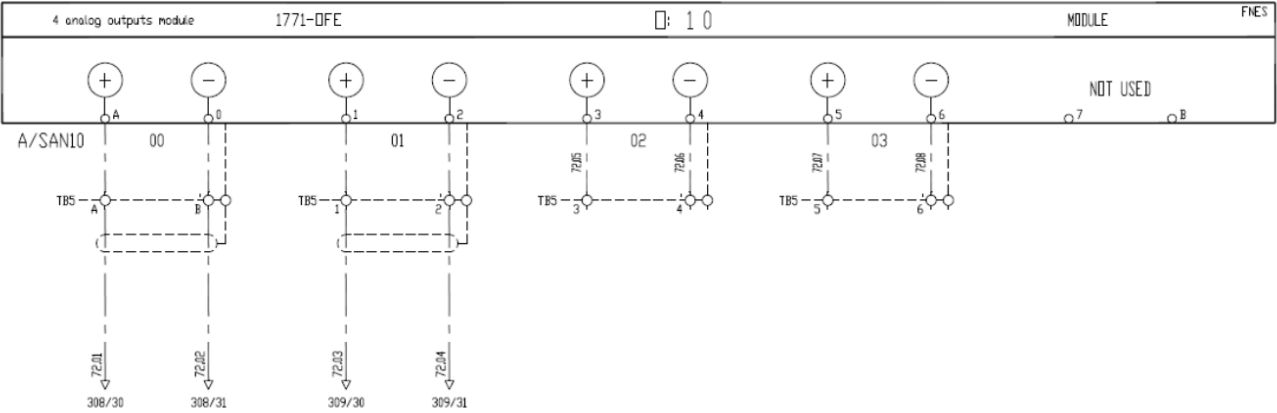
Challenge Exercise 3 Drawing Package

Parameter Settings for TT70B Drive		
Parameter	Description	Value
P 041	Motor NP Volts	460
P 042	Motor NP FLA	7
P 043	Motor NP Hz	60
P 055	Max Frequency	80
P 081	Min Drive Hz	0
P 082	Max Drive Hz	80
P 090	Speed Ref A	2
P 100	Jog Speed 1	30
P 101	Preset 1 Hz	20
P 102	Preset 2 Hz	30
P 103	Preset 3 Hz	40
P 104	Preset 4 Hz	50
P 105	Preset 5 Hz	60
P 106	Preset 6 Hz	70
P 107	Preset 7 Hz	80
P 140	Accel 1	1.5
P 142	Decel 1	2.0
P 155	Stop Mode	2
P 361	Digital Input 1	8
P 362	Digital Input 2	9
P 363	Digital Input 3	0
P 364	Digital Input 4	0
P 365	Digital Input 5	0
P 366	Digital Input 6	0
P 380	Digital Output 1	9

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Challenge Exercise Questions

#	Using Chapter 3 and the Powerflex 70 Appendix chapter as resources, answer the following questions:	
1	Which best describes the input wiring configuration to the drive?	a. Three wire, external supply, Sourcing, Fwd/Rev b. Two wire, internal supply, Sourcing, Fwd/Rev c. Two wire, external supply, Sinking, Fwd/Rev d. Three wire, internal supply, Sinking, Fwd/Rev
2	What drive terminals does the drive receive its speed reference?	
3	What is the purpose of 351CR13?	
4	Which parameters determine the Volts/Hz Ratio for the drive? What is the <u>actual</u> V/Hz ratio for the drive?	
5	When will the PLC input I:51/05 be "ON"?	
6	Which PLC output will command the drive to run in reverse?	
7	What does the PLC output O:54/10 control?	
8	What function does the jumper between drive terminals 7 and 8 perform?	
9	What voltage would be measured between terminal 12 and ground when the output from the drive is closed?	
10	What is the acceleration rate for the drive?	
11	The drive has been commanded to run forward at a frequency of 45 Hz. What should be the approximate rotational speed of the motor? What should the voltage at terminals U, V and W on the drive measure, approximately? What should be the voltage measured at terminals 18 and 19 on the drive?	
12	If you want to verify the voltage for run reverse, which drive terminals would you place your meter leads?	

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Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

Start Time:	Stop Time:		
Machine #			
Symptom			
Root Cause of Fault			
T/S Procedure Step #			

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Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

Start Time:	Stop Time:		
Machine #			
Symptom			
Root Cause of Fault			
T/S Procedure Step #			

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Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

Start Time:	Stop Time:	Start Time:	Stop Time:
Machine #		Machine #	
Symptom		Symptom	
Root Cause of Fault		Root Cause of Fault	
T/S Procedure Step #		T/S Procedure Step #	

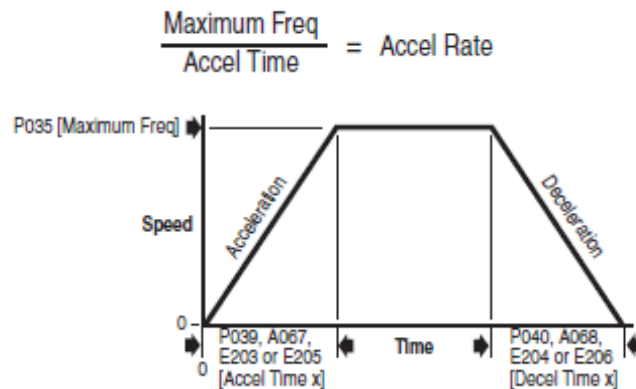
Start Time:	Stop Time:		
Machine #			
Symptom			
Root Cause of Fault			
T/S Procedure Step #			

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AC Drives Glossary of Terms

Accel/Decel (Ramping)

The acceleration time defines the time duration in which the AC Drive reaches its maximum frequency after a start signal is issued. Short acceleration times are usually for light loads, and long acceleration times for heavy loads, or in applications requiring soft start such as a bottle conveyor. The Acceleration time function is also known as ramping.



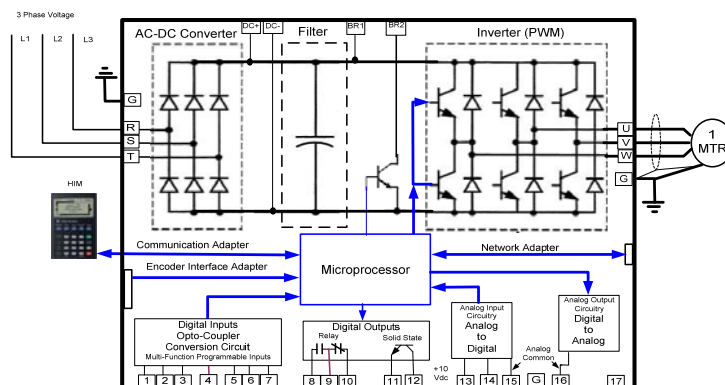
Conversely, the deceleration time defines the time duration in which the AC Drive reduces the output frequency from the maximum frequency to 0 Hz after a stop signal. If the equipment connected to a motor has low friction and a lot of inertia, it could coast for a long time. The Deceleration time function allows the load to be stopped more quickly.

Adjustable Speed

The concept of varying the speed of a motor, manually or automatically is known as adjustable speed. The desired operating speed is relatively constant regardless of load.

AC Drive Controller

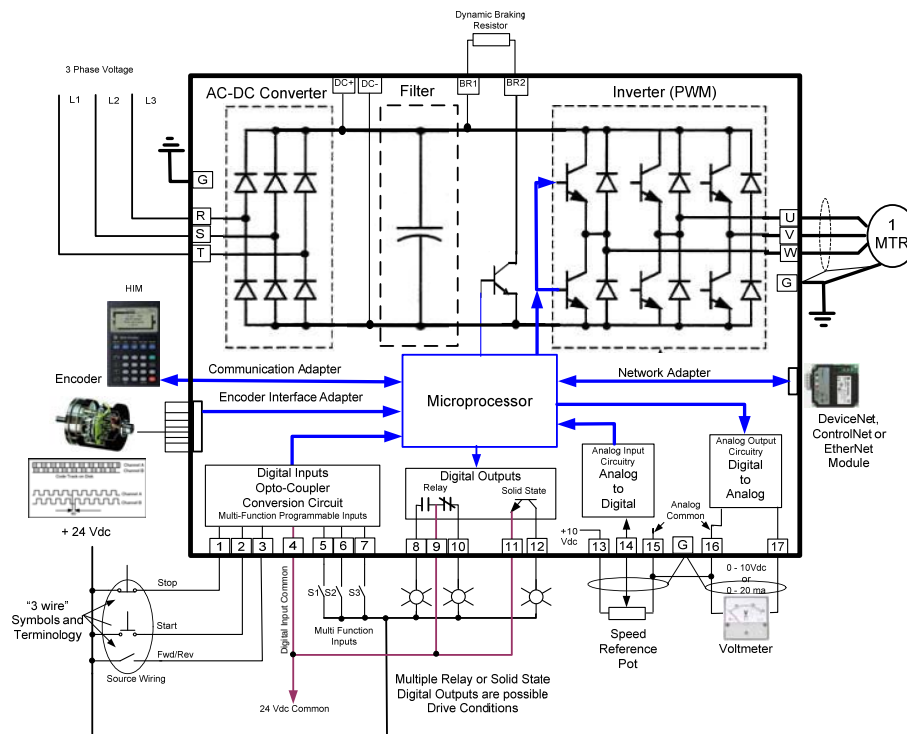
An AC Drive controller is an electronic device that can control the speed, torque, horsepower and direction of an AC motor. It consists of a Microprocessor and a Power Conversion Unit (Converter, Filter and Inverter). It varies the voltage and frequency to the motor to accomplish this.



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AC Drive System

The adjustable speed drive is comprised of the motor, AC Drive Controller and operator's controls (either manual or automatic).



Ambient Temperature

Temperature of the surrounding atmosphere into which the heat of any electrical product is dissipated.

Analog Input (Differential Input)

Generally used for linear speed reference. The types of signals may be:

- differentially isolated 0 to 10V / 4 to 20mA input
- differentially isolated bipolar -10 to +10V or 4-20 mA input

Analog Output

A single ended output capable of either 0 – 10 Vdc or 4 – 20mA; programmable for a wide variety of process parameters including output frequency, output current, output power and others. Programming is available to select either absolute or signed values of these parameters, in addition to offset and gain adjustments for matching signal requirements such as 2 – 10V DC.

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Auto Restart

Auto Restart provides automatic fault reset and restarts following a fault condition before locking out and requiring manual restart. The automatic mode is not applicable to a ground fault, shorted output faults and other internal microprocessor faults. The time between restarts is adjustable from 0.5 seconds to 30.0 seconds

Autotune Provides an automatic method for setting [IR Voltage Drop] and [Flux Current Ref], which greatly affect sensorless vector performance.

“Static Tune”: A temporary command that initiates a non-rotational motor stator resistance test for the best possible automatic setting of [IR Voltage Drop]. A start command is required following initiation of this setting. The parameter returns to “Ready” (0) following the test, at which time another start transition is required operate the drive in normal mode. Used when motor cannot be uncoupled from the load.

“Rotate Tune”: A temporary command that initiates a “Static Tune” followed by a rotational test for the best possible automatic setting of [Flux Current Ref]. A start command is required following initiation of this setting. The parameter returns to “Ready/Idle” (0) following the test, at which time another start transition is required to operate the drive in normal mode. **Important:** Used when motor is uncoupled from the load. Results may not be valid if a load is coupled to the motor during this procedure.

Base Speed

Base speed is the manufacturer’s nameplate rating where the motor will develop rated HP at rated load and voltage. It is commonly the point where 60 Hz is applied to the induction motor.

Bipolar (Speed Reference)

Bipolar refers to a two polarity voltage source (-10 Vdc to + 10 Vdc) which may provide a speed reference and directional reference to an AC Drive. The polarity of the signal will determine the direction of the motor and the magnitude will determine the speed.

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Braking (Stop Mode)

Braking provides a means of stopping an AC motor and can be accomplished in several ways:

Dynamic Braking (Resistor) --- Since AC motors do not have separate field excitation, dynamic braking is accomplished by continuing to excite the motor from the drive. This causes a regenerative current to the drive's DC Intermediate Bus Circuit. The Dynamic Brake resistors are then placed across the DC bus to dissipate the power returned. The brake resistor is usually switched by a transistor or other power switch controlled by the drive.

Regenerative Braking --- is similar to Dynamic Braking, but is accomplished electronically. The generated power is returned to the line through the power converter. It may also be dissipated as losses in the converter (within its limitations).

Motor Mounted or Separately Mounted Brake --- is a positive action, mechanical, friction device. Normal configuration is such that when the power is removed, the brake is set. This can be used as a holding brake. (Note: A Separately Mounted Brake is one which is located on some part of the mechanical drive train other than the motor.)

DC Injection --- This method of braking actually connects the motor to a DC Voltage/Current Source. This will immediately stop the motor.

Ramp to Hold --- With this method of braking the drive acknowledges the Stop command by decreasing or "ramping" the output voltage and frequency to zero in a programmed period (Decel Time), maintaining control of the motor until the drive output reaches zero.

Once the drive has reached zero output hertz on a Ramp-to-Stop and both parameters [DC Hold Time] and [DC Hold Level] are not zero, the drive applies DC to the motor producing current at the DC Hold Level for the DC Hold Time.

Breakaway Torque

The torque required to start a machine from standstill. It is always greater than the torque needed to maintain motion.

Breakdown Torque

The breakdown torque of an AC motor is the maximum torque which it will develop with rated voltage applied at the rated frequency.

Bridge Rectifier (Converter)

A full wave rectifier that conducts current in only one direction of the input current. AC applied to the input results in approximate DC at the output.

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Closed Loop

Closed loop refers to a regulation circuit in which the actual value of the controlled variable (e.g. speed) is sensed and a feedback signal proportional to the speed is compared with the speed reference. The difference between these signals (error signal) causes the actual value for the speed reference to change in the direction that will reduce the difference in signals to zero.

Comm Port (Control Source/Speed Reference)

This method is used when control of the drive's start, stop, and speed reference is from an external communications port. This would be used for DeviceNet, ControlNet or EtherNet operation of the drive. All speed and control inputs are issued from an external PLC through the communications port.

Constant Horsepower (see Chart Below)

A constant horsepower load is when the motor torque required is above the motor's base speed (60 Hz). With a constant horsepower type of load, the torque loading is a function of the changing physical dimensions of the load. These types of applications would include grinders, turret lathes and winding reels. Constant horsepower loads require high torque at low speeds and low torque at high speed. While the torque and speed changes the horsepower remains constant.

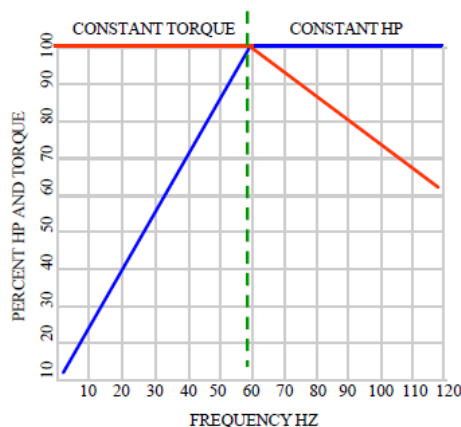
Constant Torque Loads (see Chart Below)

Constant torque loads are where applications call for the same amount of driving torque throughout the entire operating speed range. In other words, as the speed changes the load torque remains the same.

Constant Torque

A speed range in which the motor is capable of delivering a constant torque. The speed range is normally between 0 Hz and 60 Hz. The Volts/Hertz Ratio is controlling the magnetizing flux to the motor. Thus, the current in the motor remains at a constant value.

$$\text{Motor HP} = \text{Torque} * \text{rpm}$$



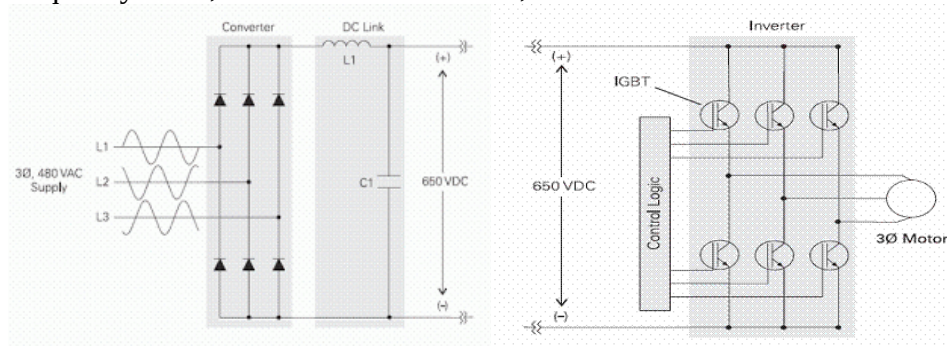
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ControlNet

ControlNet is an open industrial network protocol for industrial automation applications, also known as a fieldbus.

Converter

The process of changing AC to DC is accomplished through the use of a diode rectifier or thyristor rectifier circuit. A “frequency converter”, such as that found in an adjustable frequency drive, consists of a Converter, a DC Bus and an Inverter.



DC Bus (DC Link)

This is a high level of DC voltage which has been filtered after the Converter section of an AC Drive Controller. This Bus voltage is used to create the Three Phase Alternating Voltage on the output (Inverter) of an AC Drive Controller.

DeviceNet

DeviceNet is a network system used in the automation industry to interconnect control devices for data exchange. It uses Controller Area Network as the backbone technology and defines an application layer to cover a range of device profiles. Typical applications include information exchange, safety devices, and large I/O control networks.

Digital Input

A programmable (multi-function) input on an AC Drive controller which can be configured to perform a variety of functions based on the user application (eg. Start, Stop, Jog, Preset Select). The term “Digital” refers to the state is either On or Off.

Digital (Relay) Output

A programmable Relay output from an AC Drive Controller which may be configured for a variety of operations based on the user application (eg. Drive Ready, At Speed). The term “Digital” refers to the state is either On or Off.

di/dt

A line reactor is an inductor wired between a power source and a load. In addition to the current limiting function, the device serves to filter out spikes of current and may also reduce injection of harmonic currents into the power supply.

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DriveGuard (Safe Off)

The DriveGuard Safe Torque Off option provides the “Safe torque off (STO)” function defined in EN/IEC 61800-5-2. It blocks gate firing signals from reaching the IGBT output power devices of the drive. This prevents the IGBT’s from switching in the sequence necessary to generate torque in the connected motor.

Droop Control

Droop Control is intended to improve load sharing in applications where several inverters / motors drive a mechanically coupled load. For example, where several inverters each drive a motor, and the motors are distributed along a conveyer. This function reduces output frequency in response to load, allowing other drives to share the load. Function is adjustable in amount of RPM difference at full load.

Duty Cycle

Duty cycle is the relationship between the operating and rest times or repeatable operation at different loads.

dv/dt

This is the rate of change in voltage versus a rate of change in time. Specially designed Resistor-Capacitor networks can help protect the SCRs from excessive dv/dt which can result from line voltage spikes, line disturbances and circuit configurations with extreme forward conducting or reverse blocking requirements.

EMC

Another important environmental feature is electromagnetic compatibility (EMC). It is very important that a drive system fulfills the EMC directives. This means that the drive system can bear conductive and radiative disturbances, and that it does not send any conductive or radiative disturbances itself either to the electrical supply or the surrounding environment.

Enable

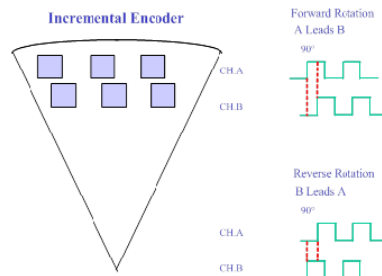
To allow the drive to operate by applying an appropriate signal to the appropriate input (eg. Enable an AC Drive Controller to Operate). An Enable may be hardwired to a digital input or internally jumpered on most AC Drive Controllers.

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Encoder

A rotary encoder, also called a shaft encoder, is an electro-mechanical device that converts the angular position or motion of a shaft or axle to a digital code or pulses. There are two main types: absolute and incremental.

- The output of absolute encoders indicates the current position of the shaft, making them angle transducers.
- The output of incremental encoders provides information about the motion of the shaft, which is typically further processed elsewhere into information such as speed, distance, and position.



Error

Error is a difference between the set-point (speed/torque reference) and the actual feedback speed/torque.

Factory Default Parameter

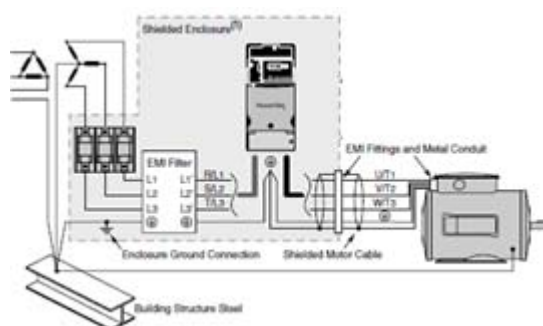
This feature will automatically reset all of the software parameters back to a factory default condition.

Feedback

This is a signal representing the actual controlled variable (Speed/Torque). The element of a control system that provides an actual operation signal for comparison with the set-point to establish an error signal used by the regulator circuit.

Filter (Line Conditioning)

This is common in AC Drive controllers. Using an EMI Filter reduces the potential of stray signals affecting the AC Line which, in turn, may cause drive faults.



Force

This is the tendency to change the motion or position of an object with a push or pull. Force

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is measured in ounces or pounds.

Four-Quadrant Operation

The four combinations of forward/reverse rotation and forward/reverse torque of which a regenerative drive is capable. The four combinations are:

- Forward rotation/forward torque (motoring)
- Forward rotation/reverse torque (regeneration)
- Reverse rotation/reverse torque (motoring)
- Reverse rotation/forward torque (regeneration)

Frame Size (Motors)

This is the physical size of a motor, usually consisting of NEMA defined “D” and “F” dimensions at a minimum. The “D” dimension is the distance in quarter inches from the center of the motor shaft to the bottom of the mounting feet. The “F” dimensions relates to the distance between the centers of the mounting feet holes.

Frequency

This is the number of cycles per second for an AC electric system; the number of times per second that the current flow changes direction.

FLC (FLA)

This is the electrical current required during normal motor operation to generate its designed horsepower. Full load current is also known as full load amps (FLA). A motor’s full load current is used when selecting motor overload protection devices.

Flying Start

The flying start feature is used when a motor is connected to a flywheel or a high inertia load. The flying start works even without speed feedback. In case of rotating motor, the inverter is first started with a reduced voltage and then synchronized to the rotating rotor. After synchronized, the voltage and the speed are increased to the corresponding levels.

Full-Load Torque

The full-load torque of a motor is the torque necessary to produce rated horsepower at full load speed.

Hertz (Hz)

A unit of frequency equal to one cycle per second. Abbreviated Hz.

Horsepower

This represents the amount of work done by a machine. Relative to motors, horsepower indicates the power of the motor. Motor horsepower is a selection criterion for motor control products such as manual and magnetic motor starters (***Motor HP = Torque * rpm***).

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Human Interface Module (HIM)

The PowerFlex AC drives provide common Human Interface tools that are easy to use.

LCD Human Interface modules provide:

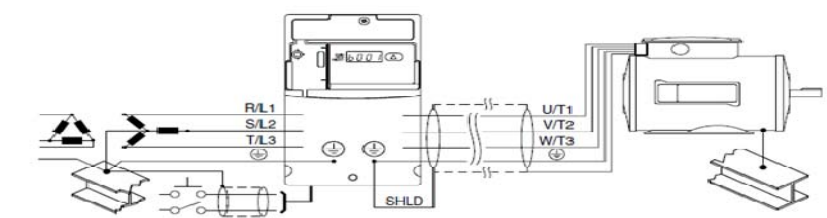
- Large and easy to read 7 line x 21 character backlit display
- Variety of languages (English, French, German, Italian, Spanish, Portuguese, Dutch)
- Alternate function keys for shortcuts to common tasks
- “Calculator-like” number pad for fast and easy data entry (Full Numeric version only)
- Control keys for local start, stop, speed, and direction
- Remote versions for panel mount application

Inertia Ride Through

Responds to a loss of AC input power by adjusting the output frequency to create a regenerative situation in the motor allowing the drive to retain control of the motor during a power outage. Performance is based on the amount of system inertia and the length of the outage.

Grounding

Because of the EMI noise associated with AC Drives, proper grounding of the AC Drive is very important. Also, the use of shielded cables is generally required. Here is a typical grounding and shielding scheme that should be followed. This scheme applies to practically all AC Drives used in Michelin today.



IGBT (Insulated Gate Bipolar Transistor)

This is a type of electronic power device frequently used in inverter sections of drives. The IGBT is noted for its ease in switching on and off and high switching frequencies. This device is also used for switching on/off the external braking resistor and is an integral part of the Inverter section of an AC Drive Controller.

Internal Frequency (MOP)

This method of speed reference utilizes a single programmed internal frequency setting in the drive. The motor would run at this frequency at all times.

Inverter

A term commonly used for an AC adjustable frequency drive. An inverter is also a term used to describe a particular section of an AC drive. This section uses the DC voltage from a previous stage (DC Bus) to produce an AC current or voltage having the desired frequency.

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Jogging

Jogging is a means of accomplishing momentary motor movement by repetitive closure of a circuit using a single push-button or contact element.

Line Reactor (see di/dt)

Locked-Rotor Current

This is a steady state current taken from the line with the rotor at standstill (at rated voltage and frequency). This is the inrush current when starting the motor.

Locked-Rotor Torque

The minimum torque that a motor will develop at rest for all angular positions of the rotor (with rated voltage applied at rated frequency).

Maximum Frequency

The parameter setting on the drive which determines the highest frequency the drive will output.

Minimum Frequency

The parameter setting on the drive which determines the lowest frequency the drive will output.

MOP (Internal Frequency) This feature lets you control the speed reference of a drive using an internal frequency. Using digital inputs configured as “MOP Inc/MOP Dec”, the operator may increase or decrease the speed of the drive in a controlled rate (MOP Rate) by these inputs. The Internal Frequency is what is incremented/decremented.

MOV (Metal Oxide Varistor)

This is an electronic component with a "diode-like" nonlinear current–voltage characteristic. At low voltage it is a high resistance, and at high voltage it changes and becomes a low resistance. They are used as protection devices for electronic drives to protect the drive from high voltage spikes and transients.

Opto Isolator (Coupler)

An opto-isolator, also called an opto-coupler, photo-coupler, or optical isolator, is an electronic component that transfers electrical signals between two isolated circuits by using light. It is commonly found in interface circuits for PLCs and drives.

Parameters

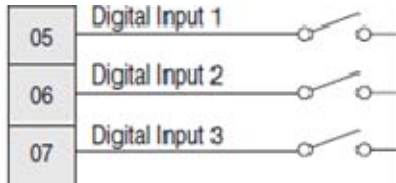
All AC Drive controllers have parameters for matching the drive to the motor along with adjusting and communicating with the drives. Many parameters do not require adjustments to be made and will operate the drive based on the factory default settings. Other parameters must be set based on the application.

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Preset Speed Select (Speed Reference)

This method of speed reference allows the user to program a block of frequencies into the drive that are selectable by digital inputs. Some drives may support up to eight (8) preset speeds and these speeds are selected by a binary number equal to the preset speed reference.

Digital input #1 off, #2 on, and #3 off would represent 010_2 and the drive would select Preset Speed 2 (20 Hz) as the commanded speed reference.



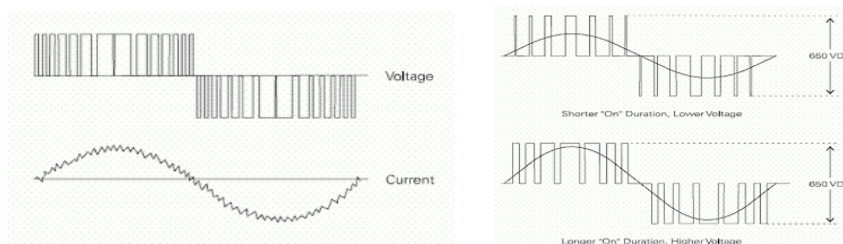
Preset #	Frequency	Input 3	Input 2	Input 1
Preset 0 Hz	5 Hz	0	0	0
Preset 1 Hz	10 Hz	0	0	1
Preset 2 Hz	20 Hz	0	1	0
Preset 3 Hz	30 Hz	0	1	1
Preset 4 Hz	40 Hz	1	0	0
Preset 5 Hz	70 Hz	1	0	1
Preset 6 Hz	80 Hz	1	1	0
Preset 7 Hz	90 Hz	1	1	1

Preset Speed (Speed Reference)

Preset speed refers to one or more fixed speeds at which the drive will operate. This method of speed reference usually requires digital inputs to be configured as a binary-decimal selection.

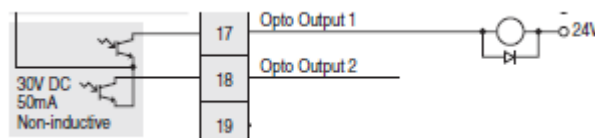
Pulse Width Modulation (PWM)

A PWM is a type of AC adjustable frequency drive that accomplishes frequency and voltage control at the output section (inverter) of the drive. The drive's output voltage is always at a constant amplitude and by "chopping" (pulse width modulating), the average voltage is controlled.



Recovery Diode (Flyback)

When using a Solid State Output (Opto) with an inductive load (relay or solenoid), a recovery diode must be installed to protect the solid state component from high voltage spikes that occur whenever the inductive load is de-energized.



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Regulation (Speed)

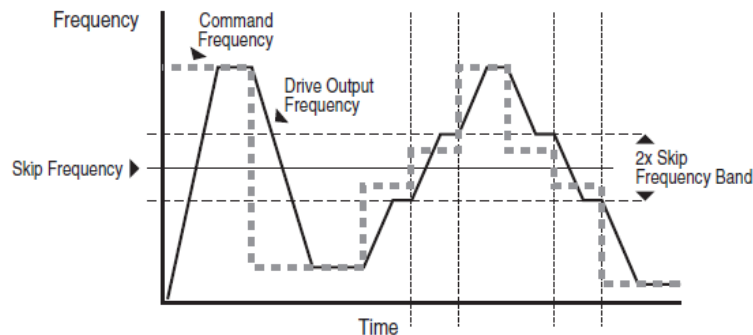
The ability of a control system to hold a speed once it has been set. Regulation is given in percentages of either base speed or set speed. Regulation is the percentage of speed change with a defined change in load, assuming all other parameters to be constant. Regulation of 0.001% is possible using digital feedback schemes utilizing incremental encoders in AC Drive controllers.

Setpoint (Speed Reference)

Setpoint is the desired value or speed reference of the controlled variable for a closed loop application. For an AC Drive Controller, it sets the desired speed reference for the motor.

Skip Frequency (Skip Band)

This feature is used when certain frequencies may have undesirable effects on the machine. It may allow the drive to accelerate from min to max speeds, skipping certain frequencies that may create excessive harmonics or vibrations which may be detrimental to the machine. Usually, the Skip Frequency is programmed in conjunction with a Skip Frequency Band to avoid a specific range of frequencies.



Slip Compensation

This is a method of increasing the output frequency (beyond maximum speed) to maintain motor speed as the load on the motor increases. This is used to attempt to improve speed regulation without adding feedback devices (Open Loop).

Soft Start

A soft start device reduces the voltage, thus reducing the current, at startup to relieve the stress on the motor and machinery. There are many applications where it is critical to do just this. In addition, the soft start can allow a motor to have smooth acceleration up to 100% operating speed and can control a motor's smooth deceleration back down to zero as well.

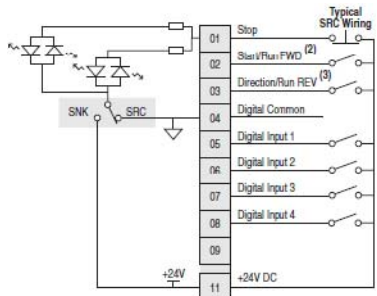
Solid State Output

A programmable electronic (*Opto*) output from an AC Drive Controller which may be configured for a variety of operations based on the user application (eg. Drive Ready, At Speed).

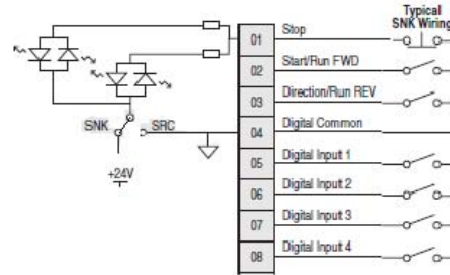
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Source/Sink Input Wiring

AC Drive Controller input interfaces normally start, stop or limit the drive function in some way. These interfaces may be configured as Sink or Source inputs. Sourcing inputs are more typical in Michelin applications with input interfacing. A sourcing input requires a positive voltage at the input terminal with respect to common to activate that input. Another term for Source input wiring is Positive Logic. Sink input wiring requires a negative voltage to activate the input, also referred to as Negative Logic.



Source Input Wiring Scheme



Sink Input Wiring Scheme

Speed Range

The speed minimum and maximum at which a motor must operate under constant or variable torque load conditions. A 50:1 speed range for a motor with top speed of 1800 RPM means the motor must operate as low as 36 RPM and still remain within regulation specifications. Controllers are capable of wider controllable speed ranges than motors because there is no thermal limitation, only electrical. Controllable speed range of a motor is limited by the ability to deliver 100% torque below base speed without additional cooling.

Speed Reference

A Speed Reference sets the reference output frequency for an AC Drive Controller. The most common manual configuration is a three-wire pot used to manually set the speed reference. Some are connected as 0v to +10v, others -10v to +10v and even 4ma- 20ma current loops. Most AC Drive Controllers can obtain a speed reference from a variety of configurations such as:

- Digital MOP (Used in conjunction with MOP Increment and decrement inputs)
- Jog Frequency
- Analog Input signals (Unipolar Voltage/Current or Bipolar Voltage)
- Preset Speeds (Generally, most drives will allow up to seven and sometimes eight)
- Network Communications (DNET, CNET, ENET)
- Encoder (pulse) signal
- PID Control
- HIM

Speed Regulation

The numerical measure in percent, of how accurately the motor speed can be maintained. It is the percentage of change in speed between full load and no load.

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Starting Torque:

Starting torque, also referred to as locked rotor torque, is the torque that the motor develops each time it is started at rated voltage and frequency. When voltage is initially applied to the motor's stator, there is an instant before the rotor turns.

Stop Mode: (see Braking)

Synchronous Speed

The speed of an AC induction motor's rotating magnetic field. It is determined by the frequency applied to the stator and the number of magnetic poles present in each phase of the stator windings. Mathematically, it is expressed as:

$$\text{Sync Speed (RPM)} = 120 \times \text{Applied Freq. (Hz)} / \text{Number of poles per phase}$$

Thermistor (Motor Thermistor)

A thermistor is a non-linear resistance temperature device, made from semi-conductor material. Each specific thermistor has its own unique resistance vs. temperature characteristic.

There are two general types of Thermistors:

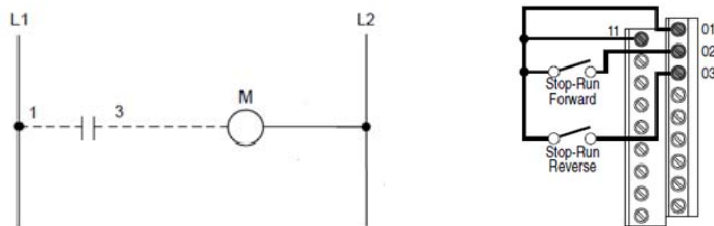
- The positive temperature coefficient (PTC) type has a resistance that increases with increasing temperature. The PTC thermistor has a sharp rise in resistance when the temperature reaches the 'knee', or switching point.
- The negative temperature coefficient (NTC) type thermistor has a resistance that decreases with increasing temperature.

Torque (motor)

A turning force applied to a shaft, tending to cause rotation. Torque is normally measured in ounce-inches or pound-feet and is equal to the force applied, times the radius through which it acts. Torque, Slip and Current are directly related in a Squirrel Cage Motor.

Two Wire Control

Below is a 2-wire control scheme, using a maintained contact pilot device in series with the starter coil.

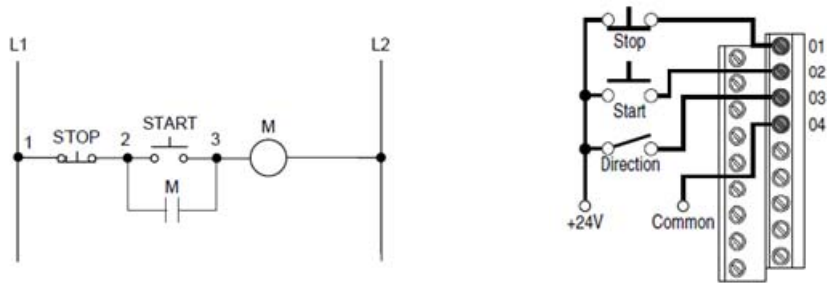


The term “2-wire” control is derived from the fact that in the basic circuit, only two wires are required to connect the pilot device to the starter. This scheme requires **maintained** contacts in order for the load to stay energized.

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Three Wire Control

Below is a 3-wire control scheme, using momentary contact push buttons or similar pilot devices to energize the starter coil.



The term “3-wire” control is derived from the fact that in the basic circuit, at least three wires are required to connect the pilot devices to the starter.

This scheme utilizes momentary contacts to energize the motor starter and a set of normally open contacts to seal itself in.

Under-voltage

This is a fault condition that occurs when the input voltage to the drive is below the trip value. Under-voltage is generally not a parameter that can be adjusted.

Unipolar (Speed Reference)

Unipolar refers to a single polarity voltage source (0-10 Vdc) which may provide a speed reference to an AC Drive or an Analog Output signal.

Variable Volts/Hertz

When the output voltage varies at a different rate than the rate at which the output frequency varies. Variable Volts/Hertz is sometimes desired to decrease motor noise and reduce motor core losses.

Variable Torque Loads

There are only two kinds of variable torque loads: centrifugal pumps and fans. With a variable torque load, the loading is a function of the speed. Variable torque loads generally require low torque at low speeds and higher torque at higher speeds.

Vector

A vector is a quantity that has magnitude, direction and sense. This quantity is commonly represented by a directed line segment whose length represents the magnitude and whose orientation in space represents the direction.

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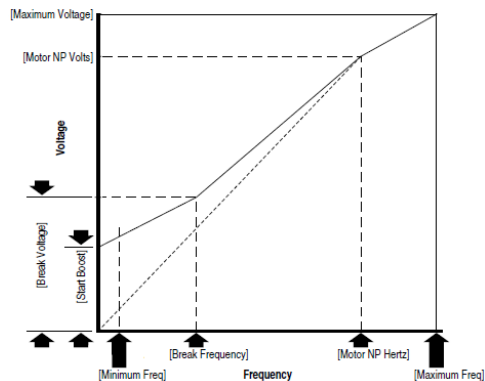
Vector Control Drive (VCD)

The Vector Control AC Drive borrows its name from the fact that a vector (from math class) has both a magnitude and direction. The VCD uses a more sophisticated regulator to control both the magnitude and direction of the magnetic flux in an AC motor through independent control loops.

Two separate control loops have been added making it more complex. The control loops allow independent control of the speed and strength of the motor's magnetic field. They may also allow the regulator to measure the actual speed of the motor (Encoder Feedback) and to calculate the amount of torque being produced.

Voltage Boost

This feature will increase the Volts/Hertz ratio of drives at low speeds to compensate for resistance losses in the motor core. This compensation allows the motor to develop rated torque at very low speeds.



Volts per Hertz Ratio (V/Hz)

This relationship exists in AC drives where the output voltage is varied directly proportional to frequency. This type of operation is required to allow the motor to produce constant rated torque as speed is varied at speeds less than nameplate base frequency. This ratio is based on the Motor Nameplate Voltage / Motor Nameplate Frequency.

